Galois Cohomology

— *EYNTKA* Series —

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Introduction

Group cohomology assigns to a group G acting on a module M a sequence of abelian groups $H^n(G, M)$ that measure the failure of exactness, the obstructions to lifting, and the extensions of M by G. When G is a Galois group $\mathrm{Gal}(L/K)$ and M is an arithmetic object—the multiplicative group of a field, the points of an algebraic variety, the idèle class group of a number field—these cohomology groups encode the deepest structural information in algebraic number theory. The Brauer group of a field is H^2. The Hilbert 90 theorem is the vanishing of H^1. The local and global reciprocity laws of class field theory are consequences of the cup product structure on Tate cohomology. The Selmer group controlling the Mordell–Weil rank of an elliptic curve is a subgroup of H^1 defined by local conditions.

This book develops the theory of group cohomology for profinite groups and applies it systematically to the Galois groups of local and global fields. The treatment is self-contained given the prerequisites listed below, and proceeds from the abstract foundations (Part I) through the structural theory of cohomological dimension and the Brauer group (Part II) to the arithmetic applications: local and global Tate duality, the Poitou–Tate exact sequence, class field theory via class formations, Selmer groups, and the connections to descent, non-abelian cohomology, and étale cohomology (Part III).

Prerequisites

The reader is assumed to be familiar with the material in the following books from the *Everything You Need To Know About* series:

- [6]: groups, rings, modules, tensor products, categories. In particular: the structure of finitely generated modules over PIDs.
- [7]: the Artin-Wedderburn theorem for semisimple algebras and the basic theory of central simple algebras.
- [4]: Dedekind domains, prime splitting, the class group, the Dirichlet unit theorem, and the ideal-theoretic formulation of class field theory (the Artin symbol, ray class groups). The

cohomological reformulation of class field theory is developed in this book (Chapters 8 and 9) and verified in [17].

- [10]: places and completions of number fields, the structure of non-archimedean local fields (valuation rings, ramification, the local norm), the product formula, and strong approximation.
- [9]: chain complexes, derived functors (Ext, Tor), spectral sequences (Grothendieck, Hochschild–Serre), and the universal coefficient theorem.

Class field theory ([17]) is *not* a prerequisite. On the contrary, this book provides the cohomological tools that [17] will use: the local reciprocity law is proved here (Chapter 8), and the global class formation axioms are verified here (Chapter 9), with the first inequality and the existence theorem deferred to [17].

Structure

The book has three parts. Part I (Chapters **??**–4) develops the abstract theory: profinite groups, cohomology of discrete and profinite groups, Tate cohomology, and cup products. No fields or arithmetic appear; the results are purely algebraic and topological. Part II (Chapters **??**–**??**) develops the structural theory: cohomological dimension, pro-p-groups and the Golod–Shafarevich inequality, and the Brauer group. These chapters bridge the abstract theory of Part I and the arithmetic of Part III. Part III (Chapters 8–10, with an appendix on adèles and idèles) applies the machinery to local and global fields: Tate duality, the Poitou–Tate sequence, class formations, reciprocity, Selmer groups, descent, non-abelian cohomology, and connections to étale cohomology.

Conventions

All groups are written multiplicatively unless otherwise stated. Profinite groups carry the profinite topology, and all homomorphisms between profinite groups are continuous. Modules over profinite groups are discrete (every element has an open stabilizer) unless explicitly marked as profinite. The notation $\varprojlim$ denotes the (categorical) limit and $\varinjlim$ the colimit; "inverse limit" and "direct limit" are avoided. For a number field K: the ring of integers is $\mathcal{O}_K$, the absolute Galois group is $G_K = \mathrm{Gal}(K^{\mathrm{sep}}/K)$, and the completion at a place v is K_v. Cross-references use `\cref` throughout. Internal references to other books in the series are by citation code (e.g., [6]).

Part I

Foundation

The cohomology of a group G with coefficients in a G-module M is a sequence of functors $H^n(G,-)$ that extend the fixed-point functor $H^0(G,M) = M^G$ to higher degrees. The theory has two sources: the algebraic, rooted in the homological algebra of group rings (resolutions, derived functors, Ext), and the topological, rooted in the cohomology of classifying spaces and the structure of profinite groups as limits of finite groups. Part I develops both perspectives and fuses them into a single framework.

Chapter **??** constructs the category of profinite groups and establishes the foundational results: the characterization as compact, Hausdorff, totally disconnected topological groups; Sylow theory; and the profinite completion. Chapter 2 develops cohomology for discrete groups via resolutions, establishing the long exact sequence, the Hochschild–Serre spectral sequence, and the interpretation of H^1 and H^2 in terms of extensions and outer actions. Chapter 3 passes to the profinite setting, where cohomology is computed as a colimit of finite-group cohomology, and Shapiro's lemma and inflation-restriction become the primary computational tools. Chapter 4 introduces Tate cohomology (extending the theory to negative degrees for finite groups) and the cup product, the multiplicative structure that will drive the duality theorems and reciprocity laws of Part III.

No fields or number-theoretic objects appear in Part I. The results are purely algebraic and topological, applicable to any profinite group acting on any discrete module. The arithmetic content begins in Part II, where the abstract machine acquires the characteristic shapes (cohomological dimension, pro-p-structure, Brauer classes) that govern its behavior over fields.

1

Profinite Groups

The absolute Galois group $G_K = \mathrm{Gal}(K^{\mathrm{sep}}/K)$ is the central object of modern algebraic number theory. For a finite extension L/K, the Galois group $\mathrm{Gal}(L/K)$ is a finite group whose structure is studied in [6]. For the separable closure, G_K is an infinite group carrying a natural topology—the Krull topology—inherited from the finite quotients $\mathrm{Gal}(L/K)$ as L ranges over finite Galois extensions of K. Extracting cohomological information from G_K requires first understanding this topology.

This chapter develops the theory of profinite groups: topological groups arising as limits of finite discrete groups. The central result is that a topological group is profinite if and only if it is compact, Hausdorff, and totally disconnected (theorem 1.1.10). From there, the structure of profinite groups (open and closed subgroups, Sylow theory, supernatural numbers) is developed, and the absolute Galois groups of many common fields are examined as profinite groups, establishing the concrete objects on which the cohomological machinery of the rest of the book will act.

1.1 Limits and Profinite Groups

Galois theory produces, for each number field K, a tower of finite groups $\mathrm{Gal}(L/K)$ indexed by the finite Galois extensions L/K, connected by restriction maps. The "total" Galois group G_K should assemble from this tower of compatible group—elements in each $\mathrm{Gal}(L/K)$, respecting the restriction maps. Making this precise requires the formalism of limits over cofiltered diagrams.

Throughout this section, all topological groups are assumed Hausdorff unless stated otherwise. The reader is assumed familiar with the basic theory of topological groups from [11].

Definition 1.1.1: Cofiltered System of Topological Groups

A *cofiltered system* of topological groups consists of:

1. a directed partially ordered set $(I, \leq)$,
2. for each $i \in I$, a topological group G_i, and
3. for each pair $i \leq j$ in I, a continuous group homomorphism $\varphi_{ji} : G_j \to G_i$,

subject to $\varphi_{ii} = \mathrm{id}_{G_i}$ for all i, and $\varphi_{ki} = \varphi_{ji} \circ \varphi_{kj}$ for all $i \leq j \leq k$.

The directedness condition on I (every pair of elements has an upper bound) ensures that the system is "connected"—any two groups in the system can be compared via a common third.

1.1.2 Remark: Categorical Vocabulary This text uses modern categorical terminology throughout. What many classical references—including [18] and [21]—call an "inverse limit" or "projective limit" is a *limit*, specifically a limit over a cofiltered index category. Dually, a "direct limit" or "injective limit" is a *colimit* (a filtered colimit). The notation $\varprojlim$ for limits and $\varinjlim$ for colimits is standard and retained here.

Definition 1.1.3: Limit of a Cofiltered System

Let $(G_i, \varphi_{ji})_{i\in I}$ be a cofiltered system of topological groups. The *limit* of the system is the topological group:

$$\varprojlim_{i\in I} G_i = \left\{ (g_i)_{i\in I} \in \prod_{i\in I} G_i \;\middle|\; \varphi_{ji}(g_j) = g_i \text{ for all } i \leq j \right\}$$

equipped with the subspace topology from $\prod_{i\in I} G_i$, where each G_i carries its given topology and the product carries the product topology.

An element of $\varprojlim G_i$ is a *compatible sequence*, a choice of element $g_i \in G_i$ for each i, such that the transition maps carry later choices to earlier ones. The limit is a closed subgroup of the product, being the intersection of the closed sets $\{(g_i) \mid \varphi_{ji}(g_j) = g_i\}$ as (i, j) ranges over pairs with $i \leq j$. Each such set is closed as the map $(g_i) \mapsto \varphi_{ji}(g_j) - g_i$ is continuous (as a composition of a projection and φ_{ji}) and $\{0\}$ is closed in the Hausdorff group G_i.

Proposition 1.1.4: Limits of Compact Groups

If $(G_i, \varphi_{ji})_{i\in I}$ is a cofiltered system of compact Hausdorff topological groups, then $\varprojlim G_i$ is a compact Hausdorff topological group.

***Proof*:**
The product $\prod_{i\in I} G_i$ is compact by Tychonoff's theorem (see [11, Tychonoff's Theorem]) and Hausdorff as a product of Hausdorff spaces. Since $\varprojlim G_i$ is a closed subgroup of this product, it is compact (a closed subset of a compact space is compact) and Hausdorff (a subspace of a

Hausdorff space is Hausdorff).

With the notion of limit in hand, the central definition of this chapter can be stated.

Definition 1.1.5: Profinite Group

A *profinite group* is a topological group isomorphic to the limit of a cofiltered system of finite groups, where each finite group carries the discrete topology.

Since every finite discrete group is compact and Hausdorff, proposition 1.1.4 shows that every profinite group is compact and Hausdorff. The converse question—which compact Hausdorff groups are profinite—is answered by theorem 1.1.10 below.

Any group gives rise to a profinite group through the following construction.

Definition 1.1.6: Profinite Completion

Let G be a group. The *profinite completion* of G is the limit:

$$\widehat{G} = \varprojlim_{N \trianglelefteq_{\mathrm{fin}} G} G/N$$

where N ranges over all normal subgroups of finite index in G, ordered by reverse inclusion ($N_1 \leq N_2$ in the poset when $N_1 \supseteq N_2$), and the transition maps are the natural projections $G/N_1 \to G/N_2$ for $N_1 \subseteq N_2$.

The natural map $\iota\colon G \to \widehat{G}$, sending g to the tuple of cosets $(gN)_N$, is a group homomorphism. Its kernel is $\bigcap_N N$, the intersection of all finite-index normal subgroups. A group for which this intersection is trivial—so that ι is injective—is called *residually finite*.

Example 1.1.7: Profinite Groups and Completions

1. Every finite group G, equipped with the discrete topology, is profinite: it is the limit of the constant system consisting of G alone. The profinite completion satisfies $\widehat{G} \cong G$.

2. The profinite completion of $\mathbb{Z}$ is:
$$\widehat{\mathbb{Z}} = \varprojlim_{n \geq 1} \mathbb{Z}/n\mathbb{Z} \cong \prod_p \mathbb{Z}_p$$
where $\mathbb{Z}_p = \varprojlim_{k \geq 1} \mathbb{Z}/p^k\mathbb{Z}$ is the ring of p-adic integers. The isomorphism follows from the Chinese Remainder Theorem: for $n = \prod p_i^{a_i}$, the canonical map $\mathbb{Z}/n\mathbb{Z} \xrightarrow{\sim} \prod \mathbb{Z}/p_i^{a_i}\mathbb{Z}$ is an isomorphism, and passing to the limit over n gives the decomposition. The group $\mathbb{Z}$ is residually finite (the intersection $\bigcap_{n \geq 1} n\mathbb{Z} = \{0\}$) and embeds into $\widehat{\mathbb{Z}}$ with dense image.

3. The additive group $\mathbb{Q}$ has no proper subgroups of finite index: if $H \leq \mathbb{Q}$ satisfies $[\mathbb{Q} : H] = n$, then $n\mathbb{Q} \subseteq H$, and $n\mathbb{Q} = \mathbb{Q}$ forces $H = \mathbb{Q}$. The profinite completion is therefore trivial: $\widehat{\mathbb{Q}} = \{0\}$.

4. The p-adic integers $\mathbb{Z}_p$ form a profinite group by construction. The open subgroups are precisely $p^k\mathbb{Z}_p$ for $k \geq 0$, and the finite quotients $\mathbb{Z}_p/p^k\mathbb{Z}_p \cong \mathbb{Z}/p^k\mathbb{Z}$ are all p-groups. This is the prototypical example of a *pro-p-group* (section 1.3).

5. (If the reader has read [2]) For a connected scheme X with geometric point $\overline{x}$, the étale fundamental group $\pi_1^{\text{ét}}(X, \overline{x})$ is a profinite group: it is the limit of the automorphism groups of finite étale covers of X, ordered by refinement. When $X = \operatorname{Spec} K$ for a field K, this recovers the absolute Galois group G_K.

The definition of a profinite group presents it as a limit of some system of finite groups, but there is a canonical such system: the finite quotients of the group itself. The following proposition shows that every profinite group is determined by its own open normal subgroups.

Proposition 1.1.8: Profinite Groups as Limits of Finite Quotients

Let G be a profinite group. Then the open normal subgroups of G form a cofiltered system (ordered by reverse inclusion), every such subgroup has finite index, and the natural map:

$$G \xrightarrow{\sim} \varprojlim_{U \trianglelefteq_{\text{open}} G} G/U$$

is an isomorphism of topological groups.

***Proof*:**
Write $G = \varprojlim_{i \in I} G_i$ with each G_i finite and discrete, and let $\pi_i \colon G \to G_i$ denote the projection. Each $\ker \pi_i$ is open in G (the preimage of $\{e\}$ under the continuous map π_i to a discrete space) and normal with $G/\ker \pi_i$ finite (it embeds into G_i). Conversely, any open normal subgroup U of G has finite index: the cosets of U form an open cover of the compact space G, so finitely many suffice, giving $[G : U] < \infty$.

The natural continuous homomorphism $\alpha \colon G \to \varprojlim_U G/U$ sends g to the tuple $(gU)_U$. It is injective: if $g \in \bigcap_U U$, then $g \in \ker \pi_i$ for every i, hence $g_i = e$ for all i, so $g = e$ in $G \subseteq \prod G_i$. It is surjective: an element $(g_U U)_U \in \varprojlim_U G/U$ determines, for each i, an element $g_{\ker \pi_i} \cdot \ker \pi_i \in G/\ker \pi_i$, and the compatibility conditions ensure that these assemble into a compatible thread in $\varprojlim G_i = G$. Finally, α is a continuous bijection from the compact space G to the Hausdorff space $\varprojlim_U G/U$, hence a homeomorphism.

This proposition makes precise the sense in which a profinite group "is" the limit of its finite quotients. No external system of finite groups is needed—the group's own topology determines the system canonically.

The profinite completion now acquires a clean universal property.

Proposition 1.1.9: Universal Property of Profinite Completion

Let G be a group and $\iota\colon G \to \widehat{G}$ the natural map to its profinite completion. For any profinite group H and any group homomorphism $\varphi\colon G \to H$ whose image is dense, there exists a unique continuous surjection $\tilde{\varphi}\colon \widehat{G} \to H$ such that $\tilde{\varphi} \circ \iota = \varphi$. More generally, for any group homomorphism $\varphi\colon G \to H$ with H profinite, there exists a unique continuous homomorphism $\tilde{\varphi}\colon \widehat{G} \to H$ with $\tilde{\varphi} \circ \iota = \varphi$:

$$\begin{tikzcd} G \arrow[r, "\iota"] \arrow[dr, "\varphi"'] & \widehat{G} \arrow[d, dashed, "\exists!\,\tilde{\varphi}"] \\ & H \end{tikzcd}$$

***Proof*:**
By proposition 1.1.8, write $H \cong \varprojlim_U H/U$ where U ranges over the open normal subgroups of H. For each such U, the composition $G \xrightarrow{\varphi} H \to H/U$ is a homomorphism to a finite group, so it factors through $G/\varphi^{-1}(U)$. Since U is open and has finite index in H, the preimage $\varphi^{-1}(U)$ is a finite-index normal subgroup of G, hence appears in the system defining $\widehat{G}$. The resulting maps $G/\varphi^{-1}(U) \to H/U$ are compatible as U varies, and the universal property of the limit $\widehat{G} = \varprojlim_N G/N$ produces a unique continuous homomorphism $\tilde{\varphi}\colon \widehat{G} \to H$. That $\tilde{\varphi} \circ \iota = \varphi$ is immediate from the construction.

The universal property says that the profinite completion is the best approximation of G by a profinite group: every homomorphism from G to a profinite group factors uniquely through $\widehat{G}$. In categorical language, the profinite completion functor is left adjoint to the forgetful functor from profinite groups to groups.

The stage is set for the main structural result of this section. The following theorem characterizes profinite groups intrinsically, without reference to any particular presentation as a limit.

Theorem 1.1.10: Characterizing Profinite Groups

A topological group G is profinite if and only if it is compact, Hausdorff, and totally disconnected.

***Proof*:**
Step 1: Profinite groups are compact, Hausdorff, and totally disconnected. Write $G = \varprojlim_{i \in I} G_i$ with each G_i finite and discrete. Each G_i is compact (finite), Hausdorff (discrete), and totally disconnected (discrete). The product $\prod G_i$ inherits all three properties: compactness by Tychonoff's theorem, and the Hausdorff and total disconnectedness properties because they are preserved by arbitrary products and pass to subspaces. Since G is a closed subgroup of $\prod G_i$, it is compact, Hausdorff, and totally disconnected.

Step 2: Compact, Hausdorff, totally disconnected groups are profinite. The goal is to show that the open normal subgroups of G form a neighborhood basis of the identity and have finite index, so that the natural map $G \to \varprojlim_{N \trianglelefteq_{\text{open}} G} G/N$ is an isomorphism.

Step 2a: Separating points by finite quotients. Let $g \in G$ with $g \neq e$. Since G is compact, Hausdorff, and totally disconnected, the clopen (simultaneously open and closed) subsets form a basis for the topology. Choose a clopen set $C \subseteq G$ with $e \in C$ and $g \notin C$. The stabilizer $\mathrm{Stab}_G(C) = \{h \in G \mid hC = C\}$ is open: for the inclusion $\{h \mid hC \subseteq C\}$, if $h_0 C \subseteq C$, then for each $c \in C$ the continuity of the map $h \mapsto hc$ gives an open neighborhood U_c of h_0 with $U_c \cdot c \subseteq C$; since C is compact (closed in the compact space G), finitely many $c_1, \ldots, c_m$ give $U = U_{c_1} \cap \cdots \cap U_{c_m}$ with $UC \subseteq C$; the same argument applied to $G \setminus C$ shows $\{h \mid h(G \setminus C) \subseteq G \setminus C\}$ is open; intersecting gives that $\mathrm{Stab}_G(C)$ is open.

Since $\mathrm{Stab}_G(C)$ is open in the compact group G, it has finite index, and the set $X = G/\mathrm{Stab}_G(C)$ of left cosets (equivalently, the set of distinct left translates of C) is finite. The left multiplication action of G on X gives a continuous homomorphism $\rho\colon G \to \mathrm{Sym}(X)$ to a finite symmetric group with the discrete topology. The kernel $N = \ker \rho$ is open (preimage of $\{\mathrm{id}\}$ in a discrete group), normal, and has finite index (since G/N embeds into $\mathrm{Sym}(X)$). If $h \in N$, then h fixes every translate of C; in particular $hC = C$, and since $e \in C$ this gives $h = he \in hC = C$. Thus $N \subseteq C$, and since $g \notin C$, it follows that $g \notin N$.

Step 2b: Completing the characterization. Step 2a shows that for every $g \neq e$, there exists an open normal subgroup N of finite index with $g \notin N$. Equivalently, the intersection of all open normal subgroups is trivial. Moreover, for every open neighborhood U of e, choosing any clopen C with $e \in C \subseteq U$ and applying the construction of Step 2a produces an open normal subgroup $N \subseteq C \subseteq U$. The open normal subgroups therefore form a neighborhood basis of e.

The natural map $\alpha\colon G \to \varprojlim_N G/N$ is continuous, injective (the intersection of all open normal subgroups is trivial), and surjective (by a compactness argument identical to that in proposition 1.1.8). As a continuous bijection from a compact space to a Hausdorff space, α is a homeomorphism. Since each G/N is finite, G is profinite.

The characterization theorem reduces checking that a topological group is profinite to verifying three topological properties, none of which mentions limits or cofiltered systems. In practice, compactness and the Hausdorff property are often immediate, and the essential content is total disconnectedness—the absence of nontrivial connected subsets. For Galois groups, total disconnectedness is built into the Krull topology by construction, as will be seen in section 1.4.

Exercise 1.1.1

1. Let G be a group and $\iota\colon G \to \widehat{G}$ the natural map to its profinite completion. Show that $\iota(G)$ is dense in $\widehat{G}$.
2. Show that $\iota\colon G \to \widehat{G}$ is injective if and only if G is residually finite, and give an example of a group that is not residually finite.
3. Show that the profinite completion commutes with finite direct products: for any groups G and H, there is a canonical isomorphism $\widehat{G \times H} \cong \widehat{G} \times \widehat{H}$.
4. Let G be a profinite group and D a discrete group. Show that every continuous homomorphism $\varphi\colon G \to D$ has open kernel, and conclude that $\varphi(G)$ is finite.
5. Show that $\mathbb{Z}_p$ is profinite by verifying the three conditions of theorem 1.1.10 directly, without

using the definition of $\mathbb{Z}_p$ as a limit.

1.2 Topological Properties of Profinite Groups

The characterization theorem (theorem 1.1.10) reduces the study of profinite groups to the study of compact, Hausdorff, totally disconnected groups. The next task is understanding the internal structure: which subgroups inherit the profinite property, how quotients behave, and what the profinite analogue of Sylow theory looks like. The key theme is that the topology and algebra of a profinite group are tightly coupled—for closed subgroups, being open and having finite index are the same condition.

The characterization theorem gives an immediate criterion for recognizing profinite subgroups.

Proposition 1.2.1: Closed Subgroups Are Profinite

Every closed subgroup of a profinite group is profinite.

***Proof*:**
Let H be a closed subgroup of a profinite group G. By theorem 1.1.10, G is compact, Hausdorff, and totally disconnected. As a closed subset of a compact space, H is compact. As a subspace of a Hausdorff space, H is Hausdorff. As a subspace of a totally disconnected space, H is totally disconnected. Applying theorem 1.1.10 again, H is profinite.

Non-closed subgroups, by contrast, need not be profinite—they may fail to be compact. For instance, $\mathbb{Z}$ embeds densely into $\widehat{\mathbb{Z}}$ as a non-closed subgroup, and $\mathbb{Z}$ with the subspace topology is not compact.

The relationship between the topological and algebraic notions of "smallness" for subgroups is governed by the following.

Proposition 1.2.2: Open and Finite-Index Subgroups

Let G be a profinite group and $H \leq G$. Then:

1. If H is open, then H is closed and has finite index in G.
2. If H is closed and has finite index, then H is open.
3. A closed subgroup is open if and only if it has finite index.

***Proof*:**
For (1), if H is open, its complement $G \setminus H = \bigcup_{g \notin H} gH$ is a union of open cosets (each gH is open since left multiplication is a homeomorphism), hence open; so H is closed. The cosets of H form an open cover of the compact space G, so finitely many suffice, giving $[G : H] < \infty$.

For (2), if H is closed with $[G : H] = n$, the complement $G \setminus H = g_2H \cup \cdots \cup g_nH$ is a finite union of closed sets (each g_iH is closed as a translate of the closed set H), hence closed. So H is open.

Part (3) combines (1) and (2).

Part (3) fails without the closure hypothesis: a dense subgroup of a profinite group can have finite index without being open (see exercise 1.2.1 (5)). The following result of Nikolov and Segal shows that for finitely generated profinite groups, this pathology does not arise.

Theorem 1.2.3: Nikolov–Segal Theorem

Let G be a topologically finitely generated profinite group. Then every subgroup of finite index in G is open.

The proof, which uses the classification of finite simple groups, is beyond the scope of this text; see [19]. The theorem implies that for finitely generated profinite groups, the algebraic structure (finite-index subgroups) completely determines the topology (open subgroups). In particular, the topology on such a group is the unique one making it profinite.

Quotients of profinite groups by closed normal subgroups remain profinite.

Proposition 1.2.4: Quotients of Profinite Groups

Let G be a profinite group and $N \trianglelefteq G$ a closed normal subgroup. Then G/N with the quotient topology is profinite, and the projection $\pi\colon G \to G/N$ is open.

***Proof*:**
The quotient G/N is compact (as the continuous image of the compact space G under π) and Hausdorff (a standard result: the quotient of a compact Hausdorff space by a closed equivalence relation is Hausdorff; see [11]). For total disconnectedness: given $gN \neq hN$ in G/N, the element $g^{-1}h \notin N$. Since N is the intersection of all open normal subgroups containing it (by proposition 1.1.8, G is determined by its open normal subgroups), there exists an open normal subgroup $U \trianglelefteq G$ with $g^{-1}h \notin UN$. Then UN/N is an open normal subgroup of G/N separating gN from hN, and since UN/N is clopen (its complement is a finite union of open cosets), this separates the two points by a clopen set. Thus G/N is totally disconnected, and by theorem 1.1.10 it is profinite.

For the second claim: the projection π is open because for any open subset $V \subseteq G$, the set $\pi^{-1}(\pi(V)) = VN = \bigcup_{n \in N} Vn$ is a union of open sets (each Vn is open), hence open; since π is a quotient map, $\pi(V)$ is open.

A continuous homomorphism between profinite groups always has profinite image.

Corollary 1.2.5: Images of Continuous Homomorphisms

Let $f\colon G \to H$ be a continuous homomorphism of profinite groups. Then $f(G)$ is a closed subgroup of H, hence profinite.

***Proof*:**
The image $f(G)$ is compact (continuous image of the compact space G) and hence closed in the Hausdorff space H. By proposition 1.2.1, $f(G)$ is profinite.

The remainder of this section develops the profinite analogue of the Sylow theorems. The proofs illustrate a technique that recurs throughout the theory of profinite groups: reducing to the finite case and passing to the limit. The following observation makes this passage possible.

Lemma 1.2.6: Limits of Non-Empty Finite Sets

The limit of a cofiltered system of non-empty finite sets is non-empty.

***Proof*:**
Let $(X_i, f_{ji})_{i\in I}$ be such a system. Give each X_i the discrete topology, so that $\prod_{i\in I} X_i$ is compact by Tychonoff's theorem. For each finite subset $F \subseteq I$, define

$$Y_F = \left\{ (x_i) \in \prod X_i \;\middle|\; f_{ji}(x_j) = x_i \text{ for all } i \leq j \text{ with } i, j \in F \right\}$$

. Each Y_F is closed (a finite intersection of closed conditions) and non-empty: by directedness, there exists $k \in I$ with $k \geq i$ for all $i \in F$, and any $x_k \in X_k$ determines compatible values $x_i = f_{ki}(x_k)$ for $i \in F$. Since $Y_{F_1 \cup F_2} \subseteq Y_{F_1} \cap Y_{F_2}$, the family $\{Y_F\}$ has the finite intersection property. By compactness, $\bigcap_F Y_F = \varprojlim X_i$ is non-empty.

With this tool, the classical Sylow theorems extend to profinite groups.

Theorem 1.2.7: Sylow Theory for Profinite Groups

Let G be a profinite group and p a prime. Then:

1. G has a maximal pro-p-subgroup (a *Sylow pro-p-subgroup*).
2. Any two Sylow pro-p-subgroups of G are conjugate.
3. Every pro-p-subgroup of G is contained in a Sylow pro-p-subgroup.

***Proof*:**
Write $G \cong \varprojlim_U G/U$ where U ranges over the open normal subgroups, with projections $\pi_U\colon G \to G/U$ and transition maps $\pi_{VU}\colon G/U \to G/V$ for $U \subseteq V$. Each G/U is a finite group. The image of a Sylow p-subgroup of G/U under the surjection π_{VU} is a Sylow p-subgroup of G/V: if P is Sylow in G/U with $|P| = |G/U|_p$ and $K = \ker \pi_{VU} = V/U$, then $|P \cap K| = |K|_p$ (since $K \trianglelefteq G/U$ and a Sylow subgroup of G/U intersects any normal subgroup in a Sylow subgroup of that normal subgroup), giving $|\pi_{VU}(P)| = |P|/|P \cap K| = |G/U|_p/|K|_p = |G/V|_p$. So the assignment $P \mapsto \pi_{VU}(P)$ defines a map $\mathrm{Syl}_p(G/U) \to \mathrm{Syl}_p(G/V)$, and the sets $\mathrm{Syl}_p(G/U)$ form a cofiltered system of non-empty finite sets.

Step 1: Existence. By lemma 1.2.6, the limit $\varprojlim_U \mathrm{Syl}_p(G/U)$ is non-empty. A compatible family $(P_U)_U$ with $P_U \in \mathrm{Syl}_p(G/U)$ and $\pi_{VU}(P_U) = P_V$ for all $U \subseteq V$ defines a closed subgroup

$S = \varprojlim_U P_U \subseteq G$. Each P_U is a p-group, so S is a pro-p-group. To see that S is maximal: if $S \subseteq T$ with T a pro-p-subgroup, then $\pi_U(T)$ is a p-subgroup of G/U containing the Sylow subgroup P_U, forcing $\pi_U(T) = P_U$ for all U. Since $T \subseteq \varprojlim_U \pi_U(T) = \varprojlim_U P_U = S$, maximality follows.

Step 2: Conjugacy. Let S_1, S_2 be Sylow pro-p-subgroups. For each U, the Sylow p-subgroups $\pi_U(S_1)$ and $\pi_U(S_2)$ of G/U are conjugate by finite Sylow theory. The set

$$C_U = \left\{ g \in G/U \;\middle|\; g\, \pi_U(S_1)\, g^{-1} = \pi_U(S_2) \right\}$$

is non-empty and finite (it is a coset of the normalizer of $\pi_U(S_1)$). The projection π_{VU} maps C_U into C_V, so $(C_U)_U$ is a cofiltered system of non-empty finite sets. By lemma 1.2.6, there exists $g = (g_U)_U \in \varprojlim_U C_U \subseteq G$ with $g\, S_1\, g^{-1} = S_2$.

Step 3: Containment. Let T be a pro-p-subgroup of G. For each U, the set

$$D_U = \left\{ P \in \mathrm{Syl}_p(G/U) \;\middle|\; \pi_U(T) \subseteq P \right\}$$

is non-empty (by finite Sylow theory). The transition maps send D_U into D_V: if $\pi_U(T) \subseteq P$, then $\pi_V(T) = \pi_{VU}(\pi_U(T)) \subseteq \pi_{VU}(P)$, and $\pi_{VU}(P) \in D_V$. By lemma 1.2.6, the limit of the system $(D_U)_U$ is non-empty, and a compatible family $(P_U)_U$ with $\pi_U(T) \subseteq P_U$ defines a Sylow pro-p-subgroup $S = \varprojlim_U P_U$ containing T.

The Sylow theorem for profinite groups recovers the finite Sylow theorems when G is finite (the only open normal subgroup is the trivial group), and demonstrates the power of the limit-of-finite-sets technique: each part of the proof reduces to the corresponding finite statement, then passes to the limit using lemma 1.2.6. This technique will reappear in Chapters 3 and 5.

Example 1.2.8: Sylow Structure

1. The profinite group $\widehat{\mathbb{Z}} \cong \prod_p \mathbb{Z}_p$ decomposes as a product of its Sylow pro-p-subgroups: the Sylow pro-p-subgroup is $\mathbb{Z}_p$, embedded as the factor corresponding to p. Since each Sylow subgroup is normal (in any abelian group), the decomposition $\widehat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$ is the unique way to write $\widehat{\mathbb{Z}}$ as a product of its Sylow subgroups. This mirrors the decomposition $\mathbb{Z}/n\mathbb{Z} \cong \prod_{p|n} \mathbb{Z}/p^{v_p(n)}\mathbb{Z}$ from the Chinese Remainder Theorem.
2. (If the reader has read [8]) Let E/K be an elliptic curve over a number field and p a prime. The Galois action on the p-power torsion gives a continuous homomorphism $\rho_{E,p}\colon G_K \to \mathrm{GL}_2(\mathbb{Z}_p)$. The image $\rho_{E,p}(G_K)$ is a closed subgroup of $\mathrm{GL}_2(\mathbb{Z}_p)$ by corollary 1.2.5, hence profinite. The structure of this image—and in particular its Sylow subgroups—encodes arithmetic information about E.

Exercise 1.2.1

1. Let G be a profinite group and $H \leq G$ a subgroup (not necessarily closed). Show that:

$$\overline{H} = \bigcap_{U \trianglelefteq_{\text{open}} G} HU$$

2. Determine all open subgroups of $\mathbb{Z}_p$ and describe their inclusion relations.
3. Let $f : G \to H$ be a continuous surjective homomorphism of profinite groups. Show that f is an open map.
4. Let G be a profinite group and $N \trianglelefteq G$ a closed normal subgroup. Show that the open normal subgroups of G/N are exactly the images UN/N where U ranges over the open normal subgroups of G containing N.
5. Show that the closure hypothesis in proposition 1.2.2(2) is necessary: give an example of a profinite group G with a dense subgroup H of finite index that is not open. *Hint:* use the axiom of choice to construct a discontinuous $\mathbb{F}_2$-linear functional on $(\mathbb{Z}/2\mathbb{Z})^{\mathbb{N}}$.

1.3 Pro-p-Groups and Supernatural Numbers

Among profinite groups, those built entirely from p-groups for a single prime p form the most tractable and best-understood class. These pro-p-groups are the objects on which the cohomological dimension theory of Chapter 5 and the Golod–Shafarevich inequality of Chapter 6 are best understood. To measure the "size" of a profinite group—which may have uncountably many elements—the classical notion of group order must be generalized. Supernatural numbers provide this generalization.

Definition 1.3.1: Pro-p-Group

Let p be a prime. A *pro-p-group* is a profinite group in which every finite quotient is a p-group.

Equivalently, a pro-p-group is a limit of a cofiltered system of finite p-groups. The two formulations are interchangeable: by proposition 1.1.8, any profinite group is the limit of its finite quotients G/U, and if each G/U is a p-group, the limit is clearly a pro-p-group.

Proposition 1.3.2: Characterizations of Pro-p-Groups

Let G be a profinite group and p a prime. The following are equivalent:

1. G is a pro-p-group.
2. Every open normal subgroup $U \trianglelefteq G$ has p-power index.
3. G is its own Sylow pro-p-subgroup: the only Sylow pro-p-subgroup of G is G itself.

***Proof*:**
The equivalence (1) $\Leftrightarrow$ (2) is immediate: the finite quotients of G are exactly the groups G/U for U open normal, and a finite group is a p-group if and only if its order is a power of p.

For (1) $\Rightarrow$ (3): if G is a pro-p-group, then G is a pro-p-subgroup of itself. Any Sylow pro-p-subgroup S contains G (by maximality and the fact that G is already pro-p), so $S = G$. For (3) $\Rightarrow$ (1): if G is its own Sylow pro-p-subgroup, then G is a pro-p-group by definition of Sylow.

Example 1.3.3: Pro-p-Groups

1. The p-adic integers $\mathbb{Z}_p = \varprojlim_k \mathbb{Z}/p^k\mathbb{Z}$ form a pro-p-group, with all finite quotients being cyclic p-groups.
2. For $n \geq 1$, the kernel of the reduction map $\mathrm{GL}_n(\mathbb{Z}_p) \to \mathrm{GL}_n(\mathbb{F}_p)$ is the subgroup $K_n = \{A \in \mathrm{GL}_n(\mathbb{Z}_p) \mid A \equiv I \pmod{p}\}$. Each finite quotient of K_n (obtained by reducing modulo p^k) is a p-group: the kernel of $\mathrm{GL}_n(\mathbb{Z}/p^k\mathbb{Z}) \to \mathrm{GL}_n(\mathbb{F}_p)$ has order $p^{(k-1)n^2}$. So K_n is a pro-p-group. The full group $\mathrm{GL}_n(\mathbb{Z}_p)$ is *not* a pro-p-group: the quotient $\mathrm{GL}_n(\mathbb{F}_p)$ has order $\prod_{i=0}^{n-1}(p^n - p^i)$, which is divisible by primes other than p for $n \geq 2$.
3. The profinite completion $\widehat{\mathbb{Z}} \cong \prod_\ell \mathbb{Z}_\ell$ is not a pro-p-group for any prime p, since it has quotients of every finite order. Its Sylow pro-p-subgroup is $\mathbb{Z}_p$, embedded as the p-component of the product.

To express statements like "$G_{\mathbb{Q}_p}$ has pro-p part of infinite p-rank" or "the index $[G : H]$ is divisible by p^∞," a generalization of the natural numbers is needed.

Definition 1.3.4: Supernatural Numbers

A *supernatural number* is a formal product:

$$\alpha = \prod_p p^{n_p}$$

where p ranges over all primes and each exponent $n_p \in \{0, 1, 2, \ldots\} \cup \{\infty\}$. The supernatural numbers form a partially ordered set under divisibility: $\alpha \big| \beta$ if $n_p(\alpha) \leq n_p(\beta)$ for all p. The gcd and lcm of any family of supernatural numbers are defined by taking the componentwise infimum and supremum of exponents, respectively.

Every positive integer is a supernatural number (with all but finitely many exponents equal to zero, and none equal to ∞). The supernatural number $\prod_p p^\infty$ is the largest, divisible by every supernatural number. Unlike positive integers, supernatural numbers do not form a group under multiplication: there is no cancellation law.

The order and index of profinite groups take values in supernatural numbers.

Definition 1.3.5: Order and Index

Let G be a profinite group. The *order* of G is the supernatural number:

$$\#G = \operatorname{lcm}\{|G/U| \mid U \trianglelefteq_{\text{open}} G\}$$

If $H \leq G$ is a closed subgroup, the *index* of H in G is:

$$[G : H] = \operatorname{lcm}\{[G/U : \pi_U(H)] \mid U \trianglelefteq_{\text{open}} G\}$$

where $\pi_U : G \to G/U$ is the projection.

When G is finite, $\#G$ is the ordinary group order: the only open normal subgroup is $\{e\}$, and $|G/\{e\}| = |G|$. For a closed subgroup H of a finite group, $[G : H]$ is the ordinary index.

Proposition 1.3.6: Lagrange's Theorem for Profinite Groups

Let G be a profinite group and $H \leq G$ a closed subgroup. Then:

$$\#G = [G : H] \cdot \#H$$

as supernatural numbers.

***Proof*:**
The equality is verified prime by prime. Fix a prime p. For each open normal $U \trianglelefteq G$, the projection $\pi_U(H) = HU/U \cong H/(H \cap U)$ gives:

$$v_p(|G/U|) = v_p([G/U : \pi_U(H)]) + v_p(|H/(H \cap U)|)$$

Both terms on the right are non-decreasing as U shrinks (a finer quotient G/U' with $U' \subseteq U$ gives $[G/U' : \pi_{U'}(H)] \geq [G/U : \pi_U(H)]$ and $|H/(H \cap U')| \geq |H/(H \cap U)|$). Taking the supremum over U: the supremum of a sum of non-decreasing functions equals the sum of the suprema, giving $v_p(\#G) = v_p([G : H]) + v_p(\#H)$.

Example 1.3.7: Supernatural Orders

1. $\#\widehat{\mathbb{Z}} = \prod_p p^\infty$, since $\widehat{\mathbb{Z}}/n\widehat{\mathbb{Z}} \cong \mathbb{Z}/n\mathbb{Z}$ for every $n \geq 1$, and $\operatorname{lcm}\{n : n \geq 1\} = \prod_p p^\infty$.
2. $\#\mathbb{Z}_p = p^\infty$ and $[\widehat{\mathbb{Z}} : \mathbb{Z}_p] = \prod_{\ell \neq p} \ell^\infty$, consistent with Lagrange: $\prod_\ell \ell^\infty = \prod_{\ell \neq p} \ell^\infty \cdot p^\infty$.
3. For a finite group G of order n, $\#G = n$. This recovers the classical Lagrange theorem from proposition 1.3.6.

Supernatural numbers give a clean characterization of pro-p-groups in terms of the order.

Corollary 1.3.8: Pro-p-Groups via Order

A profinite group G is a pro-p-group if and only if $\#G$ is a power of p (i.e., $v_\ell(\#G) = 0$ for all primes $\ell \neq p$).

***Proof*:**
By proposition 1.3.2, G is pro-p if and only if every open normal subgroup has p-power index. This holds if and only if $|G/U|$ is a power of p for every open normal U, which is equivalent to $v_\ell(\#G) = \sup_U v_\ell(|G/U|) = 0$ for all $\ell \neq p$.

The section closes with a structural result about pro-p-groups that will be essential in Chapter 6: the Frattini subgroup controls the minimal number of topological generators.

Definition 1.3.9: Frattini Subgroup

Let G be a pro-p-group. The *Frattini subgroup* $\Phi(G)$ is the intersection of all maximal open subgroups of G. Equivalently, $\Phi(G)$ is the closure of the subgroup generated by all pth powers and commutators: $\Phi(G) = \overline{G^p[G,G]}$.

Each maximal open subgroup has index p in G (since maximal subgroups of a finite p-group have index p), so $\Phi(G)$ is a closed normal subgroup and $G/\Phi(G)$ is a quotient in which every element has order dividing p.

Proposition 1.3.10: Burnside Basis Theorem for Pro-p-Groups

Let G be a pro-p-group. Then:

1. The quotient $G/\Phi(G)$ is an elementary abelian p-group: there exists a cardinal d with $G/\Phi(G) \cong (\mathbb{Z}/p\mathbb{Z})^d$.
2. Elements $s_1, \ldots, s_n \in G$ topologically generate G (i.e., the closure $\overline{\langle s_1, \ldots, s_n \rangle} = G$) if and only if their images $\bar{s}_1, \ldots, \bar{s}_n$ generate $G/\Phi(G)$ as an $\mathbb{F}_p$-vector space.
3. The minimal number of topological generators of G is $d(G) = \dim_{\mathbb{F}_p} G/\Phi(G)$.

***Proof*:**
For (1): the quotient $G/\Phi(G)$ embeds into $\prod_M G/M$ where M ranges over maximal open subgroups. Each G/M has order p, so the product is an elementary abelian p-group, and a subgroup of an elementary abelian p-group is elementary abelian.

For (2): the forward direction is immediate—if $\{s_i\}$ generates G topologically, its image generates any quotient. For the converse, suppose $\{\bar{s}_i\}$ generates $G/\Phi(G)$, and let $H = \overline{\langle s_1, \ldots, s_n \rangle}$. Then $H\Phi(G) = G$ (since the images of the s_i generate $G/\Phi(G)$). For any open normal $U \trianglelefteq G$, the projection $\pi_U(H) \cdot \Phi(G/U) = G/U$, where $\Phi(G/U)$ is the Frattini subgroup of the finite p-group G/U (using $\pi_U(\Phi(G)) \subseteq \Phi(G/U)$, which follows from the correspondence between maximal subgroups of G containing U and maximal subgroups of G/U). The classical Burnside basis theorem for finite p-groups (see [6]) gives $\pi_U(H) = G/U$. Since this holds for all U, and H is closed, proposition 1.1.8 gives $H = G$.

Part (3) follows from (2): $d(G) = \dim_{\mathbb{F}_p} G/\Phi(G)$ because any generating set must span $G/\Phi(G)$, and any spanning set of $G/\Phi(G)$ lifts to a topological generating set.

The Burnside basis theorem reduces the question of topological generation to linear algebra over $\mathbb{F}_p$. For instance, $\mathbb{Z}_p$ has $\Phi(\mathbb{Z}_p) = p\mathbb{Z}_p$ and $\mathbb{Z}_p/p\mathbb{Z}_p \cong \mathbb{F}_p$, so $d(\mathbb{Z}_p) = 1$: the group $\mathbb{Z}_p$ is topologically generated by any element not divisible by p.

Exercise 1.3.1

1. Compute the supernatural order $\#\mathrm{GL}_n(\mathbb{Z}_p)$ for $n \geq 1$.
2. Let G be an abelian profinite group. Show that G is isomorphic to the direct product of its Sylow pro-p-subgroups: $G \cong \prod_p G_p$, where G_p denotes the unique Sylow pro-p-subgroup of G.
3. Let G be a pro-p-group and $N \trianglelefteq G$ a closed normal subgroup. Show that both N and G/N are pro-p-groups.
4. Compute the Frattini subgroup and the minimal number of topological generators for: (a) $\mathbb{Z}_p$, (b) $\mathbb{Z}_p \times \mathbb{Z}_p$, (c) $\mathbb{Z}_p^n$ for general $n \geq 1$.
5. (If the reader has read [10]) Let K be a finite extension of $\mathbb{Q}_p$ with absolute Galois group G_K, inertia group I_K, and wild inertia group P_K. Verify that P_K is a pro-p-group and that I_K/P_K is a pro-p'-group (i.e., has order coprime to p).

1.4 The Absolute Galois Group

The preceding sections developed profinite groups as abstract topological objects. The reason this machinery matters for algebraic number theory is that Galois groups of infinite extensions are profinite. This section makes the connection precise, establishes the profinite structure of the absolute Galois group, and computes it for the fields that serve as running examples throughout the book.

Theorem 1.4.1: Infinite Galois Theory

Let L/K be a Galois extension (possibly infinite), and let $\mathcal{F}$ denote the set of finite Galois extensions F/K contained in L, directed by inclusion. Then:

1. The restriction maps assemble into a topological isomorphism:

$$\mathrm{Gal}(L/K) \xrightarrow{\sim} \varprojlim_{F\in\mathcal{F}} \mathrm{Gal}(F/K)$$

   where the right side carries the limit topology (each $\mathrm{Gal}(F/K)$ discrete). The induced topology on $\mathrm{Gal}(L/K)$ is called the *Krull topology.* In particular, $\mathrm{Gal}(L/K)$ is a profinite group.

2. There is an inclusion-reversing bijection:

$$\{\text{closed subgroups of } \mathrm{Gal}(L/K)\} \longleftrightarrow \{\text{intermediate fields } K \subseteq M \subseteq L\}$$

   given by $H \mapsto L^H$ and $M \mapsto \mathrm{Gal}(L/M)$. Under this bijection, *open* subgroups correspond to *finite* extensions of K, and open normal subgroups to finite Galois extensions of K contained in L.

***Proof*:**
Step 1: The isomorphism. Define $\varphi\colon \mathrm{Gal}(L/K) \to \varprojlim_F \mathrm{Gal}(F/K)$ by $\varphi(\sigma) = (\sigma|_F)_F$. Injectivity: if $\sigma|_F = \mathrm{id}$ for all $F \in \mathcal{F}$, then σ fixes every element of L (since $L = \bigcup_F F$), so $\sigma = \mathrm{id}$. Surjectivity: a compatible family $(\sigma_F)_F$ defines a K-automorphism σ of L by setting $\sigma(x) = \sigma_F(x)$ where F is any element of $\mathcal{F}$ containing x; compatibility ensures this is well-defined, and the resulting map is a field automorphism fixing K. Since φ is a continuous bijection from the compact space $\varprojlim \mathrm{Gal}(F/K)$ to itself (with the Krull topology, where $\mathrm{Gal}(L/F)$ for $F \in \mathcal{F}$ form a neighborhood basis of the identity), it is a homeomorphism.

Step 2: The Galois correspondence. Let $H \leq \mathrm{Gal}(L/K)$ be closed. The fixed field $L^H = \{x \in L \mid hx = x \text{ for all } h \in H\}$ satisfies L/L^H Galois with $\mathrm{Gal}(L/L^H) = H$ (closure is used here: every $\sigma \in \mathrm{Gal}(L/K)$ fixing L^H pointwise agrees with some element of H on each finite subextension, so σ lies in the closure of H, which equals H). Conversely, $M \mapsto \mathrm{Gal}(L/M)$ is closed (it is $\varprojlim_F \mathrm{Gal}(F/F \cap M)$, a limit of finite groups). The correspondence is inclusion-reversing by construction. For the final claim: H is open if and only if $[G : H] < \infty$ (by proposition 1.2.2), and $[G : H] = [\mathrm{Gal}(L/K) : \mathrm{Gal}(L/L^H)] = [L^H : K]$ by the degree formula, so open subgroups correspond to finite extensions.

The infinite Galois theorem says that the Galois theory of L/K is *entirely* encoded in the profinite group $\mathrm{Gal}(L/K)$—but only through its *closed* subgroups. Non-closed subgroups exist and have no field-theoretic meaning. This distinction between closed and non-closed is absent in finite Galois theory (where all subgroups are closed) and is the fundamental reason why the profinite topology must be taken seriously.

Definition 1.4.2: Absolute Galois Group

Let K be a field. The *absolute Galois group* of K is the profinite group:

$$G_K = \mathrm{Gal}(K^{\mathrm{sep}}/K)$$

where K^{sep} is a fixed separable closure of K.

The group G_K depends (up to inner automorphism) on the choice of separable closure, but all structural invariants of G_K—its closed subgroups, its quotients, and (as developed in Chapter 2) its cohomology—are independent of this choice.

The cyclotomic character, which records the action of Galois on roots of unity, is the first and most fundamental Galois representation.

Definition 1.4.3: Cyclotomic Character

Let K be a field, and for $n \geq 1$ let $\mu_n \subseteq K^{\mathrm{sep}}$ denote the group of nth roots of unity. The action of $\sigma \in G_K$ on μ_n is determined by a unique element $\chi_n(\sigma) \in (\mathbb{Z}/n\mathbb{Z})^\times$ satisfying $\sigma(\zeta) = \zeta^{\chi_n(\sigma)}$ for all $\zeta \in \mu_n$. The maps χ_n assemble into a continuous homomorphism:

$$\chi \colon G_K \to \widehat{\mathbb{Z}}^\times = \varprojlim_n (\mathbb{Z}/n\mathbb{Z})^\times$$

called the *cyclotomic character* of K.

The cyclotomic character is surjective when $K = \mathbb{Q}$ (since $\mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times$ for all n), but not in general: if K contains all roots of unity, then χ is trivial. The kernel of χ is the absolute Galois group of $K(\mu_\infty) = K(\zeta_n : n \geq 1)$, the field obtained by adjoining all roots of unity to K.

Example 1.4.4: Running Examples: Absolute Galois Groups

The following Galois groups serve as running examples throughout the book. Each illustrates different features of the profinite theory, and their cohomology will be computed explicitly as the machinery is developed.

1. The only algebraic extension of $\mathbb{R}$ is $\mathbb{C}$, so $G_{\mathbb{R}} = \mathrm{Gal}(\mathbb{C}/\mathbb{R}) \cong \mathbb{Z}/2\mathbb{Z}$, generated by complex conjugation. This is the simplest nontrivial profinite group: finite, cyclic of order 2. The cyclotomic character $\chi \colon G_{\mathbb{R}} \to \widehat{\mathbb{Z}}^\times$ sends complex conjugation to -1 (since $\overline{\zeta} = \zeta^{-1}$ for any root of unity ζ on the unit circle). The cohomology of $G_{\mathbb{R}}$ is the periodic cohomology of $\mathbb{Z}/2\mathbb{Z}$, which is nontrivial in every even degree—a reflection of the fact that passage from $\mathbb{C}$ to $\mathbb{R}$ involves genuine arithmetic obstructions.

2. The absolute Galois group $G_{\mathbb{Q}_p}$ has a rich internal structure governed by the ramification theory of [10]. The maximal unramified extension $\mathbb{Q}_p^{\mathrm{nr}} = \mathbb{Q}_p(\zeta_{p^n-1} : n \geq 1)$ gives a short exact sequence:
$$1 \to I_p \to G_{\mathbb{Q}_p} \xrightarrow{\mathrm{res}} \mathrm{Gal}(\mathbb{Q}_p^{\mathrm{nr}}/\mathbb{Q}_p) \to 1$$
where $I_p = \mathrm{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p^{\mathrm{nr}})$ is the *inertia group* and $\mathrm{Gal}(\mathbb{Q}_p^{\mathrm{nr}}/\mathbb{Q}_p) \cong G_{\mathbb{F}_p} \cong \widehat{\mathbb{Z}}$ is topologically generated by the *Frobenius* automorphism $\mathrm{Frob}_p \colon x \mapsto x^p$. The inertia group itself

decomposes via the ramification filtration:

$$1 \to P_p \to I_p \to I_p/P_p \to 1$$

where P_p is the *wild inertia* (a pro-p-group by exercise 1.3.1 (5)) and $I_p/P_p \cong \prod_{\ell \neq p} \mathbb{Z}_\ell$ is the *tame inertia*. The supernatural order is $\#G_{\mathbb{Q}_p} = \prod_\ell \ell^\infty$: every prime appears with infinite multiplicity, reflecting the existence of extensions of $\mathbb{Q}_p$ of every degree.

3. The absolute Galois group $G_\mathbb{Q}$ is the most complex and least understood of the running examples. No explicit presentation by generators and relations is known. Its structure is constrained by the local Galois groups: for each prime p, a choice of embedding $\overline{\mathbb{Q}} \hookrightarrow \overline{\mathbb{Q}}_p$ determines a continuous injection $G_{\mathbb{Q}_p} \hookrightarrow G_\mathbb{Q}$ (a *decomposition group* at p, well-defined up to conjugacy), and the real embedding gives an inclusion $G_\mathbb{R} \hookrightarrow G_\mathbb{Q}$ (the decomposition group at the archimedean place). The collection of all these local subgroups constrains $G_\mathbb{Q}$ but does not determine it—the additional structure is captured by the global duality theorems developed in Chapter 9.

4. (If the reader has read [8]) An elliptic curve $E/\mathbb{Q}$ gives a continuous Galois representation $\rho_{E,p}\colon G_\mathbb{Q} \to \mathrm{GL}_2(\mathbb{Z}_p)$ via the action on the Tate module $T_p(E) = \varprojlim_n E[p^n]$. The image $\rho_{E,p}(G_\mathbb{Q})$ is a closed subgroup of $\mathrm{GL}_2(\mathbb{Z}_p)$ (by corollary 1.2.5), hence profinite. By Serre's open image theorem, this image is open in $\mathrm{GL}_2(\mathbb{Z}_p)$ for all but finitely many p when E has no complex multiplication. The cohomological study of such representations is a central application of the theory developed in this book.

The following criterion detects when the absolute Galois group is a pro-p-group.

Proposition 1.4.5: Pro-p Absolute Galois Groups

The absolute Galois group G_K is a pro-p-group if and only if K has no separable extensions of degree coprime to p (other than K itself).

***Proof*:**
By proposition 1.3.2, G_K is pro-p if and only if every finite quotient is a p-group. The finite quotients of G_K are the groups $\mathrm{Gal}(F/K)$ for finite Galois extensions F/K, by theorem 1.4.1. If G_K is pro-p, every finite Galois extension has p-power degree, and by taking normal closures, every finite separable extension has p-power degree. Conversely, if every finite separable extension has p-power degree, every finite Galois extension has p-power degree (since the degree of a Galois extension equals the order of its Galois group), and G_K is pro-p.

Exercise 1.4.1

1. Show that G_K is trivial if and only if K is separably closed.
2. Compute $G_{\mathbb{F}_q}$ for a finite field $\mathbb{F}_q$ and show it is isomorphic to $\widehat{\mathbb{Z}}$, topologically generated by the Frobenius automorphism $\mathrm{Frob}_q\colon x \mapsto x^q$.
3. Show that $\mathrm{Gal}(\mathbb{Q}(\zeta_{p^\infty})/\mathbb{Q}) \cong \mathbb{Z}_p^\times$, where $\mathbb{Q}(\zeta_{p^\infty}) = \bigcup_{n\geq 1} \mathbb{Q}(\zeta_{p^n})$. Decompose this as $\mathbb{Z}_p^\times \cong \mu_{p-1} \times \mathbb{Z}_p$ for odd p (where μ_{p-1} is the group of $(p-1)$th roots of unity in $\mathbb{Z}_p^\times$) and describe the corresponding decomposition of $\mathrm{Gal}(\mathbb{Q}(\zeta_{p^\infty})/\mathbb{Q})$ into Galois groups of explicit sub-extensions.

4. Compute the supernatural order $\#G_{\mathbb{Q}_p}$ and verify that it equals $\prod_\ell \ell^\infty$.
5. (If the reader has read [4]) The Kronecker–Weber theorem states that every abelian extension of $\mathbb{Q}$ is contained in a cyclotomic field $\mathbb{Q}(\zeta_n)$. Use this to show that $G_{\mathbb{Q}}^{\mathrm{ab}} \cong \widehat{\mathbb{Z}}^\times$, where $G_{\mathbb{Q}}^{\mathrm{ab}} = G_{\mathbb{Q}}/\overline{[G_{\mathbb{Q}}, G_{\mathbb{Q}}]}$ is the Galois group of the maximal abelian extension of $\mathbb{Q}$. Identify the isomorphism with the cyclotomic character χ.

2

Cohomology of Discrete Groups

Given a group G acting on an abelian group M, the fixed-point subgroup M^G is natural invariant of the action. The functor $M \mapsto M^G$ is left-exact but not exact: a surjection $M \twoheadrightarrow N$ of G-modules need not induce a surjection $M^G \twoheadrightarrow N^G$. The right-derived functors of $(-)^G$, denoted $H^n(G, M)$, measure this failure. These are the *group cohomology* groups, and they encode arithmetic information ranging from the structure of unit groups (Hilbert's Theorem 90) to the classification of central simple algebras (the Brauer group).

This chapter develops group cohomology for discrete groups, first through derived functors (connecting to the general theory in [9]), then through explicit cocycle descriptions that make the low-degree groups computable and interpretable. The profinite refinement—where continuity of cochains becomes essential—is developed in Chapter 3.

2.1 G-Modules and the Invariant Functor

To define cohomology of a group G, a category of coefficients is needed. For a topological space, cohomology takes coefficients in abelian groups (or sheaves); for a group, the natural coefficients are abelian groups equipped with a compatible G-action.

Definition 2.1.1: G-Module

Let G be a group. A *G-module* is an abelian group M equipped with a left G-action by group automorphisms: a map $G \times M \to M$, written $(g, m) \mapsto g \cdot m$, satisfying $g \cdot (m + m') = g \cdot m + g \cdot m'$, $1 \cdot m = m$, and $(gh) \cdot m = g \cdot (h \cdot m)$ for all $g, h \in G$ and $m, m' \in M$. Equivalently, a G-module is a left module over the group ring $\mathbb{Z}[G]$.

The equivalence between G-actions on abelian groups and $\mathbb{Z}[G]$-module structures is immediate: the group ring $\mathbb{Z}[G] = \bigoplus_{g \in G} \mathbb{Z} \cdot g$ acts by $(\sum a_g g) \cdot m = \sum a_g (g \cdot m)$, and conversely any $\mathbb{Z}[G]$-module structure restricts to a G-action via the embedding $G \hookrightarrow \mathbb{Z}[G]^\times$. This perspective identifies the category of G-modules with the module category $\mathbb{Z}[G]$-Mod. Every k-linear representation of G (a $k[G]$-module as studied in [7]) is a G-module with the action restricted to the $\mathbb{Z}[G]$-structure, but the G-module framework is more general: it accommodates non-semisimple and torsion phenomena that are invisible in the representation-theoretic setting.

A morphism of G-modules is a group homomorphism $\varphi \colon M \to N$ that commutes with the G-action: $\varphi(g \cdot m) = g \cdot \varphi(m)$ for all $g \in G$ and $m \in M$. Equivalently, φ is a $\mathbb{Z}[G]$-module homomorphism. The set of all such morphisms is denoted $\mathrm{Hom}_G(M, N)$.

Proposition 2.1.2: The Category of G-Modules

The category G-Mod of G-modules is an abelian category with enough injectives and enough projectives.

***Proof*:**
Since G-Mod is isomorphic to the module category $\mathbb{Z}[G]$-Mod, this follows from the general theory of module categories over rings (see [9, Category With And Without Projective/Injective Resolutions]).

The existence of enough injectives and projectives guarantees that derived functors can be computed, which is the foundation for defining group cohomology in section 2.2.

Example 2.1.3: G-Modules

1. For any abelian group A, the *trivial G-module* structure on A is defined by $g \cdot a = a$ for all $g \in G$ and $a \in A$. The most common trivial modules are $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{Q}/\mathbb{Z}$, and $\mathbb{Z}/n\mathbb{Z}$.

2. The group ring $\mathbb{Z}[G]$ itself is a G-module via left multiplication: $g \cdot (\sum a_h h) = \sum a_h (gh)$. This is the *free G-module of rank* 1: for any G-module M, the map $\mathrm{Hom}_G(\mathbb{Z}[G], M) \xrightarrow{\sim} M$ given by $\varphi \mapsto \varphi(1)$ is an isomorphism.

3. Let L/K be a Galois extension with $G = \mathrm{Gal}(L/K)$. The additive group L^+, the multiplicative group $L^\times$, and the group of units $\mathcal{O}_L^\times$ of the ring of integers are all G-modules via the natural Galois action. The fixed-point subgroups recover K-arithmetic: $(L^\times)^G = K^\times$ and $(\mathcal{O}_L^\times)^G = \mathcal{O}_K^\times$.

4. For a subgroup $H \leq G$, the *permutation module* $\mathbb{Z}[G/H]$ is the free abelian group on the left cosets G/H, with G acting by left multiplication on cosets. When $H = \{1\}$, this recovers $\mathbb{Z}[G]$.

5. (If the reader has read [8]) For an elliptic curve E over a field K, the group $E(K^{\mathrm{sep}})$ of K^{sep}-rational points is a G_K-module, and $E(K^{\mathrm{sep}})^{G_K} = E(K)$. The m-torsion subgroup $E[m] \subseteq E(K^{\mathrm{sep}})$ is a finite G_K-submodule isomorphic to $(\mathbb{Z}/m\mathbb{Z})^2$ as an abelian group (for $\mathrm{char} K \nmid m$), with a nontrivial G_K-action that encodes the arithmetic of E over K.

A fundamental piece of structure on $\mathbb{Z}[G]$ is the augmentation map, whose kernel plays a central role throughout the cohomological theory.

Definition 2.1.4: Augmentation Map and Augmentation Ideal

The *augmentation map* is the ring homomorphism $\epsilon\colon \mathbb{Z}[G] \to \mathbb{Z}$ defined by:

$$\epsilon\left(\sum_{g\in G} a_g\, g\right) = \sum_{g\in G} a_g$$

It is a G-module homomorphism when $\mathbb{Z}$ carries the trivial action. The *augmentation ideal* $I_G = \ker\epsilon$ is the kernel of this map, giving a short exact sequence of G-modules:

$$0 \to I_G \to \mathbb{Z}[G] \xrightarrow{\epsilon} \mathbb{Z} \to 0$$

As a $\mathbb{Z}$-module, I_G is free with basis $\{g - 1 : g \in G \setminus \{1\}\}$.

The augmentation ideal consists of formal $\mathbb{Z}$-linear combinations of group elements whose coefficients sum to zero. The generators $g-1$ measure the deviation of each group element from the identity: they vanish when g acts trivially and are nonzero otherwise. The exact sequence $0 \to I_G \to \mathbb{Z}[G] \to \mathbb{Z} \to 0$ expresses $\mathbb{Z}$ (with trivial action) as the quotient of the free module $\mathbb{Z}[G]$ by the "non-trivial part" of the group action, and will appear repeatedly—in the standard resolution (section 2.3), in the computation of H_1 (section 2.5), and in many explicit cohomological calculations.

The two fundamental functors on G-modules extract, respectively, the elements fixed by the action and the elements modulo the action.

Definition 2.1.5: G-Invariants

Let M be a G-module. The subgroup of G-*invariants* (or G-*fixed points*) is:

$$M^G = \{m \in M \mid g \cdot m = m \text{ for all } g \in G\}$$

This is the largest submodule of M on which G acts trivially.

Corollary 2.1.6: Hom Representation of Invariants

$M^G \cong \mathrm{Hom}_G(\mathbb{Z}, M)$, where $\mathbb{Z}$ carries the trivial G-action.

***Proof*:**
A G-module homomorphism $\varphi\colon \mathbb{Z} \to M$ is determined by $\varphi(1) \in M$, and the G-equivariance condition $\varphi(g \cdot 1) = g \cdot \varphi(1)$ becomes $\varphi(1) = g \cdot \varphi(1)$ for all g, i.e., $\varphi(1) \in M^G$.

This identification is the reason group cohomology can be expressed as Ext: since $H^n(G, M) = R^n((-)^G)(M)$ and $(-)^G = \mathrm{Hom}_G(\mathbb{Z}, -)$, the derived functors give $H^n(G, M) = \mathrm{Ext}^n_G(\mathbb{Z}, M)$.

Definition 2.1.7: G-Coinvariants

Let M be a G-module. The *G-coinvariants* of M is the quotient:

$$M_G = \frac{M}{\langle g \cdot m - m \big| g \in G,\, m \in M \rangle}$$

This is the largest quotient of M on which G acts trivially.

Proposition 2.1.8: Tensor Representation of Coinvariants

$M_G \cong \mathbb{Z} \otimes_{\mathbb{Z}[G]} M$, where $\mathbb{Z}$ is a right $\mathbb{Z}[G]$-module via the augmentation.

***Proof*:**
The tensor product $\mathbb{Z} \otimes_{\mathbb{Z}[G]} M$ satisfies $1 \otimes (g \cdot m) = g^{-1} \cdot 1 \otimes m = 1 \otimes m$ (since g^{-1} acts trivially on $\mathbb{Z}$), so the relation $g \cdot m \sim m$ is imposed. The universal property of the tensor product identifies $\mathbb{Z} \otimes_{\mathbb{Z}[G]} M$ with M_G.

The invariant and coinvariant functors form an adjoint pair with the trivial-action functor, and this adjunction determines their exactness properties.

Proposition 2.1.9: Exactness of Invariants and Coinvariants

The functor $(-)^G \colon G\text{-Mod} \to \mathbf{Ab}$ is left-exact, and the functor $(-)_G \colon G\text{-Mod} \to \mathbf{Ab}$ is right-exact.

***Proof*:**
Let $T \colon \mathbf{Ab} \to G\text{-Mod}$ denote the trivial-action functor, which sends an abelian group A to A with trivial G-action. This functor is exact (a sequence in $\mathbf{Ab}$ is exact if and only if it is exact as a sequence of G-modules with trivial action). For any G-module M and abelian group A:

$$\mathrm{Hom}_G(T(A), M) \cong \mathrm{Hom}_{\mathbb{Z}}(A, M^G) \qquad \text{and} \qquad \mathrm{Hom}_G(M, T(A)) \cong \mathrm{Hom}_{\mathbb{Z}}(M_G, A)$$

The first adjunction shows $(-)^G$ is right adjoint to T, hence left-exact. The second shows $(-)_G$ is left adjoint to T, hence right-exact.

Neither functor is exact in general—this is the fundamental fact that motivates group cohomology. The following example makes the failure concrete.

Example 2.1.10: Failure of Exactness
Let $G = C_2 = \langle \sigma \rangle$ and consider the augmentation sequence:

$$0 \to I_G \to \mathbb{Z}[G] \xrightarrow{\epsilon} \mathbb{Z} \to 0$$

Here $I_G = \mathbb{Z} \cdot (\sigma - 1)$ (free of rank 1 as a $\mathbb{Z}$-module). Taking $(-)^G$: the module $\mathbb{Z}[G]^G = \mathbb{Z} \cdot (1 + \sigma)$ consists of elements $a + a\sigma$ for $a \in \mathbb{Z}$. The augmentation restricted to this subgroup sends $a(1 + \sigma) \mapsto 2a$, so the image is $2\mathbb{Z} \subsetneq \mathbb{Z} = \mathbb{Z}^G$. The functor $(-)^G$ fails to preserve surjectivity.

The same sequence demonstrates that $(-)_G$ is not left-exact. Applying $(-)_G$: the coinvariants $\mathbb{Z}[G]_G = \mathbb{Z}[G]/(\sigma-1)\mathbb{Z}[G] \cong \mathbb{Z}$ (via ϵ), while $(I_G)_G = I_G/(\sigma-1)I_G$. Since $(\sigma-1)^2 = \sigma^2 - 2\sigma + 1 = -2(\sigma-1)$ in $\mathbb{Z}[G]$, the coinvariants $(I_G)_G \cong \mathbb{Z}/2\mathbb{Z}$, and the induced map $(I_G)_G \to \mathbb{Z}[G]_G$ sends the generator to 0. The injection $I_G \hookrightarrow \mathbb{Z}[G]$ does not remain injective after taking coinvariants.

Exercise 2.1.1

1. Verify the two adjunctions in the proof of proposition 2.1.9 by constructing explicit natural bijections.
2. Let G be a finite group. Show that $\mathbb{Z}[G]^G$ is the free $\mathbb{Z}$-module of rank 1 generated by the *norm element* $N_G = \sum_{g\in G} g$, and that $\mathbb{Z}[G]_G \cong \mathbb{Z}$ via the augmentation.
3. Let $G = C_3 = \langle\sigma\rangle$ act on $M = \mathbb{Z}^2$ by $\sigma \cdot (a,b) = (-b, a-b)$. Compute M^G and M_G.
4. Show that the functor $(-)^G\colon G\text{-Mod} \to \mathbf{Ab}$ is exact if and only if $G = \{1\}$.
5. Let G be a group. Show that the map $G^{\text{ab}} = G/[G,G] \to I_G/I_G^2$ defined by $g[G,G] \mapsto (g-1) + I_G^2$ is a well-defined isomorphism of abelian groups.

2.2 Group Cohomology via Derived Functors

The failure of $(-)^G$ to preserve surjections, made concrete in example 2.1.10, induces the right-derived functors of $(-)^G$ measuring the obstruction to exactness in each degree:

Definition 2.2.1: Group Cohomology and Homology

Let G be a group and M a G-module. The *nth cohomology group* of G with coefficients in M is:

$$H^n(G,M) = R^n((-)^G)(M) = \operatorname{Ext}^n_{\mathbb{Z}[G]}(\mathbb{Z}, M)$$

where $\mathbb{Z}$ carries the trivial G-action. The *nth homology group* of G with coefficients in M is:

$$H_n(G,M) = L_n((-)_G)(M) = \operatorname{Tor}_n^{\mathbb{Z}[G]}(\mathbb{Z}, M)$$

The two expressions for cohomology agree because $M^G = \operatorname{Hom}_G(\mathbb{Z}, M)$ (by corollary 2.1.6): deriving the right-hand side via an injective resolution of M gives $R^n\operatorname{Hom}_G(\mathbb{Z}, -)$, and deriving via a projective resolution of $\mathbb{Z}$ gives $\operatorname{Ext}^n_G(\mathbb{Z}, -)$. Both compute the same derived functor (see [9]).

The Ext description is often more practical: rather than resolving each coefficient module M individually, one resolves $\mathbb{Z}$ once and applies $\operatorname{Hom}_G(-, M)$ to the resolution. The standard resolution of section 2.3 provides a canonical choice.

Proposition 2.2.2: Long Exact Sequence in Cohomology

A short exact sequence of G-modules $0 \to A \to B \to C \to 0$ induces a long exact sequence in cohomology:

$$0 \to A^G \to B^G \to C^G \xrightarrow{\delta^0} H^1(G, A) \to H^1(G, B) \to H^1(G, C) \xrightarrow{\delta^1} H^2(G, A) \to \cdots$$

and a long exact sequence in homology:

$$\cdots \to H_1(G, C) \xrightarrow{\partial_1} H_0(G, A) \to H_0(G, B) \to H_0(G, C) \to 0$$

The connecting homomorphisms δ^n and ∂_n are natural in the short exact sequence.

***Proof*:**
This is the long exact sequence associated to a left-exact functor applied to a short exact sequence (for cohomology) and a right-exact functor (for homology). The construction is given in [9, Long Exact Sequence From a Right-Derived Functor, Long Exact Sequence From Derived Functor]. The connecting homomorphisms arise from the snake lemma applied to the cochain (resp. chain) complexes obtained from injective (resp. projective) resolutions.

The long exact sequence is the primary computational tool in group cohomology: given information about two of the three modules in a short exact sequence, the connecting homomorphisms transfer it to the third. The concrete description of δ^0 will be made explicit through cocycles in section 2.4.

Proposition 2.2.3: Degree Zero

For any group G and G-module M: $H^0(G, M) = M^G$ and $H_0(G, M) = M_G$.

***Proof*:**
$H^0(G, M) = R^0((-)^G)(M) = M^G$ since the zeroth derived functor of a left-exact functor is the functor itself. Similarly, $H_0(G, M) = L_0((-)_G)(M) = M_G$.

The higher cohomology groups $H^n(G, M)$ for $n \geq 1$ therefore represent the "higher-order obstructions" beyond the fixed-point functor: H^1 measures the failure of surjectivity (as will be made precise in section 2.4), H^2 classifies extensions, and so on. A module for which these obstructions all vanish is called acyclic.

To produce acyclic modules, the following construction is fundamental.

Definition 2.2.4: Induced and Coinduced Modules

Let G be a group and M an abelian group (with no G-action). The *coinduced module* is:

$$\mathrm{CoInd}^G(M) = \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}[G], M)$$

with the G-action $(g \cdot f)(x) = f(xg)$ for $g \in G$, $x \in \mathbb{Z}[G]$, and $f \in \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}[G], M)$. The *induced module* is:

$$\mathrm{Ind}^G(M) = \mathbb{Z}[G] \otimes_{\mathbb{Z}} M$$

with the G-action $g \cdot (x \otimes m) = (gx) \otimes m$.

When G is infinite, $\mathrm{Ind}^G(M)$ and $\mathrm{CoInd}^G(M)$ are distinct: the former consists of finite formal sums $\sum g_i \otimes m_i$, while the latter consists of all functions $G \to M$ (not necessarily finitely supported). When G is finite, the two are isomorphic as G-modules—the isomorphism $\alpha\colon \mathrm{Ind}^G(M) \xrightarrow{\sim} \mathrm{CoInd}^G(M)$ sends $g \otimes m$ to the function $f_{g,m}$ defined by $f_{g,m}(x) = m$ if $x = g^{-1}$ and $f_{g,m}(x) = 0$ otherwise (the reader should verify G-equivariance in exercise 2.2.1 (4)).

The coinduced module satisfies an adjunction that makes it the "right" notion for cohomological vanishing.

Proposition 2.2.5: Adjunction for Coinduced Modules

For any G-module N and abelian group M:

$$\mathrm{Hom}_G(N, \mathrm{CoInd}^G(M)) \cong \mathrm{Hom}_{\mathbb{Z}}(N, M)$$

The isomorphism is natural in both N and M, and sends $\psi \in \mathrm{Hom}_G(N, \mathrm{CoInd}^G(M))$ to the map $n \mapsto \psi(n)(1)$.

***Proof*:**
Given $\psi\colon N \to \mathrm{CoInd}^G(M) = \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}[G], M)$, define $\Phi(\psi)\colon N \to M$ by $\Phi(\psi)(n) = \psi(n)(1)$. This is a $\mathbb{Z}$-module homomorphism. Conversely, given $\varphi\colon N \to M$, define $\Psi(\varphi)\colon N \to \mathrm{CoInd}^G(M)$ by $\Psi(\varphi)(n)(x) = \varphi(x \cdot n)$ for $x \in \mathbb{Z}[G]$. The G-equivariance of $\Psi(\varphi)$ follows from: $(g \cdot \Psi(\varphi)(n))(x) = \Psi(\varphi)(n)(xg) = \varphi(xg \cdot n)$ and $\Psi(\varphi)(g \cdot n)(x) = \varphi(x \cdot (g \cdot n)) = \varphi((xg) \cdot n)$, which agree. The maps Φ and Ψ are inverse to each other: $\Phi(\Psi(\varphi))(n) = \Psi(\varphi)(n)(1) = \varphi(1 \cdot n) = \varphi(n)$ and $\Psi(\Phi(\psi))(n)(x) = \Phi(\psi)(x \cdot n) = \psi(x \cdot n)(1) = (x \cdot \psi(n))(1) = \psi(n)(x)$ (using G-equivariance of ψ for the third equality).

Theorem 2.2.6: Coinduced Modules Are Acyclic

For any abelian group M: $H^n(G, \mathrm{CoInd}^G(M)) = 0$ for all $n \geq 1$.

***Proof*:**
Let $P_\bullet \to \mathbb{Z}$ be a projective resolution of $\mathbb{Z}$ in G-Mod. The cohomology $H^n(G, \mathrm{CoInd}^G(M))$ is the nth cohomology of the complex $\mathrm{Hom}_G(P_\bullet, \mathrm{CoInd}^G(M))$. By proposition 2.2.5, each term satisfies $\mathrm{Hom}_G(P_i, \mathrm{CoInd}^G(M)) \cong \mathrm{Hom}_{\mathbb{Z}}(P_i, M)$, and this isomorphism is compatible with the

differentials. The complex $\mathrm{Hom}_{\mathbb{Z}}(P_\bullet, M)$ computes $\mathrm{Ext}^n_{\mathbb{Z}}(\mathbb{Z}, M)$: each P_i is projective over $\mathbb{Z}[G]$, and since $\mathbb{Z}[G]$ is free as a $\mathbb{Z}$-module, each P_i is projective (in fact free) as a $\mathbb{Z}$-module. So $P_\bullet \to \mathbb{Z}$ is a projective resolution of $\mathbb{Z}$ over $\mathbb{Z}$. Since $\mathbb{Z}$ is itself free over $\mathbb{Z}$, the resolution splits, giving $\mathrm{Ext}^n_{\mathbb{Z}}(\mathbb{Z}, M) = 0$ for $n \geq 1$.

Coinduced modules are the group-theoretic analogue of flasque sheaves in sheaf cohomology: they are "large enough" to absorb all cohomological obstructions. The corresponding statement for induced modules and homology holds by a dual argument: $H_n(G, \mathrm{Ind}^G(M)) = 0$ for $n \geq 1$. Rather than an adjunction, this follows from the associativity of the tensor product: the complex $P_\bullet \otimes_{\mathbb{Z}[G]} (\mathbb{Z}[G] \otimes_{\mathbb{Z}} M)$ is isomorphic to $P_\bullet \otimes_{\mathbb{Z}} M$, which computes $\mathrm{Tor}^{\mathbb{Z}}_n(\mathbb{Z}, M) = 0$.

Example 2.2.7: Acyclicity in Practice

1. For the trivial group $G = \{1\}$, every G-module M satisfies $M^G = M$, and the functor $(-)^G$ is exact. Thus $H^n(\{1\}, M) = 0$ for all $n \geq 1$ and all M.
2. Let G be a finite group. Since $\mathrm{Ind}^G(\mathbb{Z}) = \mathbb{Z}[G] \cong \mathrm{CoInd}^G(\mathbb{Z})$ (the isomorphism holding because G is finite), theorem 2.2.6 gives $H^n(G, \mathbb{Z}[G]) = 0$ for all $n \geq 1$. The group ring is cohomologically trivial. More generally, $H^n(G, \mathbb{Z}[G] \otimes_{\mathbb{Z}} A) = 0$ for any abelian group A and $n \geq 1$, since $\mathbb{Z}[G] \otimes_{\mathbb{Z}} A \cong \mathrm{Ind}^G(A) \cong \mathrm{CoInd}^G(A)$.
3. For any group G (possibly infinite), $\mathrm{CoInd}^G(\mathbb{Q}/\mathbb{Z}) = \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}[G], \mathbb{Q}/\mathbb{Z})$ is acyclic and injective as a $\mathbb{Z}[G]$-module (since $\mathbb{Q}/\mathbb{Z}$ is injective over $\mathbb{Z}$ and $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}[G], -)$ preserves injectivity).

Exercise 2.2.1

1. Show that $\{H^n(G, -)\}_{n \geq 0}$ is a universal δ-functor (as defined in [9]), using the fact that every G-module embeds into a coinduced module (hence an acyclic module).
2. Let $0 \to A \to B \to C \to 0$ and $0 \to A' \to B' \to C' \to 0$ be short exact sequences of G-modules, and let (f, g, h) be a morphism between them. Show that the connecting homomorphisms satisfy $\delta^n \circ h^* = f^* \circ \delta^n$ for all $n \geq 0$, where $h^*\colon H^n(G, C) \to H^n(G, C')$ and $f^*\colon H^{n+1}(G, A) \to H^{n+1}(G, A')$ are the induced maps.
3. Verify directly (without using derived functors) that $\mathrm{CoInd}^G(M)^G \cong M$ for any abelian group M, and describe the isomorphism explicitly.
4. Let G be a finite group. Show that the map $\alpha\colon \mathrm{Ind}^G(M) \to \mathrm{CoInd}^G(M)$ defined by $\alpha(g \otimes m)(x) = \begin{cases} m & x = g^{-1} \\ 0 & x \neq g^{-1} \end{cases}$ (extended $\mathbb{Z}$-linearly) is a G-module isomorphism.
5. Show that cohomology commutes with arbitrary direct products: for any family $\{M_i\}_{i \in I}$ of G-modules, the natural map $H^n(G, \prod_i M_i) \to \prod_i H^n(G, M_i)$ is an isomorphism for all $n \geq 0$.

2.3 The Standard Resolution and Explicit Cocycles

The derived functor definition of $H^n(G, M)$ is conceptually clean but not directly computable: it requires choosing a resolution and taking cohomology. This section provides a canonical projective resolution of $\mathbb{Z}$ over $\mathbb{Z}[G]$—the standard (or bar) resolution—and translates the abstract cohomology into explicit formulas for cocycles and coboundaries. These formulas make the low-degree cohomology groups computable by hand and are the foundation for the interpretations developed in section 2.4.

Definition 2.3.1: Standard Resolution

Let G be a group. The *standard resolution* (or *bar resolution*) of $\mathbb{Z}$ by free $\mathbb{Z}[G]$-modules is the complex:

$$\cdots \to \mathbb{Z}[G^{n+1}] \xrightarrow{d_n} \mathbb{Z}[G^n] \to \cdots \xrightarrow{d_1} \mathbb{Z}[G] \xrightarrow{\epsilon} \mathbb{Z} \to 0$$

where $\mathbb{Z}[G^{n+1}]$ is the free abelian group on $(n+1)$-tuples $(g_0, \ldots, g_n)$ of elements of G, with G acting diagonally: $g \cdot (g_0, \ldots, g_n) = (gg_0, \ldots, gg_n)$. The differentials are the alternating face maps:

$$d_n(g_0, \ldots, g_n) = \sum_{i=0}^{n} (-1)^i (g_0, \ldots, \widehat{g_i}, \ldots, g_n)$$

where $\widehat{g_i}$ indicates omission, and ϵ is the augmentation from definition 2.1.4.

Each $\mathbb{Z}[G^{n+1}]$ is a free $\mathbb{Z}[G]$-module: writing $(g_0, \ldots, g_n) = g_0 \cdot (1, g_0^{-1}g_1, \ldots, g_0^{-1}g_n)$, the module $\mathbb{Z}[G^{n+1}]$ is free on the set $\{(1, h_1, \ldots, h_n) : h_i \in G\}$, which is a copy of G^n.

Lemma 2.3.2: Exactness of the Standard Resolution

The standard resolution is exact.

***Proof*:**
Define $\mathbb{Z}$-module maps $s_n : \mathbb{Z}[G^{n+1}] \to \mathbb{Z}[G^{n+2}]$ by $s_n(g_0, \ldots, g_n) = (1, g_0, \ldots, g_n)$, and $s_{-1} : \mathbb{Z} \to \mathbb{Z}[G]$ by $s_{-1}(1) = (1)$. A direct computation gives $d_{n+1} \circ s_n + s_{n-1} \circ d_n = \mathrm{id}$ for all $n \geq 0$, and $\epsilon \circ s_{-1} = \mathrm{id}_{\mathbb{Z}}$. This exhibits $\{s_n\}$ as a contracting homotopy for the augmented complex, proving exactness.

The identity $d_{n+1}s_n + s_{n-1}d_n = \mathrm{id}$ is verified on generators: the term $d_{n+1}s_n(g_0, \ldots, g_n) = d_{n+1}(1, g_0, \ldots, g_n)$ produces $(g_0, \ldots, g_n)$ from the $i = 0$ face plus alternating terms $(1, g_0, \ldots, \widehat{g_j}, \ldots, g_n)$ for $j \geq 0$. The term $s_{n-1}d_n(g_0, \ldots, g_n)$ produces the same alternating terms with opposite signs, and the two cancel, leaving only $(g_0, \ldots, g_n)$.

The homotopy s_n is *not* G-equivariant (it prepends the identity element 1, which is not preserved by the G-action). This is expected: a contracting homotopy of G-module maps would split the resolution, forcing all cohomology to vanish.

Applying $\mathrm{Hom}_G(-, M)$ to the standard resolution produces a cochain complex whose cohomology is $H^*(G, M)$. The resulting cochains admit a concrete description.

Definition 2.3.3: Inhomogeneous Cochains

Let G be a group and M a G-module. An *inhomogeneous n-cochain* is a function $f\colon G^n \to M$ (with no equivariance condition). The set of all n-cochains is $C^n(G,M) = \mathrm{Map}(G^n, M)$, an abelian group under pointwise addition. The *coboundary map* $d^n\colon C^n(G,M) \to C^{n+1}(G,M)$ is:

$$\begin{aligned}(d^n f)(g_1,\dots,g_{n+1}) = g_1 \cdot f(g_2,\dots,g_{n+1}) \\ &+ \sum_{i=1}^{n}(-1)^i f(g_1,\dots,g_i g_{i+1},\dots,g_{n+1}) \\ &+ (-1)^{n+1} f(g_1,\dots,g_n)\end{aligned}$$

The *n-cocycles* are $Z^n(G,M) = \ker d^n$ and the *n-coboundaries* are $B^n(G,M) = \mathrm{im} d^{n-1}$ (with $B^0 = 0$).

The inhomogeneous cochains arise from the isomorphism $\mathrm{Hom}_G(\mathbb{Z}[G^{n+1}], M) \cong \mathrm{Map}(G^n, M)$ that sends a G-equivariant map $\varphi\colon \mathbb{Z}[G^{n+1}] \to M$ to the function $\bar{\varphi}(g_1,\dots,g_n) = \varphi(1, g_1, g_1g_2, \dots, g_1\cdots g_n)$. The coboundary formula is the transport of the dual differential $d_n^* = \mathrm{Hom}_G(d_n, M)$ through this isomorphism.

Proposition 2.3.4: Cocycle Description of Cohomology

For any group G and G-module M:

$$H^n(G,M) \cong \frac{Z^n(G,M)}{B^n(G,M)}$$

***Proof*:**
The standard resolution is a free (hence projective) resolution of $\mathbb{Z}$ by definition 2.3.1 and lemma 2.3.2. Applying $\mathrm{Hom}_G(-,M)$ gives the cochain complex $(C^*(G,M), d^*)$ via the inhomogeneous identification. Its cohomology is $\mathrm{Ext}^n_G(\mathbb{Z}, M) = H^n(G,M)$.

The cocycle description makes the low-degree coboundary maps explicit. In degree 0: $C^0(G,M) = M$ (functions from $G^0 = \{*\}$ to M), and $(d^0 m)(g) = g\cdot m - m$. So $Z^0(G,M) = M^G$ and $B^0 = 0$, recovering $H^0(G,M) = M^G$. The content of the cocycle description begins in degree 1.

Definition 2.3.5: Crossed Homomorphism

A *crossed homomorphism* (or 1-*cocycle*) from G to M is a function $f\colon G \to M$ satisfying:

$$f(gh) = g \cdot f(h) + f(g)$$

for all $g, h \in G$. The group of all crossed homomorphisms is $Z^1(G,M)$.

When G acts trivially on M, the cocycle condition reduces to $f(gh) = f(g) + f(h)$: ordinary group homomorphisms $G \to M$.

Definition 2.3.6: Principal Crossed Homomorphism

A *principal crossed homomorphism* (or 1-*coboundary*) is a crossed homomorphism of the form $f_m(g) = g \cdot m - m$ for some fixed $m \in M$. The group of all principal crossed homomorphisms is $B^1(G, M)$.

A principal crossed homomorphism is the "trivial" part of a cocycle: it measures how far an element m is from being G-fixed, and contributes nothing to $H^1(G, M)$.

In degree 2, the cocycle condition encodes the associativity constraint for group extensions.

Definition 2.3.7: 2-Cocycle

A 2-*cocycle* from G to M is a function $f\colon G \times G \to M$ satisfying the *cocycle condition*:

$$g_1 \cdot f(g_2, g_3) - f(g_1g_2, g_3) + f(g_1, g_2g_3) - f(g_1, g_2) = 0$$

for all $g_1, g_2, g_3 \in G$. The group of 2-cocycles is $Z^2(G, M)$, and the 2-coboundaries $B^2(G, M)$ consist of functions of the form $(g_1, g_2) \mapsto g_1 \cdot \psi(g_2) - \psi(g_1g_2) + \psi(g_1)$ for a 1-cochain $\psi\colon G \to M$.

The cocycle condition for $f \in Z^2(G, M)$ is the precise analogue of the associativity axiom for a group multiplication twisted by f. The interpretation of $H^2(G, M)$ as classifying group extensions with this structure is developed in section 2.4.

The standard resolution is universal but not always efficient. For specific groups, smaller resolutions yield faster computations. The most important case is cyclic groups, where the relation $\sigma^n = 1$ produces a periodic resolution.

Example 2.3.8: Cohomology of Cyclic Groups

Let $G = C_n = \langle \sigma \rangle$ be cyclic of order n. The elements $\sigma - 1$ and $N = \sum_{i=0}^{n-1} \sigma^i$ (the norm) satisfy $N(\sigma - 1) = \sigma^n - 1 = 0$ and $(\sigma - 1)N = 0$ in $\mathbb{Z}[G]$. This yields a periodic free resolution:

$$\cdots \xrightarrow{\sigma-1} \mathbb{Z}[G] \xrightarrow{N} \mathbb{Z}[G] \xrightarrow{\sigma-1} \mathbb{Z}[G] \xrightarrow{\epsilon} \mathbb{Z} \to 0$$

Applying $\mathrm{Hom}_G(-, M)$ and using $\mathrm{Hom}_G(\mathbb{Z}[G], M) \cong M$ gives the complex:

$$0 \to M \xrightarrow{\sigma-1} M \xrightarrow{N} M \xrightarrow{\sigma-1} M \xrightarrow{N} \cdots$$

where $(\sigma - 1)(m) = \sigma \cdot m - m$ and $N(m) = \sum_{i=0}^{n-1} \sigma^i \cdot m$. The cohomology is 2-periodic for $k \geq 1$:

$$H^k(C_n, M) = \begin{cases} M^G/NM & k \text{ even, } k \geq 2 \\ {}_NM/(\sigma - 1)M & k \text{ odd, } k \geq 1 \end{cases}$$

where ${}_NM = \ker(N\colon M \to M)$ is the *norm kernel*.

As a concrete application, take $M = \mathbb{Z}$ with trivial action. Then $\sigma - 1$ acts as 0 and N acts as multiplication by n. The cohomology is:

$$H^k(C_n, \mathbb{Z}) = \begin{cases} \mathbb{Z}/n\mathbb{Z} & k \text{ even, } k \geq 2 \\ 0 & k \text{ odd, } k \geq 1 \end{cases}$$

This periodicity is special to cyclic groups and will be generalized by Tate cohomology in Chapter 4.

The periodic description has a companion that the colimit computations of chapter 3 depend on: what the inflation maps do as the cyclic group grows. Surjecting a larger cyclic group onto a smaller one and inflating turns out to be multiplication by a power of the index, and the power grows with the degree, which is what makes cohomology in high degrees die in the limit.

Proposition 2.3.9: Inflation Between Cyclic Groups

Let $C = \langle\sigma\rangle$ be cyclic of order N, let $C' = \langle\sigma'\rangle$ be cyclic of order dN, and let $\pi\colon C' \to C$ be the surjection with $\pi(\sigma') = \sigma$. Regard a C-module M as a C'-module through π, so that $M^{C'} = M^C$ and the norm element of C' acts on M as $d\,N_C$. Under the identifications of example 2.3.8 for C and for C', the inflation map

$$\mathrm{Inf}\colon H^i(C, M) \longrightarrow H^i(C', M)$$

is induced by multiplication by d^j on M, where $j = \lfloor i/2 \rfloor$.

***Proof*:**
Inflation is computed by lifting $\mathrm{id}_{\mathbb{Z}}$ to a morphism from the periodic resolution of $\mathbb{Z}$ over $\mathbb{Z}[C']$ to the one over $\mathbb{Z}[C]$, the latter regarded as a complex of C'-modules through π. Write $D = \sigma - 1$ and N_C for the two elements of $\mathbb{Z}[C]$ appearing in proposition 4.3.2, and $D' = \sigma' - 1$, $N_{C'}$ for those of $\mathbb{Z}[C']$. Note $\pi(D') = D$ and

$$\pi(N_{C'}) = \sum_{t=0}^{dN-1} \sigma^t = d\,N_C$$

since π hits each power of σ exactly d times.

A C'-map $f_k\colon \mathbb{Z}[C'] \to \mathbb{Z}[C]$ is determined by $a_k := f_k(1)$, namely $f_k(x) = \pi(x)\,a_k$. The lifting conditions read off one square at a time. Degree 0 needs $\epsilon(a_0) = 1$, so $a_0 = 1$. The square $f_0 D' = D f_1$ gives $D = D\,a_1$, so $a_1 = 1$. The square $f_1 N_{C'} = N_C f_2$ gives $d\,N_C = N_C\,a_2$, so $a_2 = d$. The next two squares give $D\,d = D\,a_3$ and $d\,N_C\,d = N_C\,a_4$, so $a_3 = d$ and $a_4 = d^2$. Inductively the factor is picked up once per period, at each passage through $N_{C'}$:

$$a_{2j} = a_{2j+1} = d^j$$

Applying $\mathrm{Hom}_{C'}(-, M)$ and using $\mathrm{Hom}_{C'}(\mathbb{Z}[C'], M) \cong M$ and $\mathrm{Hom}_C(\mathbb{Z}[C], M) \cong M$, the map induced by f_k in degree k is multiplication by a_k on M. Passing to cohomology identifies Inf in degree i with the map induced by multiplication by $d^{\lfloor i/2 \rfloor}$, as we sought to show.

Example 2.3.10: Cohomology of Free Groups

Let $F = F_S$ be the free group on a set S. The augmentation ideal I_F is a free $\mathbb{Z}[F]$-module with basis $\{s - 1 : s \in S\}$ (this can be verified using the universal property of free groups, or by the Fox calculus). The augmentation sequence $0 \to I_F \to \mathbb{Z}[F] \xrightarrow{\epsilon} \mathbb{Z} \to 0$ is therefore a free resolution of length 1, giving:

$$H^n(F_S, M) = 0 \quad \text{for all } n \geq 2$$

and $H^1(F_S, M) \cong \mathrm{Hom}_G(I_F, M) \cong \prod_{s\in S} M$ (choosing the image of each $s-1$ independently). Free groups have *cohomological dimension* 1—a concept developed systematically in Chapter 5.

2.3.11 Remark: Connection to Singular Cohomology (If the reader has read [16]) The standard resolution has a topological origin. The classifying space BG of a discrete group G satisfies $\pi_1(BG) = G$ and $\pi_n(BG) = 0$ for $n \geq 2$. Its singular chain complex, computed via the bar construction on $EG \to BG$, recovers the standard resolution. The resulting isomorphism:

$$H^n(G, M) \cong H^n_{\mathrm{Sing}}(BG, M)$$

holds for any G-module M (with local coefficients when the action is nontrivial). This identifies group cohomology with the singular cohomology of a topological space and explains the simplicial character of the cocycle formulas.

Exercise 2.3.1

1. Verify the identity $d_{n+1} \circ s_n + s_{n-1} \circ d_n = \mathrm{id}$ from the proof of lemma 2.3.2 by direct computation for $n = 1$.
2. Write out the coboundary maps d^0, d^1, d^2, and d^3 in inhomogeneous form explicitly, and verify that $d^1 \circ d^0 = 0$ and $d^2 \circ d^1 = 0$.
3. Compute $H^n(C_2, \mathbb{Z}/2\mathbb{Z})$ for all $n \geq 0$, where $C_2 = \langle \sigma \rangle$ acts trivially on $\mathbb{Z}/2\mathbb{Z}$.
4. Let $G = \mathbb{Z}$ (as a discrete group) and M a G-module, with the generator $1 \in \mathbb{Z}$ acting by an automorphism $\varphi\colon M \to M$. Using the resolution $0 \to \mathbb{Z}[\mathbb{Z}] \xrightarrow{t-1} \mathbb{Z}[\mathbb{Z}] \xrightarrow{\epsilon} \mathbb{Z} \to 0$ (where t is the generator), show that $H^0(\mathbb{Z}, M) = M^{\mathbb{Z}} = \ker(\varphi - \mathrm{id})$, $H^1(\mathbb{Z}, M) = M/(\varphi - \mathrm{id})M$, and $H^n(\mathbb{Z}, M) = 0$ for $n \geq 2$.
5. (If the reader has read [9]) Show that for groups G_1, G_2 and a trivial $G_1 \times G_2$-module M, there is a Künneth isomorphism:

   $$H^n(G_1 \times G_2, M) \cong \bigoplus_{p+q=n} H^p(G_1, \mathbb{Z}) \otimes_{\mathbb{Z}} H^q(G_2, M) \oplus \bigoplus_{p+q=n+1} \mathrm{Tor}_1^{\mathbb{Z}}(H^p(G_1, \mathbb{Z}), H^q(G_2, M))$$

   Verify this for $G_1 = G_2 = C_2$ and $M = \mathbb{Z}$.

2.4 Low-Degree Cohomology: Interpretations

The abstract definition of $H^n(G, M)$ as a derived functor acquires concrete meaning in low degrees. The group H^0 is the invariant subgroup (already known). The group H^1 classifies crossed homomorphisms modulo principal ones, which amounts to measuring an obstruction to lifting fixed points. The group H^2 classifies group extensions. These interpretations connect the cohomological formalism to explicit algebraic problems.

The first interpretation makes the connecting homomorphism from the long exact sequence (proposition 2.2.2) concrete through the cocycle language of section 2.3.

Proposition 2.4.1: The Connecting Homomorphism via Cocycles

Let $0 \to A \xrightarrow{\iota} B \xrightarrow{\pi} C \to 0$ be a short exact sequence of G-modules, and let $c \in C^G$. Choose any lift $b \in B$ with $\pi(b) = c$. Then the function $\lambda\colon G \to A$ defined by $\iota(\lambda(g)) = g \cdot b - b$ is a crossed homomorphism (definition 2.3.5), and the connecting homomorphism sends:

$$\delta^0(c) = [\lambda] \in H^1(G, A)$$

The class $[\lambda]$ is independent of the choice of lift b.

***Proof*:**
Since $c \in C^G$, the element $g \cdot b - b$ satisfies $\pi(g \cdot b - b) = g \cdot c - c = 0$, so $g \cdot b - b \in \ker \pi = \operatorname{im} \iota$. The function λ is well-defined because ι is injective. The crossed homomorphism condition follows from:

$$\begin{aligned} \iota(\lambda(gh)) &= gh \cdot b - b = g \cdot (h \cdot b - b) + (g \cdot b - b) \\ &= g \cdot \iota(\lambda(h)) + \iota(\lambda(g)) = \iota(g \cdot \lambda(h) + \lambda(g)) \end{aligned}$$

so $\lambda(gh) = g \cdot \lambda(h) + \lambda(g)$. A different lift $b' = b + \iota(a)$ for some $a \in A$ changes the cocycle to $\lambda'(g) = \lambda(g) + g \cdot a - a$, which differs from λ by the principal crossed homomorphism $g \mapsto g \cdot a - a$ (definition 2.3.6). So $[\lambda'] = [\lambda]$ in $H^1(G, A)$.

The connecting homomorphism δ^0 measures a precise obstruction: the element $c \in C^G$ lifts to a G-fixed element of B if and only if $\delta^0(c) = 0$. The lift exists precisely when the crossed homomorphism λ is principal—when b can be adjusted by an element of A to become G-fixed.

Example 2.4.2: The Connecting Homomorphism in Action

Let $G = C_2 = \langle \sigma \rangle$ act on $\mathbb{Z}$ by negation ($\sigma \cdot m = -m$), and consider the exact sequence $0 \to \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0$, where the first map is multiplication by 2. All three modules carry the negation action. Since $-1 \equiv 1 \pmod 2$, the action on $\mathbb{Z}/2\mathbb{Z}$ is trivial, giving $(\mathbb{Z}/2\mathbb{Z})^{C_2} = \mathbb{Z}/2\mathbb{Z}$. Meanwhile $\mathbb{Z}^{C_2} = 0$ (the only integer fixed by negation is 0).

The connecting homomorphism $\delta^0\colon \mathbb{Z}/2\mathbb{Z} \to H^1(C_2, \mathbb{Z})$ sends $\bar{1} \in \mathbb{Z}/2\mathbb{Z}$ to the cocycle class computed as follows: lift $\bar{1}$ to $b = 1 \in \mathbb{Z}$. The cocycle is $\lambda(\sigma) = \frac{1}{2}(\sigma \cdot 1 - 1) = \frac{1}{2}(-1 - 1) = -1$ (using ι^{-1}, i.e., dividing by 2). Using the periodic resolution, $H^1(C_2, \mathbb{Z}) = {}_N\mathbb{Z}/(\sigma - 1)\mathbb{Z} = \mathbb{Z}/2\mathbb{Z}$ (with the negation action: $(\sigma - 1)m = -2m$ and $N(m) = m + (-m) = 0$). The cocycle $\lambda(\sigma) = -1$ represents the nontrivial class, so δ^0 is an isomorphism $\mathbb{Z}/2\mathbb{Z} \xrightarrow{\sim} \mathbb{Z}/2\mathbb{Z}$.

When G acts trivially, the crossed homomorphism condition simplifies and H^1 reduces to ordinary homomorphisms.

Proposition 2.4.3: H^1 with Trivial Action

If G acts trivially on M, then $H^1(G, M) \cong \operatorname{Hom}(G, M)$.

***Proof*:**
With trivial action, a crossed homomorphism $f\colon G \to M$ satisfies $f(gh) = f(g) + f(h)$ (by definition 2.3.5 with $g \cdot f(h) = f(h)$), so $Z^1(G, M) = \mathrm{Hom}(G, M)$. A principal crossed homomorphism $f_m(g) = g \cdot m - m = 0$ (trivial action), so $B^1(G, M) = 0$. The quotient gives $H^1(G, M) = \mathrm{Hom}(G, M)$.

For finite G, this gives $H^1(G, \mathbb{Z}) = \mathrm{Hom}(G, \mathbb{Z}) = 0$ (no nontrivial homomorphisms from a finite group to a torsion-free group). In contrast, $H^1(G, \mathbb{Q}/\mathbb{Z}) = \mathrm{Hom}(G, \mathbb{Q}/\mathbb{Z})$ is the Pontryagin dual of G^{ab} and is typically nonzero.

The identification $H^1(G, M) = \mathrm{Hom}(G, M)$ for trivial action applies to the Galois setting: $H^1(G_K, \mathbb{Z}/n\mathbb{Z}) = \mathrm{Hom}(G_K, \mathbb{Z}/n\mathbb{Z})$ classifies cyclic extensions of K of degree dividing n (by theorem 1.4.1, continuous homomorphisms $G_K \to \mathbb{Z}/n\mathbb{Z}$ correspond to Galois extensions with cyclic Galois group). The nontrivial action case—where H^1 captures genuine arithmetic obstructions—is exemplified by Hilbert's Theorem 90 ($H^1(\mathrm{Gal}(L/K), L^\times) = 0$), which will be proved in Chapter 8.

The second cohomology group classifies group extensions. This is the analogue, for the group G, of the classification of principal bundles in topology or of torsors in algebraic geometry.

Definition 2.4.4: Group Extension

Let G be a group and M a G-module. A *group extension* of G by M (with the given action) is a short exact sequence of groups:

$$1 \to M \xrightarrow{\iota} E \xrightarrow{\pi} G \to 1$$

such that the conjugation action of E on $\iota(M)$ agrees with the G-action on M: for all $e \in E$ and $m \in M$, $e\,\iota(m)\,e^{-1} = \iota(\pi(e) \cdot m)$. Two extensions E and E' are *equivalent* if there exists a group isomorphism $\varphi\colon E \xrightarrow{\sim} E'$ making the diagram commute:

$$\begin{tikzcd}
1 \arrow[r] & M \arrow[r, "\iota"] \arrow[d, equal] & E \arrow[r, "\pi"] \arrow[d, "\varphi", "\sim"'] & G \arrow[r] \arrow[d, equal] & 1 \\
1 \arrow[r] & M \arrow[r, "\iota'"] & E' \arrow[r, "\pi'"] & G \arrow[r] & 1
\end{tikzcd}$$

The set of equivalence classes of extensions is denoted $\mathrm{Ext}(G, M)$.

The split extension $M \rtimes G$ (semidirect product with the given action) is always an extension, serving as the "trivial" class.

Theorem 2.4.5: Classification of Extensions by H^2

Let G be a group and M a G-module. There is a bijection:

$$H^2(G, M) \xrightarrow{\sim} \mathrm{Ext}(G, M)$$

Under this bijection, the zero element of $H^2(G, M)$ corresponds to the split extension $M \rtimes G$.

***Proof*:**

Step 1: From extensions to cocycles. Given an extension $1 \to M \xrightarrow{\iota} E \xrightarrow{\pi} G \to 1$, choose a set-theoretic section $s\colon G \to E$ with $\pi \circ s = \mathrm{id}_G$ and $s(1) = 1$. For $g_1, g_2 \in G$, both $s(g_1)s(g_2)$ and $s(g_1g_2)$ map to g_1g_2 under π, so their ratio lies in $\ker \pi = \iota(M)$. Define $f\colon G \times G \to M$ by:

$$\iota(f(g_1, g_2)) = s(g_1)\, s(g_2)\, s(g_1g_2)^{-1}$$

The function f is a 2-cocycle: the associativity $(s(g_1)s(g_2))s(g_3) = s(g_1)(s(g_2)s(g_3))$ in E, combined with the relation $s(g)^{-1}\iota(m)s(g) = \iota(g \cdot m)$, yields the cocycle condition of definition 2.3.7.

Step 2: Independence of section. A different section $s'(g) = s(g)\,\iota(\psi(g))$ for some function $\psi\colon G \to M$ produces a new cocycle:

$$f'(g_1, g_2) = f(g_1, g_2) + g_1 \cdot \psi(g_2) - \psi(g_1g_2) + \psi(g_1)$$

so $f' - f$ is a 2-coboundary and $[f'] = [f]$ in $H^2(G, M)$.

Step 3: From cocycles to extensions. Given a 2-cocycle $f\colon G \times G \to M$, define a group $E_f = M \times G$ (as a set) with multiplication:

$$(m_1, g_1) \cdot (m_2, g_2) = (m_1 + g_1 \cdot m_2 + f(g_1, g_2),\ g_1g_2)$$

The cocycle condition ensures associativity (the computation is a direct expansion of both sides of $(a\cdot b)\cdot c = a\cdot(b\cdot c)$, using the cocycle identity to match the M-components). The normalization $f(1, g) = f(g, 1) = 0$ (which can always be arranged by modifying f by a coboundary) ensures $(0, 1)$ is the identity. The maps $\iota(m) = (m, 1)$ and $\pi(m, g) = g$ give an extension.

Step 4: Bijection. Steps 1–3 define inverse maps between $H^2(G, M)$ and $\mathrm{Ext}(G, M)$. Equivalent extensions produce cohomologous cocycles (Step 2), and cohomologous cocycles produce equivalent extensions (the isomorphism $E_f \to E_{f'}$ is $(m, g) \mapsto (m + \psi(g), g)$ where $f' - f = d^1\psi$). The split extension $M \rtimes G$ corresponds to $f = 0$.

The classification shows that $H^2(G, M)$ is the complete obstruction to splitting an extension: a group E mapping onto G with kernel M splits (as a semidirect product) if and only if the corresponding cohomology class vanishes.

Example 2.4.6: Extensions Classified by H^2

1. By example 2.3.8, $H^2(C_n, \mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}$ (with trivial action). The n equivalence classes of extensions $0 \to \mathbb{Z} \to E \to C_n \to 1$ are represented by $E_k = \langle a, t \,|\, tat^{-1} = a,\ t^n = a^k \rangle$ for $0 \le k < n$, where a generates $\mathbb{Z}$ and t maps to a generator of C_n. The class $k = 0$ is the split extension $\mathbb{Z} \times C_n$. For $n = 2$ and $k = 1$: $E_1 = \langle a, t \,|\, t^2 = a \rangle \cong \mathbb{Z}$ (generated by t, with $a = t^2$), giving the non-split extension $0 \to \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0$.
2. The Brauer group of a field K is $\mathrm{Br}(K) \cong H^2(G_K, (K^{\mathrm{sep}})^\times)$, where the 2-cocycle $f\colon G_K \times G_K \to (K^{\mathrm{sep}})^\times$ encodes the multiplication table of a central simple K-algebra via the crossed product construction. This identification is developed in Chapter 7.

3. (If the reader has read [16]) For a connected space X with $\pi_1(X) = G$, the group $H^2(G, A)$ (trivial action, A abelian) classifies central extensions $1 \to A \to E \to G \to 1$, which correspond to fibrations over BG with fibre $K(A, 1) = BA$ (principal A-bundles are classified by $H^1(BG; A)$, one degree lower).

In higher degrees, the cohomological interpretations become progressively more complex. The group $H^3(G, M)$ classifies certain "crossed modules" (equivalently, 2-group extensions), and appears as the obstruction to lifting a homomorphism $G \to \mathrm{Out}(E)$ to a group extension with kernel E. These higher-degree interpretations, while structurally significant, are rarely needed in the number-theoretic applications of this book.

Exercise 2.4.1

1. Classify (up to equivalence) all group extensions $1 \to \mathbb{Z}/2\mathbb{Z} \to E \to \mathbb{Z}/2\mathbb{Z} \to 1$ where $\mathbb{Z}/2\mathbb{Z}$ acts trivially on $\mathbb{Z}/2\mathbb{Z}$. Identify each extension explicitly as a group of order 4.

2. Let $0 \to M \to E \to G \to 1$ be a group extension. Show that the extension splits (i.e., the corresponding class in $H^2(G, M)$ is trivial) if and only if π admits a group homomorphism section $s\colon G \to E$ with $\pi \circ s = \mathrm{id}_G$.

3. Let $0 \to A \to B \xrightarrow{\pi} C \to 0$ be a short exact sequence of G-modules. Show that if $H^1(G, A) = 0$, then $\pi^G\colon B^G \to C^G$ is surjective. Is the converse true? Give an example where π^G is surjective but $H^1(G, A) \neq 0$.

4. Let G be a finite group. The *Schur multiplier* of G is $H^2(G, \mathbb{C}^\times)$ (with trivial action). Show that $H^2(G, \mathbb{C}^\times) \cong H^2(G, \mathbb{Q}/\mathbb{Z}) \cong H_2(G, \mathbb{Z})^*$ where $(-)^*$ denotes the Pontryagin dual. *Hint:* use the exact sequence $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$ and the universal coefficient theorem.

5. Compute $H^2(C_2, \mathbb{Z}/2\mathbb{Z})$ (trivial action) using the periodic resolution, and describe the corresponding extensions explicitly.

2.5 Homology and Coinvariants

Cohomology derives the invariant functor $(-)^G$; homology derives the coinvariant functor $(-)_G$. The two theories extract different information from the same action: where cohomology measures obstructions to lifting fixed points and classifies extensions, homology captures the abelianization and relations among relations. The reason cohomology takes precedence in this book—and in the Galois-theoretic literature more broadly—is threefold: the invariant functor $(-)^G$ is arithmetically natural (fixed fields, invariant elements), cup products endow $H^*(G, -)$ with a graded ring structure that homology lacks, and the passage to profinite groups in Chapter 3 is cleaner for cohomology with discrete coefficients. Homology is nonetheless indispensable: the results of this section are used in the structure theory of pro-p-groups (Chapter 6) and in the Tate cohomology of Chapter 4, which unifies homology and cohomology into a single theory.

Group homology was defined in definition 2.2.1 as $H_n(G, M) = \mathrm{Tor}_n^{\mathbb{Z}[G]}(\mathbb{Z}, M)$. The long exact sequence in homology (proposition 2.2.2) and the identification $H_0(G, M) = M_G$ (proposition 2.2.3) are already in hand. The first task is to compute homology explicitly for the groups whose cohomology was computed in section 2.3.

Example 2.5.1: Homology of Cyclic Groups

Let $G = C_n = \langle \sigma \rangle$. The periodic resolution from example 2.3.8, applied to homology via tensoring with $\mathbb{Z}$ over $\mathbb{Z}[G]$, gives the complex:

$$\cdots \xrightarrow{0} \mathbb{Z} \xrightarrow{n} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{n} \mathbb{Z} \xrightarrow{0} \mathbb{Z}$$

where $(\sigma - 1)$ acts as 0 on $\mathbb{Z}$ (trivial action) and the norm N acts as multiplication by n. The homology is:

$$H_k(C_n, \mathbb{Z}) = \begin{cases} \mathbb{Z} & k = 0 \\ \mathbb{Z}/n\mathbb{Z} & k \text{ odd} \\ 0 & k \text{ even}, k \geq 2 \end{cases}$$

Comparing with the cohomology $H^k(C_n, \mathbb{Z})$ from example 2.3.8: the nonzero groups appear in complementary degrees—odd for homology, even for cohomology. This complementarity reflects the periodicity that Tate cohomology (Chapter 4) makes explicit.

More generally, for a C_n-module M with arbitrary action, the periodic resolution gives $H_k(C_n, M) = M^G/NM$ for even $k \geq 2$ and $H_k(C_n, M) = {}_N M/(\sigma - 1)M$ for odd $k \geq 1$, with $H_0(C_n, M) = M_G = M/(\sigma - 1)M$. Comparing with the cohomology (example 2.3.8): the formulas for H_k and H^k coincide in degrees $k \geq 1$. The difference lies entirely in degree 0: $H_0 = M_G$ (coinvariants) while $H^0 = M^G$ (invariants). This coincidence in positive degrees is specific to cyclic groups and reflects the periodicity of the resolution.

For general groups, homology is computed by tensoring a projective resolution of $\mathbb{Z}$ with M over $\mathbb{Z}[G]$ (rather than applying $\operatorname{Hom}_G(-, M)$ as for cohomology). The standard resolution of definition 2.3.1 serves this purpose: tensoring $\mathbb{Z}[G^{n+1}]$ with M over $\mathbb{Z}[G]$ gives M^{n+1} (via $\mathbb{Z}[G^{n+1}] \otimes_{\mathbb{Z}[G]} M \cong M \otimes_{\mathbb{Z}} \mathbb{Z}[G^n] \cong M^{|G^n|}$ when G is finite, or more generally, the free M-module on G^n). The resulting chain complex produces the homology via the same face maps as in definition 2.3.1, now acting on the M-component through the G-action.

The most important structural result about H_1 is that it recovers the abelianization.

Theorem 2.5.2: H_1 and the Abelianization

For any group G: $H_1(G, \mathbb{Z}) \cong G^{\mathrm{ab}} = G/[G, G]$.

***Proof*:**

Apply the long exact sequence in homology to the augmentation sequence $0 \to I_G \to \mathbb{Z}[G] \xrightarrow{\epsilon} \mathbb{Z} \to 0$ (definition 2.1.4):

$$H_1(G, \mathbb{Z}[G]) \to H_1(G, \mathbb{Z}) \xrightarrow{\partial} (I_G)_G \xrightarrow{\bar{\iota}} (\mathbb{Z}[G])_G \xrightarrow{\bar{\epsilon}} \mathbb{Z}_G \to 0$$

Step 1: Vanishing of $H_1(G, \mathbb{Z}[G])$. Since $\mathbb{Z}[G] \cong \operatorname{Ind}^G(\mathbb{Z})$, the associativity of the tensor product gives $P_\bullet \otimes_{\mathbb{Z}[G]} \mathbb{Z}[G] \cong P_\bullet$ for any projective resolution $P_\bullet \to \mathbb{Z}$, so $H_n(G, \mathbb{Z}[G]) = \operatorname{Tor}_n^{\mathbb{Z}[G]}(\mathbb{Z}, \mathbb{Z}[G]) = 0$ for $n \geq 1$.

Step 2: Computing $(I_G)_G$. The coinvariants of I_G are $I_G/\langle g \cdot x - x : g \in G, x \in I_G \rangle$. For $x \in I_G$, the element $g \cdot x - x = (g-1) \cdot x$ lies in $I_G \cdot I_G = I_G^2$ (since $g - 1 \in I_G$). Conversely, every generator

$(g_1-1)(g_2-1)$ of I_G^2 has the form $g_1 \cdot (g_2-1)-(g_2-1)$, a coinvariant relation. So $(I_G)_G = I_G/I_G^2$.

Step 3: The map $\bar{\iota}$ vanishes. The inclusion $\iota\colon I_G \hookrightarrow \mathbb{Z}[G]$ on coinvariants gives $\bar{\iota}\colon I_G/I_G^2 \to \mathbb{Z}[G]/I_G$. Since $\iota(I_G) = I_G$ maps to 0 in $\mathbb{Z}[G]/I_G$, the map $\bar{\iota}$ is the zero map.

Step 4: Conclusion. The vanishing of $H_1(G,\mathbb{Z}[G])$ and $\bar{\iota}$ reduces the long exact sequence to $0 \to H_1(G,\mathbb{Z}) \xrightarrow{\sim} I_G/I_G^2$. The isomorphism $I_G/I_G^2 \cong G^{\mathrm{ab}}$ from exercise 2.1.1 (5) (sending $\overline{g-1} \mapsto g[G,G]$) completes the proof.

The abelianization G^{ab} is the first and coarsest invariant of G: it captures the "linear" part of the group, discarding all higher-order commutator structure. The theorem says that $H_1(G,\mathbb{Z})$ detects precisely this information. For instance, $H_1(S_n,\mathbb{Z}) = S_n^{\mathrm{ab}} = \mathbb{Z}/2\mathbb{Z}$ for $n \geq 2$ (the sign homomorphism), and $H_1(F_S,\mathbb{Z}) = F_S^{\mathrm{ab}} = \mathbb{Z}^S$ for a free group on the set S.

The universal coefficient theorem relates homology and cohomology, allowing the computation of one from the other when the coefficient module is a trivial G-module.

Theorem 2.5.3: Universal Coefficient Theorem

Let G be a group and A an abelian group with trivial G-action. For each $n \geq 0$, there is a natural short exact sequence:

$$0 \to \mathrm{Ext}^1_{\mathbb{Z}}(H_{n-1}(G,\mathbb{Z}),A) \to H^n(G,A) \to \mathrm{Hom}_{\mathbb{Z}}(H_n(G,\mathbb{Z}),A) \to 0$$

This sequence splits (non-canonically).

***Proof*:**
This is a special case of the universal coefficient theorem for the cohomology of a chain complex (see [9]), applied to the chain complex $P_\bullet \otimes_{\mathbb{Z}[G]} \mathbb{Z}$ (which computes $H_*(G,\mathbb{Z})$) and the functor $\mathrm{Hom}_{\mathbb{Z}}(-,A)$ (which, applied to $P_\bullet \otimes_{\mathbb{Z}[G]} \mathbb{Z}$, computes $H^*(G,A)$ for trivial A).

Example 2.5.4: The UCT in Action
For $G = C_m$ and $A = \mathbb{Z}/k\mathbb{Z}$ (trivial action), the UCT gives:

$$0 \to \mathrm{Ext}^1_{\mathbb{Z}}(H_{n-1}(C_m,\mathbb{Z}),\mathbb{Z}/k\mathbb{Z}) \to H^n(C_m,\mathbb{Z}/k\mathbb{Z}) \to \mathrm{Hom}_{\mathbb{Z}}(H_n(C_m,\mathbb{Z}),\mathbb{Z}/k\mathbb{Z}) \to 0$$

Using $H_n(C_m,\mathbb{Z})$ from example 2.5.1: for odd n, $H_n(C_m,\mathbb{Z}) = \mathbb{Z}/m\mathbb{Z}$ and $H_{n-1}(C_m,\mathbb{Z}) = 0$, giving $H^n(C_m,\mathbb{Z}/k\mathbb{Z}) \cong \mathrm{Hom}(\mathbb{Z}/m\mathbb{Z},\mathbb{Z}/k\mathbb{Z}) \cong \mathbb{Z}/\gcd(m,k)\mathbb{Z}$. For even $n \geq 2$, $H_n(C_m,\mathbb{Z}) = 0$ and $H_{n-1}(C_m,\mathbb{Z}) = \mathbb{Z}/m\mathbb{Z}$, giving $H^n(C_m,\mathbb{Z}/k\mathbb{Z}) \cong \mathrm{Ext}^1(\mathbb{Z}/m\mathbb{Z},\mathbb{Z}/k\mathbb{Z}) \cong \mathbb{Z}/\gcd(m,k)\mathbb{Z}$. In both cases, $H^n(C_m,\mathbb{Z}/k\mathbb{Z}) \cong \mathbb{Z}/\gcd(m,k)\mathbb{Z}$ for $n \geq 1$.

The second homology group admits an interpretation in terms of generators and relations.

Theorem 2.5.5: Hopf Formula

Let G be a group with a free presentation $1 \to R \to F \xrightarrow{\pi} G \to 1$ (i.e., F is a free group and $R = \ker \pi$). Then:

$$H_2(G, \mathbb{Z}) \cong \frac{R \cap [F, F]}{[R, F]}$$

***Proof*:**
The five-term exact sequence in homology associated to the group extension $1 \to R \to F \to G \to 1$ (a consequence of the Lyndon–Hochschild–Serre spectral sequence, developed in Chapter 3) gives:

$$H_2(F, \mathbb{Z}) \to H_2(G, \mathbb{Z}) \to R/[R, F] \to F^{\mathrm{ab}} \to G^{\mathrm{ab}} \to 0$$

Since F is free, $H_2(F, \mathbb{Z}) = 0$ (free groups have cohomological dimension 1 by example 2.3.10, and the same holds for homology). The map $R/[R, F] \to F^{\mathrm{ab}} = F/[F, F]$ sends $r[R, F] \mapsto r[F, F]$, with kernel $(R \cap [F, F])/[R, F]$. Exactness at $H_2(G, \mathbb{Z})$ gives the isomorphism.

The numerator $R \cap [F, F]$ consists of relators that are also commutators in the free group—"relations that hold for trivial reasons in the abelianization." The denominator $[R, F]$ consists of consequences of the relations that are automatically trivial. The quotient $H_2(G, \mathbb{Z})$ therefore measures the "relations among relations" that are not accounted for by formal consequences. The group $H_2(G, \mathbb{Z})$ is also called the *Schur multiplier* of G (see exercise 2.4.1 (4)).

For finite groups, the Tate cohomology of Chapter 4 merges homology and cohomology into a single sequence $\hat{H}^n(G, M)$ indexed by all integers $n \in \mathbb{Z}$, with $\hat{H}^n = H^n$ for $n \geq 1$ and $\hat{H}^{-n} = H_{n-1}$ for $n \geq 2$ (up to modification in degrees 0 and -1). The periodicity visible in the cyclic group computations then extends to a single 2-periodic sequence.

Exercise 2.5.1

1. Compute $H_n(C_m, M)$ for all $n \geq 0$ and an arbitrary C_m-module M, using the periodic resolution. Express the answer in terms of the norm map N and the action of $\sigma - 1$.
2. Compute $H_1(S_3, \mathbb{Z})$ using theorem 2.5.2, and verify the result independently by computing H_1 from a free presentation of S_3.
3. Verify the universal coefficient theorem for $G = C_2$ and $A = \mathbb{Z}/2\mathbb{Z}$ in degrees $n = 1$ and $n = 2$: check that both sides of the UCT short exact sequence agree with the direct computation from exercise 2.3.1 (3).
4. Show that $H_n(G, \mathbb{Z}[G]) = 0$ for all $n \geq 1$, using the isomorphism $P_\bullet \otimes_{\mathbb{Z}[G]} \mathbb{Z}[G] \cong P_\bullet$ for a projective resolution $P_\bullet \to \mathbb{Z}$.
5. Use the Hopf formula to compute $H_2(C_2 \times C_2, \mathbb{Z})$, starting from the presentation $\langle a, b \big| a^2, b^2, [a, b] \rangle$. Verify the result using the Künneth formula.

3

Profinite Group Cohomology

The cohomology theory developed in chapter 2 applies to any group G acting on an abelian group M: one forms the derived functors of $M \mapsto M^G$ and obtains a sequence of abelian groups $H^n(G, M)$ that measure the failure of the invariant functor to be exact. Nothing in that construction uses the topology of G, and when G is profinite, this is a serious defect. The group $G = \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, for instance, acts on $\overline{\mathbb{Q}}^{\times}$ through a continuous action, and the cohomology that captures arithmetic information must respect this continuity. Ignoring the topology produces groups that are too large, too poorly behaved, and disconnected from the Galois-theoretic applications that motivate the entire subject.

This chapter rebuilds group cohomology in the profinite setting. The essential modification is to restrict attention to *discrete* G-modules — those on which G acts continuously when the module carries the discrete topology — and to define cohomology as the derived functors of the continuous invariant functor on this restricted category. The payoff is immediate: the resulting cohomology groups are colimits of the finite-group cohomology from chapter 2, making them computable by reduction to finite quotients. With this foundation in place, the standard functorial apparatus — inflation, restriction, corestriction, the Lyndon–Hochschild–Serre spectral sequence, and Shapiro's lemma — transfers to the profinite setting with only minor modifications.

3.1 Continuous Cochains and Discrete Modules

The first task is to identify the correct category of coefficient modules for a profinite group G. The naive approach — taking all $\mathbb{Z}[G]$-modules and computing derived functors — ignores the topology of G entirely. The correct approach requires the G-action to be continuous, which forces a topological constraint on the module.

Throughout this section, G denotes a profinite group with its natural topology (compact, Hausdorff, and totally disconnected, as established in chapter 1).

Definition 3.1.1: Discrete G-Module

Let G be a profinite group. A *discrete G-module* is an abelian group M equipped with a left G-action $G \times M \to M$ such that the stabilizer $\mathrm{Stab}_G(m) = \{g \in G \mid g \cdot m = m\}$ is open in G for every $m \in M$. Equivalently, M is a discrete G-module if and only if

$$M = \bigcup_{U \trianglelefteq_o G} M^U,$$

where the union runs over all open normal subgroups U of G, and $M^U = \{m \in M \mid u \cdot m = m \text{ for all } u \in U\}$ is the submodule of U-invariants.

The equivalence in the definition is immediate: if every stabilizer is open, then for each $m \in M$ there exists an open normal subgroup $U \subseteq \mathrm{Stab}_G(m)$ (since open subgroups of a profinite group contain open normal subgroups, by proposition 1.2.2), so $m \in M^U$. Conversely, if $m \in M^U$ for some open normal U, then $U \subseteq \mathrm{Stab}_G(m)$, and a subgroup containing an open subgroup is itself open.

The terminology "discrete" refers to the topology on M: the action $G \times M \to M$ is continuous if and only if M carries the discrete topology. Indeed, continuity of the action means that for each $m \in M$, the preimage of $\{m\}$ under the map $g \mapsto g \cdot m$ is open in G. This preimage is precisely $\mathrm{Stab}_G(m)$, so continuity is equivalent to every stabilizer being open — exactly the condition in the definition.

Proposition 3.1.2: The Category of Discrete G-Modules

Let G be a profinite group. The category $\mathrm{Mod}_G^{\mathrm{disc}}$ of discrete G-modules, with G-equivariant group homomorphisms as morphisms, is an abelian category.

***Proof*:**

The category $\mathrm{Mod}_G^{\mathrm{disc}}$ is a full subcategory of the category Mod_G of all $\mathbb{Z}[G]$-modules. It is closed under taking submodules, quotients, and arbitrary direct sums. For submodules: if $N \subseteq M$ is a G-stable subgroup and M is discrete, then $\mathrm{Stab}_G(n) \geqslant \mathrm{Stab}_G(n) \cap G = \mathrm{Stab}_G(n)$ is open for each $n \in N$ since it is open in M. For quotients: if $\pi : M \to M/N$ is the projection and $\overline{m} = \pi(m)$, then $\mathrm{Stab}_G(\overline{m}) \geqslant \mathrm{Stab}_G(m)$, which is open. For direct sums: if $(M_i)_{i \in I}$ is a family of discrete G-modules, each element of $\bigoplus_i M_i$ has finite support, so its stabilizer is a finite intersection of open subgroups, hence open. Since $\mathrm{Mod}_G^{\mathrm{disc}}$ is a full abelian subcategory of Mod_G closed under kernels and cokernels, it is abelian.

The next order of business is to show that $\mathrm{Mod}_G^{\mathrm{disc}}$ has enough injectives, so that derived functors can be defined in this category. The argument mirrors the classical one for Mod_G, with a small but important modification: the coinduced module construction must land in discrete modules.

Proposition 3.1.3: Enough Injectives

The category $\mathrm{Mod}_G^{\mathrm{disc}}$ has enough injectives. Specifically, for every discrete G-module M, there exists an injective object I in $\mathrm{Mod}_G^{\mathrm{disc}}$ and a G-equivariant injection $M \hookrightarrow I$.

***Proof*:**
Let M be a discrete G-module. First, embed M into an injective abelian group J (this is possible since the category of abelian groups has enough injectives). Define the *discrete coinduction*

$$I = \mathrm{Hom}_{\mathbb{Z}}^{\mathrm{cts}}(\mathbb{Z}[G], J),$$

the group of continuous $\mathbb{Z}$-module homomorphisms from $\mathbb{Z}[G]$ (with the profinite topology inherited from G) to J (with the discrete topology). The group G acts on I by $(g \cdot f)(x) = f(xg)$ for $g, x \in G$.

Step 1: I is a discrete G-module. A homomorphism $f : \mathbb{Z}[G] \to J$ is continuous if and only if $\ker f$ is open in $\mathbb{Z}[G]$, which occurs if and only if f factors through $\mathbb{Z}[G/U]$ for some open normal subgroup U. For such an f and any $u \in U$, the action gives $(u \cdot f)(x) = f(xu) = f(x)$ for all $x \in G$ (since f is constant on U-cosets). Hence $U \subseteq \mathrm{Stab}_G(f)$, and the stabilizer is open.

Step 2: The embedding $M \hookrightarrow I$. The injection $\iota : M \hookrightarrow J$ induces a G-equivariant map $\varphi : M \to I$ defined by $\varphi(m)(g) = \iota(g^{-1} \cdot m)$. This is injective: if $\varphi(m) = 0$, then $\iota(g^{-1} \cdot m) = 0$ for all $g \in G$, and taking $g = 1$ gives $\iota(m) = 0$, hence $m = 0$. The map $\varphi(m)$ is continuous because m has open stabilizer, so $g \mapsto g^{-1} \cdot m$ is locally constant.

Step 3: I is injective in $\mathrm{Mod}_G^{\mathrm{disc}}$. Let $A \hookrightarrow B$ be an injection of discrete G-modules and $f : A \to I$ a G-equivariant map. The adjunction $\mathrm{Hom}_G(B, \mathrm{Hom}_{\mathbb{Z}}^{\mathrm{cts}}(\mathbb{Z}[G], J)) \cong \mathrm{Hom}_{\mathbb{Z}}(B, J)$ (the discrete version of the Frobenius reciprocity adjunction) reduces the extension problem to extending a $\mathbb{Z}$-linear map $A \to J$ to $B \to J$. Since J is injective as an abelian group, the extension exists.

With the abelian category structure and enough injectives established, the derived functor definition of cohomology proceeds exactly as in the discrete case.

Definition 3.1.4: Continuous Cohomology

Let G be a profinite group and M a discrete G-module. The *continuous cohomology groups* $H^n(G, M)$ for $n \geq 0$ are defined as the right derived functors of the invariant functor

$$(-)^G : \mathrm{Mod}_G^{\mathrm{disc}} \to \mathbf{Ab}, \qquad M \mapsto M^G = \{m \in M \mid g \cdot m = m \text{ for all } g \in G\},$$

applied to M. That is, choosing an injective resolution $0 \to M \to I^0 \to I^1 \to \cdots$ in $\mathrm{Mod}_G^{\mathrm{disc}}$, one sets

$$H^n(G, M) = H^n\big((I^0)^G \to (I^1)^G \to (I^2)^G \to \cdots\big).$$

By the general theory of derived functors ([9, Unique Projection Resolution, Long Exact Sequence From a Right-Derived Functor]), the groups $H^n(G, M)$ are independent of the choice of injective resolution, and the short exact sequence $0 \to M' \to M \to M'' \to 0$ of discrete G-modules induces a long exact sequence

$$0 \to (M')^G \to M^G \to (M'')^G \xrightarrow{\delta} H^1(G, M') \to H^1(G, M) \to \cdots$$

The connecting homomorphisms δ are natural in the short exact sequence.

As in section 2.3, one can compute these derived functors using cochains rather than injective resolutions. The profinite analogue of the standard cochain complex requires a continuity constraint.

Definition 3.1.5: Continuous Cochain Complex

Let G be a profinite group and M a discrete G-module. The group of *continuous n-cochains* is

$$C^n_{\text{cts}}(G, M) = \{f : G^n \to M \mid f \text{ is continuous}\},$$

where G^n carries the product topology and M the discrete topology. Since M is discrete, a map $f : G^n \to M$ is continuous if and only if it is *locally constant*: every point of G^n has an open neighbourhood on which f is constant. The coboundary maps $d^n : C^n_{\text{cts}}(G, M) \to C^{n+1}_{\text{cts}}(G, M)$ are defined by the same formula as in definition 2.3.3:

$$\begin{aligned}(d^n f)(g_1, \ldots, g_{n+1}) = g_1 \cdot f(g_2, \ldots, g_{n+1})& \\ + \sum_{i=1}^{n} (-1)^i f(g_1, \ldots, g_i g_{i+1}, \ldots, g_{n+1})& \\ + (-1)^{n+1} f(g_1, \ldots, g_n).&\end{aligned}$$

These maps preserve continuity (the action of g_1 on a locally constant function remains locally constant since the action map is continuous), so $(C^\bullet_{\text{cts}}(G, M), d^\bullet)$ is a cochain complex. The *continuous cocycles*, *continuous coboundaries*, and *continuous cochain cohomology* groups are

$$Z^n_{\text{cts}}(G, M) = \ker d^n, \qquad B^n_{\text{cts}}(G, M) = \operatorname{im} d^{n-1}, \qquad \tilde{H}^n(G, M) = Z^n_{\text{cts}}/B^n_{\text{cts}}.$$

The tilde notation is temporary: the following proposition shows that the cochain cohomology agrees with the derived functor cohomology, so the two notions coincide.

Proposition 3.1.6: Equivalence of Definitions

Let G be a profinite group and M a discrete G-module. There is a canonical isomorphism

$$\tilde{H}^n(G, M) \cong H^n(G, M)$$

for all $n \geq 0$.

***Proof*:**
The argument is analogous to the discrete case (proposition 2.3.4). One shows that the continuous cochain complex $C^\bullet_{\text{cts}}(G, M)$ computes the right derived functors of $(-)^G$ by constructing a continuous analogue of the standard resolution.

For each $n \geq 0$, define $\mathcal{C}^n(G) = \mathrm{Map}_{\text{cts}}(G^{n+1}, \mathbb{Z})$, the group of continuous (hence locally constant) functions $G^{n+1} \to \mathbb{Z}$, with G acting diagonally: $(g \cdot f)(g_0, \ldots, g_n) = f(g^{-1} g_0, \ldots, g^{-1} g_n)$. These are discrete G-modules (each function factors through a finite quotient, giving open stabilizers), and the sequence

$$\cdots \to \mathcal{C}^1(G) \to \mathcal{C}^0(G) \xrightarrow{\varepsilon} \mathbb{Z} \to 0$$

is exact, with $\varepsilon(f) = f(1)$ and differential maps defined by the standard alternating sum.

Each $\mathcal{C}^n(G)$ is acyclic for the invariant functor (the proof parallels lemma 2.3.2, using the contracting homotopy $s(f)(g_0, \ldots, g_{n+1}) = f(1, g_0, \ldots, g_n)$, which is well defined and continuous). Computing invariants gives

$$(\mathcal{C}^n(G))^G = \mathrm{Map}_{\mathrm{cts}}(G^{n+1}, \mathbb{Z})^G \cong C^n_{\mathrm{cts}}(G, \mathbb{Z}),$$

and tensoring the resolution by M (or applying Hom from it) yields $C^\bullet_{\mathrm{cts}}(G, M)$. Since the resolution is acyclic, the cohomology of the resulting complex equals the derived functor cohomology.

Henceforth, $H^n(G, M)$ refers to either the derived functor definition or the continuous cochain definition interchangeably.

The low-degree continuous cohomology groups admit the same concrete interpretations as in section 2.4. In degree zero, $H^0(G, M) = M^G$ by definition. In degree one, $H^1(G, M)$ classifies continuous crossed homomorphisms modulo principal ones: a continuous 1-cocycle is a continuous map $f : G \to M$ satisfying $f(gh) = g \cdot f(h) + f(g)$, and two such maps are cohomologous if they differ by $g \mapsto g \cdot m - m$ for some fixed $m \in M$. In degree two, $H^2(G, M)$ classifies topological group extensions of G by M (continuous sections, compatible with the profinite topology on G). These interpretations carry over verbatim from the discrete theory because the continuity constraint is built into the cochain complex.

Example 3.1.7: Continuous Cohomology of Running Examples

1. Let $G = \hat{\mathbb{Z}}$ act trivially on $M = \mathbb{Z}$. An element $m \in M$ is G-invariant if and only if $g \cdot m = m$ for all $g \in G$, which is automatic since the action is trivial. Hence $H^0(\hat{\mathbb{Z}}, \mathbb{Z}) = \mathbb{Z}$. A continuous 1-cocycle $f : \hat{\mathbb{Z}} \to \mathbb{Z}$ is a continuous group homomorphism (since the action is trivial, the cocycle condition reduces to $f(g + h) = f(g) + f(h)$). But $\hat{\mathbb{Z}}$ is compact and $\mathbb{Z}$ is discrete, so the image of f is a compact subgroup of $\mathbb{Z}$, hence $\{0\}$. Therefore $H^1(\hat{\mathbb{Z}}, \mathbb{Z}) = 0$.

2. Let $G = G_{\mathbb{Q}_p} = \mathrm{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p)$ act on $M = \overline{\mathbb{Q}}_p^\times$ by Galois automorphisms. The invariants are

$$H^0(G_{\mathbb{Q}_p}, \overline{\mathbb{Q}}_p^\times) = (\overline{\mathbb{Q}}_p^\times)^{G_{\mathbb{Q}_p}} = \mathbb{Q}_p^\times,$$

by the fundamental theorem of Galois theory (theorem 1.4.1). The group $H^1(G_{\mathbb{Q}_p}, \overline{\mathbb{Q}}_p^\times)$ will be computed later: Hilbert's Theorem 90 (chapter 4) asserts that it vanishes. The group $H^2(G_{\mathbb{Q}_p}, \overline{\mathbb{Q}}_p^\times)$ will be identified with $\mathbb{Q}/\mathbb{Z}$ via the invariant map of the Brauer group (chapter 7).

3. Let $G = G_{\mathbb{R}} = \mathrm{Gal}(\mathbb{C}/\mathbb{R}) \cong \mathbb{Z}/2\mathbb{Z}$ act on $\mathbb{C}^\times$ by complex conjugation. Since G is finite (hence discrete and profinite simultaneously), continuous cohomology coincides with the discrete group cohomology of chapter 2. The invariants are $H^0(G_{\mathbb{R}}, \mathbb{C}^\times) = \mathbb{R}^\times$. By example 2.3.8, the cohomology of a cyclic group of order 2 acting on a module M is 2-periodic in positive degrees, with

$$H^{2k+1}(G_{\mathbb{R}}, \mathbb{C}^\times) \cong \ker(\mathrm{Nm})/\mathrm{im}(g-1), \qquad H^{2k}(G_{\mathbb{R}}, \mathbb{C}^\times) \cong M^G/\mathrm{Nm}(M),$$

where $\mathrm{Nm} = 1 + \sigma$ is the norm element (σ being complex conjugation) and $g - 1 = \sigma - 1$. Since $\mathrm{Nm}(z) = z \cdot \overline{z} = |z|^2$ and $(\sigma - 1)(z) = \overline{z}/z$, the norm map $\mathrm{Nm} : \mathbb{C}^\times \to \mathbb{R}^\times$ is surjective onto $\mathbb{R}^\times_{>0}$, so $H^2(G_\mathbb{R}, \mathbb{C}^\times) \cong \mathbb{R}^\times/\mathbb{R}^\times_{>0} \cong \mathbb{Z}/2\mathbb{Z}$. The kernel of the norm consists of elements with $|z|^2 = 1$, i.e., the unit circle S^1, while $\mathrm{im}(\sigma - 1) = \{\overline{z}/z \mid z \in \mathbb{C}^\times\}$ also equals S^1 (since $|\overline{z}/z| = 1$ and every element of S^1 has the form $e^{i\theta} = \overline{e^{i\theta/2}}/e^{i\theta/2}$). Hence $H^1(G_\mathbb{R}, \mathbb{C}^\times) = 0$.

The computation for $G_\mathbb{R}$ is a preview of a general phenomenon: Hilbert's Theorem 90 asserts $H^1(G_K, \overline{K}^\times) = 0$ for *any* field K, and $H^2(G_K, \overline{K}^\times)$ is the Brauer group $\mathrm{Br}(K)$. These identifications are central to the arithmetic applications in chapters 8 and 9.

The following proposition records a basic but important point: a module that fails to be discrete can produce cohomology with no arithmetic content.

Proposition 3.1.8: Non-Discrete Modules Produce Pathological Cohomology

Let $G = \mathbb{Z}_p$ (viewed as a profinite group under addition) act on $M = \mathbb{Z}_p$ by the trivial action. Then M is *not* a discrete G-module (since $\mathbb{Z}_p$ with the p-adic topology is not discrete). If one naively defines H^1 using all (not necessarily continuous) 1-cocycles, the resulting group $\mathrm{Hom}_{\mathrm{grp}}(\mathbb{Z}_p, \mathbb{Z}_p)$ is uncountable (it is a free $\mathbb{Z}_p$-module of rank 1 as a module of continuous homomorphisms, but the group of all abstract homomorphisms $\mathbb{Z}_p \to \mathbb{Z}_p$ is far larger — it has the cardinality of the continuum, since $\mathbb{Z}_p \cong \mathbb{Z}^{(\mathfrak{c})}$ as an abstract abelian group). In contrast, the continuous cohomology $H^1(\mathbb{Z}_p, \mathbb{Z}_p)$, computed by restricting to continuous 1-cocycles (continuous homomorphisms $\mathbb{Z}_p \to \mathbb{Z}_p$), equals $\mathbb{Z}_p$ — a group with transparent arithmetic meaning.

***Proof*:**
Since the action is trivial, continuous 1-cocycles are continuous group homomorphisms $\mathbb{Z}_p \to \mathbb{Z}_p$. Such a homomorphism is determined by the image of $1 \in \mathbb{Z}_p$ (since the image of any $a \in \mathbb{Z}_p$ is $a \cdot f(1)$ by continuity and the density of $\mathbb{Z}$ in $\mathbb{Z}_p$), so $\mathrm{Hom}_{\mathrm{cts}}(\mathbb{Z}_p, \mathbb{Z}_p) \cong \mathbb{Z}_p$. The coboundaries are zero (since $g \cdot m - m = 0$ for the trivial action), giving $H^1(\mathbb{Z}_p, \mathbb{Z}_p) \cong \mathbb{Z}_p$. For the non-continuous version, one uses the fact that $\mathbb{Z}_p$ is torsion-free and divisible by all primes $\ell \neq p$ as an abelian group, and $\mathbb{Z}_p$ as an abstract abelian group has $\mathbb{Q}_p$-vector space structure on its divisible part, producing an uncountable Hom group.

This example explains why the discreteness condition on coefficients is essential: without it, cohomology detects the algebraic (but not topological) structure of G, producing groups that are too large to carry arithmetic information.

Exercise 3.1.1

1. Let G be a profinite group and M an abelian group with a G-action. Prove that the following are equivalent:

   (a) M is a discrete G-module.

   (b) The action map $G \times M \to M$ is continuous when M carries the discrete topology.

   (c) For every $m \in M$, the orbit map $G \to M$ defined by $g \mapsto g \cdot m$ is continuous.

   (d) $M = \varinjlim_{U \trianglelefteq_o G} M^U$, where the colimit is taken in the category of abelian groups over the directed system of open normal subgroups ordered by reverse inclusion.

2. Let G be a profinite group and let M, N be discrete G-modules. Show that $\mathrm{Hom}_{\mathbb{Z}}(M, N)$ (with the conjugation action $(g \cdot f)(m) = g \cdot f(g^{-1} \cdot m)$) need not be a discrete G-module in general, but that $\mathrm{Hom}_{\mathbb{Z}}(M, N)$ is discrete whenever M is finitely generated as a $\mathbb{Z}$-module.
3. Let $G = \hat{\mathbb{Z}}$ and let $M = \mathbb{Z}$ with the trivial G-action. Use the continuous cochain definition to verify directly that $H^2(\hat{\mathbb{Z}}, \mathbb{Z}) \cong \mathbb{Q}/\mathbb{Z}$. (*Hint:* A continuous 2-cocycle $f : \hat{\mathbb{Z}} \times \hat{\mathbb{Z}} \to \mathbb{Z}$ factors through some $(\mathbb{Z}/n\mathbb{Z}) \times (\mathbb{Z}/n\mathbb{Z})$. Use the result from example 2.3.8 for cyclic groups and pass to the colimit.)
4. Let G be a profinite group and M a (not necessarily discrete) $\mathbb{Z}[G]$-module. Define $M^{\mathrm{disc}} = \bigcup_{U \trianglelefteq_o G} M^U$. Show that M^{disc} is the largest discrete G-submodule of M, and that the functor $M \mapsto M^{\mathrm{disc}}$ from Mod_G to $\mathrm{Mod}_G^{\mathrm{disc}}$ is right adjoint to the inclusion functor $\mathrm{Mod}_G^{\mathrm{disc}} \hookrightarrow \mathrm{Mod}_G$.
5. Let G be a profinite group acting trivially on a discrete abelian group M. Show that $H^1(G, M) \cong \mathrm{Hom}_{\mathrm{cts}}(G, M)$, the group of continuous homomorphisms from G to M. Compute $H^1(G_{\mathbb{Q}_p}, \mathbb{Z}/n\mathbb{Z})$ for each $n \geq 1$.

3.2 Cohomology as a Colimit

The definition of continuous cohomology in section 3.1 is conceptually clean but not yet directly computable: it requires constructing injective resolutions or working with continuous cochains on a profinite group, which is an uncountable topological space. The theorem of this section eliminates both difficulties at once. It shows that the continuous cohomology of a profinite group G is the colimit of the ordinary (discrete) cohomology of its finite quotients — a reduction that makes every computation in this book possible.

The mechanism is elementary: a continuous cochain on G is locally constant, so it factors through a finite quotient. Making this precise requires showing that the entire cochain complex decomposes as a colimit.

Lemma 3.2.1: The Continuous Cochain Complex as a Colimit

Let G be a profinite group and M a discrete G-module. For each open normal subgroup U of G, the finite group G/U acts on M^U, and the ordinary cochain complex $C^\bullet(G/U, M^U)$ from definition 2.3.3 embeds into $C^\bullet_{\mathrm{cts}}(G, M)$ via inflation: a cochain $f : (G/U)^n \to M^U$ is sent to the composition $G^n \twoheadrightarrow (G/U)^n \xrightarrow{f} M^U \hookrightarrow M$. This embedding is compatible with the coboundary maps, giving an inclusion of cochain complexes $C^\bullet(G/U, M^U) \hookrightarrow C^\bullet_{\mathrm{cts}}(G, M)$. The resulting colimit map

$$\varinjlim_{U \trianglelefteq_o G} C^n(G/U, M^U) \xrightarrow{\sim} C^n_{\mathrm{cts}}(G, M)$$

is an isomorphism for each $n \geq 0$, where the colimit is taken over open normal subgroups of G ordered by reverse inclusion (so that the transition maps are inflation).

***Proof*:**
Step 1: Surjectivity. Let $f : G^n \to M$ be a continuous n-cochain. Since M is discrete, f is locally constant: for each $(g_1, \ldots, g_n) \in G^n$, there exists an open neighbourhood on which f is constant. By compactness of G^n, finitely many such neighbourhoods cover G^n. Each neighbourhood can be refined to a coset of some open normal subgroup, so there exists a single open normal subgroup U of G such that f is constant on cosets of $U^n \subseteq G^n$. This means f factors through the projection $G^n \twoheadrightarrow (G/U)^n$.

It remains to check that f takes values in M^U, possibly after shrinking U. Fix a value $m = f(g_1, \ldots, g_n)$ in the image of f. Since M is a discrete G-module, the stabilizer $\mathrm{Stab}_G(m)$ is open, so it contains some open normal subgroup V_m. The image of f is finite (as f is locally constant on the compact space G^n), so the intersection $V = \bigcap_{m \in \mathrm{im}(f)} V_m$ is a finite intersection of open subgroups and hence open. Replacing U by $U \cap V$ (which is still open and normal), the cochain f factors through $(G/(U \cap V))^n$ and every value of f is fixed by $U \cap V$. Hence f defines an element of $C^n(G/(U \cap V), M^{U \cap V})$, as required.

Step 2: Injectivity. The colimit is a filtered colimit of abelian groups, so an element of the colimit is zero if and only if it becomes zero in some term of the system. An element of $C^n(G/U, M^U)$ that maps to zero in $C^n_{\mathrm{cts}}(G, M)$ is a cochain $f : (G/U)^n \to M^U$ such that the inflated cochain $G^n \to M$ is zero. Since the projection $G^n \to (G/U)^n$ is surjective, f itself must be zero.

Step 3: Compatibility with differentials. The inflation map $C^n(G/U, M^U) \to C^n_{\mathrm{cts}}(G, M)$ commutes with the coboundary operator d^n because the coboundary formula involves only the group operation and the G-action, both of which are compatible with the projection $G \twoheadrightarrow G/U$.

The colimit formula for cohomology is now an immediate consequence.

Theorem 3.2.2: Cohomology as a Colimit

Let G be a profinite group and M a discrete G-module. For every $n \geq 0$, there is a canonical isomorphism

$$H^n(G, M) \cong \varinjlim_{U \trianglelefteq_{\mathrm{o}} G} H^n(G/U, M^U),$$

where the colimit runs over the directed system of open normal subgroups of G (ordered by reverse inclusion), and the transition maps are the inflation homomorphisms $\mathrm{Inf} : H^n(G/U, M^U) \to H^n(G/V, M^V)$ for $V \subseteq U$.

***Proof*:**
By proposition 3.1.6, $H^n(G, M)$ is the n-th cohomology of the continuous cochain complex $C^\bullet_{\mathrm{cts}}(G, M)$. By lemma 3.2.1, this complex is the colimit of the cochain complexes $C^\bullet(G/U, M^U)$:

$$H^n(G, M) = H^n(C^\bullet_{\mathrm{cts}}(G, M)) = H^n\left(\varinjlim_U C^\bullet(G/U, M^U)\right).$$

The open normal subgroups of G form a directed set under reverse inclusion (for any $U_1, U_2 \trianglelefteq_{\mathrm{o}} G$, the intersection $U_1 \cap U_2$ is again open and normal, and is contained in both). The colimit of the cochain complexes is therefore a filtered colimit. Since filtered colimits in the category of abelian groups are exact (they commute with finite limits, and in particular with kernels and cokernels), taking cohomology commutes with the colimit:

$$H^n\left(\varinjlim_U C^\bullet(G/U, M^U)\right) \cong \varinjlim_U H^n\big(C^\bullet(G/U, M^U)\big) = \varinjlim_U H^n(G/U, M^U).$$

The last equality uses the cocycle description of finite-group cohomology from proposition 2.3.4.

This theorem is the computational engine of profinite group cohomology. It converts every cohomological question about a profinite group into a question about the cohomology of its finite quotients, where the tools of chapter 2 apply directly. For Galois cohomology, it says that the cohomology of an absolute Galois group $G_K = \mathrm{Gal}(K^{\mathrm{sep}}/K)$ is computed by taking the colimit over all finite Galois extensions L/K:

$$H^n(G_K, M) = \varinjlim_{L/K} H^n(\mathrm{Gal}(L/K), M^{G_L}),$$

where L ranges over finite Galois extensions of K contained in a fixed separable closure K^{sep}, and $G_L = \mathrm{Gal}(K^{\mathrm{sep}}/L)$ is the corresponding open normal subgroup. Every element of $H^n(G_K, M)$ “lives at a finite level” — it is represented by a cocycle of some finite Galois group.

The first payoff is the complete cohomology of the simplest infinite profinite group. It is worth having in closed form, because $\widehat{\mathbb{Z}}$ is the absolute Galois group of every finite field, and the answer is what makes arithmetic over finite fields tractable: one invariant module in degree zero, one coinvariant module in degree one, and nothing above.

Theorem 3.2.3: The Cohomology of $\widehat{\mathbb{Z}}$

Let $\Gamma = \widehat{\mathbb{Z}}$ with topological generator φ, and let M be a *torsion* discrete Γ-module. Then

$$H^0(\Gamma, M) = M^\Gamma, \qquad H^1(\Gamma, M) = M_\Gamma, \qquad H^i(\Gamma, M) = 0 \quad (i \geq 2)$$

where $M^\Gamma = \ker(\varphi - 1)$ and $M_\Gamma = M/(\varphi - 1)M$ are the invariants and coinvariants of definition 2.1.7.

***Proof*:**
Reduce first to finite M. A torsion discrete Γ-module is the filtered union of its finite Γ-submodules: each element has finite orbit, since its stabilizer is open, and finite order, so it generates a finite submodule. Cohomology commutes with that colimit by corollary 3.2.5, and invariants and coinvariants do too, so it suffices to treat M finite. Let e be an exponent for M.

The open subgroups of $\widehat{\mathbb{Z}}$ are the $m\widehat{\mathbb{Z}}$ with quotient $\mathbb{Z}/m\mathbb{Z}$, cyclic of order m generated by the image of φ. Fix m with $M^{m\widehat{\mathbb{Z}}} = M$, which is possible because the action is continuous and M finite; then for every multiple the module of invariants is all of M, and theorem 3.2.2 gives

$$H^i(\Gamma, M) = \varinjlim_c H^i\big(\mathbb{Z}/cm\mathbb{Z},\ M\big)$$

the colimit over $c \geq 1$ with inflation as transition maps.

Each term is computed by example 2.3.8, and by proposition 2.3.9 the transition from level m to level cm is induced by multiplication by $c^{\lfloor i/2 \rfloor}$ on M.

For $i = 0$ and $i = 1$ the exponent $\lfloor i/2 \rfloor$ is zero, so every transition is the identity and the colimit is the common value: M^Γ in degree zero, and in degree one $\ker(N)/(\varphi - 1)M$ where N is the norm of $\mathbb{Z}/cm\mathbb{Z}$. Choosing c so that $e \big| cm$ makes N act as zero on M, since $N = \sum_t \varphi^t$ acts on the invariants as multiplication by cm; hence $\ker N = M$ and the degree-one term is $M/(\varphi - 1)M = M_\Gamma$.

For $i \geq 2$ the exponent is at least one. Every group in the system is a subquotient of M, hence killed by e, and the transition from level m to level em is multiplication by $e^{\lfloor i/2 \rfloor}$, which is zero on such a group. Every element of the colimit therefore dies at a finite stage, so the colimit vanishes, completing the proof.

The vanishing above degree one is exactly the statement that $\widehat{\mathbb{Z}}$ has cohomological dimension one, recovered independently in example 5.1.5, and the degree-one answer is the one used whenever a Frobenius acts: invariants below, coinvariants above, and the two have the same cardinality when M is finite, since $\varphi - 1$ has equal kernel and cokernel on a finite abelian group.

Corollary 3.2.4: Torsion Property

Let G be a profinite group and M a discrete G-module. For every $n \geq 1$, the group $H^n(G, M)$ is torsion: every element has finite order.

***Proof*:**

By theorem 3.2.2, every element $\alpha \in H^n(G, M)$ lies in the image of $H^n(G/U, M^U)$ for some open normal subgroup U. The group G/U is finite, and the cohomology of a finite group in positive degrees is annihilated by the order of the group: $|G/U| \cdot H^n(G/U, M^U) = 0$ for $n \geq 1$. (This follows from the norm element $N_{G/U} = \sum_{\bar{g} \in G/U} \bar{g}$ in the group ring $\mathbb{Z}[G/U]$: the identity $N_{G/U} \circ (\bar{g} - 1) = 0$ and $(\bar{g} - 1) \circ N_{G/U} = 0$ for all $\bar{g} \in G/U$ imply, via the standard resolution of definition 2.3.1, that multiplication by $|G/U|$ is null-homotopic on the cochain complex in positive degrees.) In particular, $|G/U| \cdot \alpha = 0$.

The torsion property has an immediate and important structural consequence: $H^n(G, M)$ decomposes as a direct sum of its p-primary components $H^n(G, M)\{p\} = \left\{x \in H^n(G, M) \;\middle|\; p^k x = 0 \text{ for some } k \geq 1\right\}$ over all primes p. Many arguments in Galois cohomology proceed by reducing to a single prime at a time, and the torsion property guarantees that no information is lost in doing so.

A word of caution: the torsion property does *not* imply that $H^n(G, M)$ has bounded exponent. The orders of the finite quotients G/U can grow without bound, and different elements of the colimit may be killed by different orders. For example, $H^2(\hat{\mathbb{Z}}, \mathbb{Z}) \cong \mathbb{Q}/\mathbb{Z}$ is torsion but not of bounded exponent — every positive integer appears as the order of some element.

Another consequence of the colimit formula is that continuous cohomology commutes with colimits in the coefficient module.

Corollary 3.2.5: Commutation with Coefficient Colimits

Let G be a profinite group and $(M_i)_{i \in I}$ a directed system of discrete G-modules. Then for each $n \geq 0$,

$$H^n\left(G, \varinjlim_i M_i\right) \cong \varinjlim_i H^n(G, M_i).$$

***Proof*:**
The continuous cochain complex commutes with filtered colimits in the coefficient variable: $C^n_{\text{cts}}(G, \varinjlim_i M_i) \cong \varinjlim_i C^n_{\text{cts}}(G, M_i)$, because a locally constant function into $\varinjlim_i M_i$ factors through some M_i (by compactness of G^n and discreteness of the colimit). Taking cohomology and using the exactness of filtered colimits gives the result.

This commutation property is used constantly in Galois cohomology, often without explicit mention. For instance, a discrete G_K-module M that is a torsion abelian group can be written as $M = \varinjlim M_i$ where each M_i is a finitely generated (hence finite) G_K-submodule, and the cohomology of M reduces to the cohomology of its finite submodules.

Example 3.2.6: Cohomology of $\hat{\mathbb{Z}}$ via the Colimit

1. The open normal subgroups of $\hat{\mathbb{Z}}$ are $n\hat{\mathbb{Z}}$ for $n \geq 1$, with quotients $\hat{\mathbb{Z}}/n\hat{\mathbb{Z}} \cong \mathbb{Z}/n\mathbb{Z}$. By theorem 3.2.2, for any discrete $\hat{\mathbb{Z}}$-module M with trivial action,

$$H^q(\hat{\mathbb{Z}}, M) = \varinjlim_n H^q(\mathbb{Z}/n\mathbb{Z}, M),$$

where the colimit is over positive integers ordered by divisibility. The cohomology of the cyclic group $\mathbb{Z}/n\mathbb{Z}$ with trivial coefficients in M is periodic (example 2.3.8):

$$H^q(\mathbb{Z}/n\mathbb{Z}, M) \cong \begin{cases} M & q = 0, \\ M[n] & q \text{ odd}, q \geq 1, \\ M/nM & q \text{ even}, q \geq 2, \end{cases}$$

where $M[n] = \ker(n : M \to M)$ is the n-torsion submodule and M/nM is the quotient by the image of multiplication by n.

For $M = \mathbb{Z}$: the n-torsion $\mathbb{Z}[n] = 0$, so $H^q(\hat{\mathbb{Z}}, \mathbb{Z}) = 0$ for all odd $q \geq 1$. For even $q \geq 2$, the transition maps $\mathbb{Z}/n\mathbb{Z} \to \mathbb{Z}/m\mathbb{Z}$ (for $n|m$) are the canonical embeddings $k/n \mapsto k/n$ in $\mathbb{Q}/\mathbb{Z}$ (as computed in exercise 3.1.1 (3)), giving

$$H^q(\hat{\mathbb{Z}}, \mathbb{Z}) = \varinjlim_n \mathbb{Z}/n\mathbb{Z} = \mathbb{Q}/\mathbb{Z} \qquad (q \text{ even}, q \geq 2).$$

2. For $M = \mathbb{Z}/m\mathbb{Z}$ (trivial action): the n-torsion is $(\mathbb{Z}/m\mathbb{Z})[n] = \mathbb{Z}/\gcd(m,n)\mathbb{Z}$, and the quotient is $(\mathbb{Z}/m\mathbb{Z})/n(\mathbb{Z}/m\mathbb{Z}) \cong \mathbb{Z}/\gcd(m,n)\mathbb{Z}$. Once $m|n$, both stabilize to $\mathbb{Z}/m\mathbb{Z}$. Hence

$$H^q(\hat{\mathbb{Z}}, \mathbb{Z}/m\mathbb{Z}) \cong \mathbb{Z}/m\mathbb{Z} \qquad \text{for all } q \geq 1.$$

The colimit stabilizes at any n divisible by m — a reflection of the fact that $\hat{\mathbb{Z}}$ acts on $\mathbb{Z}/m\mathbb{Z}$ through its quotient $\mathbb{Z}/m\mathbb{Z}$.

The periodicity in the $\hat{\mathbb{Z}}$ example is special to cyclic (or procyclic) groups and does not persist for general profinite groups. The cohomological dimension of a profinite group — the largest n for which $H^n(G, M)$ can be nonzero — will be a central invariant in chapter 5.

Example 3.2.7: Galois Cohomology of the Multiplicative Group

1. Let K be a field and $G_K = \mathrm{Gal}(K^{\mathrm{sep}}/K)$ its absolute Galois group. The multiplicative group $(K^{\mathrm{sep}})^\times$ is a discrete G_K-module: for any $\alpha \in (K^{\mathrm{sep}})^\times$, the stabilizer $\mathrm{Stab}_{G_K}(\alpha)$ is the open subgroup $G_{K(\alpha)} = \mathrm{Gal}(K^{\mathrm{sep}}/K(\alpha))$. The colimit formula gives
$$H^n(G_K, (K^{\mathrm{sep}})^\times) = \varinjlim_{L/K} H^n(\mathrm{Gal}(L/K), L^\times),$$
where L ranges over finite Galois extensions of K inside K^{sep}.

2. For $K = \mathbb{Q}_p$, the colimit formula applied in degree 2 gives
$$H^2(G_{\mathbb{Q}_p}, \overline{\mathbb{Q}}_p^\times) = \varinjlim_{L/\mathbb{Q}_p} H^2(\mathrm{Gal}(L/\mathbb{Q}_p), L^\times).$$
Each group $H^2(\mathrm{Gal}(L/\mathbb{Q}_p), L^\times)$ classifies central simple algebras over $\mathbb{Q}_p$ split by L (chapter 7), and the colimit assembles into the Brauer group $\mathrm{Br}(\mathbb{Q}_p)$. The identification $\mathrm{Br}(\mathbb{Q}_p) \cong \mathbb{Q}/\mathbb{Z}$ via the invariant map is one of the central results of local class field theory (chapter 8).

3. For $K = \mathbb{R}$: the only nontrivial finite Galois extension is $\mathbb{C}/\mathbb{R}$, so the colimit stabilizes immediately and $H^n(G_\mathbb{R}, \mathbb{C}^\times) = H^n(\mathbb{Z}/2\mathbb{Z}, \mathbb{C}^\times)$ for all n. By the computation in example 3.1.7, $H^1(G_\mathbb{R}, \mathbb{C}^\times) = 0$ and $H^2(G_\mathbb{R}, \mathbb{C}^\times) \cong \mathbb{Z}/2\mathbb{Z}$. The latter is $\mathrm{Br}(\mathbb{R}) \cong \mathbb{Z}/2\mathbb{Z}$, reflecting the fact that the only non-split central simple algebra over $\mathbb{R}$ is the quaternion algebra $\mathbb{H}$.

The colimit formula interacts well with the long exact sequence. If $0 \to M' \to M \to M'' \to 0$ is a short exact sequence of discrete G-modules, then for each open normal subgroup U, the sequence $0 \to (M')^U \to M^U \to (M'')^U$ is left exact (but not necessarily exact on the right — exactness on the right can fail because the invariant functor is only left exact). The long exact sequences for each finite quotient G/U assemble into a colimit, and since filtered colimits are exact, the resulting long exact sequence for G is
$$\cdots \to H^n(G, M') \to H^n(G, M) \to H^n(G, M'') \xrightarrow{\delta} H^{n+1}(G, M') \to \cdots \tag{3.1}$$
This is the same long exact sequence that the derived functor formalism produces automatically, but the colimit description provides a concrete computational advantage: the connecting homomorphism δ on a class $[\alpha] \in H^n(G, M'')$ represented by a cocycle at finite level G/U is computed by applying the connecting homomorphism for the finite group G/U and then passing to the colimit. Since the

connecting homomorphism for finite groups is computed by explicit diagram chasing in the cochain complex (proposition 2.3.4), every connecting homomorphism in profinite cohomology is computable in principle — one reduces to a finite quotient where the computation is concrete.

Exercise 3.2.1

1. Let G be a profinite group and $0 \to \mathbb{Z} \xrightarrow{\times p} \mathbb{Z} \to \mathbb{Z}/p\mathbb{Z} \to 0$ the short exact sequence of discrete G-modules (with trivial G-action on all three terms). Write out the long exact sequence in cohomology and use the colimit formula to compute the connecting homomorphism $\delta : H^1(G, \mathbb{Z}/p\mathbb{Z}) \to H^2(G, \mathbb{Z})$. Identify δ as the map $\mathrm{Hom}_{\mathrm{cts}}(G, \mathbb{Z}/p\mathbb{Z}) \to H^2(G, \mathbb{Z})$ that sends a continuous homomorphism $\chi : G \to \mathbb{Z}/p\mathbb{Z}$ to the class of the extension $0 \to \mathbb{Z} \to E_\chi \to G/\ker\chi \to 0$.

2. Let $G = \mathbb{Z}_p$ (the additive group of p-adic integers, viewed as a pro-p-group). Compute $H^n(\mathbb{Z}_p, \mathbb{Z}/p\mathbb{Z})$ for all $n \geq 0$ using the colimit formula and the cohomology of $\mathbb{Z}/p^k\mathbb{Z}$ with coefficients in $\mathbb{Z}/p\mathbb{Z}$ (trivial action). (*Hint:* The open subgroups of $\mathbb{Z}_p$ are $p^k\mathbb{Z}_p$ with quotient $\mathbb{Z}/p^k\mathbb{Z}$. The colimit stabilizes.)

3. Give a direct proof (not using the derived functor formalism) that the long exact sequence (3.1) is exact, by showing that the long exact sequences for each finite quotient G/U form a directed system of exact sequences and applying the exactness of filtered colimits.

4. Show that the torsion property (corollary 3.2.4) fails in degree zero: give an example of a profinite group G and a discrete G-module M such that $H^0(G, M) = M^G$ contains elements of infinite order.

5. Let G be a profinite group and M a *finite* discrete G-module (i.e., $|M| < \infty$). Show that $H^n(G, M)$ is a torsion abelian group whose exponent divides $|M|^n \cdot \prod |G/U|$ for suitable U. More precisely, show that every element of $H^n(G, M)$ is killed by $|M|^n$. (*Hint:* Use dimension shifting: embed M into a coinduced module and apply induction on n.)

3.3 Inflation, Restriction, and Corestriction

The colimit formula of theorem 3.2.2 expresses $H^n(G, M)$ in terms of the cohomology of finite quotients, with the transition maps being a specific instance of a general construction: *inflation*. This section develops three functorial operations — inflation, restriction, and corestriction — that relate the cohomology of a profinite group to the cohomology of its subgroups and quotients. These maps are the basic tools for computing cohomology in practice, and their interactions (particularly the inflation–restriction exact sequence and the formula $\mathrm{Cor} \circ \mathrm{Res} = [G : H]$) underlie most of the structural results in Parts II and III.

Throughout this section, G denotes a profinite group and M a discrete G-module.

Definition 3.3.1: Inflation

Let N be a closed normal subgroup of G. The quotient G/N is a profinite group (by proposition 1.2.4), and M^N is a discrete G/N-module. The *inflation map*

$$\mathrm{Inf} = \mathrm{Inf}_{G/N}^{G} : H^n(G/N, M^N) \to H^n(G, M)$$

is defined at the cochain level by precomposition with the projection $\pi : G \twoheadrightarrow G/N$. Explicitly, if $f : (G/N)^n \to M^N$ is a continuous n-cocycle, then

$$\mathrm{Inf}(f)(g_1, \ldots, g_n) = f(\pi(g_1), \ldots, \pi(g_n)) \in M^N \subseteq M.$$

Since π is a continuous group homomorphism and $M^N \hookrightarrow M$ is G-equivariant (with G acting on M^N through G/N), the inflated cochain is continuous and the coboundary of an inflated cochain is the inflation of the coboundary. Hence Inf is a well-defined homomorphism on cohomology.

Inflation is the map that "extends a cocycle from a quotient to the full group by making it constant on N-cosets." The transition maps in the colimit formula of theorem 3.2.2 are precisely inflation maps: for open normal subgroups $V \subseteq U$, the map $H^n(G/U, M^U) \to H^n(G/V, M^V)$ is the inflation along $G/V \twoheadrightarrow G/U$.

Definition 3.3.2: Restriction

Let H be a closed subgroup of G. Every discrete G-module M is also a discrete H-module (if $\mathrm{Stab}_G(m)$ is open in G, then $\mathrm{Stab}_H(m) = H \cap \mathrm{Stab}_G(m)$ is open in H). The *restriction map*

$$\mathrm{Res} = \mathrm{Res}_G^H : H^n(G, M) \to H^n(H, M)$$

is defined at the cochain level by restricting the domain: if $f : G^n \to M$ is a continuous n-cocycle, then $\mathrm{Res}(f) = f|_{H^n} : H^n \to M$ is a continuous n-cocycle for H. The coboundary formula is unchanged, so restriction is a chain map and induces a well-defined map on cohomology.

In degree zero, restriction is the inclusion $M^G \hookrightarrow M^H$: if m is fixed by all of G, it is certainly fixed by the subgroup H. In degree one, restriction sends a continuous crossed homomorphism $f : G \to M$ to its restriction $f|_H : H \to M$, which is again a continuous crossed homomorphism.

Both inflation and restriction are natural in M: they commute with the maps in the long exact sequence induced by a short exact sequence $0 \to M' \to M \to M'' \to 0$.

The first fundamental relation between inflation and restriction constrains how cohomology classes can "come from a quotient."

Proposition 3.3.3: Inflation–Restriction Exact Sequence

Let N be a closed normal subgroup of G and M a discrete G-module. The sequence

$$0 \to H^1(G/N, M^N) \xrightarrow{\text{Inf}} H^1(G, M) \xrightarrow{\text{Res}} H^1(N, M)$$

is exact. That is: inflation is injective on H^1, and a class in $H^1(G, M)$ restricts to zero in $H^1(N, M)$ if and only if it is inflated from $H^1(G/N, M^N)$.

***Proof*:**
Step 1: Inf *is injective.* Let $f : G/N \to M^N$ be a continuous 1-cocycle such that $\text{Inf}(f) = f \circ \pi$ is a coboundary: $f(\pi(g)) = g \cdot m - m$ for some $m \in M$ and all $g \in G$. For $n \in N$, this gives $f(\pi(n)) = f(\bar{1}) = 0 = n \cdot m - m$, so $m \in M^N$. Then $f(\bar{g}) = g \cdot m - m$ for all $\bar{g} \in G/N$ (where g is any lift of $\bar{g}$, and the right side depends only on $\bar{g}$ because $m \in M^N$). Hence f is a coboundary in $C^1(G/N, M^N)$.

Step 2: $\text{im}(\text{Inf}) \subseteq \ker(\text{Res})$. If $f : G/N \to M^N$ is a 1-cocycle, then $\text{Res}(\text{Inf}(f))(n) = f(\pi(n)) = f(\bar{1}) = 0$ for all $n \in N$. So $\text{Res} \circ \text{Inf} = 0$.

Step 3: $\ker(\text{Res}) \subseteq \text{im}(\text{Inf})$. Let $f : G \to M$ be a continuous 1-cocycle with $f|_N = 0$ (i.e., f is a coboundary when restricted to N, and since $\text{Res}(f) = 0$ in $H^1(N, M)$, there exists $m_0 \in M$ with $f(n) = n \cdot m_0 - m_0$ for all $n \in N$). Replacing f by $f - d^0(m_0)$ (which is cohomologous to f), assume $f|_N = 0$. The cocycle condition $f(gn) = g \cdot f(n) + f(g) = f(g)$ shows that f is constant on N-cosets, so f factors through G/N. The cocycle condition also gives $f(ng) = n \cdot f(g) + f(n) = n \cdot f(g)$, and since $f(ng) = f(g)$ (as ng and g may not lie in the same coset — this requires care). Applying the cocycle condition: $f(ng) = n \cdot f(g) + f(n) = n \cdot f(g)$. But also $f(ng) = f(g)$ since $ng \in gN$ only if $g^{-1}ng \in N$, which holds since N is normal. So $n \cdot f(g) = f(g)$ for all $n \in N$ and $g \in G$, giving $f(g) \in M^N$. The factored map $\bar{f} : G/N \to M^N$ is a continuous 1-cocycle with $\text{Inf}(\bar{f}) = f$.

The exact sequence extends to a five-term sequence involving H^2. The proof of the extended version requires the *transgression map*, which arises from the Lyndon–Hochschild–Serre spectral sequence constructed in §4 of this chapter. For now, the statement is recorded for reference.

Theorem 3.3.4: Five-Term Exact Sequence

Let N be a closed normal subgroup of G and M a discrete G-module. The sequence

$$0 \to H^1(G/N, M^N) \xrightarrow{\text{Inf}} H^1(G, M) \xrightarrow{\text{Res}} H^1(N, M)^{G/N} \xrightarrow{\text{tg}} H^2(G/N, M^N) \xrightarrow{\text{Inf}} H^2(G, M)$$

is exact, where $H^1(N, M)^{G/N}$ denotes the G/N-invariants of $H^1(N, M)$ under the conjugation action, and tg is the transgression map.

The conjugation action of G/N on $H^1(N, M)$ is defined as follows: for $\bar{g} \in G/N$ and a 1-cocycle $f : N \to M$, the conjugate cocycle is $(\bar{g} \cdot f)(n) = g \cdot f(g^{-1}ng)$. This is independent of the choice of lift g of $\bar{g}$, and a class in $H^1(N, M)$ is G/N-invariant if and only if for every lift g, the cocycle $n \mapsto g \cdot f(g^{-1}ng)$ is cohomologous to f.

The third fundamental operation moves cohomology classes in the opposite direction to restriction: from a subgroup back to the full group. It is defined only for subgroups of finite index.

Definition 3.3.5: Corestriction

Let H be an open (hence finite-index) subgroup of G, and let $\{s_1, \ldots, s_m\}$ be a set of left coset representatives for H in G (so $G = \bigsqcup_{i=1}^m s_i H$ and $m = [G : H]$). For each $g \in G$ and each index i, there is a unique index $\sigma_g(i)$ and a unique element $h_g^{(i)} \in H$ such that $gs_i = s_{\sigma_g(i)} h_g^{(i)}$. The *corestriction* (or *transfer*) map

$$\mathrm{Cor} = \mathrm{Cor}_H^G : H^n(H, M) \to H^n(G, M)$$

is defined at the cochain level as follows. For a continuous n-cochain $f : H^n \to M$, set

$$\mathrm{Cor}(f)(g_1, \ldots, g_n) = \sum_{i=1}^m s_i \cdot f\Big(h_{g_1}^{(i)},\ h_{g_2}^{(\sigma_{g_1}(i))},\ \ldots,\ h_{g_n}^{(\sigma_{g_{n-1}} \cdots g_1(i))}\Big).$$

This formula is independent of the choice of coset representatives (a different choice changes each s_i by a right multiple from H, which is absorbed by the H-equivariance of f). The map Cor is a chain map and induces a well-defined homomorphism on cohomology.

The formula is complex in general, but in low degrees it simplifies. In degree 0, the corestriction $\mathrm{Cor} : M^H \to M^G$ is the *trace* (or *norm*) map:

$$\mathrm{Cor}(m) = \sum_{i=1}^m s_i \cdot m.$$

This is independent of the coset representatives and lands in M^G (since $g \cdot \mathrm{Cor}(m) = \sum_i gs_i \cdot m = \sum_i s_{\sigma_g(i)} h_g^{(i)} \cdot m = \sum_i s_{\sigma_g(i)} \cdot m = \mathrm{Cor}(m)$, because $h_g^{(i)} \cdot m = m$ for $m \in M^H$ and σ_g is a permutation). In degree 1, if $f : H \to M$ is a continuous 1-cocycle, then

$$\mathrm{Cor}(f)(g) = \sum_{i=1}^m s_i \cdot f(s_i^{-1} g s_{\sigma_g(i)^{-1}}) = \sum_{i=1}^m s_i \cdot f(h_g^{(i)}). \tag{3.2}$$

The central algebraic identity governing the interaction of restriction and corestriction is the following.

Proposition 3.3.6: The Formula $\mathrm{Cor} \circ \mathrm{Res} = [G : H]$

Let H be an open subgroup of G with $[G : H] = m$. For every $n \geq 0$ and every $\alpha \in H^n(G, M)$,

$$\mathrm{Cor}(\mathrm{Res}(\alpha)) = m \cdot \alpha.$$

***Proof*:**
Let $f : G^n \to M$ be a continuous n-cocycle representing α. Then $\mathrm{Res}(f) = f|_{H^n}$, and

$$\mathrm{Cor}(\mathrm{Res}(f))(g_1, \ldots, g_n) = \sum_{i=1}^m s_i \cdot f(h_{g_1}^{(i)}, h_{g_2}^{(\sigma_{g_1}(i))}, \ldots, h_{g_n}^{(\sigma_{g_{n-1}} \cdots g_1(i))}).$$

The key observation is that f is a G-cocycle, so it satisfies the cocycle identity. The elements

$h_{g_j}^{(\cdot)}$ are defined by $g_j s_\cdot = s_\cdot h_{g_j}^{(\cdot)}$, and the cocycle identity applied to the decomposition $g_1 = s_{\sigma_{g_1}(i)} h_{g_1}^{(i)} s_i^{-1}$ allows one to rewrite $s_i \cdot f(h_{g_1}^{(i)}, \ldots)$ in terms of $f(g_1, \ldots, g_n)$ plus coboundary terms. Summing over i and using the fact that σ_{g_1} is a permutation of $\{1, \ldots, m\}$, the coboundary terms cancel, and the result is $m \cdot f(g_1, \ldots, g_n)$.

A cleaner argument uses the colimit formula. By theorem 3.2.2, the class α is represented at some finite level G/U for an open normal subgroup $U \subseteq H$. The formula $\mathrm{Cor} \circ \mathrm{Res} = [G : H]$ for the finite group G/U acting on M^U is a standard identity in finite-group cohomology (proved by the same cochain-level computation, which is a finite sum). Since the operations Res, Cor, and multiplication by m are all compatible with inflation, the identity lifts to the colimit.

This formula has several immediate and important consequences.

Corollary 3.3.7: Consequences of $\mathrm{Cor} \circ \mathrm{Res} = [G : H]$

Let H be an open subgroup of G with $[G : H] = m$.

1. *Restriction detects m-torsion-free elements:* if $\alpha \in H^n(G, M)$ satisfies $\mathrm{Res}(\alpha) = 0$, then $m \cdot \alpha = 0$. In particular, if $H^n(G, M)$ has no m-torsion, then Res is injective.
2. *Corestriction surjects onto m-divisible elements:* the image of Cor contains $m \cdot H^n(G, M)$. In particular, if m is invertible in $H^n(G, M)$, then Cor is surjective.
3. *p-primary decomposition:* if p is a prime not dividing $m = [G : H]$, then Res is injective on the p-primary component $H^n(G, M)\{p\}$, and the composition $\frac{1}{m}\mathrm{Cor} : H^n(H, M)\{p\} \to H^n(G, M)\{p\}$ is a left inverse of Res.
4. *Annihilation by the order:* if G is finite, then $|G| \cdot H^n(G, M) = 0$ for $n \geq 1$. (Apply the formula with $H = \{1\}$ and note that $H^n(\{1\}, M) = 0$ for $n \geq 1$.)

Part (3) is the primary tool for reducing Galois cohomology computations to one prime at a time. If H is a Sylow p-subgroup of a finite group G (or, in the profinite case, a pro-p-Sylow subgroup), then $[G : H]$ is prime to p, and restriction gives an injection on p-primary parts. This reduces many questions about $H^n(G, M)\{p\}$ to questions about the pro-p-group H, where the theory of chapter 6 applies.

Example 3.3.8: Inflation and Restriction for $G_{\mathbb{Q}_p}$

Let $K = \mathbb{Q}_p$ and consider the absolute Galois group $G_{\mathbb{Q}_p}$. The inertia subgroup $I_p = \mathrm{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p^{\mathrm{nr}})$ is a closed normal subgroup with quotient $G_{\mathbb{Q}_p}/I_p \cong \mathrm{Gal}(\mathbb{Q}_p^{\mathrm{nr}}/\mathbb{Q}_p) \cong \hat{\mathbb{Z}}$ (generated topologically by the Frobenius automorphism Frob_p).

1. For any discrete $G_{\mathbb{Q}_p}$-module M, the inflation–restriction exact sequence gives

$$0 \to H^1(\hat{\mathbb{Z}}, M^{I_p}) \xrightarrow{\mathrm{Inf}} H^1(G_{\mathbb{Q}_p}, M) \xrightarrow{\mathrm{Res}} H^1(I_p, M)^{\hat{\mathbb{Z}}}.$$

   When M is an unramified module (meaning I_p acts trivially, so $M^{I_p} = M$), the restriction map lands in $\mathrm{Hom}_{\mathrm{cts}}(I_p, M)^{\hat{\mathbb{Z}}}$, which captures the "ramified part" of $H^1(G_{\mathbb{Q}_p}, M)$. The inflation from $H^1(\hat{\mathbb{Z}}, M)$ captures the "unramified part."

2. Take $M = \mathbb{Z}/n\mathbb{Z}$ with trivial action. Since I_p acts trivially, $M^{I_p} = \mathbb{Z}/n\mathbb{Z}$, and the inflation–restriction sequence reads

$$0 \to H^1(\hat{\mathbb{Z}}, \mathbb{Z}/n\mathbb{Z}) \xrightarrow{\text{Inf}} H^1(G_{\mathbb{Q}_p}, \mathbb{Z}/n\mathbb{Z}) \xrightarrow{\text{Res}} H^1(I_p, \mathbb{Z}/n\mathbb{Z})^{\hat{\mathbb{Z}}}.$$

The group $H^1(\hat{\mathbb{Z}}, \mathbb{Z}/n\mathbb{Z}) = \mathrm{Hom}_{\text{cts}}(\hat{\mathbb{Z}}, \mathbb{Z}/n\mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}$ (generated by the homomorphism sending Frob_p to 1). This is the subgroup of $H^1(G_{\mathbb{Q}_p}, \mathbb{Z}/n\mathbb{Z})$ classifying unramified $\mathbb{Z}/n\mathbb{Z}$-extensions of $\mathbb{Q}_p$ (cyclic extensions contained in $\mathbb{Q}_p^{\text{nr}}$). The restriction map detects the ramification: $\mathrm{Res}(\chi) = 0$ if and only if the extension classified by χ is unramified.

This example illustrates a pattern that pervades the local theory of chapter 8: the inflation–restriction sequence for the pair $(G_{\mathbb{Q}_p}, I_p)$ separates the "unramified" and "ramified" contributions to Galois cohomology, and the structure of I_p (a pro-p-group times a tame quotient) provides a further decomposition.

Exercise 3.3.1

1. Let N be a closed normal subgroup of G and $0 \to M' \to M \to M'' \to 0$ a short exact sequence of discrete G-modules. Show that inflation commutes with the connecting homomorphism: the diagram
$$\begin{array}{ccc} H^n(G/N, (M'')^N) & \xrightarrow{\delta} & H^{n+1}(G/N, (M')^N) \\ \downarrow{\scriptstyle \text{Inf}} & & \downarrow{\scriptstyle \text{Inf}} \\ H^n(G, M'') & \xrightarrow{\delta} & H^{n+1}(G, M') \end{array}$$
commutes. State and prove the analogous result for restriction.

2. Let $G = \mathbb{Z}/6\mathbb{Z}$ and $H = 2\mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/3\mathbb{Z}$ (the unique subgroup of index 2). Let $M = \mathbb{Z}$ with trivial G-action. Compute $\mathrm{Cor} : H^2(H, \mathbb{Z}) \to H^2(G, \mathbb{Z})$ explicitly using the cochain formula and coset representatives $\{0, 1\}$. Verify that $\mathrm{Cor} \circ \mathrm{Res} = 2$ on $H^2(G, \mathbb{Z}) \cong \mathbb{Z}/6\mathbb{Z}$.

3. Let G be a profinite group and H an open subgroup. Show that $\mathrm{Res} \circ \mathrm{Cor} : H^n(H, M) \to H^n(H, M)$ is given by
$$\mathrm{Res}(\mathrm{Cor}(\alpha)) = \sum_{s \in H\backslash G/H} \mathrm{Cor}^H_{H \cap sHs^{-1}} \left(\mathrm{Res}^{H \cap sHs^{-1}}_H (s_* \alpha) \right),$$
where the sum runs over double coset representatives and s_* denotes the conjugation action. This is the *Mackey formula* (or double coset formula) for group cohomology.

4. Let G be a profinite group and M a discrete G-module. Use proposition 3.3.6 to give a short proof that $H^n(G, M)\{p\} = 0$ for $n \geq 1$ whenever G is a pro-p'-group (i.e., every open normal subgroup U satisfies $p \nmid [G : U]$).

5. Let $G = \mathbb{Z}/4\mathbb{Z}$, $N = 2\mathbb{Z}/4\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z}$, and $M = \mathbb{Z}$ with trivial action. Write out the five-term exact sequence (theorem 3.3.4) and compute all five terms explicitly. Identify the transgression map $\mathrm{tg} : H^1(N, \mathbb{Z})^{G/N} \to H^2(G/N, \mathbb{Z})$ and verify exactness.

3.4 The Lyndon–Hochschild–Serre Spectral Sequence

The inflation–restriction exact sequence of proposition 3.3.3 captures only the beginning of the relationship between $H^\bullet(G, M)$, $H^\bullet(N, M)$, and $H^\bullet(G/N, -)$. The five-term exact sequence (theorem 3.3.4) extends this one step further, but a complete description requires a spectral sequence. The Lyndon–Hochschild–Serre (LHS) spectral sequence, also called the Hochschild–Serre spectral sequence, is the fundamental tool for computing cohomology inductively: it assembles $H^n(G, M)$ from the cohomology of the quotient G/N with coefficients in the cohomology of the normal subgroup N.

The general framework is the Grothendieck spectral sequence for a composition of derived functors, treated in [9, The Grothendieck Spectral Sequence]. This section instantiates that framework in the profinite setting, verifies the necessary hypotheses, and then develops the concrete output: the E_2 page, the edge maps, the transgression, the recovery of the five-term exact sequence, and explicit computations.

Before stating the spectral sequence, the G/N-module structure on $H^q(N, M)$ must be made precise, since this structure determines the coefficients on the E_2 page. Let N be a closed normal subgroup of G and M a discrete G-module.

Definition 3.4.1: Conjugation Action

For $g \in G$, define the *conjugation map* $c_g : N \to N$ by $c_g(n) = gng^{-1}$ and the *twist map* $\tau_g : M \to M$ by $\tau_g(m) = g \cdot m$. Together, these induce a map on continuous cochains

$$g_\# : C^q_{\text{cts}}(N, M) \to C^q_{\text{cts}}(N, M), \qquad (g_\# f)(n_1, \ldots, n_q) = g \cdot f(g^{-1}n_1 g, \ldots, g^{-1}n_q g).$$

This map commutes with the coboundary operator, hence induces a map $g_\# : H^q(N, M) \to H^q(N, M)$. Since $n_\#$ is the identity on cohomology for every $n \in N$ (the maps c_n and τ_n are inner, and inner automorphisms act trivially on cohomology), the map $g_\#$ depends only on the coset $\bar{g} = gN \in G/N$. This defines the *conjugation action* of G/N on $H^q(N, M)$.

The claim that $n_\#$ acts trivially on cohomology for $n \in N$ deserves verification. A continuous q-cocycle $f : N^q \to M$ and its conjugate $n_\# f$ are related by

$$(n_\# f)(n_1, \ldots, n_q) = n \cdot f(n^{-1}n_1 n, \ldots, n^{-1}n_q n).$$

The difference $n_\# f - f$ is a coboundary: one constructs an explicit $(q-1)$-cochain h with $d^{q-1}h = n_\# f - f$ using the homotopy $h(n_1, \ldots, n_{q-1}) = \sum_{i=0}^{q-1}(-1)^i f(n^{-1}n_1 n, \ldots, n^{-1}n_i n, n, n_{i+1}, \ldots, n_{q-1})$ (adjusted with appropriate signs and the action of n). The key algebraic input is that $n \in N$ acts on itself by an inner automorphism, and inner automorphisms are homotopic to the identity in the bar complex.

Proposition 3.4.2: $H^q(N, M)$ as a Discrete G/N-Module

The conjugation action makes $H^q(N, M)$ into a discrete G/N-module.

***Proof*:**
By the colimit formula (theorem 3.2.2), $H^q(N, M) = \varinjlim_V H^q(N/V, M^V)$, where V ranges over open normal subgroups of N that are also normal in G (every open normal subgroup of N

contains one that is normal in G, by intersecting its conjugates). The conjugation action of G/N on $H^q(N/V, M^V)$ factors through the finite group $G/(NV') = G/N \cdot (V'/V)$ for suitable V', so each element of $H^q(N/V, M^V)$ has open stabilizer in G/N. Since $H^q(N, M)$ is the colimit of these modules with open stabilizers, it is a discrete G/N-module by definition 3.1.1.

The next ingredient is the acyclicity hypothesis required by the Grothendieck spectral sequence. The functors in question are $\mathcal{F}' = (-)^N$ and $\mathcal{G} = (-)^{G/N}$, composing to $(-)^G$, and the hypothesis is that $(-)^N$ sends injective discrete G-modules to $(-)^{G/N}$-acyclic discrete G/N-modules. The following lemma establishes a stronger statement.

Lemma 3.4.3: N-Invariants Preserve Injectives

Let N be a closed normal subgroup of G. If I is an injective object in $\mathrm{Mod}_G^{\mathrm{disc}}$, then I^N is an injective object in $\mathrm{Mod}_{G/N}^{\mathrm{disc}}$. In particular, I^N is acyclic for the functor $(-)^{G/N}$.

***Proof*:**
Let $0 \to A \to B$ be an injection of discrete G/N-modules, and let $\varphi : A \to I^N$ be a G/N-equivariant map. View A and B as discrete G-modules via the projection $G \twoheadrightarrow G/N$ (the G-action factors through G/N, so every element has open stabilizer). The composite $A \xrightarrow{\varphi} I^N \hookrightarrow I$ is G-equivariant, and since I is injective in $\mathrm{Mod}_G^{\mathrm{disc}}$, this extends to a G-equivariant map $\psi : B \to I$.

The image of ψ lies in I^N: for any $b \in B$ and $n \in N$, the element $\psi(b)$ satisfies $n \cdot \psi(b) = \psi(n \cdot b) = \psi(b)$, since N acts trivially on B (as B is a G/N-module). Hence ψ factors through I^N, giving a G/N-equivariant extension $\bar{\psi} : B \to I^N$ of φ.

With the hypotheses verified, the Grothendieck spectral sequence ([9, The Grothendieck Spectral Sequence]) applies to the composition $(-)^G = (-)^{G/N} \circ (-)^N$.

Theorem 3.4.4: The Lyndon–Hochschild–Serre Spectral Sequence

Let G be a profinite group, N a closed normal subgroup, and M a discrete G-module. There is a first-quadrant cohomological spectral sequence with

$$E_2^{p,q} = H^p(G/N,\ H^q(N, M)) \implies H^{p+q}(G, M),$$

where $H^q(N, M)$ carries the conjugation action of G/N from definition 3.4.1. The spectral sequence is natural in M and functorial in the pair (G, N).

***Proof*:**
The invariant functor $(-)^G : \mathrm{Mod}_G^{\mathrm{disc}} \to \mathbf{Ab}$ factors as the composition

$$\mathrm{Mod}_G^{\mathrm{disc}} \xrightarrow{(-)^N} \mathrm{Mod}_{G/N}^{\mathrm{disc}} \xrightarrow{(-)^{G/N}} \mathbf{Ab}.$$

Both categories have enough injectives (proposition 3.1.3 for $\mathrm{Mod}_G^{\mathrm{disc}}$; the same argument applies to $\mathrm{Mod}_{G/N}^{\mathrm{disc}}$). By lemma 3.4.3, the functor $(-)^N$ sends injective objects in $\mathrm{Mod}_G^{\mathrm{disc}}$ to injective (hence acyclic) objects in $\mathrm{Mod}_{G/N}^{\mathrm{disc}}$. The Grothendieck spectral sequence ([9, The Grothendieck

Spectral Sequence]) then gives a first-quadrant spectral sequence

$$E_2^{p,q} = (R^p(-)^{G/N})(R^q(-)^N(M)) = H^p(G/N, H^q(N, M)) \implies R^{p+q}(-)^G(M) = H^{p+q}(G, M).$$

Naturality follows from the naturality of the Grothendieck spectral sequence in the input module M and in the pair of functors.

The spectral sequence is a first-quadrant spectral sequence, meaning $E_r^{p,q} = 0$ whenever $p < 0$ or $q < 0$. This has the consequence that only finitely many differentials can affect any given term: the differentials $d_r^{p,q} : E_r^{p,q} \to E_r^{p+r,q-r+1}$ eventually land outside the first quadrant, so $E_r^{p,q}$ stabilizes to a limiting value $E_\infty^{p,q}$ for r sufficiently large.

The convergence $E_2^{p,q} \Rightarrow H^{p+q}(G, M)$ means that for each $n \geq 0$, the group $H^n(G, M)$ admits a finite descending filtration

$$H^n(G, M) = F^0 H^n \geqslant F^1 H^n \geqslant \cdots \geqslant F^n H^n \geqslant F^{n+1} H^n = 0 \tag{3.3}$$

with graded pieces $F^p H^{p+q}/F^{p+1} H^{p+q} \cong E_\infty^{p,q}$. The group $H^n(G, M)$ is thus assembled from the subquotients $E_\infty^{p,q}$ with $p + q = n$, but the extensions relating these subquotients are not determined by the spectral sequence alone.

The most directly useful features of the spectral sequence are the *edge maps* and the *transgression*, which describe the interaction between the E_2 page and the abutment in low degrees.

Proposition 3.4.5: Edge Maps

The spectral sequence of theorem 3.4.4 has the following edge maps:

1. *Inflation edge:* For each $n \geq 0$, the composition

$$H^n(G/N, M^N) = E_2^{n,0} \twoheadrightarrow E_\infty^{n,0} = F^n H^n / F^{n+1} H^n \hookrightarrow H^n(G, M)$$

(where the first map is the quotient by the images of all differentials landing on the $(n, 0)$ spot, and the second map is the inclusion of the top filtration step) coincides with the inflation map $\mathrm{Inf} : H^n(G/N, M^N) \to H^n(G, M)$ from definition 3.3.1.

2. *Restriction edge:* For each $n \geq 0$, the composition

$$H^n(G, M) = F^0 H^n \twoheadrightarrow F^0 H^n / F^1 H^n = E_\infty^{0,n} \hookrightarrow E_2^{0,n} = H^n(N, M)^{G/N}$$

(where the first map is the quotient, and the second map is the inclusion of E_∞ into E_2) coincides with the restriction map $\mathrm{Res} : H^n(G, M) \to H^n(N, M)^{G/N}$.

***Proof*:**
Both identifications follow from tracking the maps through the construction of the Grothendieck spectral sequence.

Step 1: Inflation edge. Let $0 \to M \to I^\bullet$ be an injective resolution in $\mathrm{Mod}_G^{\mathrm{disc}}$. The E_2 page is computed from the double complex $(I^p)^N \to (I^{p+1})^N$ by taking $(-)^{G/N}$-derived functors. The edge map on the bottom row ($q = 0$) is the map $H^n(((I^\bullet)^N)^{G/N}) \to H^n((I^\bullet)^G)$. Since $(I^p)^N$ is injective in $\mathrm{Mod}_{G/N}^{\mathrm{disc}}$ by lemma 3.4.3, the complex $(I^\bullet)^N$ is an injective resolution of M^N in

$\mathrm{Mod}^{\mathrm{disc}}_{G/N}$. The map from H^n of the G/N-invariants of this complex to H^n of the G-invariants of the original complex is precisely the inflation, because $(-)^{G/N} \circ (-)^N = (-)^G$ and the inflation is the map induced by the natural transformation from the derived functors of the composition to the composition of the derived functors.

Step 2: Restriction edge. The edge map on the left column ($p = 0$) sends a class in $H^n(G, M)$ to the zeroth derived functor of $(-)^{G/N}$ applied to $H^n(N, M)$, which is $H^n(N, M)^{G/N}$. Tracing through the injective resolution, this map sends a cocycle $f \in Z^n_{\mathrm{cts}}(G, M)$ to the class of its restriction $f|_{N^n} \in Z^n_{\mathrm{cts}}(N, M)$, viewed in $H^n(N, M)^{G/N}$.

Definition 3.4.6: Transgression

The *transgression* is the differential

$$\mathrm{tg} = d_2^{0,q} : E_2^{0,q} \to E_2^{2,q-1},$$

which maps $H^q(N, M)^{G/N}$ to $H^2(G/N, H^{q-1}(N, M))$. More precisely, the transgression is defined on the subgroup $E_2^{0,q} = H^0(G/N, H^q(N, M)) = H^q(N, M)^{G/N}$, but at the E_2 level its domain may be cut down by the image of $d_2^{-2,q+1}$, which vanishes since $E_2^{-2,q+1} = 0$ (we are in the first quadrant). Hence tg is defined on all of $H^q(N, M)^{G/N}$.

The transgression in degree $q = 1$ is the map

$$\mathrm{tg} : H^1(N, M)^{G/N} \to H^2(G/N, M^N)$$

that appeared in the five-term exact sequence of theorem 3.3.4. An explicit description: given a G/N-invariant class $[\alpha] \in H^1(N, M)^{G/N}$ represented by a continuous 1-cocycle $\alpha : N \to M$, the G/N-invariance means that for each $g \in G$, the cocycle $n \mapsto g \cdot \alpha(g^{-1}ng)$ is cohomologous to α. Choose, for each $\bar{g} \in G/N$, an element $\beta(\bar{g}) \in M$ such that $g \cdot \alpha(g^{-1}ng) - \alpha(n) = n \cdot \beta(\bar{g}) - \beta(\bar{g})$ for all $n \in N$. Then the function

$$c(\bar{g}_1, \bar{g}_2) = g_1 \cdot \beta(\bar{g}_2) - \beta(\overline{g_1 g_2}) + \beta(\bar{g}_1) \tag{3.4}$$

is a continuous 2-cocycle on G/N with values in M^N, and its class in $H^2(G/N, M^N)$ is $\mathrm{tg}([\alpha])$. This class is independent of the choices of lifts g and cobounding elements $\beta(\bar{g})$.

The five-term exact sequence stated in theorem 3.3.4 can now be proved as the low-degree exact sequence of the LHS spectral sequence. This derivation illustrates the general method for extracting exact sequences from spectral sequences.

Proposition 3.4.7: Five-Term Exact Sequence from the Spectral Sequence

The spectral sequence of theorem 3.4.4 yields the exact sequence

$$0 \to H^1(G/N, M^N) \xrightarrow{\mathrm{Inf}} H^1(G, M) \xrightarrow{\mathrm{Res}} H^1(N, M)^{G/N} \xrightarrow{\mathrm{tg}} H^2(G/N, M^N) \xrightarrow{\mathrm{Inf}} H^2(G, M).$$

***Proof*:**
Examine the E_2 page in the region relevant to total degree $n = 1$ and $n = 2$:

$$\begin{array}{r|ccc}
q=2 & H^0(G/N, H^2(N,M)) & \cdots & \\
q=1 & H^0(G/N, H^1(N,M)) & H^1(G/N, H^1(N,M)) & \cdots \\
q=0 & H^0(G/N, M^N) & H^1(G/N, M^N) & H^2(G/N, M^N) \\
\hline
 & p=0 & p=1 & p=2
\end{array}$$

The differential $d_2^{p,q} : E_2^{p,q} \to E_2^{p+2,q-1}$ maps two steps to the right and one step down. The relevant differentials are:

- $d_2^{0,1} : E_2^{0,1} = H^1(N, M)^{G/N} \to E_2^{2,0} = H^2(G/N, M^N)$. This is the transgression.
- $d_2^{1,0} : E_2^{1,0} = H^1(G/N, M^N) \to E_2^{3,-1} = 0$. This is the zero map.

Step 1: $E_\infty^{1,0} = E_2^{1,0}$. The term $E_2^{1,0}$ can only be affected by differentials from $E_2^{-1,1} = 0$ (incoming) and $d_2^{1,0}$ to $E_2^{3,-1} = 0$ (outgoing). Hence $E_3^{1,0} = E_2^{1,0}$. All subsequent differentials $d_r^{1,0} : E_r^{1,0} \to E_r^{1+r,1-r}$ land in the region $q < 0$ for $r \geq 2$, and differentials $d_r^{1-r,r-1} : E_r^{1-r,r-1} \to E_r^{1,0}$ originate in the region $p < 0$ for $r \geq 2$. So $E_\infty^{1,0} = E_2^{1,0} = H^1(G/N, M^N)$.

Step 2: $E_\infty^{0,1} = \ker(\mathrm{tg})$. The term $E_2^{0,1}$ can only receive differentials from $E_2^{-2,2} = 0$ and emits $d_2^{0,1} = \mathrm{tg}$. Hence $E_3^{0,1} = \ker(\mathrm{tg})$. For $r \geq 3$, all subsequent differentials to and from $E_r^{0,1}$ involve terms with $p < 0$, so $E_\infty^{0,1} = \ker(\mathrm{tg})$.

Step 3: The filtration on $H^1(G, M)$*.* The convergence gives a filtration $F^1H^1 \subseteq H^1(G, M)$ with $H^1(G, M)/F^1H^1 \cong E_\infty^{0,1} = \ker(\mathrm{tg})$ and $F^1H^1 \cong E_\infty^{1,0} = H^1(G/N, M^N)$. By the edge map identifications (proposition 3.4.5), $F^1H^1 = \mathrm{im}(\mathrm{Inf})$ and the quotient map $H^1(G, M) \twoheadrightarrow E_\infty^{0,1}$ is Res. This gives the exact sequence

$$0 \to H^1(G/N, M^N) \xrightarrow{\mathrm{Inf}} H^1(G, M) \xrightarrow{\mathrm{Res}} H^1(N, M)^{G/N} \xrightarrow{\mathrm{tg}} H^2(G/N, M^N).$$

Step 4: Exactness at $H^2(G/N, M^N)$*.* The filtration on $H^2(G, M)$ gives $E_\infty^{2,0} \cong F^2H^2 \subseteq H^2(G, M)$, and $E_\infty^{2,0} = E_2^{2,0}/\mathrm{im}(\mathrm{tg}) = H^2(G/N, M^N)/\mathrm{im}(\mathrm{tg})$ (since the only differential landing on $E_2^{2,0}$ at the E_2 level is $d_2^{0,1} = \mathrm{tg}$, and subsequent differentials from $E_r^{2-r,r-1}$ with $r \geq 3$ originate in $p < 0$). The edge map identification shows that $F^2H^2 \hookrightarrow H^2(G, M)$ is the inflation map. Hence $\ker(\mathrm{Inf} : H^2(G/N, M^N) \to H^2(G, M)) = \mathrm{im}(\mathrm{tg})$, completing the five-term exact sequence.

This proof illustrates the general method for extracting exact sequences from spectral sequences: determine which terms of the E_2 page contribute to the filtration of $H^n(G, M)$, track the differentials, and translate the resulting inclusions and quotients into maps between cohomology groups.

Beyond the five-term exact sequence, the d_2 differentials are the first and often the most important differentials on the E_2 page. The general differential

$$d_2^{p,q} : H^p(G/N, H^q(N, M)) \to H^{p+2}(G/N, H^{q-1}(N, M))$$

can be described in terms of the cup product (to be developed in chapter 4) and the connecting homomorphism. For the computations in this book, the most important cases are:

- $d_2^{0,q}$: the transgression, described explicitly via the formula (3.4).
- $d_2^{p,0}$: always zero, since the source is $H^p(G/N, M^N)$ and the target is $H^{p+2}(G/N, H^{-1}(N, M)) = 0$.

When all the differentials on a page vanish, the spectral sequence *degenerates* at that page, and the filtration on $H^n(G, M)$ is determined by the E_2 terms alone.

Example 3.4.8: A Central Extension

Let p be a prime and consider the central extension

$$1 \to N \to G \to Q \to 1$$

with $N = \mathbb{Z}/p\mathbb{Z}$, $G = \mathbb{Z}/p^2\mathbb{Z}$, and $Q = G/N \cong \mathbb{Z}/p\mathbb{Z}$. Take $M = \mathbb{Z}/p\mathbb{Z}$ with trivial G-action. Since the extension is central ($N \subseteq Z(G)$), the conjugation action of Q on $H^q(N, M)$ is trivial, which simplifies the E_2 page considerably.

1. The E_2 page. Since $N = \mathbb{Z}/p\mathbb{Z}$ and $M = \mathbb{Z}/p\mathbb{Z}$ with trivial action, the cohomology $H^q(N, M)$ is computed by the periodic cohomology of cyclic groups (example 2.3.8):

$$H^q(\mathbb{Z}/p\mathbb{Z}, \mathbb{Z}/p\mathbb{Z}) \cong \mathbb{Z}/p\mathbb{Z} \quad \text{for all } q \geq 0.$$

Since the Q-action is trivial, the E_2 page is

$$E_2^{p,q} = H^p(\mathbb{Z}/p\mathbb{Z}, \mathbb{Z}/p\mathbb{Z}) \cong \mathbb{Z}/p\mathbb{Z} \quad \text{for all } p, q \geq 0.$$

Every entry in the first quadrant is a copy of $\mathbb{Z}/p\mathbb{Z}$. This is displayed as:

$q=3$	$\mathbb{Z}/p\mathbb{Z}$	$\mathbb{Z}/p\mathbb{Z}$	$\mathbb{Z}/p\mathbb{Z}$	$\mathbb{Z}/p\mathbb{Z}$	$\cdots$
$q=2$	$\mathbb{Z}/p\mathbb{Z}$	$\mathbb{Z}/p\mathbb{Z}$	$\mathbb{Z}/p\mathbb{Z}$	$\mathbb{Z}/p\mathbb{Z}$	$\cdots$
$q=1$	$\mathbb{Z}/p\mathbb{Z}$	$\mathbb{Z}/p\mathbb{Z}$	$\mathbb{Z}/p\mathbb{Z}$	$\mathbb{Z}/p\mathbb{Z}$	$\cdots$
$q=0$	$\mathbb{Z}/p\mathbb{Z}$	$\mathbb{Z}/p\mathbb{Z}$	$\mathbb{Z}/p\mathbb{Z}$	$\mathbb{Z}/p\mathbb{Z}$	$\cdots$
	$p=0$	$p=1$	$p=2$	$p=3$	

2. The differentials. The spectral sequence converges to $H^{p+q}(G, M) = H^{p+q}(\mathbb{Z}/p^2\mathbb{Z}, \mathbb{Z}/p\mathbb{Z})$. By the periodic cohomology of cyclic groups, $H^n(\mathbb{Z}/p^2\mathbb{Z}, \mathbb{Z}/p\mathbb{Z}) \cong \mathbb{Z}/p\mathbb{Z}$ for all $n \geq 0$. So the abutment is $\mathbb{Z}/p\mathbb{Z}$ in every total degree.

On the E_2 page, the total degree n contributes $n+1$ copies of $\mathbb{Z}/p\mathbb{Z}$ (from $E_2^{0,n}, E_2^{1,n-1}, \ldots, E_2^{n,0}$). Since the abutment is a single $\mathbb{Z}/p\mathbb{Z}$ in each degree, the differentials must kill all but one copy. The differential $d_2^{p,q} : E_2^{p,q} \to E_2^{p+2,q-1}$ maps $\mathbb{Z}/p\mathbb{Z} \to \mathbb{Z}/p\mathbb{Z}$ and is therefore either zero or an isomorphism. For the dimensions to work out (each total degree must contribute exactly $\mathbb{Z}/p\mathbb{Z}$ to the abutment), the d_2 differentials must be isomorphisms along every "staircase" pattern.

3. The transgression. The differential $d_2^{0,1} : H^1(N, M)^Q \to H^2(Q, M^N)$ is a map $\mathbb{Z}/p\mathbb{Z} \to \mathbb{Z}/p\mathbb{Z}$. By the five-term exact sequence, $\ker(\mathrm{tg}) = E_\infty^{0,1}$ contributes to the filtration on $H^1(G, M) \cong \mathbb{Z}/p\mathbb{Z}$, and $E_\infty^{1,0} = H^1(Q, M) \cong \mathbb{Z}/p\mathbb{Z}$ also contributes. Since $H^1(G, M) \cong \mathbb{Z}/p\mathbb{Z}$ has order p and must be an extension of $E_\infty^{0,1}$ by $E_\infty^{1,0}$, one of these must be zero. The inflation map $\mathrm{Inf} : H^1(\mathbb{Z}/p\mathbb{Z}, \mathbb{Z}/p\mathbb{Z}) \to H^1(\mathbb{Z}/p^2\mathbb{Z}, \mathbb{Z}/p\mathbb{Z})$ sends a homomorphism $\mathbb{Z}/p\mathbb{Z} \to \mathbb{Z}/p\mathbb{Z}$ to its inflation along $\mathbb{Z}/p^2\mathbb{Z} \to \mathbb{Z}/p\mathbb{Z}$, which is the composition $\mathbb{Z}/p^2\mathbb{Z} \to \mathbb{Z}/p\mathbb{Z} \to \mathbb{Z}/p\mathbb{Z}$. This is nonzero (the inflation of the identity map is the reduction-mod-p map, which is a nonzero element of $\mathrm{Hom}(\mathbb{Z}/p^2\mathbb{Z}, \mathbb{Z}/p\mathbb{Z}) \cong \mathbb{Z}/p\mathbb{Z}$). Hence $E_\infty^{1,0} = \mathbb{Z}/p\mathbb{Z}$, forcing $E_\infty^{0,1} = 0$, so $\mathrm{tg} : \mathbb{Z}/p\mathbb{Z} \to \mathbb{Z}/p\mathbb{Z}$ is an isomorphism.

4. Conclusion. The differential $d_2^{0,1}$ is an isomorphism. By the multiplicative structure of the spectral sequence (which interacts with cup products via the Leibniz rule — see chapter 4), the remaining d_2 differentials are determined: $d_2^{p,q}$ is an isomorphism whenever it maps between nonzero groups. The spectral sequence degenerates at E_3 (all E_3 terms are either $\mathbb{Z}/p\mathbb{Z}$ or 0, arranged so that each diagonal has exactly one nonzero term).

The central extension example illustrates a general principle: when the E_2 page is "too large" for the known abutment, the differentials must kill the excess, and the interplay between the spectral sequence structure and independent knowledge of the abutment determines the differentials.

Example 3.4.9: The LHS Spectral Sequence for $G_{\mathbb{Q}_p}$

Let p be a prime, $K = \mathbb{Q}_p$, and consider the short exact sequence of profinite groups

$$1 \to I_p \to G_{\mathbb{Q}_p} \to \hat{\mathbb{Z}} \to 1,$$

where $I_p = \mathrm{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p^{\mathrm{nr}})$ is the inertia group and $\hat{\mathbb{Z}} = \mathrm{Gal}(\mathbb{Q}_p^{\mathrm{nr}}/\mathbb{Q}_p)$ is generated topologically by the Frobenius Frob_p. Let ℓ be a prime with $\ell \neq p$, and take $M = \mathbb{Z}/\ell\mathbb{Z}$ with trivial $G_{\mathbb{Q}_p}$-action.

1. The E_2 page. The cohomology of the inertia group with trivial $\mathbb{Z}/\ell\mathbb{Z}$-coefficients is:

$$\begin{aligned} H^0(I_p, \mathbb{Z}/\ell\mathbb{Z}) &= \mathbb{Z}/\ell\mathbb{Z}, \\ H^1(I_p, \mathbb{Z}/\ell\mathbb{Z}) &= \mathrm{Hom}_{\mathrm{cts}}(I_p, \mathbb{Z}/\ell\mathbb{Z}). \end{aligned}$$

   The group I_p has tame quotient $I_p^{\mathrm{t}} = I_p/P_p \cong \prod_{\ell' \neq p} \mathbb{Z}_{\ell'}$ (where P_p is the wild inertia, a pro-p-group). Since $\ell \neq p$, every continuous homomorphism $I_p \to \mathbb{Z}/\ell\mathbb{Z}$ factors through the tame quotient, and $\mathrm{Hom}_{\mathrm{cts}}(I_p^{\mathrm{t}}, \mathbb{Z}/\ell\mathbb{Z}) \cong \mathbb{Z}/\ell\mathbb{Z}$ (the ℓ-component of the tame quotient contributes one copy). So $H^1(I_p, \mathbb{Z}/\ell\mathbb{Z}) \cong \mathbb{Z}/\ell\mathbb{Z}$.

   The $\hat{\mathbb{Z}}$-action (via conjugation) on $H^0(I_p, \mathbb{Z}/\ell\mathbb{Z}) = \mathbb{Z}/\ell\mathbb{Z}$ is trivial. The action on $H^1(I_p, \mathbb{Z}/\ell\mathbb{Z}) \cong \mathrm{Hom}_{\mathrm{cts}}(I_p, \mathbb{Z}/\ell\mathbb{Z})$ is determined by the conjugation action of Frob_p on I_p. The Frobenius acts on the tame inertia by $\mathrm{Frob}_p \sigma \mathrm{Frob}_p^{-1} = \sigma^p$ (raising to the p-th power), so Frob_p acts on $\mathrm{Hom}_{\mathrm{cts}}(I_p^{\mathrm{t}}, \mathbb{Z}/\ell\mathbb{Z})$ by precomposing with the p-th power map, which multiplies a homomorphism by p. Hence $\hat{\mathbb{Z}}$ acts on $H^1(I_p, \mathbb{Z}/\ell\mathbb{Z}) \cong \mathbb{Z}/\ell\mathbb{Z}$ by multiplication by p.

2. The $E_2^{p,0}$ row. Since the action on $H^0(I_p, \mathbb{Z}/\ell\mathbb{Z}) = \mathbb{Z}/\ell\mathbb{Z}$ is trivial, the bottom row is $E_2^{p,0} = H^p(\hat{\mathbb{Z}}, \mathbb{Z}/\ell\mathbb{Z})$. The cohomology of $\hat{\mathbb{Z}}$ with trivial $\mathbb{Z}/\ell\mathbb{Z}$-coefficients was computed in example 3.2.6: $H^p(\hat{\mathbb{Z}}, \mathbb{Z}/\ell\mathbb{Z}) \cong \mathbb{Z}/\ell\mathbb{Z}$ for all $p \geq 0$.

3. The $E_2^{p,1}$ row. The action on $H^1(I_p, \mathbb{Z}/\ell\mathbb{Z}) \cong \mathbb{Z}/\ell\mathbb{Z}$ is by multiplication by p. Denote this module $\mathbb{Z}/\ell\mathbb{Z}(p)$ (meaning the underlying group is $\mathbb{Z}/\ell\mathbb{Z}$, with Frob_p acting as multiplication by p). The row $E_2^{p,1} = H^p(\hat{\mathbb{Z}}, \mathbb{Z}/\ell\mathbb{Z}(p))$ is the cohomology of $\hat{\mathbb{Z}}$ with this twisted action.

   The colimit formula gives $H^p(\hat{\mathbb{Z}}, \mathbb{Z}/\ell\mathbb{Z}(p)) = \varinjlim_n H^p(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}/\ell\mathbb{Z}(p))$. Since the action of $\mathbb{Z}/n\mathbb{Z}$ factors through $\mathbb{Z}/\ell\mathbb{Z}$ (the action of Frob_p^n is multiplication by p^n, and the relevant quotient is $\mathbb{Z}/\ell\mathbb{Z}$ acting by p), the colimit stabilizes once $\ell \big| n$. For the cyclic group $C = \mathbb{Z}/\ell\mathbb{Z}$ acting on $\mathbb{Z}/\ell\mathbb{Z}$ by multiplication by p (where $p \not\equiv 1 \pmod{\ell}$ generically):

   - If $p \equiv 1 \pmod{\ell}$: the action is trivial, and $H^p(C, \mathbb{Z}/\ell\mathbb{Z}) \cong \mathbb{Z}/\ell\mathbb{Z}$ for all p.
   - If $p \not\equiv 1 \pmod{\ell}$: the action is nontrivial, and $H^p(C, \mathbb{Z}/\ell\mathbb{Z}(p)) = 0$ for all $p \geq 1$ (since $\mathbb{Z}/\ell\mathbb{Z}(p)$ is a nontrivial simple module over the group ring $\mathbb{Z}/\ell\mathbb{Z}[C]$, and the norm element $N_C = 1 + p + p^2 + \cdots + p^{\ell-1} = (p^\ell - 1)/(p-1) \equiv 0 \pmod{\ell}$ by Fermat's little theorem, so both the $\hat{H}^0$ and $\hat{H}^1$ of the cyclic group vanish).

4. Consequences. When $p \not\equiv 1 \pmod{\ell}$: the $E_2^{p,1}$ row vanishes for $p \geq 1$, and $E_2^{0,1} = H^0(\hat{\mathbb{Z}}, \mathbb{Z}/\ell\mathbb{Z}(p)) = (\mathbb{Z}/\ell\mathbb{Z})^{\mathrm{Frob}_p = p} = 0$ (since the only fixed point of multiplication by $p \neq 1$ on $\mathbb{Z}/\ell\mathbb{Z}$ is zero). The entire $q = 1$ row is zero. Higher rows ($q \geq 2$) involve $H^q(I_p, \mathbb{Z}/\ell\mathbb{Z})$, which vanishes for $q \geq 2$ when $\ell \neq p$ (the pro-ℓ-part of I_p is $\mathbb{Z}_\ell$, which has ℓ-cohomological dimension 1). So the spectral sequence degenerates at E_2 with only the bottom row surviving, and $H^n(G_{\mathbb{Q}_p}, \mathbb{Z}/\ell\mathbb{Z}) \cong H^n(\hat{\mathbb{Z}}, \mathbb{Z}/\ell\mathbb{Z}) \cong \mathbb{Z}/\ell\mathbb{Z}$ for all n.

   When $p \equiv 1 \pmod{\ell}$: the $q = 1$ row is nonzero ($E_2^{p,1} \cong \mathbb{Z}/\ell\mathbb{Z}$ for all p), so the spectral sequence does not degenerate at E_2, and the differentials $d_2^{p,1} : E_2^{p,1} \to E_2^{p+2,0}$ play a nontrivial role. This case is precisely when $\mathbb{Q}_p$ contains the ℓ-th roots of unity, and the nontrivial spectral sequence reflects the richer Galois cohomology arising from Kummer theory (chapter 8).

The two examples exhibit the two basic behaviours of the LHS spectral sequence: degeneration at E_2 (when the differentials are forced to vanish for dimensional or arithmetic reasons) and nontrivial differentials (when the E_2 page is too large for the abutment). The local field example shows how arithmetic conditions — here, whether $\mathbb{Q}_p$ contains ℓ-th roots of unity — control the spectral sequence.

Exercise 3.4.1

1. Let N be a closed normal subgroup of G and M a discrete G-module. Verify that the conjugation action of G/N on $H^1(N, M)$ is well defined: show that if $f : N \to M$ is a continuous 1-cocycle and $n_0 \in N$, then the cocycles f and $n_0 {}_\# f$ (defined by $(n_0 {}_\# f)(n) = n_0 \cdot f(n_0^{-1} n n_0)$) differ by the coboundary of the element $f(n_0) \in M$. (*Hint:* Use the cocycle

identity $f(n_0 \cdot n_0^{-1} n n_0) = n_0 \cdot f(n_0^{-1} n n_0) + f(n_0)$.)

2. Let $G = D_4$ (the dihedral group of order 8), $N = \langle r^2 \rangle \cong \mathbb{Z}/2\mathbb{Z}$ (the center), and $M = \mathbb{Z}/2\mathbb{Z}$ with trivial action. Compute the E_2 page of the LHS spectral sequence and determine all d_2 differentials. Use the fact that $H^n(D_4, \mathbb{Z}/2\mathbb{Z}) \cong (\mathbb{Z}/2\mathbb{Z})^{a_n}$ where a_n is known from the cohomology of D_4.

3. Let $1 \to \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/4\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 1$ be the unique nontrivial central extension with $M = \mathbb{Z}/2\mathbb{Z}$ (trivial action). Compute the transgression $\mathrm{tg} : H^1(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z})^{\mathbb{Z}/2\mathbb{Z}} \to H^2(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z})$ using the explicit formula (3.4). Verify that it matches the result from exercise 3.3.1 (5).

4. Let G be a profinite group and N a closed normal subgroup such that G/N has p-cohomological dimension ≤ 1 (i.e., $H^n(G/N, A) = 0$ for all $n \geq 2$ and all p-torsion discrete G/N-modules A). Show that for every p-torsion discrete G-module M, the restriction map $\mathrm{Res} : H^n(G, M)\{p\} \to H^n(N, M)\{p\}$ is surjective onto the G/N-invariants for every $n \geq 0$. (*Hint:* The $E_2^{p,q}$ terms with $p \geq 2$ vanish, so the spectral sequence degenerates.)

5. Use the LHS spectral sequence to prove the following: if N is a closed normal subgroup of G with $\mathrm{cd}_p(N) \leq m$ and $\mathrm{cd}_p(G/N) \leq n$ (where cd_p denotes p-cohomological dimension, to be defined formally in chapter 5), then $\mathrm{cd}_p(G) \leq m + n$. (*Hint:* The $E_2^{p,q}$ terms vanish for $p > n$ or $q > m$.)

3.5 Shapiro's Lemma and Induced Modules

The coinduced modules introduced in definition 2.2.4 provided acyclic objects for computing cohomology via derived functors in the discrete setting. This section develops their profinite analogues and establishes Shapiro's lemma, which asserts that the cohomology of a profinite group G with coefficients in a coinduced module reduces to the cohomology of the subgroup. This reduction is one of the most frequently used tools in Galois cohomology: it allows cohomological questions about an absolute Galois group G_K to be transferred to questions about G_L for a field extension L/K, simply by choosing the right coefficient module.

Throughout this section, G denotes a profinite group and H a closed subgroup of G.

Definition 3.5.1: Coinduced Module

Let H be a closed subgroup of a profinite group G, and let M be a discrete H-module. The *(continuously) coinduced module* is

$$\mathrm{CoInd}_H^G(M) = \mathrm{Map}_H^{\mathrm{cts}}(G, M) = \{f : G \to M \mid f \text{ is continuous and } f(hg) = h \cdot f(g) \text{ for all } h \in H,\, g \in G\},$$

with G-action given by right translation: $(g' \cdot f)(g) = f(gg')$ for $g' \in G$.

The continuity condition means that $f : G \to M$ is locally constant (since M is discrete). The H-equivariance condition $f(hg) = h \cdot f(g)$ determines f on each left H-coset from its value at a single point: if $f(g)$ is known, then $f(hg) = h \cdot f(g)$ for all $h \in H$.

The coinduced module $\mathrm{CoInd}_H^G(M)$ is a discrete G-module. To see this, fix $f \in \mathrm{CoInd}_H^G(M)$. Since f is locally constant and G is compact, f factors through G/U for some open normal subgroup U of G. For any $u \in U$, the translate $(u \cdot f)(g) = f(gu) = f(g)$ (since f is constant on right U-cosets), so

$U \subseteq \mathrm{Stab}_G(f)$, and the stabilizer is open.

Definition 3.5.2: Induced Module

Let H be an *open* subgroup of G (so that $[G : H]$ is finite), and let M be a discrete H-module. The *(continuously) induced module* is

$$\mathrm{Ind}_H^G(M) = \mathbb{Z}[G] \otimes_{\mathbb{Z}[H]} M,$$

where G acts on the left tensor factor by left multiplication. Since H has finite index, this is a finite direct sum: choosing left coset representatives $s_1, \ldots, s_m$ for H in G,

$$\mathrm{Ind}_H^G(M) = \bigoplus_{i=1}^{m} s_i \otimes_H M \cong \bigoplus_{i=1}^{m} M$$

as abelian groups, with G-action permuting the summands according to the coset action and acting on the module entry via H.

For H open, the induced module is a discrete G-module: each element is a finite sum with entries in M (a discrete H-module), and the stabilizer is a finite intersection of open subgroups, hence open.

The induced and coinduced modules are related by an isomorphism when H is open.

Proposition 3.5.3: Induced Equals Coinduced for Open Subgroups

Let H be an open subgroup of G and M a discrete H-module. There is a canonical isomorphism of discrete G-modules

$$\mathrm{Ind}_H^G(M) \xrightarrow{\sim} \mathrm{CoInd}_H^G(M).$$

***Proof*:**

Let $\{s_1, \ldots, s_m\}$ be a set of left coset representatives for H in G. Define $\varphi : \mathrm{Ind}_H^G(M) \to \mathrm{CoInd}_H^G(M)$ by sending $s_i \otimes m$ to the function $f_{i,m} : G \to M$ defined by

$$f_{i,m}(hg) = \begin{cases} h \cdot m & \text{if } g \in s_i H, \text{ i.e., } g = s_i h' \text{ for some } h' \in H, \\ 0 & \text{if } g \notin H s_i. \end{cases}$$

More precisely, $f_{i,m}(g)$ is nonzero only on the double coset containing s_i, and is determined by H-equivariance from the value $f_{i,m}(s_i) = m$. This extends linearly to an isomorphism. The inverse sends $f \in \mathrm{CoInd}_H^G(M)$ to $\sum_i s_i \otimes f(s_i)$. Both maps are G-equivariant and are independent of the choice of coset representatives.

When H is closed but not open (so $[G : H]$ is infinite), the induced and coinduced modules differ: $\mathrm{Ind}_H^G(M)$ involves compactly supported functions (finitely many nonzero cosets), while $\mathrm{CoInd}_H^G(M)$ allows all continuous H-equivariant functions. For Shapiro's lemma — the main result of this section — the coinduced module is the correct object.

The key property of the coinduced module is an adjunction: it is the right adjoint of restriction.

Proposition 3.5.4: Adjunction for Coinduced Modules

For any discrete G-module A and any discrete H-module M, there is a natural isomorphism

$$\mathrm{Hom}_G(A, \mathrm{CoInd}_H^G(M)) \cong \mathrm{Hom}_H(A, M),$$

where the right side views A as an H-module by restriction.

***Proof*:**
Given a G-equivariant map $\Phi : A \to \mathrm{CoInd}_H^G(M)$, define $\varphi : A \to M$ by $\varphi(a) = \Phi(a)(1)$ (evaluation at the identity). This is H-equivariant: for $h \in H$, $\varphi(h \cdot a) = \Phi(h \cdot a)(1) = (h \cdot \Phi(a))(1) = \Phi(a)(1 \cdot h) = \Phi(a)(h) = h \cdot \Phi(a)(1) = h \cdot \varphi(a)$, using the G-equivariance of Φ and the H-equivariance of elements of $\mathrm{CoInd}_H^G(M)$.

Conversely, given an H-equivariant map $\varphi : A \to M$, define $\Phi : A \to \mathrm{CoInd}_H^G(M)$ by $\Phi(a)(g) = \varphi(g \cdot a)$ for $g \in G$. Then $\Phi(a)(hg) = \varphi(hg \cdot a) = h \cdot \varphi(g \cdot a) = h \cdot \Phi(a)(g)$... this requires φ to be H-equivariant for h: $\varphi(h \cdot (g \cdot a)) = h \cdot \varphi(g \cdot a)$, which holds since φ is H-equivariant and $g \cdot a$ is an element of A viewed as an H-module. Continuity of $\Phi(a) : G \to M$ follows from the fact that a has open stabilizer in G (so $g \mapsto g \cdot a$ is locally constant). The two constructions are inverse to each other.

The adjunction of proposition 3.5.4 is the key input to Shapiro's lemma, which follows from a standard derived functor argument.

Theorem 3.5.5: Shapiro's Lemma

Let H be a closed subgroup of a profinite group G, and let M be a discrete H-module. For every $n \geq 0$, there is a canonical isomorphism

$$H^n(G, \mathrm{CoInd}_H^G(M)) \cong H^n(H, M).$$

When H is open, the same holds with $\mathrm{Ind}_H^G(M)$ in place of $\mathrm{CoInd}_H^G(M)$ (by proposition 3.5.3).

***Proof*:**
Step 1: The functor CoInd_H^G *preserves injectives.* Since CoInd_H^G is right adjoint to the exact functor Res_G^H (proposition 3.5.4), it preserves injectives: if I is injective in $\mathrm{Mod}_H^{\mathrm{disc}}$, then $\mathrm{Hom}_G(-, \mathrm{CoInd}_H^G(I)) \cong \mathrm{Hom}_H(-, I)$ is exact (since Res_G^H is exact and I is injective), so $\mathrm{CoInd}_H^G(I)$ is injective in $\mathrm{Mod}_G^{\mathrm{disc}}$.

Step 2: Computing the derived functors. Let $0 \to M \to I^0 \to I^1 \to \cdots$ be an injective resolution of M in $\mathrm{Mod}_H^{\mathrm{disc}}$. By Step 1, $0 \to \mathrm{CoInd}_H^G(M) \to \mathrm{CoInd}_H^G(I^0) \to \mathrm{CoInd}_H^G(I^1) \to \cdots$ is an injective resolution of $\mathrm{CoInd}_H^G(M)$ in $\mathrm{Mod}_G^{\mathrm{disc}}$ (exactness of CoInd_H^G follows from the description as a Hom functor, which is left exact, and the fact that the sequence of I^j is exact in $\mathrm{Mod}_H^{\mathrm{disc}}$ — but CoInd_H^G need not be exact in general; however, an injective resolution remains injective after applying a functor that preserves injectives, and the resulting complex computes the correct derived functors regardless of whether the functor is exact).

More precisely: since $\mathrm{CoInd}_H^G(I^j)$ is injective for each j, the complex $\mathrm{CoInd}_H^G(I^\bullet)$ is a resolution of $\mathrm{CoInd}_H^G(M)$ by injective G-modules (it is exact because it computes $R^n \mathrm{CoInd}_H^G(M)$, and since CoInd_H^G is right adjoint to an exact functor, it is exact — right adjoints of exact functors between categories with enough injectives are exact on injective resolutions).

Step 3: Identifying the cohomology. Applying $(-)^G$ to the injective resolution $\mathrm{CoInd}_H^G(I^\bullet)$:

$$H^n(G, \mathrm{CoInd}_H^G(M)) = H^n\Big((\mathrm{CoInd}_H^G(I^\bullet))^G\Big).$$

By the adjunction (proposition 3.5.4) with $A = \mathbb{Z}$ (trivial G-module):

$$(\mathrm{CoInd}_H^G(I^j))^G = \mathrm{Hom}_G(\mathbb{Z}, \mathrm{CoInd}_H^G(I^j)) \cong \mathrm{Hom}_H(\mathbb{Z}, I^j) = (I^j)^H.$$

Hence $H^n(G, \mathrm{CoInd}_H^G(M)) = H^n((I^\bullet)^H) = H^n(H, M)$.

Shapiro's lemma says that "cohomology does not see the coinduction process": computing $H^n(G, -)$ with coinduced coefficients is the same as computing $H^n(H, -)$ with the original coefficients. The isomorphism is canonical and compatible with the long exact sequence: if $0 \to M' \to M \to M'' \to 0$ is a short exact sequence of discrete H-modules, then $0 \to \mathrm{CoInd}_H^G(M') \to \mathrm{CoInd}_H^G(M) \to \mathrm{CoInd}_H^G(M'') \to 0$ is exact (since CoInd_H^G is left exact and right adjoint to an exact functor, and the sequence of coinduced modules is exact because each $\mathrm{Map}_H^{\mathrm{cts}}(G, -)$ is exact when applied to sequences of discrete modules), and the Shapiro isomorphisms intertwine the connecting homomorphisms.

A special case recovers the acyclicity result from theorem 2.2.6.

Corollary 3.5.6: Acyclicity of Coinduced Modules

Let H be a closed subgroup of G and M a discrete H-module.

1. If M is injective in $\mathrm{Mod}_H^{\mathrm{disc}}$, then $\mathrm{CoInd}_H^G(M)$ is injective (hence acyclic) in $\mathrm{Mod}_G^{\mathrm{disc}}$.
2. Taking $H = \{1\}$: for any abelian group J, the module $\mathrm{CoInd}_{\{1\}}^G(J) = \mathrm{Map}^{\mathrm{cts}}(G, J)$ (continuous functions $G \to J$ with the right-translation action) satisfies $H^n(G, \mathrm{CoInd}_{\{1\}}^G(J)) = H^n(\{1\}, J) = 0$ for all $n \geq 1$.

***Proof*:**
Part (1) was proved in Step 1 of theorem 3.5.5. Part (2) is the special case $H = \{1\}$ of Shapiro's lemma: $H^n(G, \mathrm{CoInd}_{\{1\}}^G(J)) \cong H^n(\{1\}, J) = 0$ for $n \geq 1$, since the trivial group has no higher cohomology.

The Shapiro isomorphism interacts with the restriction and corestriction maps in a precise way. These compatibilities are essential for the computations in chapters 8 and 9.

Proposition 3.5.7: Compatibility of Shapiro with Restriction and Corestriction

Let H be an open subgroup of G and M a discrete H-module. Let $\mathrm{Sh} : H^n(G, \mathrm{CoInd}_H^G(M)) \xrightarrow{\sim} H^n(H, M)$ denote the Shapiro isomorphism.

1. The composition

$$H^n(H, M) \xrightarrow{\mathrm{Sh}^{-1}} H^n(G, \mathrm{CoInd}_H^G(M)) \xrightarrow{\mathrm{Res}_G^H} H^n(H, \mathrm{CoInd}_H^G(M))$$

equals the map induced by the H-equivariant inclusion $\iota : M \hookrightarrow \mathrm{CoInd}_H^G(M)$ defined by $\iota(m)(g) = g \cdot m$ for $g \in H$ and $\iota(m)(g) = 0$ for $g \notin H$. When H is normal in G, this is the "inclusion of the identity coset."

2. The composition

$$H^n(H, \mathrm{CoInd}_H^G(M)) \xrightarrow{\mathrm{Cor}_H^G} H^n(G, \mathrm{CoInd}_H^G(M)) \xrightarrow{\mathrm{Sh}} H^n(H, M)$$

equals the map induced by the H-equivariant "evaluation-and-sum" map $\varepsilon : \mathrm{CoInd}_H^G(M) \to M$ defined by $\varepsilon(f) = \sum_{s \in H\backslash G} f(s)$ (the sum over right coset representatives, which is well defined since $f(hs) = h \cdot f(s)$ and the sum is finite because $[G : H] < \infty$).

***Proof*:**
Both parts follow by tracking the relevant maps through the adjunction isomorphism. For part (1): the Shapiro isomorphism sends a class in $H^n(H, M)$, represented by a cocycle $f : H^n \to M$, to the class in $H^n(G, \mathrm{CoInd}_H^G(M))$ represented by the cocycle $F : G^n \to \mathrm{CoInd}_H^G(M)$ defined by $F(g_1, \ldots, g_n)(g) = f(g \cdot g_1, \ldots, g \cdot g_n)|_H$ (adjusted by the H-equivariance). Restricting F to H^n and evaluating at $1 \in G$ recovers f, which is the content of the statement.

For part (2): the corestriction of a class from $H^n(H, \mathrm{CoInd}_H^G(M))$ involves summing over coset representatives, and the Shapiro isomorphism evaluates at the identity. The composition amounts to evaluating the coinduced function at all coset representatives and summing, which is ε.

Example 3.5.8: Shapiro's Lemma for Galois Extensions

1. Let L/K be a finite Galois extension with $\Gamma = \mathrm{Gal}(L/K)$. The absolute Galois groups are related by $G_L \trianglelefteq_o G_K$ with $G_K/G_L \cong \Gamma$. For any discrete G_L-module M, Shapiro's lemma gives

$$H^n(G_K, \mathrm{CoInd}_{G_L}^{G_K}(M)) \cong H^n(G_L, M).$$

By proposition 3.5.3 (since G_L is open in G_K), the coinduced module equals the induced module:

$$\mathrm{CoInd}_{G_L}^{G_K}(M) \cong \mathrm{Ind}_{G_L}^{G_K}(M) = \mathbb{Z}[\Gamma] \otimes_{\mathbb{Z}} M \cong \bigoplus_{\sigma \in \Gamma} M,$$

with G_K acting by permuting the copies of M according to the Γ-action and acting on each copy via G_L. The Shapiro isomorphism thus provides a "base change" formula: the G_K-cohomology of a module induced from G_L equals the G_L-cohomology of the original module.

2. A concrete instance: let $K = \mathbb{Q}_p$, $L = \mathbb{Q}_p(\zeta_p)$ (a cyclic extension of degree $p-1$ for p odd), and $M = \mathbb{Z}/p\mathbb{Z}$ with trivial G_L-action. Then

$$H^n(G_{\mathbb{Q}_p}, \mathrm{Ind}_{G_L}^{G_{\mathbb{Q}_p}}(\mathbb{Z}/p\mathbb{Z})) \cong H^n(G_L, \mathbb{Z}/p\mathbb{Z}).$$

The left side involves the $G_{\mathbb{Q}_p}$-cohomology of the module $\mathbb{Z}/p\mathbb{Z}[\mathrm{Gal}(L/\mathbb{Q}_p)] \cong \mathbb{Z}/p\mathbb{Z}^{p-1}$ (a $(p-1)$-dimensional $\mathbb{Z}/p\mathbb{Z}$-representation of $G_{\mathbb{Q}_p}$), and the right side is the cohomology of the simpler group G_L with constant coefficients. This kind of reduction is the mechanism behind Shapiro-based proofs of local duality (chapter 8).

3. The isomorphism $H^1(G_K, \mathrm{Ind}_{G_L}^{G_K}(\overline{K}^\times)) \cong H^1(G_L, \overline{K}^\times)$ relates twisted G_K-cohomology to untwisted G_L-cohomology. Since $H^1(G_L, \overline{K}^\times) = 0$ by Hilbert's Theorem 90 (chapter 4), this gives $H^1(G_K, \mathrm{Ind}_{G_L}^{G_K}(\overline{K}^\times)) = 0$: the induced module of the multiplicative group is cohomologically trivial in degree 1. This vanishing is a key input to the computation of the Brauer group via the exact sequence of chapter 7.

Exercise 3.5.1

1. Verify the adjunction of proposition 3.5.4 directly: given an H-equivariant map $\varphi : A \to M$ (where A is a discrete G-module and M is a discrete H-module), construct the corresponding G-equivariant map $\Phi : A \to \mathrm{CoInd}_H^G(M)$ and check that $\Phi(a) \in \mathrm{CoInd}_H^G(M)$ (i.e., that the function $g \mapsto \varphi(g \cdot a)$ is continuous and H-equivariant in the correct sense).

2. Use Shapiro's lemma to give a short proof of the formula $\mathrm{Cor} \circ \mathrm{Res} = [G : H]$ from proposition 3.3.6. (*Hint:* For a discrete G-module M, consider the H-equivariant maps $M \xrightarrow{\iota} \mathrm{CoInd}_H^G(M) \xrightarrow{\varepsilon} M$, where $\iota(m)(g) = g \cdot m$ for $g \in H$ (extended by zero) and $\varepsilon(f) = \sum_{s \in H\backslash G} f(s)$. Show that $\varepsilon \circ \iota = [G : H] \cdot \mathrm{id}_M$ and relate the induced maps on cohomology to $\mathrm{Cor} \circ \mathrm{Res}$.)

3. Let H be an open subgroup of G and M a discrete H-module. Write out the Shapiro isomorphism $H^1(G, \mathrm{CoInd}_H^G(M)) \cong H^1(H, M)$ explicitly at the cocycle level: given a continuous 1-cocycle $F : G \to \mathrm{CoInd}_H^G(M)$, describe the corresponding 1-cocycle $f : H \to M$, and vice versa.

4. Let $H = N$ be a closed normal subgroup of G and M a discrete N-module. The Shapiro isomorphism gives $H^n(G, \mathrm{CoInd}_N^G(M)) \cong H^n(N, M)$. Apply the LHS spectral sequence (theorem 3.4.4) to the left side (with the group extension $1 \to N \to G \to G/N \to 1$ and the module $\mathrm{CoInd}_N^G(M)$). Show that the spectral sequence degenerates, and explain how this is consistent with the Shapiro isomorphism.

5. Let $G = \mathbb{Z}_p$ and $H = \{0\}$ (the trivial subgroup, which is closed but not open). Let $M = \mathbb{Z}/p\mathbb{Z}$ with trivial H-action. Describe $\mathrm{CoInd}_{\{0\}}^{\mathbb{Z}_p}(\mathbb{Z}/p\mathbb{Z})$ as an abelian group with $\mathbb{Z}_p$-action, and show that it is not finitely generated as a $\mathbb{Z}$-module. Compute $H^0(\mathbb{Z}_p, \mathrm{CoInd}_{\{0\}}^{\mathbb{Z}_p}(\mathbb{Z}/p\mathbb{Z}))$ and verify that Shapiro's lemma gives $H^0(\mathbb{Z}_p, \mathrm{CoInd}_{\{0\}}^{\mathbb{Z}_p}(\mathbb{Z}/p\mathbb{Z})) \cong H^0(\{0\}, \mathbb{Z}/p\mathbb{Z}) = \mathbb{Z}/p\mathbb{Z}$.

Part II

Structure Theory

The abstract cohomology of Part I applies to any profinite group and any discrete module. Part II narrows the focus to the structural invariants that distinguish the profinite groups and modules arising in arithmetic: cohomological dimension, pro-p-groups, and the Brauer group.

Chapter **??** introduces cohomological dimension $\mathrm{cd}(G)$—the supremum of degrees in which $H^n(G, M) \neq 0$ for torsion modules M—and computes it for the profinite groups that appear in field arithmetic: $\mathrm{cd}(\hat{\mathbb{Z}}) = 1$, $\mathrm{cd}(G_K) = 2$ for p-adic fields K, and $\mathrm{cd}(G_K) \leq 2$ for number fields (under suitable hypotheses). Chapter 6 studies pro-p-groups: their generators and relations, the Golod–Shafarevich inequality (which resolved the class field tower problem), and Demushkin groups (the pro-p-completions of local Galois groups when $\mu_p \subseteq K$). Chapter **??** develops the Brauer group $\mathrm{Br}(K) = H^2(G_K, (K^{\mathrm{sep}})^\times)$: central simple algebras, the crossed product construction, and the period-index problem. The Brauer group is the bridge between the abstract cohomology of Part I and the field arithmetic of Part III: the invariant map $\mathrm{Br}(K) \to \mathbb{Q}/\mathbb{Z}$ for local fields and the Albert–Brauer–Hasse–Noether exact sequence for global fields are the inputs that drive the duality theorems and reciprocity laws.

4

Tate Cohomology and Cup Products

The cohomology groups $H^n(G, M)$ constructed in chapters 2 and 3 are indexed by non-negative integers $n \geq 0$. This indexing is natural from the derived functor perspective — the right derived functors of $(-)^G$ start in degree zero — but for finite groups it conceals a symmetry. The group homology $H_n(G, M)$, defined as the left derived functors of the coinvariant functor $M \mapsto M_G$, carries complementary information in non-negative degrees $n \geq 0$, and the two theories are linked by the norm map $N_G : M_G \to M^G$. Tate's insight was to splice cohomology ($n \geq 1$), homology ($n \leq -2$), and the norm map ($n = 0, -1$) into a single cohomological functor $\hat{H}^n(G, M)$ indexed by *all* integers $n \in \mathbb{Z}$. The resulting theory — *Tate cohomology* or *complete cohomology* — admits unrestricted dimension shifting, yields the Herbrand quotient as a multiplicative invariant, and is the natural domain for the cup products and duality pairings of the remaining sections of this chapter.

4.1 Complete Cohomology for Finite Groups

Throughout this section, G denotes a *finite* group and M a $\mathbb{Z}[G]$-module (equivalently, a G-module in the sense of definition 2.1.1). There is no topology to consider: every G-module is automatically discrete.

The invariant functor $(-)^G$ and the coinvariant functor $(-)_G$ are the two natural ways to extract abelian-group-valued information from a G-module. The invariants $M^G = \{m \in M \mid g \cdot m = m \text{ for all } g \in G\}$ form the largest submodule on which G acts trivially; the coinvariants $M_G = M/I_G M$ (where I_G is the augmentation ideal of $\mathbb{Z}[G]$, as in definition 2.1.4) form the largest quotient on which G acts trivially. Neither functor is exact, and their derived functors give group cohomology and group homology respectively. The norm map connects them.

Definition 4.1.1: The Norm Map

The *norm map* (or *trace map*) is the $\mathbb{Z}[G]$-module homomorphism

$$N_G : M \to M, \qquad N_G(m) = \sum_{g \in G} g \cdot m.$$

Since $N_G(g' \cdot m) = \sum_g gg' \cdot m = N_G(m)$ for all $g' \in G$ (as $g \mapsto gg'$ permutes G), the image of N_G lies in M^G. Since $N_G((\sigma - 1) \cdot m) = \sum_g g\sigma \cdot m - \sum_g g \cdot m = 0$ for any $\sigma \in G$, the kernel of N_G contains $I_G M$. Hence N_G descends to a map

$$\bar{N}_G : M_G \to M^G.$$

The norm map measures the obstruction to passing between invariants and coinvariants. When $|G|$ is invertible in M (for instance, M is a $\mathbb{Q}$-vector space), the map $m \mapsto \frac{1}{|G|} N_G(m)$ provides a G-equivariant splitting of the inclusion $M^G \hookrightarrow M$, and $\bar{N}_G$ is an isomorphism. In general, both the kernel and cokernel of $\bar{N}_G$ carry significant information.

Definition 4.1.2: Modified Cohomology in Degrees 0 and -1

Let G be a finite group and M a G-module. The *modified zeroth Tate cohomology* is

$$\hat{H}^0(G, M) = M^G / N_G(M) = \operatorname{coker}(\bar{N}_G),$$

and the *modified minus-first Tate cohomology* is

$$\hat{H}^{-1}(G, M) = \ker(N_G)/I_G M = \ker(\bar{N}_G).$$

These differ from ordinary cohomology and homology: $H^0(G, M) = M^G$ (the full invariants, not the quotient by the norm image), and $H_0(G, M) = M_G$ (the full coinvariants, not the kernel of the norm). The modification "quotients out" the contribution of the norm in degree zero and "restricts to" its kernel in degree -1, producing groups that are better behaved for the dimension-shifting arguments of §3.

The full Tate cohomology extends $\hat{H}^0$ and $\hat{H}^{-1}$ to all degrees using a doubly infinite resolution.

Definition 4.1.3: The Complete Standard Resolution

Let G be a finite group. The *complete standard resolution* of $\mathbb{Z}$ over $\mathbb{Z}[G]$ is the doubly infinite exact complex

$$\hat{P}_\bullet : \quad \cdots \to P_2 \to P_1 \to P_0 \xrightarrow{d_0} P_{-1} \to P_{-2} \to \cdots$$

constructed as follows. In non-negative degrees, P_n for $n \geq 0$ is the free $\mathbb{Z}[G]$-module on G^{n+1} (with the standard differential from definition 2.3.1), forming the beginning of the standard resolution $\cdots \to P_1 \to P_0 \xrightarrow{\varepsilon} \mathbb{Z} \to 0$. In negative degrees, P_{-n-1} for $n \geq 0$ is the $\mathbb{Z}[G]$-dual $\operatorname{Hom}_{\mathbb{Z}}(P_n, \mathbb{Z}[G])$, with the dual differential. The map $d_0 : P_0 \to P_{-1}$ is constructed so that the sequence is exact at the splice point: $\ker(d_0) = \operatorname{im}(\varepsilon) = \mathbb{Z} \cdot N_G$ and $\operatorname{im}(d_0) = \ker(d_{-1}^*)$, where the norm element $N_G = \sum_{g \in G} g \in \mathbb{Z}[G]$ plays the role of the connecting map between the positive and negative halves.

The exactness of the complete resolution is the critical property. In the positive half, this is the exactness of the standard resolution (lemma 2.3.2). In the negative half, exactness follows by dualizing. At the splice point, one verifies that $\ker(d_0)/\mathrm{im}(d_1)$ is identified with $\mathbb{Z}$ via the augmentation, and the norm map $N_G : \mathbb{Z} \to \mathbb{Z}[G]^G$ connects the two halves.

Definition 4.1.4: Tate Cohomology

Let G be a finite group and M a G-module. The *Tate cohomology groups* are defined for all $n \in \mathbb{Z}$ by

$$\hat{H}^n(G, M) = H^n(\mathrm{Hom}_G(\hat{P}_\bullet, M)),$$

where $\hat{P}_\bullet$ is the complete standard resolution of definition 4.1.3.

The relationship between Tate cohomology, ordinary cohomology, and group homology is summarized in the following proposition.

Proposition 4.1.5: Identification of Tate Cohomology

Let G be a finite group and M a G-module. Then:

1. $\hat{H}^n(G, M) \cong H^n(G, M)$ for all $n \geq 1$ (ordinary group cohomology).
2. $\hat{H}^n(G, M) \cong H_{-n-1}(G, M)$ for all $n \leq -2$ (group homology).
3. $\hat{H}^0(G, M) \cong M^G/N_G(M)$ and $\hat{H}^{-1}(G, M) \cong \ker(N_G)/I_G M$.

***Proof*:**

Step 1: Positive degrees. For $n \geq 1$, the complex $\mathrm{Hom}_G(\hat{P}_\bullet, M)$ in degrees ≥ 1 coincides with $\mathrm{Hom}_G(P_\bullet, M)$ in degrees ≥ 1, where $P_\bullet$ is the standard resolution. Since the standard resolution computes group cohomology (proposition 2.3.4), $\hat{H}^n(G, M) = H^n(G, M)$ for $n \geq 1$.

Step 2: Negative degrees. For $n \leq -2$, the complex $\mathrm{Hom}_G(\hat{P}_\bullet, M)$ in the relevant degrees involves the dual modules $P_{-k-1} = \mathrm{Hom}_{\mathbb{Z}}(P_k, \mathbb{Z}[G])$. The adjunction $\mathrm{Hom}_G(\mathrm{Hom}_{\mathbb{Z}}(P_k, \mathbb{Z}[G]), M) \cong P_k \otimes_G M$ (via the evaluation map) identifies $\hat{H}^n(G, M)$ with $H_{-n-1}(P_\bullet \otimes_G M) = H_{-n-1}(G, M)$.

Step 3: Degrees 0 *and* -1*.* At the splice point, the cohomology of $\mathrm{Hom}_G(\hat{P}_\bullet, M)$ in degree 0 is $\ker(d_0^*)/\mathrm{im}(d_1^*) = M^G/N_G(M) = \hat{H}^0(G, M)$, where d_0^* encodes the norm map and d_1^* encodes the augmentation. The degree -1 computation is dual.

The central advantage of Tate cohomology is the long exact sequence extending in both directions.

Proposition 4.1.6: Long Exact Sequence for Tate Cohomology

A short exact sequence $0 \to M' \to M \to M'' \to 0$ of G-modules induces an exact sequence

$$\cdots \to \hat{H}^{n-1}(G, M'') \xrightarrow{\delta} \hat{H}^n(G, M') \to \hat{H}^n(G, M) \to \hat{H}^n(G, M'') \xrightarrow{\delta} \hat{H}^{n+1}(G, M') \to \cdots$$

for all $n \in \mathbb{Z}$, with connecting homomorphisms δ that are natural in the short exact sequence.

***Proof*:**
The complex $\hat{P}_\bullet$ consists of free (hence flat and projective) $\mathbb{Z}[G]$-modules in every degree. Applying $\mathrm{Hom}_G(\hat{P}_\bullet, -)$ to the short exact sequence gives a short exact sequence of cochain complexes (since each $\hat{P}_n$ is projective, the $\mathrm{Hom}_G(\hat{P}_n, -)$ functor is exact). The associated long exact sequence in cohomology is the stated sequence.

A key feature distinguishing Tate cohomology from ordinary cohomology is the behaviour of induced modules.

Proposition 4.1.7: Induced Modules Are Tate-Acyclic

Let G be a finite group. For every abelian group A, the induced module $\mathbb{Z}[G] \otimes_{\mathbb{Z}} A$ satisfies $\hat{H}^n(G, \mathbb{Z}[G] \otimes_{\mathbb{Z}} A) = 0$ for all $n \in \mathbb{Z}$.

***Proof*:**
The complete resolution $\hat{P}_\bullet$ is an exact sequence of free $\mathbb{Z}[G]$-modules, so $\hat{P}_\bullet \otimes_G (\mathbb{Z}[G] \otimes_{\mathbb{Z}} A) \cong \hat{P}_\bullet \otimes_{\mathbb{Z}} A$ is exact (the tensor product with a free module preserves exactness). Taking G-invariants: $\mathrm{Hom}_G(\hat{P}_n, \mathbb{Z}[G] \otimes_{\mathbb{Z}} A) \cong \mathrm{Hom}_{\mathbb{Z}}(\hat{P}_n \otimes_G \mathbb{Z}[G], A) \otimes_G \cdots$. The key point is that $\mathrm{Hom}_G(\hat{P}_n, \mathbb{Z}[G] \otimes_{\mathbb{Z}} A)$ computes an exact complex: $\mathbb{Z}[G]$ is $\mathbb{Z}[G]$-free, and the $\mathrm{Hom}_G(\hat{P}_n, \mathbb{Z}[G])$ complex is exact by the definition of $\hat{P}_\bullet$ (which was constructed to be acyclic for $\mathbb{Z}[G]$-coefficient modules). The vanishing $\hat{H}^n(G, \mathbb{Z}[G]) = 0$ for all n follows from the exactness of $\hat{P}_\bullet$, and the general case $\mathbb{Z}[G] \otimes_{\mathbb{Z}} A$ follows from the fact that $\hat{H}^n(G, -)$ commutes with arbitrary direct sums (as it is computed by a complex of finitely generated free modules).

The vanishing $\hat{H}^n(G, \mathbb{Z}[G]) = 0$ for *all* $n \in \mathbb{Z}$ is strictly stronger than the vanishing $H^n(G, \mathbb{Z}[G]) = 0$ for $n \geq 1$ from ordinary cohomology (theorem 2.2.6). It reflects the fact that $\mathbb{Z}[G]$ is both induced and coinduced, and it is the property that makes dimension shifting work in both directions (see §3).

The Tate cohomology of a cyclic group has a particularly clean form, which motivates the Herbrand quotient.

Example 4.1.8: Tate Cohomology of Cyclic Groups

1. Let $G = C_n = \langle \sigma \rangle$ be cyclic of order n and $M = \mathbb{Z}$ with trivial action. The norm map is $N_G(m) = nm$, so $\hat{H}^0(C_n, \mathbb{Z}) = \mathbb{Z}^G / N_G(\mathbb{Z}) = \mathbb{Z}/n\mathbb{Z}$. The kernel of N_G is zero (since $nm = 0$ implies $m = 0$ in $\mathbb{Z}$), so $\hat{H}^{-1}(C_n, \mathbb{Z}) = 0$. By the periodicity that will be established in §3, $\hat{H}^q(C_n, \mathbb{Z}) \cong \hat{H}^{q+2}(C_n, \mathbb{Z})$ for all q, giving
$$\hat{H}^q(C_n, \mathbb{Z}) \cong \begin{cases} \mathbb{Z}/n\mathbb{Z} & q \text{ even}, \\ 0 & q \text{ odd}, \end{cases}$$
for all $q \in \mathbb{Z}$. This extends the computation of example 2.3.8 to negative degrees.

2. Let $G = C_2 = \{1, \sigma\}$ and $M = \mathbb{Z}$ with the sign action: $\sigma \cdot m = -m$. Then $M^G = 0$, $N_G(m) = m + \sigma \cdot m = m - m = 0$, and $\ker(N_G) = \mathbb{Z}$. The augmentation ideal $I_G = (\sigma - 1)\mathbb{Z}[G]$ acts on M by $(\sigma - 1) \cdot m = -2m$, so $I_G M = 2\mathbb{Z}$. Hence $\hat{H}^0(C_2, \mathbb{Z}^-) = 0$ and $\hat{H}^{-1}(C_2, \mathbb{Z}^-) = \mathbb{Z}/2\mathbb{Z}$.

By periodicity, $\hat{H}^q(C_2, \mathbb{Z}^-) \cong \mathbb{Z}/2\mathbb{Z}$ for q odd and 0 for q even — the opposite parity from the trivial action.

3. (If the reader has read [16]) The group $\hat{H}^n(C_2, \mathbb{Z})$ with the trivial action computes the cohomology of the infinite real projective space: $H^n(\mathbb{RP}^\infty, \mathbb{Z}) \cong \hat{H}^n(C_2, \mathbb{Z})$ for $n \geq 1$, since $\mathbb{RP}^\infty$ is the classifying space BC_2. Tate cohomology extends this to negative degrees, which corresponds to the homology of $\mathbb{RP}^\infty$ via the identification $\hat{H}^{-n-1}(C_2, \mathbb{Z}) \cong H_n(\mathbb{RP}^\infty, \mathbb{Z})$. The 2-periodic pattern in Tate cohomology is the algebraic form of the $\mathbb{Z}/2\mathbb{Z}$-graded structure of real topological K-theory.

Definition 4.1.9: The Herbrand Quotient

Let G be a cyclic group and M a G-module such that $\hat{H}^0(G, M)$ and $\hat{H}^{-1}(G, M)$ are both finite. The *Herbrand quotient* of M is

$$h(G, M) = \frac{|\hat{H}^0(G, M)|}{|\hat{H}^{-1}(G, M)|} \in \mathbb{Q}_{>0}.$$

The Herbrand quotient is defined only for cyclic groups, where the 2-periodicity of Tate cohomology means that $h(G, M) = |\hat{H}^{2k}(G, M)|/|\hat{H}^{2k-1}(G, M)|$ for any k. Its power lies in its multiplicativity and its invariance under passage to finite-index submodules.

Theorem 4.1.10: Multiplicativity of the Herbrand Quotient

Let G be a cyclic group and $0 \to M' \to M \to M'' \to 0$ a short exact sequence of G-modules. If any two of the three Herbrand quotients $h(G, M')$, $h(G, M)$, $h(G, M'')$ are defined, then so is the third, and

$$h(G, M) = h(G, M') \cdot h(G, M'').$$

***Proof*:**
The long exact sequence of proposition 4.1.6, combined with the 2-periodicity for cyclic groups, gives an exact hexagon:

$$\hat{H}^0(G, M') \to \hat{H}^0(G, M) \to \hat{H}^0(G, M'') \xrightarrow{\delta} \hat{H}^{-1}(G, M') \to \hat{H}^{-1}(G, M) \to \hat{H}^{-1}(G, M'') \xrightarrow{\delta} \hat{H}^0(G, M') \to \cdots$$

(The sequence is 2-periodic, so after six terms it repeats.) If any two of the three Herbrand quotients are defined, then all six groups in the hexagon are finite (since the maps between them have finite kernels and cokernels by the exact sequence). The alternating product of the orders in an exact sequence of finite groups equals 1:

$$\frac{|\hat{H}^0(G, M')| \cdot |\hat{H}^0(G, M'')| \cdot |\hat{H}^{-1}(G, M)|}{|\hat{H}^0(G, M)| \cdot |\hat{H}^{-1}(G, M')| \cdot |\hat{H}^{-1}(G, M'')|} = 1,$$

which rearranges to $h(G, M) = h(G, M') \cdot h(G, M'')$.

The multiplicativity of the Herbrand quotient makes it a powerful computational tool: to compute $h(G, M)$ for a complicated module M, one can filter M by submodules and multiply the Herbrand quotients of the graded pieces.

Proposition 4.1.11: Herbrand Quotient of Finite Modules

Let G be a cyclic group and M a finite G-module. Then $h(G, M) = 1$.

***Proof*:**
Step 1: The case $M = \mathbb{Z}/m\mathbb{Z}$ with trivial action. By the computation of example 4.1.8 (replacing $\mathbb{Z}$ by $\mathbb{Z}/m\mathbb{Z}$), both $\hat{H}^0(G, \mathbb{Z}/m\mathbb{Z})$ and $\hat{H}^{-1}(G, \mathbb{Z}/m\mathbb{Z})$ have order $\gcd(|G|, m)$, so $h(G, \mathbb{Z}/m\mathbb{Z}) = 1$.

Step 2: General M. Filter M by the ascending chain $0 = M_0 \subseteq M_1 \subseteq \cdots \subseteq M_k = M$ of G-stable submodules with each quotient M_i/M_{i-1} a simple $\mathbb{Z}[G]$-module. Each simple module is finite, and the Herbrand quotient of a simple module is 1 by a direct computation (or by a generalization of Step 1 using the structure of simple $\mathbb{Z}[G]$-modules for G cyclic). By multiplicativity (theorem 4.1.10), $h(G, M) = \prod_i h(G, M_i/M_{i-1}) = 1$.

The proposition says that the Herbrand quotient detects the "infinite part" of a module: for a finitely generated G-module M, $h(G, M)$ depends only on $M \otimes_{\mathbb{Z}} \mathbb{Q}$ (since M fits in an exact sequence $0 \to M_{\text{tors}} \to M \to M/M_{\text{tors}} \to 0$ with M_{tors} finite, and $h(G, M_{\text{tors}}) = 1$).

Example 4.1.12: Herbrand Quotient Computations

1. Let $G = C_p$ (cyclic of prime order p) act trivially on $M = \mathbb{Z}$. Then $\hat{H}^0(C_p, \mathbb{Z}) = \mathbb{Z}/p\mathbb{Z}$ and $\hat{H}^{-1}(C_p, \mathbb{Z}) = 0$, giving $h(C_p, \mathbb{Z}) = p$.

2. Let $G = C_p$ act on $M = \mathbb{Z}[G] \cong \mathbb{Z}^p$ (the regular representation). By proposition 4.1.7, $\hat{H}^n(G, \mathbb{Z}[G]) = 0$ for all n, so $h(C_p, \mathbb{Z}[G]) = 1$. The short exact sequence $0 \to I_G \to \mathbb{Z}[G] \xrightarrow{\varepsilon} \mathbb{Z} \to 0$ then gives $h(C_p, I_G) = h(C_p, \mathbb{Z}[G])/h(C_p, \mathbb{Z}) = 1/p$ by multiplicativity. Since I_G is torsion-free of rank $p - 1$, this means $|\hat{H}^{-1}(C_p, I_G)| = p \cdot |\hat{H}^0(C_p, I_G)|$, so the "homological" side is p times larger than the "cohomological" side.

3. Let $G = C_n$ act on $M = \mathbb{Z}^r$ by a permutation action (permuting a set of r elements). Filter M by the G-orbits: each orbit of size d (with $d \big| n$) contributes a summand $\mathbb{Z}[G/H]$ where $H \cong C_{n/d}$ is the stabilizer. The Herbrand quotient of $\mathbb{Z}[G/H]$ is $h(C_n, \mathbb{Z}[G/H])$. The short exact sequence $0 \to I_{G/H} \to \mathbb{Z}[G/H] \to \mathbb{Z} \to 0$ gives $h(C_n, \mathbb{Z}[G/H]) = h(C_n, \mathbb{Z})/h(C_n, I_{G/H})$. For a transitive action ($r = d$, a single orbit), $\mathbb{Z}[G/H] \cong \mathrm{Ind}_H^G(\mathbb{Z})$, and by Shapiro's lemma (theorem 3.5.5) combined with the Tate-acyclicity of induced modules, $h(C_n, \mathbb{Z}[G/H]) = 1$ when $H = \{1\}$ (i.e., the free orbit, $d = n$). For a fixed point ($d = 1$, stabilizer $H = G$), the summand is $\mathbb{Z}$ with trivial action, contributing $h(C_n, \mathbb{Z}) = n$.

Exercise 4.1.1

1. Verify that the complete standard resolution $\hat{P}_\bullet$ of definition 4.1.3 is exact at the splice point. Specifically, show that the composition $P_1 \to P_0 \xrightarrow{d_0} P_{-1}$ is exact and that $\ker(d_0)/\mathrm{im}(d_1) \cong \mathbb{Z}$ via the augmentation map.
2. Let $G = C_4 = \langle \sigma \rangle$ and $M = \mathbb{Z}$ with trivial action. Compute $\hat{H}^{-2}(C_4, \mathbb{Z})$ using the identification $\hat{H}^{-2} \cong H_1$ and the formula $H_1(G, \mathbb{Z}) \cong G^{\mathrm{ab}}$. Verify that the result matches the periodicity prediction $\hat{H}^{-2}(C_4, \mathbb{Z}) \cong \hat{H}^0(C_4, \mathbb{Z}) \cong \mathbb{Z}/4\mathbb{Z}$.
3. Let $G = C_p$ (cyclic of prime order) and consider the G-module $M = \mathbb{Z}[G]/(N_G)$, where (N_G) is the ideal generated by the norm element. Show that M is torsion-free of rank $p-1$ and compute $h(C_p, M)$.
4. Give a direct proof that $h(G, M) = 1$ for any finite G-module M (with G cyclic) using only the long exact sequence: show that the six-term exact hexagon has all groups finite and apply the alternating-product formula for exact sequences of finite abelian groups.
5. Let $G = C_3 = \langle \sigma \rangle$ and let $M = \mathbb{Z}^2$ with the action $\sigma \cdot (a, b) = (b, -a - b)$ (the standard representation of C_3 on the root lattice of type A_2). Compute $\hat{H}^0(C_3, M)$, $\hat{H}^{-1}(C_3, M)$, and the Herbrand quotient $h(C_3, M)$. (*Hint:* Compute N_G and $I_G M$ explicitly using the matrix of σ.)

4.2 Cup Products

The cohomology groups $H^n(G, M)$ are abelian groups, but they carry additional multiplicative structure: a bilinear pairing

$$H^p(G, M) \times H^q(G, N) \to H^{p+q}(G, M \otimes_{\mathbb{Z}} N)$$

called the *cup product*. This pairing turns the graded group $H^\bullet(G, M)$ (with $N = M$) into a graded ring, and it extends to Tate cohomology for finite groups. The cup product is the mechanism behind the periodicity isomorphism of §3, the duality pairings of §4, and the proof of the local reciprocity law in chapter 8.

Definition 4.2.1: Cup Product on Cochains

Let G be a group, M and N two G-modules. For inhomogeneous cochains $f \in C^p(G, M)$ and $g \in C^q(G, N)$, the *cup product* $f \smile g \in C^{p+q}(G, M \otimes_{\mathbb{Z}} N)$ is defined by

$$(f \smile g)(g_1, \ldots, g_{p+q}) = f(g_1, \ldots, g_p) \otimes (g_1 \cdots g_p) \cdot g(g_{p+1}, \ldots, g_{p+q}). \tag{4.1}$$

The tensor product $m \otimes n$ lies in $M \otimes_{\mathbb{Z}} N$, which is a G-module with diagonal action $g \cdot (m \otimes n) = (g \cdot m) \otimes (g \cdot n)$.

The formula records the "first p variables go to f, the last q variables go to g" splitting of the $(p+q)$-tuple, with the G-action on g's output twisted by the product of the first p group elements. This twist ensures compatibility with the coboundary operator.

Proposition 4.2.2: The Leibniz Rule

The cup product satisfies the *graded Leibniz rule*: for $f \in C^p(G,M)$ and $g \in C^q(G,N)$,

$$d^{p+q}(f \smile g) = (d^p f) \smile g + (-1)^p f \smile (d^q g).$$

In particular, the cup product of two cocycles is a cocycle, the cup product of a cocycle with a coboundary is a coboundary, and the cup product descends to a well-defined bilinear pairing

$$\smile : H^p(G,M) \times H^q(G,N) \to H^{p+q}(G, M \otimes_{\mathbb{Z}} N).$$

***Proof*:**
The verification is a direct computation from the coboundary formula (definition 2.3.3) and the cup product formula (4.1). One expands $d(f \smile g)(g_1, \dots, g_{p+q+1})$ using the coboundary formula, which has $p+q+2$ terms (one for the action of g_1, $p+q$ terms for the "face" insertions, and one for the last variable). Separating these into terms where the insertion occurs in the first $p+1$ positions (which contribute to $(df) \smile g$) and terms where it occurs in the last $q+1$ positions (which contribute to $(-1)^p f \smile (dg)$, with the sign coming from the offset) yields the stated identity.

The consequences for cohomology: if $df = 0$ and $dg = 0$, then $d(f \smile g) = 0$, so the cup product of cocycles is a cocycle. If $f = dh$ is a coboundary and $dg = 0$, then $f \smile g = (dh) \smile g = d(h \smile g) - (-1)^{p-1} h \smile (dg) = d(h \smile g)$, so the cup product is a coboundary. The case $df = 0$ and $g = dk$ is analogous with a sign.

For finite groups, the cup product extends to Tate cohomology: a pairing $\hat{H}^p(G,M) \times \hat{H}^q(G,N) \to \hat{H}^{p+q}(G, M \otimes_{\mathbb{Z}} N)$ defined for all $p, q \in \mathbb{Z}$.

Definition 4.2.3: Cup Product on Tate Cohomology

Let G be a finite group, M and N two G-modules. The *cup product on Tate cohomology*

$$\smile : \hat{H}^p(G,M) \times \hat{H}^q(G,N) \to \hat{H}^{p+q}(G, M \otimes_{\mathbb{Z}} N)$$

is defined via a *diagonal approximation* $\Delta : \hat{P}_\bullet \to \hat{P}_\bullet \otimes_{\mathbb{Z}} \hat{P}_\bullet$ on the complete standard resolution (definition 4.1.3). This is a G-equivariant chain map (with G acting diagonally on the tensor product) that is unique up to chain homotopy by the acyclicity of $\hat{P}_\bullet$. Given cocycles $f \in \mathrm{Hom}_G(\hat{P}_p, M)$ and $g \in \mathrm{Hom}_G(\hat{P}_q, N)$, the cup product $f \smile g = (f \otimes g) \circ \Delta_p \in \mathrm{Hom}_G(\hat{P}_{p+q}, M \otimes_{\mathbb{Z}} N)$, where Δ_p is the component of Δ landing in bidegree (p,q). The result is independent of the choice of Δ on the level of cohomology.

In non-negative degrees ($p, q \geq 0$), the diagonal approximation can be taken to be the Alexander–Whitney map, and the cup product on Tate cohomology restricts to the cochain-level cup product of definition 4.2.1. In negative degrees, the diagonal approximation on the dual part of the complete resolution requires a construction using the duality between P_n and P_{-n-1}, but the result is canonical.

Theorem 4.2.4: Properties of the Cup Product

The cup product on $H^\bullet(G,-)$ (and on $\hat{H}^\bullet(G,-)$ for G finite) satisfies:

1. *Bilinearity.* For fixed $\beta \in H^q(G,N)$, the map $\alpha \mapsto \alpha \smile \beta$ is a group homomorphism, and similarly in the second variable.
2. *Associativity.* For $\alpha \in H^p(G,L)$, $\beta \in H^q(G,M)$, $\gamma \in H^r(G,N)$:
$$(\alpha \smile \beta) \smile \gamma = \alpha \smile (\beta \smile \gamma) \in H^{p+q+r}(G, L \otimes M \otimes N).$$
3. *Graded-commutativity.* For $\alpha \in H^p(G,M)$ and $\beta \in H^q(G,N)$:
$$\alpha \smile \beta = (-1)^{pq}\, \tau_*(\beta \smile \alpha) \in H^{p+q}(G, M \otimes N),$$
where $\tau : N \otimes M \to M \otimes N$ is the twist isomorphism $n \otimes m \mapsto m \otimes n$.
4. *Unit.* The element $1 \in H^0(G,\mathbb{Z}) = \mathbb{Z}^G = \mathbb{Z}$ acts as a two-sided identity: $1 \smile \alpha = \alpha = \alpha \smile 1$ under the identifications $\mathbb{Z} \otimes M \cong M \cong M \otimes \mathbb{Z}$.

***Proof*:**
Step 1: Bilinearity. The cup product formula (4.1) is $\mathbb{Z}$-bilinear in f and g at the cochain level, so it is bilinear on cohomology.

Step 2: Associativity. At the cochain level, $((f \smile g) \smile h)(g_1,\ldots,g_{p+q+r})$ and $(f \smile (g \smile h))(g_1,\ldots,g_{p+q+r})$ both equal $f(g_1,\ldots,g_p) \otimes (g_1 \cdots g_p) \cdot g(g_{p+1},\ldots,g_{p+q}) \otimes (g_1 \cdots g_{p+q}) \cdot h(g_{p+q+1},\ldots,g_{p+q+r})$, using the associativity of the tensor product $L \otimes M \otimes N$.

Step 3: Graded-commutativity. This is more involved. One constructs, for each pair (p,q), an explicit chain homotopy between the Alexander–Whitney diagonal approximation and its twist by the transposition of tensor factors. The homotopy introduces the sign $(-1)^{pq}$. The classical argument uses the acyclic models theorem or an explicit cup-i-product construction; see [9] for the general framework.

Step 4: Unit. The element $1 \in H^0(G,\mathbb{Z})$ is represented by the cocycle $f_1 \in C^0(G,\mathbb{Z})$ with $f_1 = 1$ (the constant function). The cup product $f_1 \smile g = g$ for any $g \in C^q(G,M)$, since $(f_1 \smile g)(g_1,\ldots,g_q) = 1 \otimes g(g_1,\ldots,g_q) = g(g_1,\ldots,g_q)$ under $\mathbb{Z} \otimes M \cong M$.

The cup product interacts with the three functorial maps from section 3.3 in the following way.

Proposition 4.2.5: Compatibility with Restriction, Corestriction, and Inflation

Let G be a profinite group, M and N discrete G-modules.

1. *Restriction.* For $H \leq G$ closed and $\alpha \in H^p(G, M)$, $\beta \in H^q(G, N)$:
$$\operatorname{Res}^H_G(\alpha \smile \beta) = \operatorname{Res}^H_G(\alpha) \smile \operatorname{Res}^H_G(\beta).$$
2. *Projection formula.* For $H \leq G$ open, $\alpha \in H^p(G, M)$, and $\beta \in H^q(H, N)$:
$$\operatorname{Cor}^G_H(\operatorname{Res}^H_G(\alpha) \smile \beta) = \alpha \smile \operatorname{Cor}^G_H(\beta).$$
3. *Inflation.* For $N \trianglelefteq G$ closed normal and $\alpha \in H^p(G/N, M^N)$, $\beta \in H^q(G/N, (N')^N)$:
$$\operatorname{Inf}^G_{G/N}(\alpha \smile \beta) = \operatorname{Inf}^G_{G/N}(\alpha) \smile \operatorname{Inf}^G_{G/N}(\beta).$$

***Proof*:**
Part (1) is immediate from the cochain-level formula: restriction acts by restricting the domain of cochains from G^n to H^n, and the cup product formula involves only the group operation and action, which are unchanged.

Part (2), the projection formula, is the most substantial. At the cochain level, $\operatorname{Cor}(\operatorname{Res}(\alpha) \smile \beta)(g_1, \ldots, g_{p+q})$ involves a sum over coset representatives of the cup product of $\alpha|_H$ with β, and regrouping the sum using the cocycle property of α on G produces $\alpha \smile \operatorname{Cor}(\beta)$.

Part (3) follows from the cochain formula for inflation: $\operatorname{Inf}(f)(g_1, \ldots, g_n) = f(\bar{g}_1, \ldots, \bar{g}_n)$, and the cup product formula applied to inflated cochains produces the inflation of the cup product.

The projection formula in part (2) is particularly important: it says that corestriction is a module map over the cup product ring $H^\bullet(G, -)$, viewing $H^\bullet(H, -)$ as an $H^\bullet(G, -)$-module via restriction.

The cup product also interacts with the connecting homomorphisms of the long exact sequence.

Proposition 4.2.6: Cup Product and Connecting Homomorphisms

Let $0 \to M' \xrightarrow{\iota} M \xrightarrow{\pi} M'' \to 0$ be a short exact sequence of G-modules, and let N be a G-module.

1. For $\alpha \in H^p(G, M'')$ and $\beta \in H^q(G, N)$:
$$\delta(\alpha) \smile \beta = \delta(\alpha \smile \beta) \in H^{p+q+1}(G, M' \otimes N),$$
where δ on the left is the connecting homomorphism from the sequence $0 \to M' \to M \to M'' \to 0$, and on the right from the induced sequence $0 \to M' \otimes N \to M \otimes N \to M'' \otimes N \to 0$.

2. For $\alpha \in H^p(G, N)$ and $\beta \in H^q(G, M'')$:
$$\alpha \smile \delta(\beta) = (-1)^p \delta(\alpha \smile \beta) \in H^{p+q+1}(G, N \otimes M'),$$
with the analogous connecting homomorphism from $0 \to N \otimes M' \to N \otimes M \to N \otimes M'' \to 0$.

The same identities hold for Tate cohomology when G is finite.

***Proof*:**
Both identities are verified by a diagram chase at the cochain level. For part (1): represent α by a cochain $\tilde{f} \in C^p(G, M)$ lifting $f \in Z^p(G, M'')$ (so $\pi \circ \tilde{f} = f$ and $d\tilde{f}$ takes values in M'). Then $\delta(\alpha)$ is represented by $d\tilde{f} \in Z^{p+1}(G, M')$. The Leibniz rule gives $d(\tilde{f} \smile g) = (d\tilde{f}) \smile g + (-1)^p \tilde{f} \smile dg = (d\tilde{f}) \smile g$ (since $dg = 0$). Hence $\delta(\alpha) \smile \beta$ is the class of $d(\tilde{f} \smile g)$, which is δ applied to the class of $f \smile g$ in $H^{p+q}(G, M'' \otimes N)$.

Part (2) uses the same argument with the roles of the two factors interchanged, producing the sign $(-1)^p$ from the Leibniz rule.

This compatibility is the algebraic engine of dimension shifting: given a short exact sequence $0 \to M \to I \to Q \to 0$ with I acyclic, the connecting homomorphism $\delta : H^n(G, Q) \xrightarrow{\sim} H^{n+1}(G, M)$ is an isomorphism for $n \geq 1$. The cup product identity $\delta(\alpha \smile \beta) = \delta(\alpha) \smile \beta$ shows that this isomorphism is compatible with the cup product: shifting the degree of one factor by 1 via δ shifts the degree of the cup product by 1.

Example 4.2.7: Cup Products in Low Degrees and for Cyclic Groups

1. In Tate cohomology, the cup product $\hat{H}^0(G, M) \times \hat{H}^0(G, N) \to \hat{H}^0(G, M \otimes N)$ is the map induced by $m \otimes n \mapsto m \otimes n$ on $M^G / N_G(M) \times N^G / N_G(N) \to (M \otimes N)^G / N_G(M \otimes N)$. This is well defined because $N_G(m) \otimes n = N_G(m \otimes n)$ for $n \in N^G$ (as $g \cdot (m \otimes n) = (g \cdot m) \otimes n$ when n is G-invariant).

2. Let $G = C_n = \langle \sigma \rangle$ and consider the periodicity generator $u \in \hat{H}^2(G, \mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}$ (corresponding to $1 \bmod n$; its construction will be made explicit in §3). The cup product

$- \smile u : \hat{H}^q(G, M) \to \hat{H}^{q+2}(G, M \otimes \mathbb{Z}) \cong \hat{H}^{q+2}(G, M)$ is an isomorphism for all $q \in \mathbb{Z}$ — this is the content of the periodicity theorem of §3, and the cup product is the mechanism that produces it.

3. Let $G = C_2 = \{1, \sigma\}$ and $M = \mathbb{Z}/2\mathbb{Z}$ with trivial action. The group $H^1(C_2, \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$ is generated by the unique nontrivial homomorphism $\chi : C_2 \to \mathbb{Z}/2\mathbb{Z}$. The cup product $\chi \smile \chi \in H^2(C_2, \mathbb{Z}/2\mathbb{Z} \otimes \mathbb{Z}/2\mathbb{Z}) \cong H^2(C_2, \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$ is computed via the cochain formula: $(\chi \smile \chi)(\sigma^a, \sigma^b) = \chi(\sigma^a) \cdot \sigma^a \cdot \chi(\sigma^b) = a \cdot b \pmod 2$ (where $a, b \in \{0, 1\}$ and the action is trivial). This is the cocycle $c(a, b) = ab \bmod 2$, which represents the nonzero element of $H^2(C_2, \mathbb{Z}/2\mathbb{Z})$. By induction, $\chi^n \neq 0$ for all $n \geq 1$, and the cup product ring is

$$H^\bullet(C_2, \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}[\chi], \qquad |\chi| = 1.$$

This is a polynomial ring in one variable of degree 1.

4. (If the reader has read [16]) The identification $H^\bullet(C_2, \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}[\chi]$ is the algebraic counterpart of the cohomology ring $H^\bullet(\mathbb{RP}^\infty; \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}[w_1]$, where w_1 is the first Stiefel–Whitney class of the tautological line bundle. The classifying space $BC_2 = \mathbb{RP}^\infty$ realizes group cohomology as singular cohomology, and the cup product on group cohomology corresponds to the topological cup product. This correspondence extends to all finite groups via the classifying space construction.

Exercise 4.2.1

1. Verify the associativity of the cup product at the cochain level: for $f \in C^p(G, L)$, $g \in C^q(G, M)$, $h \in C^r(G, N)$, show that $((f \smile g) \smile h)(g_1, \ldots, g_{p+q+r}) = (f \smile (g \smile h))(g_1, \ldots, g_{p+q+r})$ by expanding both sides using the formula (4.1).
2. Let $G = C_4 = \langle \sigma \rangle$, $H = C_2 = \langle \sigma^2 \rangle$, and $M = N = \mathbb{Z}$ with trivial action. Compute the cup product $\mathrm{Res}(\alpha) \smile \beta$ for α a generator of $H^2(C_4, \mathbb{Z}) \cong \mathbb{Z}/4\mathbb{Z}$ and β a generator of $H^2(C_2, \mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$, and verify the projection formula $\mathrm{Cor}(\mathrm{Res}(\alpha) \smile \beta) = \alpha \smile \mathrm{Cor}(\beta)$ by computing both sides independently.
3. Let $G = C_p$ (cyclic of prime order p) and $M = \mathbb{Z}/p\mathbb{Z}$ with trivial action. Compute the cup product ring $H^\bullet(C_p, \mathbb{Z}/p\mathbb{Z})$.
   (a) Show that $H^\bullet(C_p, \mathbb{Z}/p\mathbb{Z}) \cong \mathbb{Z}/p\mathbb{Z}[\chi, u]/(\chi^2)$ where $|\chi| = 1$ and $|u| = 2$, for p odd. (*Hint:* $\chi^2 = 0$ by graded-commutativity since p is odd, and $\chi \smile u \neq 0$ generates H^3.)
   (b) Show that $H^\bullet(C_2, \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}[\chi]$ where $|\chi| = 1$ (the polynomial ring from example 4.2.7). Why does $\chi^2 \neq 0$ in this case?
4. Let G be a finite group and $0 \to \mathbb{Z} \xrightarrow{\times n} \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z} \to 0$ the short exact sequence with trivial G-action. Use the compatibility of the cup product with connecting homomorphisms (proposition 4.2.6) to express the Bockstein homomorphism $\beta : H^q(G, \mathbb{Z}/n\mathbb{Z}) \to H^{q+1}(G, \mathbb{Z}/n\mathbb{Z})$

(defined as the composition of $\delta : H^q(G, \mathbb{Z}/n\mathbb{Z}) \to H^{q+1}(G, \mathbb{Z})$ with reduction mod n) in terms of the cup product with a class in $H^1(G, \mathbb{Z}/n\mathbb{Z})$ when G is cyclic of order n.

5. Let G be a profinite group of p-cohomological dimension $\leq d$ (foreshadowing chapter 5: $H^n(G, M) = 0$ for all $n > d$ and all p-torsion discrete G-modules M). Show that the cup product $H^p(G, M) \times H^q(G, N) \to H^{p+q}(G, M \otimes N)$ is zero whenever $p + q > d$ and $M \otimes N$ is p-torsion.

4.3 Dimension Shifting and Periodicity

Dimension shifting is the technique of computing cohomology in one degree by embedding the coefficient module into an acyclic module and using the connecting homomorphism to shift the degree. In ordinary cohomology, this works only in the positive direction (from degree n to degree $n + 1$), because the connecting homomorphism $\delta : H^n(G, M'') \to H^{n+1}(G, M')$ requires $n \geq 0$. In Tate cohomology, the long exact sequence extends to all $n \in \mathbb{Z}$ (proposition 4.1.6), so dimension shifting works in both directions without restriction. This unrestricted shifting is the key to the periodicity theorem for cyclic groups, which asserts that cup product with a canonical class $u \in \hat{H}^2(G, \mathbb{Z})$ gives an isomorphism $\hat{H}^q(G, M) \cong \hat{H}^{q+2}(G, M)$ for all $q \in \mathbb{Z}$.

Proposition 4.3.1: Dimension Shifting

Let G be a finite group and M a G-module.

1. *Upward shift.* Let $I = \mathrm{CoInd}_{\{1\}}^G(M) = \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}[G], M)$ be the coinduced module, with the natural inclusion $\iota : M \hookrightarrow I$ (sending m to the map $g \mapsto g \cdot m$). Let $Q = I/\iota(M)$. Since I is Tate-acyclic (proposition 4.1.7), the connecting homomorphism in the long exact sequence for $0 \to M \to I \to Q \to 0$ gives isomorphisms
$$\delta : \hat{H}^n(G, Q) \xrightarrow{\sim} \hat{H}^{n+1}(G, M) \qquad \text{for all } n \in \mathbb{Z}.$$

2. *Downward shift.* Let $J = \mathbb{Z}[G] \otimes_{\mathbb{Z}} M$ be the induced module, with the natural surjection $\varepsilon : J \twoheadrightarrow M$ (sending $g \otimes m$ to $g \cdot m$). Let $K = \ker(\varepsilon)$. Since J is Tate-acyclic, the connecting homomorphism for $0 \to K \to J \to M \to 0$ gives isomorphisms
$$\delta : \hat{H}^{n-1}(G, M) \xrightarrow{\sim} \hat{H}^n(G, K) \qquad \text{for all } n \in \mathbb{Z}.$$

***Proof*:**
Both statements follow from the long exact sequence of proposition 4.1.6 applied to the respective short exact sequences, using $\hat{H}^n(G, I) = \hat{H}^n(G, J) = 0$ for all n.

Dimension shifting replaces the study of $\hat{H}^{n+1}(G, M)$ with the study of $\hat{H}^n(G, Q)$, where Q depends on M but is often more tractable. Iterating: $\hat{H}^{n+k}(G, M) \cong \hat{H}^n(G, Q_k)$ for a module Q_k obtained by k successive shifts. The downward version similarly gives $\hat{H}^{n-k}(G, M) \cong \hat{H}^n(G, K_k)$. The interplay between upward and downward shifting is possible only in Tate cohomology, where the long exact sequence has no lower bound on the degree.

For cyclic groups, dimension shifting has a dramatic consequence: the Tate cohomology is 2-periodic, and this periodicity is realized by the cup product with a canonical class. The first step is to exhibit the periodic structure of the free resolution.

Proposition 4.3.2: The Periodic Resolution for Cyclic Groups

Let $G = C_n = \langle \sigma \rangle$ be cyclic of order n. Define the elements $D = \sigma - 1$ and $N = N_G = 1 + \sigma + \cdots + \sigma^{n-1}$ in $\mathbb{Z}[G]$. The sequence

$$\cdots \xrightarrow{N} \mathbb{Z}[G] \xrightarrow{D} \mathbb{Z}[G] \xrightarrow{N} \mathbb{Z}[G] \xrightarrow{D} \mathbb{Z}[G] \xrightarrow{\varepsilon} \mathbb{Z} \to 0 \tag{4.2}$$

is a free resolution of $\mathbb{Z}$ over $\mathbb{Z}[G]$, called the *periodic resolution*. The differentials alternate between multiplication by $D = \sigma - 1$ and multiplication by $N = N_G$.

***Proof*:**
The identities $D \cdot N = (\sigma - 1)(1 + \sigma + \cdots + \sigma^{n-1}) = \sigma^n - 1 = 0$ and $N \cdot D = 0$ (by the same computation) show that consecutive maps compose to zero. To verify exactness:

Step 1: At the augmentation. The kernel of $\varepsilon : \mathbb{Z}[G] \to \mathbb{Z}$ is the augmentation ideal $I_G = (D) = (\sigma - 1) \cdot \mathbb{Z}[G]$, which is the image of $D : \mathbb{Z}[G] \to \mathbb{Z}[G]$.

Step 2: At D-to-N transitions. The kernel of $D : \mathbb{Z}[G] \to \mathbb{Z}[G]$ consists of elements $f \in \mathbb{Z}[G]$ with $(\sigma - 1)f = 0$. Writing $f = \sum_{i=0}^{n-1} a_i \sigma^i$, the condition $\sigma f = f$ means $a_0 = a_1 = \cdots = a_{n-1}$, so $f = a \cdot N$ for some $a \in \mathbb{Z}$. Hence $\ker(D) = \mathbb{Z} \cdot N = \operatorname{im}(N)$.

Step 3: At N-to-D transitions. The kernel of $N : \mathbb{Z}[G] \to \mathbb{Z}[G]$ consists of elements f with $N \cdot f = 0$. Since $\mathbb{Z}[G] \cong \mathbb{Z}[x]/(x^n - 1)$ and $N = (x^n - 1)/(x - 1)$, the condition $N \cdot f = 0$ in $\mathbb{Z}[G]$ means $(x - 1) \big| f$ in $\mathbb{Z}[x]/(x^n - 1)$, i.e., $f = (x - 1)g$ for some g. Hence $\ker(N) = (\sigma - 1) \cdot \mathbb{Z}[G] = \operatorname{im}(D)$.

The periodic resolution has period 2: the pattern of differentials $(D, N, D, N, \ldots)$ repeats every two steps. The complete resolution (definition 4.1.3) for a cyclic group inherits this periodicity, splicing the periodic resolution with its $\mathbb{Z}[G]$-dual. This periodicity at the level of resolutions translates into periodicity in Tate cohomology via the cup product with the following class.

Definition 4.3.3: The Periodicity Generator

Let $G = C_n$ be cyclic of order n. The *periodicity generator* $u_G \in \hat{H}^2(G, \mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}$ is the class corresponding to $1 \bmod n$ under the canonical identification. Concretely, u_G is the extension class of the short exact sequence

$$0 \to \mathbb{Z} \xrightarrow{N_G} \mathbb{Z}[G] \xrightarrow{\sigma - 1} I_G \to 0, \tag{4.3}$$

where the first map sends $1 \mapsto N_G$ and the second is multiplication by $\sigma - 1$. (The sequence is exact by the proof of proposition 4.3.2: $\ker(\sigma - 1) = \mathbb{Z} \cdot N_G = \operatorname{im}(N_G)$, and $\operatorname{im}(\sigma - 1) = I_G = \ker(\varepsilon)$.)

The extension (4.3) splices two consecutive steps of the periodic resolution, and its class in $H^2(G, \mathbb{Z}) = \mathrm{Ext}^1_{\mathbb{Z}[G]}(I_G, \mathbb{Z})$ measures the "twisting" between the D- and N-steps. The identification $H^2(C_n, \mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}$ sends this class to 1 mod n — this can be verified using the explicit cocycle representing the extension (the "carry cocycle" from example 2.3.8).

Theorem 4.3.4: Periodicity for Cyclic Groups

Let $G = C_n$ be cyclic of order n, M a G-module, and $u_G \in \hat{H}^2(G, \mathbb{Z})$ the periodicity generator. The cup product map

$$- \smile u_G : \hat{H}^q(G, M) \xrightarrow{\sim} \hat{H}^{q+2}(G, M)$$

(using the identification $M \otimes_{\mathbb{Z}} \mathbb{Z} \cong M$) is an isomorphism for every $q \in \mathbb{Z}$.

***Proof*:**

Step 1: The case $M = \mathbb{Z}$. The periodicity of the resolution (proposition 4.3.2) gives $\hat{H}^q(G, \mathbb{Z}) \cong \hat{H}^{q+2}(G, \mathbb{Z})$ for all q (applying $\mathrm{Hom}_G(-, \mathbb{Z})$ to the complete resolution, the 2-periodic pattern of differentials produces a 2-periodic cochain complex). The cup product with u_G implements this shift: the map $- \smile u_G$ on $\hat{H}^q(G, \mathbb{Z})$ is an endomorphism of $\mathbb{Z}/n\mathbb{Z}$ (for q even) or of 0 (for q odd). For $q = 0$: the cup product $\hat{H}^0(G, \mathbb{Z}) \times \hat{H}^2(G, \mathbb{Z}) \to \hat{H}^2(G, \mathbb{Z})$ sends $(1 \bmod n, u_G) \mapsto u_G$, which is the generator of $\hat{H}^2(G, \mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}$. Hence $- \smile u_G$ is multiplication by 1 on $\mathbb{Z}/n\mathbb{Z}$, which is an isomorphism.

Step 2: Reduction to $M = \mathbb{Z}$ via dimension shifting. For a general M, embed M into the Tate-acyclic module $I = \mathbb{Z}[G] \otimes_{\mathbb{Z}} M$ with quotient $Q = I/M$. The short exact sequence $0 \to M \to I \to Q \to 0$ gives connecting isomorphisms $\delta : \hat{H}^q(G, Q) \xrightarrow{\sim} \hat{H}^{q+1}(G, M)$. By the compatibility of the cup product with connecting homomorphisms (proposition 4.2.6), the diagram

$$\begin{array}{ccc} \hat{H}^q(G, Q) & \xrightarrow{- \smile u_G} & \hat{H}^{q+2}(G, Q) \\ \downarrow_{\wr}^{\delta} & & \downarrow_{\wr}^{\delta} \\ \hat{H}^{q+1}(G, M) & \xrightarrow{- \smile u_G} & \hat{H}^{q+3}(G, M) \end{array}$$

commutes (up to sign, which does not affect whether the map is an isomorphism). Hence if $- \smile u_G$ is an isomorphism on Q, it is an isomorphism on M (shifted by one degree). By iterating dimension shifts, the problem reduces to the case where M is replaced by a module of the form Q_k obtained after k shifts, and for k large enough the computation reduces to the $M = \mathbb{Z}$ case (since the dimension-shifted modules eventually have the same Tate cohomology pattern as $\mathbb{Z}$ in the relevant degrees). A cleaner argument: the cup product with u_G commutes with the connecting homomorphisms from *any* short exact sequence (by proposition 4.2.6), so it is a natural transformation of cohomological functors. By Step 1, this natural transformation is an isomorphism on $\mathbb{Z}$, hence on all free $\mathbb{Z}$-modules, hence on all $\mathbb{Z}$-modules by a presentation argument.

Step 3: The natural transformation argument. The cup product $- \smile u_G : \hat{H}^q(G, -) \to \hat{H}^{q+2}(G, -)$ is a morphism of δ-functors (it commutes with connecting homomorphisms by proposition 4.2.6). The source and target are both universal δ-functors on the category of G-modules (since Tate cohomology is effaceable: every module embeds into a Tate-acyclic module). A morphism between universal δ-functors that is an isomorphism in one degree (here, the computation for $M = \mathbb{Z}$ in Step 1) is an isomorphism in all degrees. This completes the proof.

The periodicity theorem says that all of the Tate cohomology of a cyclic group is encoded in the two groups $\hat{H}^0(G, M) = M^G/N_G(M)$ and $\hat{H}^{-1}(G, M) = \ker(N_G)/I_G M$, repeating with period 2. The periodicity generator u_G is the algebraic mechanism behind this repetition: cupping with u_G shifts the degree by 2, and the fact that this is an isomorphism reflects the 2-periodic structure of the free resolution for cyclic groups.

Periodicity is special to cyclic groups. For a non-cyclic finite group G, the Tate cohomology $\hat{H}^n(G, M)$ can exhibit growth in n (for instance, $|\hat{H}^n(C_p \times C_p, \mathbb{Z}/p\mathbb{Z})| = p^{n+1}$ for $n \geq 0$, which is unbounded). The cohomological dimension and structure theory of profinite groups developed in chapter 5 provide the correct framework for understanding the behaviour of $\hat{H}^n$ as n grows for non-cyclic groups.

The Herbrand quotient of definition 4.1.9 is a direct consequence of periodicity: since $\hat{H}^q(G, M)$ depends only on the parity of q (for G cyclic), the ratio $h(G, M) = |\hat{H}^0|/|\hat{H}^{-1}|$ captures all the information about the "size" of the cohomology, and its multiplicativity (theorem 4.1.10) follows from the long exact sequence combined with the 2-periodic collapse.

The periodicity theorem has a powerful converse: Tate's criterion characterizes modules with vanishing Tate cohomology.

Theorem 4.3.5: Tate's Theorem on Cohomological Triviality

Let G be a finite group and M a G-module. The following are equivalent:

1. $\hat{H}^n(H, M) = 0$ for all $n \in \mathbb{Z}$ and all subgroups $H \leq G$.
2. $\hat{H}^{-1}(H, M) = \hat{H}^0(H, M) = 0$ for all subgroups $H \leq G$.
3. $\hat{H}^{-1}(H, M) = \hat{H}^0(H, M) = 0$ for all cyclic subgroups $H \leq G$.

A module satisfying these equivalent conditions is called *cohomologically trivial*.

***Proof*:**
The implication (1) $\Rightarrow$ (2) $\Rightarrow$ (3) is immediate. The content is (3) $\Rightarrow$ (1).

Step 1: G cyclic. Assume G is cyclic and $\hat{H}^0(G, M) = \hat{H}^{-1}(G, M) = 0$. By the periodicity theorem (theorem 4.3.4), $\hat{H}^q(G, M) \cong \hat{H}^{q+2}(G, M)$ for all q. Since $\hat{H}^0 = 0$ and $\hat{H}^{-1} = 0$, the periodicity gives $\hat{H}^q(G, M) = 0$ for all $q \in \mathbb{Z}$.

Step 2: Reduction to Sylow subgroups. If $\hat{H}^{-1}(H, M) = \hat{H}^0(H, M) = 0$ for all cyclic subgroups $H \leq G$, then in particular for every prime p dividing $|G|$ and every cyclic p-subgroup H, the vanishing holds. By Step 1, $\hat{H}^n(H, M) = 0$ for all n and all cyclic p-subgroups. Since every element of $\hat{H}^n(P, M)$ for a p-Sylow subgroup P restricts injectively to $\hat{H}^n(H, M)$ for some cyclic subgroup $H \leq P$ (the restriction to a cyclic subgroup of order p generating the element's support), it follows that $\hat{H}^n(P, M) = 0$ for all n and all Sylow subgroups P.

Step 3: From Sylow subgroups to G. By the formula $\mathrm{Cor} \circ \mathrm{Res} = [G : P]$ (proposition 3.3.6), the map $\mathrm{Res} : \hat{H}^n(G, M)\{p\} \to \hat{H}^n(P, M)$ is injective on the p-primary part. Since $\hat{H}^n(P, M) = 0$

for every Sylow P, the p-primary part of $\hat{H}^n(G, M)$ vanishes for every prime p. By the torsion property (corollary 3.2.4 and the p-primary decomposition), $\hat{H}^n(G, M) = 0$ for all $n \geq 1$. The same argument applied to $\hat{H}^n$ for $n \leq -2$ (which are group homology) extends the vanishing to all n.

Tate's theorem is a reduction principle: to verify cohomological triviality, it suffices to check two degrees (0 and -1) on cyclic subgroups, rather than all degrees on all subgroups. The theorem is optimal in two senses. First, a single degree does not suffice: exercise 4.1.1 (3) provides a module with $\hat{H}^0 = 0$ but $\hat{H}^{-1} \neq 0$. Second, restricting to cyclic subgroups is essential because the restriction–corestriction formula (proposition 3.3.6) reduces the p-primary part of $\hat{H}^n(G, M)$ to the p-Sylow subgroup, and cyclic groups of prime power order generate all Sylow subgroups. The theorem will be applied in chapter 8 to verify cohomological triviality of unit groups in local field extensions — there, the abstract criterion reduces a difficult arithmetic question to two explicit norm computations.

Example 4.3.6: Cohomologically Trivial Modules

1. The module $M = \mathbb{Z}[G]$ is cohomologically trivial by proposition 4.1.7: $\hat{H}^n(G, \mathbb{Z}[G]) = 0$ for all n. This can also be verified via Tate's theorem: $\hat{H}^0(H, \mathbb{Z}[G]) = (\mathbb{Z}[G])^H / N_H(\mathbb{Z}[G])$. The invariants $(\mathbb{Z}[G])^H \cong \mathbb{Z}[G/H]$ (the formal sums over cosets), and $N_H(\mathbb{Z}[G]) = N_H \cdot \mathbb{Z}[G]$. One checks that $N_H \cdot \mathbb{Z}[G] = (\mathbb{Z}[G])^H$ by observing that $N_H \cdot g = \sum_{h \in H} hg$, which is the sum over the left H-coset of g. Hence $\hat{H}^0(H, \mathbb{Z}[G]) = 0$. A similar computation gives $\hat{H}^{-1}(H, \mathbb{Z}[G]) = 0$.

2. The module $M = \mathbb{Z}/n\mathbb{Z}$ with trivial C_n-action is *not* cohomologically trivial: $\hat{H}^0(C_n, \mathbb{Z}/n\mathbb{Z}) = (\mathbb{Z}/n\mathbb{Z})/n(\mathbb{Z}/n\mathbb{Z}) = \mathbb{Z}/n\mathbb{Z} \neq 0$. The periodicity theorem gives $\hat{H}^q(C_n, \mathbb{Z}/n\mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}$ for all $q \in \mathbb{Z}$.

3. For $G = C_n$ and $M = \mathbb{Z}[G]/(N_G)$: the computation of exercise 4.1.1 (3) gives $\hat{H}^0(C_n, M) = 0$ and $\hat{H}^{-1}(C_n, M) \cong \mathbb{Z}/n\mathbb{Z} \neq 0$, so M is not cohomologically trivial. In fact, M fails the criterion in degree -1 alone — one vanishing is not enough.

4. (If the reader has read [6]) Let G be a finite group and M a finitely generated $\mathbb{Z}[G]$-module that is $\mathbb{Z}$-free. Then M is cohomologically trivial if and only if M is a projective $\mathbb{Z}[G]$-module. The forward direction follows from homological algebra: if $\hat{H}^n(H, M) = 0$ for all H and n, then $\mathrm{Ext}^1_{\mathbb{Z}[G]}(\mathbb{Z}, M) = H^1(G, M) = 0$, and by Rim's theorem, M is projective. The converse is immediate since projective modules are direct summands of free modules, which are Tate-acyclic.

Exercise 4.3.1

1. Let $G = C_6$ and $M = \mathbb{Z}$ with trivial action. Use dimension shifting (with the sequence

$0 \to \mathbb{Z} \to \mathbb{Z}[C_6] \to \mathbb{Z}[C_6]/\mathbb{Z} \to 0$) to compute $\hat{H}^{-2}(C_6, \mathbb{Z})$. Verify that the result matches the periodicity prediction $\hat{H}^{-2}(C_6, \mathbb{Z}) \cong \hat{H}^0(C_6, \mathbb{Z}) \cong \mathbb{Z}/6\mathbb{Z}$.

2. For $G = C_4 = \langle\sigma\rangle$, write down the periodic resolution (4.2) explicitly (with $D = \sigma - 1$ and $N = 1 + \sigma + \sigma^2 + \sigma^3$). Construct the extension (4.3) and verify that its class in $H^2(C_4, \mathbb{Z}) \cong \mathbb{Z}/4\mathbb{Z}$ corresponds to 1 mod 4 by computing the associated 2-cocycle.

3. Let G be a finite group and $H \leq G$ a subgroup. Show that the permutation module $\mathbb{Z}[G/H]$ is cohomologically trivial by verifying the conditions of theorem 4.3.5 for all cyclic subgroups. (*Hint:* Use Shapiro's lemma (theorem 3.5.5) to relate $\hat{H}^n(K, \mathbb{Z}[G/H])$ to $\hat{H}^n(K \cap gHg^{-1}, \mathbb{Z})$ for various conjugates.)

4. Show that the condition in theorem 4.3.5 cannot be weakened to a single degree: give an example of a cyclic group G and a G-module M with $\hat{H}^0(G, M) = 0$ but $\hat{H}^{-1}(G, M) \neq 0$. (*Hint:* The module $\mathbb{Z}[G]/(N_G)$ from exercise 4.1.1 (3) works.)

5. Let $G = C_p$ (p prime) and $M = \mathbb{Z}/p\mathbb{Z}$ with trivial action. Let $u \in \hat{H}^2(C_p, \mathbb{Z})$ be the periodicity generator and $\bar{u}$ its image under the reduction map $\hat{H}^2(C_p, \mathbb{Z}) \to \hat{H}^2(C_p, \mathbb{Z}/p\mathbb{Z})$. Show that $\bar{u}$ equals the class u from the cup product ring $H^\bullet(C_p, \mathbb{Z}/p\mathbb{Z}) \cong \mathbb{Z}/p\mathbb{Z}[\chi, u]/(\chi^2)$ of exercise 4.2.1 (3) (for p odd). Explain how the periodicity isomorphism $- \smile u_G$ on $\hat{H}^\bullet(C_p, \mathbb{Z})$ descends to the periodicity $- \smile \bar{u}$ on $\hat{H}^\bullet(C_p, \mathbb{Z}/p\mathbb{Z})$.

4.4 Duality for Finite Groups

The periodicity theorem of theorem 4.3.4 shows that for a cyclic group G, the Tate cohomology $\hat{H}^p(G, M)$ in positive degrees is mirrored by $\hat{H}^{-p-1}(G, M)$ in negative degrees (with a shift). This mirroring becomes a precise duality when the cup product is used to pair $\hat{H}^p(G, M)$ with $\hat{H}^{-p-1}(G, M^*)$, where M^* is the Pontryagin dual. The resulting pairing is perfect for cyclic groups acting on finite modules — a purely group-theoretic duality that will be the model for the arithmetic duality theorems of chapters 8 and 9.

The section also establishes Nakayama's criterion, which connects the vanishing of Tate cohomology to the module-theoretic property of projectivity. This criterion is the tool that converts cohomological computations ("$\hat{H}^0$ and $\hat{H}^{-1}$ vanish") into structural conclusions ("the module is projective"), and it is used throughout Part III.

Definition 4.4.1: Pontryagin Dual with G-Action

Let G be a finite group and M a G-module. The *Pontryagin dual* of M is

$$M^* = \mathrm{Hom}_{\mathbb{Z}}(M, \mathbb{Q}/\mathbb{Z})$$

equipped with the *contragredient* G-action: $(g \cdot f)(m) = f(g^{-1} \cdot m)$ for $g \in G$, $f \in M^*$, $m \in M$. When M is finite, M^* is also finite with $|M^*| = |M|$.

The contragredient action ensures that M^* transforms "dually" to M: if G acts on M by a representation $\rho : G \to \mathrm{Aut}(M)$, then G acts on M^* by the contragredient representation $\rho^*(g) = \rho(g^{-1})^T$ (the transpose-inverse). For M finite, the canonical isomorphism $M \xrightarrow{\sim} M^{**}$ is G-equivariant, so the double dual recovers M with its original action.

The evaluation map $\mathrm{ev} : M \otimes_{\mathbb{Z}} M^* \to \mathbb{Q}/\mathbb{Z}$, defined by $\mathrm{ev}(m \otimes f) = f(m)$, is G-equivariant: $\mathrm{ev}(g \cdot m \otimes g \cdot f) = (g \cdot f)(g \cdot m) = f(g^{-1} \cdot g \cdot m) = f(m) = \mathrm{ev}(m \otimes f)$. When M is finite, the evaluation is a perfect pairing of abelian groups: the induced map $M \to (M^*)^*$ is an isomorphism. The map ev induces a homomorphism $\mathrm{ev}_* : \hat{H}^n(G, M \otimes M^*) \to \hat{H}^n(G, \mathbb{Q}/\mathbb{Z})$ for all n, and combining this with the cup product produces the duality pairing.

Definition 4.4.2: The Tate Duality Pairing

Let G be a finite group, M a G-module. For $p \in \mathbb{Z}$, the *Tate duality pairing*

$$\langle -, - \rangle : \hat{H}^p(G, M) \times \hat{H}^{-p-1}(G, M^*) \to \mathbb{Q}/\mathbb{Z}$$

is the composition

$$\hat{H}^p(G, M) \times \hat{H}^{-p-1}(G, M^*) \xrightarrow{\smile} \hat{H}^{-1}(G, M \otimes M^*) \xrightarrow{\mathrm{ev}_*} \hat{H}^{-1}(G, \mathbb{Q}/\mathbb{Z}) \xrightarrow{\mathrm{inv}} \mathbb{Q}/\mathbb{Z}$$

where the first map is the cup product (definition 4.2.3), the second is induced by the evaluation, and inv is defined as follows. For G cyclic of order n acting trivially on $\mathbb{Q}/\mathbb{Z}$: the norm map N_G acts on $\mathbb{Q}/\mathbb{Z}$ by multiplication by n, and $\hat{H}^{-1}(G, \mathbb{Q}/\mathbb{Z}) = \ker(N_G : \mathbb{Q}/\mathbb{Z} \to \mathbb{Q}/\mathbb{Z})/I_G(\mathbb{Q}/\mathbb{Z}) = \frac{1}{n}\mathbb{Z}/\mathbb{Z} \cong \mathbb{Z}/n\mathbb{Z}$ (since $\ker(n) = \frac{1}{n}\mathbb{Z}/\mathbb{Z}$ and $I_G(\mathbb{Q}/\mathbb{Z}) = 0$ for the trivial action). The *invariant map* $\mathrm{inv} : \hat{H}^{-1}(G, \mathbb{Q}/\mathbb{Z}) \to \mathbb{Q}/\mathbb{Z}$ is the natural inclusion $\frac{1}{n}\mathbb{Z}/\mathbb{Z} \hookrightarrow \mathbb{Q}/\mathbb{Z}$.

The target $\mathbb{Q}/\mathbb{Z}$ is an injective abelian group, so the pairing $\langle -, - \rangle$ induces a homomorphism $\hat{H}^p(G, M) \to \mathrm{Hom}(\hat{H}^{-p-1}(G, M^*), \mathbb{Q}/\mathbb{Z}) = \hat{H}^{-p-1}(G, M^*)^*$ (the Pontryagin dual of the Tate cohomology group). The pairing is $\mathbb{Z}$-bilinear in both variables (by the bilinearity of the cup product, theorem 4.2.4) and natural in M: a G-equivariant map $\varphi : M \to M'$ induces maps $\hat{H}^p(G, M) \to \hat{H}^p(G, M')$ and $\hat{H}^{-p-1}(G, (M')^*) \to \hat{H}^{-p-1}(G, M^*)$ (via φ^*), and the pairings for M and M' are compatible. The duality theorem asserts that the induced homomorphism is an isomorphism for cyclic groups.

Theorem 4.4.3: Tate Duality for Cyclic Groups

Let G be a cyclic group of order n and M a finite G-module. The Tate duality pairing

$$\langle -, - \rangle : \hat{H}^p(G, M) \times \hat{H}^{-p-1}(G, M^*) \to \mathbb{Q}/\mathbb{Z}$$

is a perfect pairing of finite abelian groups for every $p \in \mathbb{Z}$. That is, the induced maps

$$\hat{H}^p(G, M) \to \hat{H}^{-p-1}(G, M^*)^* \quad \text{and} \quad \hat{H}^{-p-1}(G, M^*) \to \hat{H}^p(G, M)^*$$

are isomorphisms.

***Proof*:**

Step 1: Reduction via periodicity. By the periodicity theorem (theorem 4.3.4), $\hat{H}^p(G, M) \cong \hat{H}^{p+2}(G, M)$ and $\hat{H}^{-p-1}(G, M^*) \cong \hat{H}^{-p+1}(G, M^*)$. The cup product with the periodicity generator u_G shifts both factors compatibly, so the pairing in degree p is isomorphic to the pairing in degree $p + 2$. It therefore suffices to prove perfectness for $p = 0$ and $p = -1$.

Step 2: Reduction to simple modules. Both sides of the pairing are additive in M: a short exact sequence $0 \to M' \to M \to M'' \to 0$ induces (by the long exact sequence and the compatibility of the cup product with connecting homomorphisms, proposition 4.2.6) a commutative ladder connecting the pairings for M', M, and M''. By the five lemma, if the pairing is perfect for M' and M'', it is perfect for M. Since M is finite, it has a G-stable filtration with simple subquotients (simple $\mathbb{Z}[G]$-modules killed by some prime p), and by induction on $|M|$ it suffices to verify perfectness for simple modules.

Step 3: The case $M = \mathbb{Z}/p\mathbb{Z}$ with trivial action. The Pontryagin dual is $(\mathbb{Z}/p\mathbb{Z})^* = \operatorname{Hom}(\mathbb{Z}/p\mathbb{Z}, \mathbb{Q}/\mathbb{Z}) \cong \mathbb{Z}/p\mathbb{Z}$ with trivial action (the contragredient of the trivial action is trivial). If $p|n$: both $\hat{H}^0(G, \mathbb{Z}/p\mathbb{Z}) = (\mathbb{Z}/p\mathbb{Z})/n(\mathbb{Z}/p\mathbb{Z}) = \mathbb{Z}/p\mathbb{Z}$ (since $n \equiv 0$ in $\mathbb{Z}/p\mathbb{Z}$, the norm image is zero) and $\hat{H}^{-1}(G, \mathbb{Z}/p\mathbb{Z}) = \ker(n : \mathbb{Z}/p\mathbb{Z} \to \mathbb{Z}/p\mathbb{Z}) = \mathbb{Z}/p\mathbb{Z}$. The cup product pairing $\hat{H}^0 \times \hat{H}^{-1} \to \hat{H}^{-1}(G, \mathbb{Q}/\mathbb{Z})$ factors through the evaluation $\mathbb{Z}/p\mathbb{Z} \otimes \mathbb{Z}/p\mathbb{Z} \to \mathbb{Q}/\mathbb{Z}$ (sending $a \otimes b \mapsto ab/p$). The pairing sends the generators $(\bar{1}, \bar{1})$ to $1/p \in \mathbb{Q}/\mathbb{Z}$, which is nonzero. Hence the pairing is a nondegenerate bilinear form on $\mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$, which is perfect.

If $p \nmid n$: then n is invertible in $\mathbb{Z}/p\mathbb{Z}$, so $\hat{H}^0(G, \mathbb{Z}/p\mathbb{Z}) = (\mathbb{Z}/p\mathbb{Z})/n(\mathbb{Z}/p\mathbb{Z}) = 0$ (since multiplication by n is surjective) and $\hat{H}^{-1}(G, \mathbb{Z}/p\mathbb{Z}) = \ker(n : \mathbb{Z}/p\mathbb{Z} \to \mathbb{Z}/p\mathbb{Z}) = 0$ (since multiplication by n is injective). Both sides vanish, and the pairing is trivially perfect.

Step 4: The case M simple with nontrivial action. If M is a simple $\mathbb{Z}/p\mathbb{Z}[G]$-module with nontrivial G-action (for G cyclic), then $M^G = 0$ and $M_G = 0$, giving $\hat{H}^0(G, M) = 0$ and $\hat{H}^{-1}(G, M) = 0$ (the norm and augmentation ideal act with trivial kernel and image on simple modules with nontrivial action). By the Herbrand quotient $h(G, M) = 1$ (proposition 4.1.11), both sides of the pairing vanish, and the pairing is trivially perfect.

The restriction to cyclic groups is essential for this proof (periodicity reduces the problem to finitely many degrees). For non-cyclic finite groups, the pairing remains well defined but need not be perfect, and the correct generalization requires additional structure on the module M. The local and global duality theorems of chapters 8 and 9 provide the arithmetic analogues, where the role of the cyclic group is played by the decomposition group of a prime and the role of M^* is played by an arithmetically defined dual (the Cartier dual for finite flat group schemes, or the Pontryagin dual with a Tate twist for Galois modules).

An immediate consequence of Tate duality is that the duality pairing provides a second proof of the Herbrand quotient identity $h(G, M) = 1$ for finite modules (see exercise 4.4.1 (3)): the perfect pairing forces $|\hat{H}^0(G, M)| = |\hat{H}^{-1}(G, M^*)|$, and since $M^{**} \cong M$, the identity follows.

The duality pairing interacts with the Nakayama lemma for group rings to produce a criterion for cohomological triviality in terms of module-theoretic projectivity. This connection is the algebraic foundation for the cohomological approach to local class field theory.

Theorem 4.4.4: Nakayama's Criterion

Let G be a p-group and M a finitely generated $\mathbb{Z}_p[G]$-module that is $\mathbb{Z}_p$-free. The following are equivalent:

1. M is a projective $\mathbb{Z}_p[G]$-module.
2. M is cohomologically trivial: $\hat{H}^n(H, M) = 0$ for all $n \in \mathbb{Z}$ and all subgroups $H \leq G$.
3. $\hat{H}^{-1}(G, M) = \hat{H}^0(G, M) = 0$.

***Proof*:**
The implication (1) $\Rightarrow$ (2) is immediate: projective $\mathbb{Z}_p[G]$-modules are direct summands of free modules $\mathbb{Z}_p[G]^r$, and $\hat{H}^n(H, \mathbb{Z}_p[G]) = 0$ for all n and H (since $\mathbb{Z}_p[G]$ is induced from the trivial subgroup). The implication (2) $\Rightarrow$ (3) is trivial.

(3) $\Rightarrow$ (1): The key tools are Tate's theorem (theorem 4.3.5) and the Nakayama lemma for local rings.

Since G is a p-group, every subgroup is also a p-group. The condition $\hat{H}^0(G, M) = 0$ means $M^G = N_G(M)$ (the norm is surjective onto invariants). The condition $\hat{H}^{-1}(G, M) = 0$ means $\ker(N_G) = I_G M$ (every element in the norm kernel is a sum of $(g-1) \cdot m$ terms).

By Tate's theorem (applied with the cyclic subgroup criterion), the vanishing of $\hat{H}^0$ and $\hat{H}^{-1}$ for all cyclic subgroups of G implies cohomological triviality. For a p-group, the condition for all cyclic subgroups follows from the condition for G itself: if $H \leq G$ is cyclic and $\hat{H}^0(G, M) = \hat{H}^{-1}(G, M) = 0$, then by the restriction–corestriction formula (proposition 3.3.6), $[G : H] \cdot \hat{H}^n(G, M) = \mathrm{Cor}(\mathrm{Res}(\hat{H}^n(G, M)))$, and since $[G : H]$ is a power of p and $\hat{H}^n(H, M)$ is p-torsion, the vanishing for G implies vanishing for H (by a descending induction on $|H|$, using the transfer and the p-group structure).

Once cohomological triviality is established, the module M/pM is a cohomologically trivial $\mathbb{Z}/p\mathbb{Z}[G]$-module. Since M is $\mathbb{Z}_p$-free, M/pM is a finite-dimensional $\mathbb{Z}/p\mathbb{Z}$-vector space, and cohomological triviality of a finite module over a p-group ring $\mathbb{Z}/p\mathbb{Z}[G]$ implies freeness (by the Nakayama lemma: M/pM is free over $\mathbb{Z}/p\mathbb{Z}[G/\Phi(G)]$ where $\Phi(G)$ is the Frattini subgroup, and this lifts to M being projective over $\mathbb{Z}_p[G]$ by the local Nakayama lemma).

Nakayama's criterion is the bridge between the cohomological triviality studied in theorem 4.3.5 and the module-theoretic notion of projectivity. For a general finite group G (not necessarily a p-group), the criterion generalizes as follows: a finitely generated $\mathbb{Z}[G]$-module M that is $\mathbb{Z}$-free is projective if and only if it is cohomologically trivial. The proof reduces to p-Sylow subgroups via the restriction–corestriction formula: M is projective over $\mathbb{Z}[G]$ if and only if M is projective over $\mathbb{Z}_p[G_p]$ for every prime p and every p-Sylow subgroup $G_p \leq G$, and the criterion above applies to each $\mathbb{Z}_p[G_p]$.

This algebraic criterion has a concrete computational payoff: rather than constructing a projective resolution of M (which can be difficult), one checks two cohomological vanishing conditions at each

Sylow subgroup. In the applications of chapter 8, the module in question is the unit group of a local field extension, and the Tate cohomology is computed using the filtration by higher unit groups — a computation that is tractable because the graded pieces are modules for the residue field.

Example 4.4.5: The Duality Pairing in Practice

1. Let $G = C_n$ and $M = \mathbb{Z}/n\mathbb{Z}$ with trivial action. The Pontryagin dual $M^* = \mathrm{Hom}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Q}/\mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}$ (also with trivial action, since the contragredient of the trivial action is trivial). The Tate cohomology is $\hat{H}^q(C_n, \mathbb{Z}/n\mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}$ for all q (by example 4.3.6(2)). The duality pairing in degree $p = 0$:
$$\langle -, - \rangle : \hat{H}^0(C_n, \mathbb{Z}/n\mathbb{Z}) \times \hat{H}^{-1}(C_n, \mathbb{Z}/n\mathbb{Z}) \to \mathbb{Q}/\mathbb{Z}.$$
Here $\hat{H}^0(C_n, \mathbb{Z}/n\mathbb{Z}) = (\mathbb{Z}/n\mathbb{Z})/n(\mathbb{Z}/n\mathbb{Z}) = \mathbb{Z}/n\mathbb{Z}$ (since $n \cdot \bar{a} = 0$ for all $\bar{a}$, the norm image is zero) and $\hat{H}^{-1}(C_n, \mathbb{Z}/n\mathbb{Z}) = \ker(n : \mathbb{Z}/n\mathbb{Z} \to \mathbb{Z}/n\mathbb{Z})/0 = \mathbb{Z}/n\mathbb{Z}$. The cup product sends $(\bar{a}, \bar{b}) \in \mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$ to $\bar{a} \cdot \bar{b} \in \hat{H}^{-1}(C_n, \mathbb{Q}/\mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}$, composed with the inclusion $\mathbb{Z}/n\mathbb{Z} \hookrightarrow \mathbb{Q}/\mathbb{Z}$ sending $\bar{1} \mapsto 1/n$. Hence $\langle \bar{a}, \bar{b} \rangle = ab/n \in \mathbb{Q}/\mathbb{Z}$. This is the standard perfect pairing $\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z} \to \mathbb{Q}/\mathbb{Z}$.

2. Let $G = C_p$ (p odd prime) and $M = (\mathbb{Z}/p\mathbb{Z})^2$ with the action of a generator σ given by the matrix $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ (a unipotent action). The invariants are $M^G = \ker(A - I) = \ker \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \langle e_1 \rangle \cong \mathbb{Z}/p\mathbb{Z}$ (where e_1, e_2 is the standard basis). The norm $N_G = I + A + \cdots + A^{p-1}$. Since $A = I + N$ with $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $N^2 = 0$, $A^k = I + kN$, so $N_G = pI + \frac{p(p-1)}{2} N = \frac{p(p-1)}{2} N$ (since $pI = 0$ in $(\mathbb{Z}/p\mathbb{Z})^2$). For p odd, $\frac{p-1}{2}$ is an integer and $\frac{p(p-1)}{2} \equiv 0 \pmod{p}$, so $N_G = 0$. Hence $\hat{H}^0(G, M) = M^G/N_G(M) = \mathbb{Z}/p\mathbb{Z}/0 = \mathbb{Z}/p\mathbb{Z}$.
The augmentation ideal acts by $(\sigma - 1) \cdot (a, b) = N(a, b) = (b, 0)$, so $I_G M = \langle e_1 \rangle$. Since $\ker(N_G) = M$ (as $N_G = 0$), $\hat{H}^{-1}(G, M) = M/I_G M = (\mathbb{Z}/p\mathbb{Z})^2/\langle e_1 \rangle \cong \mathbb{Z}/p\mathbb{Z}$.
The Pontryagin dual $M^* \cong (\mathbb{Z}/p\mathbb{Z})^2$ with the contragredient action $\sigma \cdot f = f \circ \sigma^{-1}$. Since $A^{-1} = I - N = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$, the matrix of σ on M^* (in the dual basis) is $A^{-T} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$.
The invariants $(M^*)^G = \ker(A^{-T} - I) = \ker \begin{pmatrix} 0 & 0 \\ -1 & 0 \end{pmatrix} = \langle e_2^* \rangle \cong \mathbb{Z}/p\mathbb{Z}$, and by the same computation $\hat{H}^{-1}(G, M^*) \cong \mathbb{Z}/p\mathbb{Z}$. The duality pairing $\hat{H}^0(G, M) \times \hat{H}^{-1}(G, M^*) \to \mathbb{Q}/\mathbb{Z}$ is a pairing $\mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z} \to \mathbb{Q}/\mathbb{Z}$, which is perfect by theorem 4.4.3.

The duality pairing is functorial in the group: for $H \leq G$ a subgroup, the restriction map Res^H_G on the first factor and the corestriction map Cor^G_H on the second factor are adjoint with respect to the pairing. That is, $\langle \mathrm{Res}(\alpha), \beta \rangle_H = \langle \alpha, \mathrm{Cor}(\beta) \rangle_G$ for $\alpha \in \hat{H}^p(G, M)$ and $\beta \in \hat{H}^{-p-1}(H, M^*)$. This adjunction is a consequence of the projection formula (proposition 4.2.5(2)) and the compatibility of the invariant map with restriction. The functoriality ensures that the duality pairings for different subgroups are compatible, which is essential for the inductive arguments in Part III.

Exercise 4.4.1

1. Let $G = C_4 = \langle \sigma \rangle$ and $M = \mathbb{Z}/4\mathbb{Z}$ with the action $\sigma \cdot \bar{a} = -\bar{a}$. Compute $\hat{H}^0(G, M)$, $\hat{H}^{-1}(G, M)$, and the Pontryagin dual M^* with its G-action. Verify that $|\hat{H}^0(G, M)| = |\hat{H}^{-1}(G, M^*)|$ and $|\hat{H}^{-1}(G, M)| = |\hat{H}^0(G, M^*)|$, as predicted by theorem 4.4.3.

2. Let $G = C_p$ and $M = \mathbb{Z}_p^2$ with the action $\sigma \cdot (a, b) = (a + pb, b)$ (a $\mathbb{Z}_p$-linear action). Verify Nakayama's criterion (theorem 4.4.4) by:

   (a) Computing $\hat{H}^0(G, M)$ and $\hat{H}^{-1}(G, M)$.

   (b) Determining whether M is a projective $\mathbb{Z}_p[C_p]$-module.

   (*Hint:* The norm N_G and augmentation ideal I_G can be computed using the matrix $A = I + pN$ where $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$.)

3. Use theorem 4.4.3 to give a second proof that $h(G, M) = 1$ for G cyclic and M finite (proposition 4.1.11). (*Hint:* The perfect pairing gives $|\hat{H}^0(G, M)| = |\hat{H}^{-1}(G, M^*)|$ and $|\hat{H}^{-1}(G, M)| = |\hat{H}^0(G, M^*)|$. Use $|M| = |M^*|$ and the fact that $h(G, M)$ depends only on $|M|$.)

4. Let $G = C_n$ and M a finite G-module. Show that the duality pairing in degree p is related to the pairing in degree $p + 2$ by the periodicity generator: $\langle \alpha \smile u_G, \beta \rangle = \langle \alpha, u_G^{-1} \smile \beta \rangle$, where u_G^{-1} denotes the inverse of u_G under the periodicity isomorphism. (This is a formal consequence of the associativity of the cup product.)

5. A G-module M is *self-dual* if $M \cong M^*$ as G-modules. Let $G = C_2$ and $M = \mathbb{Z}/2\mathbb{Z}$ with trivial action. Show that M is self-dual and that the duality pairing $\hat{H}^0(G, M) \times \hat{H}^{-1}(G, M) \to \mathbb{Q}/\mathbb{Z}$ becomes a *symmetric* pairing $\hat{H}^0 \times \hat{H}^{-1} \to \mathbb{Q}/\mathbb{Z}$. Compute this pairing explicitly.

5

Cohomological Dimension

The Tate cohomology and cup product machinery of chapter 4 provides the tools; the question now is how to measure the *complexity* of a profinite group's cohomology. For many groups of arithmetic interest — absolute Galois groups of fields, decomposition groups at primes, pro-p-completions of fundamental groups — the cohomology groups $H^n(G, M)$ vanish beyond a certain degree n, at least when the coefficients are restricted to p-torsion modules for a fixed prime p. This vanishing threshold is the *p-cohomological dimension* of G, and it is the single most important numerical invariant of a profinite group from the cohomological perspective. A group of p-cohomological dimension 1 has free pro-p-quotients; a group of dimension 2 admits Poincaré-type duality; and the passage from dimension 2 to higher dimensions marks the boundary of what the methods of class field theory can reach.

This chapter develops cohomological dimension as an abstract invariant of profinite groups (§1), computes it for the absolute Galois groups of familiar fields (§2), and introduces the duality groups that satisfy a cohomological analogue of Poincaré duality (§3).

5.1 p-Cohomological Dimension

Let G be a profinite group and p a prime. The colimit formula of theorem 3.2.2 shows that $H^n(G, M)$ is a torsion group for $n \geq 1$, so it decomposes into p-primary components. The p-cohomological dimension measures how far in degree the p-primary part of the cohomology can reach.

Definition 5.1.1: p-Cohomological Dimension

Let G be a profinite group and p a prime. The *p-cohomological dimension* of G is

$$\mathrm{cd}_p(G) = \sup\{n \geq 0 \mid H^n(G, M) \neq 0 \text{ for some } p\text{-torsion discrete } G\text{-module } M\},$$

with the convention $\sup \emptyset = 0$. The *(strict) cohomological dimension* is $\mathrm{cd}(G) = \sup_p \mathrm{cd}_p(G)$.

A module M is *p-torsion* if every element of M is annihilated by a power of p (equivalently, $M = \bigcup_k M[p^k]$). The restriction to p-torsion coefficients is essential: without it, the group $H^2(\hat{\mathbb{Z}}, \mathbb{Z}) \cong \mathbb{Q}/\mathbb{Z}$ (computed in example 3.2.6) would force $\mathrm{cd}(\hat{\mathbb{Z}}) \geq 2$, even though $\hat{\mathbb{Z}}$ has cohomological dimension 1 when measured with p-torsion coefficients at each prime p.

The first major result reduces the computation of $\mathrm{cd}_p(G)$ from checking all p-torsion modules to checking a single module.

Theorem 5.1.2: Serre's Criterion

Let G be a profinite group, p a prime, and $n \geq 0$. The following are equivalent:

1. $\mathrm{cd}_p(G) \leq n$.
2. $H^{n+1}(G, M) = 0$ for every p-torsion discrete G-module M.
3. $H^{n+1}(G, \mathbb{Z}/p\mathbb{Z}) = 0$, where $\mathbb{Z}/p\mathbb{Z}$ carries the trivial G-action.

***Proof*:**
The implication (1) $\Rightarrow$ (2) is immediate from the definition, and (2) $\Rightarrow$ (3) is a special case. The content is (3) $\Rightarrow$ (1).

Step 1: $H^{n+1}(G, M) = 0$ for all simple p-torsion modules. A simple p-torsion discrete G-module M is a simple $\mathbb{Z}/p\mathbb{Z}[G/U]$-module for some open normal subgroup U (since M is discrete and killed by p). The module $\mathbb{Z}/p\mathbb{Z}$ with trivial action is one such simple module, and the others are nontrivial simple $\mathbb{Z}/p\mathbb{Z}[G/U]$-modules. For any simple $\mathbb{Z}/p\mathbb{Z}[G/U]$-module M, there exists a surjection $\mathbb{Z}/p\mathbb{Z}[G/U]^r \twoheadrightarrow M$ for some r, and therefore M appears as a subquotient of a module built from $\mathbb{Z}/p\mathbb{Z}$ by extensions and restrictions to subgroups. Using dimension shifting: embed $\mathbb{Z}/p\mathbb{Z}$ into the coinduced module $I = \mathrm{CoInd}_{\{1\}}^G(\mathbb{Z}/p\mathbb{Z})$ (which is acyclic, by corollary 3.5.6), giving the short exact sequence $0 \to \mathbb{Z}/p\mathbb{Z} \to I \to Q \to 0$. The connecting homomorphism $\delta : H^n(G, Q) \xrightarrow{\sim} H^{n+1}(G, \mathbb{Z}/p\mathbb{Z})$ is an isomorphism for $n \geq 1$, so $H^{n+1}(G, \mathbb{Z}/p\mathbb{Z}) = 0$ implies $H^n(G, Q) = 0$. By induction on the dimension shift, $H^{n+1}(G, M) = 0$ for every simple module M that can be reached by finitely many such shifts.

Step 2: Extension to all p-torsion modules. A general p-torsion module M is a colimit of its finitely generated submodules, and by corollary 3.2.5, $H^{n+1}(G, M) = \varinjlim H^{n+1}(G, M_i)$. Each finitely generated p-torsion module M_i has a finite filtration by simple p-torsion modules, and the long exact sequence shows $H^{n+1}(G, M_i) = 0$ by induction on the length of the filtration. Hence $H^{n+1}(G, M) = 0$.

Step 3: From degree $n+1$ to all degrees $\geq n+1$. Having established $H^{n+1}(G, M) = 0$ for all p-torsion M, apply this to the module Q from Step 1 (which is p-torsion since $\mathbb{Z}/p\mathbb{Z}$ and I are). Then $H^{n+1}(G, Q) = 0$, and the dimension-shifting isomorphism $H^{n+2}(G, \mathbb{Z}/p\mathbb{Z}) \cong H^{n+1}(G, Q) = 0$ gives $H^{n+2}(G, \mathbb{Z}/p\mathbb{Z}) = 0$. Repeating, $H^m(G, \mathbb{Z}/p\mathbb{Z}) = 0$ for all $m \geq n+1$, and then Step 2 gives $H^m(G, M) = 0$ for all $m \geq n+1$ and all p-torsion M.

Serre's criterion is a powerful computational tool: to show $\mathrm{cd}_p(G) \leq n$, one need only verify that a single cohomology group — $H^{n+1}(G, \mathbb{Z}/p\mathbb{Z})$ with trivial action — vanishes. The next results establish the basic functorial properties of cd_p.

Proposition 5.1.3: Monotonicity

Let G be a profinite group and p a prime.

1. If H is a closed subgroup of G, then $\mathrm{cd}_p(H) \leq \mathrm{cd}_p(G)$.
2. If N is a closed normal subgroup of G, then $\mathrm{cd}_p(G/N) \leq \mathrm{cd}_p(G)$.
3. If N is a closed normal subgroup of G, then $\mathrm{cd}_p(G) \leq \mathrm{cd}_p(N) + \mathrm{cd}_p(G/N)$.

***Proof*:**
Part (1): If $\mathrm{cd}_p(G) \leq n$, then $H^{n+1}(G, \mathbb{Z}/p\mathbb{Z}) = 0$ by Serre's criterion. The corestriction–restriction formula $\mathrm{Cor} \circ \mathrm{Res} = [G : H]$ (proposition 3.3.6) applied to an open subgroup H' containing H shows that $\mathrm{Res} : H^{n+1}(G, \mathbb{Z}/p\mathbb{Z}) \to H^{n+1}(H', \mathbb{Z}/p\mathbb{Z})$ is injective on p-primary parts when $p \nmid [G : H']$. For a general closed subgroup H, write $H = \bigcap U_i$ as an intersection of open subgroups, so $H^{n+1}(H, \mathbb{Z}/p\mathbb{Z}) = \varinjlim H^{n+1}(U_i, \mathbb{Z}/p\mathbb{Z})$. Each $H^{n+1}(U_i, \mathbb{Z}/p\mathbb{Z}) = 0$ (by Serre's criterion for U_i, since $\mathrm{cd}_p(U_i) \leq \mathrm{cd}_p(G)$ for open subgroups by the same restriction argument). Hence $H^{n+1}(H, \mathbb{Z}/p\mathbb{Z}) = 0$, giving $\mathrm{cd}_p(H) \leq n$.

Part (2): If $\mathrm{cd}_p(G) \leq n$, then $H^{n+1}(G, \mathbb{Z}/p\mathbb{Z}) = 0$ by Serre's criterion. The inflation map $\mathrm{Inf} : H^{n+1}(G/N, \mathbb{Z}/p\mathbb{Z}) \to H^{n+1}(G, \mathbb{Z}/p\mathbb{Z})$ is injective (by the inflation–restriction exact sequence of proposition 3.3.3 for $n = 0$, and by the colimit formula for general n: every class in $H^{n+1}(G/N, \mathbb{Z}/p\mathbb{Z})$ inflates to a class in $H^{n+1}(G, \mathbb{Z}/p\mathbb{Z})$ through the surjection $G \twoheadrightarrow G/N$). Since the target vanishes, $H^{n+1}(G/N, \mathbb{Z}/p\mathbb{Z}) = 0$, giving $\mathrm{cd}_p(G/N) \leq n$.

Part (3): Set $m = \mathrm{cd}_p(N)$ and $n = \mathrm{cd}_p(G/N)$; if either is infinite there is nothing to prove. Let M be a p-torsion discrete G-module and consider the LHS spectral sequence (theorem 3.4.4) $E_2^{p,q} = H^p(G/N, H^q(N, M)) \Rightarrow H^{p+q}(G, M)$. The coefficient module $H^q(N, M)$ is a p-torsion discrete G/N-module (cochains valued in a p-torsion module are p-torsion, and the conjugation action of definition 3.4.1 is discrete), so $E_2^{p,q} = 0$ for $p > n$; and $E_2^{p,q} = 0$ for $q > m$, since already $H^q(N, M) = 0$. For total degree $k > m + n$, every term $E_2^{p,q}$ on the diagonal $p + q = k$ has $p > n$ or $q > m$, so the whole diagonal vanishes. Each $E_\infty^{p,q}$ is a subquotient of $E_2^{p,q}$, so the filtration of $H^k(G, M)$ has all graded pieces zero, and $H^k(G, M) = 0$. Hence $\mathrm{cd}_p(G) \leq m + n$.

Proposition 5.1.4: Characterization of $\mathrm{cd}_p(G) = 0$

Let G be a profinite group. Then $\mathrm{cd}_p(G) = 0$ if and only if G has no elements of order p, i.e., G is a pro-p'-group (the pro-p-Sylow subgroup of G is trivial).

***Proof*:**
If G has no element of order p, then the pro-p-Sylow subgroup is trivial, and restriction to a p-Sylow subgroup (combined with $\mathrm{Cor} \circ \mathrm{Res} = [G : P]$ with $p \nmid [G : P]$) shows $H^n(G, M)\{p\} = 0$ for all $n \geq 1$ and all M. In particular $H^1(G, \mathbb{Z}/p\mathbb{Z}) = 0$, so $\mathrm{cd}_p(G) = 0$.

Conversely, if G has an element of order p, then G contains a closed subgroup $H \cong \mathbb{Z}/p\mathbb{Z}$. Since $H^1(\mathbb{Z}/p\mathbb{Z}, \mathbb{Z}/p\mathbb{Z}) \cong \mathbb{Z}/p\mathbb{Z} \neq 0$ (the group of homomorphisms), $\mathrm{cd}_p(H) \geq 1$, and by monotonicity $\mathrm{cd}_p(G) \geq 1$.

The next example computes cd_p for the procyclic groups that have appeared throughout Parts I and II.

Example 5.1.5: Cohomological Dimension of Procyclic Groups

1. The group $\hat{\mathbb{Z}} = \varprojlim_n \mathbb{Z}/n\mathbb{Z}$ satisfies $\mathrm{cd}_p(\hat{\mathbb{Z}}) = 1$ for every prime p, and therefore $\mathrm{cd}(\hat{\mathbb{Z}}) = 1$. This is immediate from theorem 3.2.3: the module $\mathbb{Z}/p\mathbb{Z}$ is torsion, so $H^i(\hat{\mathbb{Z}}, \mathbb{Z}/p\mathbb{Z}) = 0$ for $i \geq 2$, and Serre's criterion needs no more. The lower bound is $H^1(\hat{\mathbb{Z}}, \mathbb{Z}/p\mathbb{Z}) = (\mathbb{Z}/p\mathbb{Z})_{\hat{\mathbb{Z}}} = \mathbb{Z}/p\mathbb{Z} \neq 0$, the action being trivial.

   The vanishing of $H^2(\hat{\mathbb{Z}}, \mathbb{Z})$ at p-torsion coefficients contrasts with $H^2(\hat{\mathbb{Z}}, \mathbb{Z}) \cong \mathbb{Q}/\mathbb{Z} \neq 0$. The module $\mathbb{Z}$ is not p-torsion, so this nonvanishing does not affect cd_p.

2. The pro-p-group $\mathbb{Z}_p$ satisfies $\mathrm{cd}_p(\mathbb{Z}_p) = 1$ by the same argument (the open subgroups are $p^k\mathbb{Z}_p$ with quotient $\mathbb{Z}/p^k\mathbb{Z}$, and the colimit of $H^2(\mathbb{Z}/p^k\mathbb{Z}, \mathbb{Z}/p\mathbb{Z}) \cong \mathbb{Z}/p\mathbb{Z}$ stabilizes to zero). For $\ell \neq p$, $\mathrm{cd}_\ell(\mathbb{Z}_p) = 0$ since $\mathbb{Z}_p$ is a pro-p-group with no elements of order ℓ.

3. More generally, $\mathrm{cd}_p(\mathbb{Z}_p^d) = d$ for any $d \geq 1$. This follows from the Künneth formula: $H^n(\mathbb{Z}_p^d, \mathbb{Z}/p\mathbb{Z}) \cong \bigoplus_{n_1+\cdots+n_d=n} \bigotimes_{i=1}^d H^{n_i}(\mathbb{Z}_p, \mathbb{Z}/p\mathbb{Z})$. Since $H^0(\mathbb{Z}_p, \mathbb{Z}/p\mathbb{Z}) = \mathbb{Z}/p\mathbb{Z}$ and $H^1(\mathbb{Z}_p, \mathbb{Z}/p\mathbb{Z}) = \mathbb{Z}/p\mathbb{Z}$ (and $H^n = 0$ for $n \geq 2$), the nonzero contributions have each $n_i \in \{0, 1\}$, so $H^n(\mathbb{Z}_p^d, \mathbb{Z}/p\mathbb{Z}) \neq 0$ exactly when $0 \leq n \leq d$. The top nonvanishing degree is $n = d$ (with $H^d(\mathbb{Z}_p^d, \mathbb{Z}/p\mathbb{Z}) \cong \mathbb{Z}/p\mathbb{Z}$, the "top class"), giving $\mathrm{cd}_p(\mathbb{Z}_p^d) = d$.

4. (If the reader has read [16]) The equality $\mathrm{cd}_p(\mathbb{Z}_p^d) = d$ is the profinite analogue of the fact that $H^n(T^d; \mathbb{Z}/p\mathbb{Z}) \neq 0$ for $0 \leq n \leq d$ and $= 0$ for $n > d$, where $T^d = (\mathbb{R}/\mathbb{Z})^d$ is the d-torus. The classifying space $B\mathbb{Z}_p^d$ is the p-completion of T^d, and profinite cohomological dimension equals the topological cohomological dimension of the classifying space.

The characterization of $\mathrm{cd}_p = 1$ is especially important because of its connection to free pro-p-groups.

Proposition 5.1.6: Free Pro-p-Groups Have $\mathrm{cd}_p = 1$

Let F be a free pro-p-group of rank ≥ 1. Then $\mathrm{cd}_p(F) = 1$.

***Proof*:**
A free pro-p-group F on a set S of generators satisfies $H^1(F, \mathbb{Z}/p\mathbb{Z}) \cong (\mathbb{Z}/p\mathbb{Z})^{|S|}$ (continuous homomorphisms $F \to \mathbb{Z}/p\mathbb{Z}$ correspond to functions $S \to \mathbb{Z}/p\mathbb{Z}$), so $\mathrm{cd}_p(F) \geq 1$ when $|S| \geq 1$. For the upper bound, the key fact is that free pro-p-groups have a "tree-like" structure that forces $H^2(F, M) = 0$ for all p-torsion M. This follows from the characterization of free pro-p-groups as those pro-p-groups for which every embedding problem with p-group kernel has a solution: any extension $0 \to M \to E \to F \to 1$ with M a p-torsion F-module splits (because F maps onto any finite p-quotient of E), so the obstruction class in $H^2(F, M)$ vanishes. By Serre's criterion, $\mathrm{cd}_p(F) \leq 1$.

The converse — that a pro-p-group with $\mathrm{cd}_p = 1$ is free — is also true and will be proved in chapter 6. Together, the two directions give a cohomological characterization of free pro-p-groups: a pro-p-group G with $G \neq 1$ is free if and only if $\mathrm{cd}_p(G) = 1$.

The following proposition shows how cd_p behaves under products, complementing the spectral sequence bound of proposition 5.1.3(3).

Proposition 5.1.7: Cohomological Dimension of Products

Let G_1 and G_2 be profinite groups. Then $\mathrm{cd}_p(G_1 \times G_2) = \mathrm{cd}_p(G_1) + \mathrm{cd}_p(G_2)$.

***Proof*:**
The inequality $\mathrm{cd}_p(G_1 \times G_2) \leq \mathrm{cd}_p(G_1) + \mathrm{cd}_p(G_2)$ follows from the LHS spectral sequence applied to the normal subgroup $N = G_2$ of $G = G_1 \times G_2$ with quotient $G/N = G_1$ (proposition 5.1.3(3)).

For the reverse inequality, the Künneth formula in group cohomology gives a natural map

$$\bigoplus_{i+j=n} H^i(G_1, \mathbb{Z}/p\mathbb{Z}) \otimes H^j(G_2, \mathbb{Z}/p\mathbb{Z}) \to H^n(G_1 \times G_2, \mathbb{Z}/p\mathbb{Z}).$$

When G_1 and G_2 are pro-p-groups, this map is an isomorphism (since the cohomology groups are $\mathbb{Z}/p\mathbb{Z}$-vector spaces and the Künneth formula holds over a field with no Tor correction term). In particular, $H^{\mathrm{cd}_p(G_1)}(G_1, \mathbb{Z}/p\mathbb{Z}) \neq 0$ and $H^{\mathrm{cd}_p(G_2)}(G_2, \mathbb{Z}/p\mathbb{Z}) \neq 0$, so $H^{\mathrm{cd}_p(G_1)+\mathrm{cd}_p(G_2)}(G_1 \times G_2, \mathbb{Z}/p\mathbb{Z}) \neq 0$.

For general profinite groups, the argument reduces to pro-p-Sylow subgroups via restriction–corestriction: $\mathrm{cd}_p(G) = \mathrm{cd}_p(G_p)$ where G_p is a pro-p-Sylow subgroup (proposition 5.1.3(1) gives $\leq$, and restriction to the Sylow gives $\geq$ since Res is injective on p-primary parts by $\mathrm{Cor} \circ \mathrm{Res} = [G : G_p]$ with $p \nmid [G : G_p]$). Then $\mathrm{cd}_p(G_1 \times G_2) = \mathrm{cd}_p((G_1)_p \times (G_2)_p) = \mathrm{cd}_p((G_1)_p) + \mathrm{cd}_p((G_2)_p) = \mathrm{cd}_p(G_1) + \mathrm{cd}_p(G_2)$.

Exercise 5.1.1

1. Let G be a profinite group with $\mathrm{cd}_p(G) \leq n$. Use Serre's criterion and the long exact sequence for $0 \to \mathbb{Z}/p\mathbb{Z} \to \mathbb{Z}/p^2\mathbb{Z} \to \mathbb{Z}/p\mathbb{Z} \to 0$ to show that $H^{n+1}(G, \mathbb{Z}/p^k\mathbb{Z}) = 0$ for all $k \geq 1$.
2. Let $G = N \rtimes Q$ be a semidirect product of profinite groups. Show that $\mathrm{cd}_p(G) \leq \mathrm{cd}_p(N) + \mathrm{cd}_p(Q)$. Give an example showing this inequality can be strict. (*Hint:* For strictness, consider $G = S_3 = C_3 \rtimes C_2$ and $p = 3$.)
3. Let G be a profinite group and H an open subgroup. Show that $\mathrm{cd}_p(H) = \mathrm{cd}_p(G)$ (equality, not just inequality). (*Hint:* Restriction is injective on p-primary parts when $p \nmid [G : H]$, and the colimit formula reduces the general case to this one.)
4. Let G be a profinite group and G_p a pro-p-Sylow subgroup. Prove that $\mathrm{cd}_p(G) = \mathrm{cd}_p(G_p)$.
5. Show that $\mathrm{cd}_p(G) = \sup_U \mathrm{cd}_p(G/U)$, where U ranges over open normal subgroups of G. (*Hint:* Use the colimit formula for cohomology and the fact that $H^n(G, \mathbb{Z}/p\mathbb{Z}) = \varinjlim_U H^n(G/U, \mathbb{Z}/p\mathbb{Z})$. One direction is immediate; for the other, if $H^n(G/U, \mathbb{Z}/p\mathbb{Z}) \neq 0$, show that $H^n(G, \mathbb{Z}/p\mathbb{Z}) \neq 0$ using inflation.)

5.2 Cohomological Dimension of Fields

The p-cohomological dimension of a profinite group becomes an arithmetic invariant when applied to the absolute Galois group of a field. For a field K with separable closure K^{sep} and absolute Galois group $G_K = \mathrm{Gal}(K^{\mathrm{sep}}/K)$, the number $\mathrm{cd}_p(K) := \mathrm{cd}_p(G_K)$ encodes fundamental properties of the arithmetic and geometry of K. Separably closed fields have cd $= 0$; finite fields and function fields of curves over algebraically closed fields have cd $= 1$; local fields have cd $= 2$. This section computes $\mathrm{cd}_p(K)$ for the fields that have appeared as running examples throughout the book, and introduces the C_r-property as a geometric condition that controls cohomological dimension.

Definition 5.2.1: Cohomological Dimension of a Field

Let K be a field. The *p-cohomological dimension of K* is $\mathrm{cd}_p(K) = \mathrm{cd}_p(G_K)$, and the *cohomological dimension of K* is $\mathrm{cd}(K) = \mathrm{cd}(G_K)$.

The definition depends only on the separable closure K^{sep} (or equivalently, on the isomorphism class of G_K as a profinite group), not on any embedding of K into an algebraically closed field. Since G_K is determined up to inner automorphism by the field K (by theorem 1.4.1), $\mathrm{cd}_p(K)$ is a well-defined invariant of the field.

Proposition 5.2.2: $\mathrm{cd}(K) = 0$ Characterizes Separably Closed Fields

A field K has $\mathrm{cd}(K) = 0$ if and only if K is separably closed.

***Proof*:**
The field K is separably closed if and only if $G_K = 1$ (there are no nontrivial separable extensions). The trivial group has $\mathrm{cd}_p = 0$ for all p, so $\mathrm{cd}(K) = 0$. Conversely, if K is not separably closed,

then G_K is nontrivial and contains an element of some prime order p (every nontrivial profinite group has an element of prime order, by theorem 1.2.7). By proposition 5.1.4, $\mathrm{cd}_p(G_K) \geq 1$, so $\mathrm{cd}(K) \geq 1$.

The next case, $\mathrm{cd}(K) \leq 1$, has a cohomological characterization involving the Brauer group, which will be constructed in chapter 7. The following proposition uses the identification $H^2(G_K, (K^{\mathrm{sep}})^\times) = \mathrm{Br}(K)$ (the Brauer group of K) as a black box; the identification itself is the content of chapter 7, §2.

Proposition 5.2.3: Characterization of $\mathrm{cd}_p(K) \leq 1$

Let K be a field and p a prime. The following are equivalent:

1. $\mathrm{cd}_p(K) \leq 1$.
2. $H^2(G_L, (L^{\mathrm{sep}})^\times)\{p\} = 0$ for every finite separable extension L/K.
3. $\mathrm{Br}(L)\{p\} = 0$ for every finite separable extension L/K.

***Proof*:**
The equivalence (2) $\Leftrightarrow$ (3) is the identification $H^2(G_L, (L^{\mathrm{sep}})^\times) = \mathrm{Br}(L)$ from chapter 7.

(1) $\Rightarrow$ (2): If $\mathrm{cd}_p(K) \leq 1$, then $\mathrm{cd}_p(L) \leq 1$ for every finite separable L/K (by proposition 5.1.3(1): G_L is an open, hence closed, subgroup of G_K, and only the inequality is needed). By Serre's criterion (theorem 5.1.2), $H^2(G_L, M) = 0$ for every p-torsion module M. The group $(L^{\mathrm{sep}})^\times$ is not p-torsion, but its p-primary cohomology $H^2(G_L, (L^{\mathrm{sep}})^\times)\{p\}$ is computed by the long exact sequence from $0 \to \mu_{p^k} \to (L^{\mathrm{sep}})^\times \xrightarrow{(\cdot)^{p^k}} (L^{\mathrm{sep}})^\times$ (the Kummer sequence, when L contains the p^k-th roots of unity), and the vanishing of H^2 on p-torsion modules forces $H^2(G_L, (L^{\mathrm{sep}})^\times)\{p\} = 0$.

(2) $\Rightarrow$ (1): Every p-torsion discrete G_K-module M is a colimit of finite p-torsion modules, and each finite p-torsion G_K-module is a successive extension of simple modules, each of which is a $\mathbb{Z}/p\mathbb{Z}$-vector space with G_K acting through a finite quotient $\mathrm{Gal}(L/K)$. The vanishing $H^2(G_L, (L^{\mathrm{sep}})^\times)\{p\} = 0$ for all L, combined with the connection between $\mathrm{Br}(L)\{p\}$ and $H^2(G_L, \mathbb{Z}/p\mathbb{Z})$ (via the Kummer sequence and the p-torsion of $(L^{\mathrm{sep}})^\times$), implies $H^2(G_K, \mathbb{Z}/p\mathbb{Z}) = 0$, and Serre's criterion gives $\mathrm{cd}_p(K) \leq 1$.

Example 5.2.4: Fields of Cohomological Dimension ≤ 1
The condition $\mathrm{cd}_p(K) \leq 1$ says that the Galois group of K is "one-dimensional" from the perspective of p-torsion cohomology: all higher obstructions vanish. The equivalence with the vanishing of the Brauer group makes this concrete — there are no nontrivial central simple algebras of p-power degree over K or its extensions. The following fields satisfy this condition.

1. *Finite fields.* Let $K = \mathbb{F}_q$ be a finite field with q elements. The absolute Galois group is $G_{\mathbb{F}_q} \cong \hat{\mathbb{Z}}$ (example 1.4.4), and $\mathrm{cd}_p(\hat{\mathbb{Z}}) = 1$ for all primes p (example 5.1.5). Hence $\mathrm{cd}(\mathbb{F}_q) = 1$.

2. *Function fields of curves over algebraically closed fields.* Let k be algebraically closed and

$K = k(C)$ the function field of a smooth projective curve C over k. The absolute Galois group G_K is a free profinite group (this is a theorem of Douady and Harbater, building on work of Shafarevich). By proposition 5.1.6 (extended to free profinite groups from free pro-p-groups: a free profinite group has $\mathrm{cd}_p = 1$ for every prime p, since its pro-p-completion is a free pro-p-group), $\mathrm{cd}(K) \leq 1$. Since K is not separably closed (it has nontrivial extensions, e.g., the extension obtained by adjoining a root of an irreducible polynomial of degree ≥ 2), $\mathrm{cd}(K) = 1$. This can also be seen from the C_1-property: k is C_0 (algebraically closed), so $K = k(C)$ is C_1 by proposition 5.2.6(3), and theorem 5.2.7 gives $\mathrm{cd}(K) \leq 1$.

3. *p-adic fields and completions (preview).* Let $K = \mathbb{Q}_p$. The absolute Galois group $G_{\mathbb{Q}_p}$ has $\mathrm{cd}(\mathbb{Q}_p) = 2$. This will be proved in chapter 8 using the structure of the inertia group and the cohomology of local fields. The key point is that $H^2(G_{\mathbb{Q}_p}, \mathbb{Z}/p\mathbb{Z}) \neq 0$ (so $\mathrm{cd}_p \geq 2$) but $H^3(G_{\mathbb{Q}_p}, M) = 0$ for all p-torsion M (so $\mathrm{cd}_p \leq 2$).

4. *Number fields (preview).* Let K be a number field. For p odd, $\mathrm{cd}_p(K) = 2$ (under mild conditions; specifically, when K has no real embeddings, or for the "virtual" cohomological dimension). For $p = 2$, the situation is complicated by the presence of real places: $\mathrm{cd}_2(\mathbb{R}) = \infty$ (see below), which propagates to $\mathrm{cd}_2(K) = \infty$ when K has a real embedding. Full details are in chapter 9.

The fields of $\mathrm{cd} \leq 1$ include finite fields and function fields of curves over algebraically closed fields. These examples share a geometric property: every homogeneous polynomial equation of "low" degree in "many" variables has a nontrivial solution. This is the C_r-property, introduced by Lang and studied by Tsen.

Definition 5.2.5: C_r-Fields

A field K is a *C_r-field* (for $r \geq 0$) if every homogeneous polynomial $f \in K[x_1, \ldots, x_n]$ of degree d with $n > d^r$ has a nontrivial zero in K^n.

A C_0-field is algebraically closed (every nonconstant polynomial has a root, and the homogeneous condition extends this). The condition becomes weaker as r increases: C_r-fields allow polynomials to have nontrivial zeros provided the number of variables exceeds a degree-dependent threshold. The classical results are:

Proposition 5.2.6: Tsen–Lang Theorem

The following fields are C_r-fields:

1. Algebraically closed fields are C_0.
2. Finite fields are C_1 (Chevalley–Warning theorem).
3. If K is C_r, then $K(t)$ (the rational function field in one variable) is C_{r+1} (Tsen's theorem for $r = 0$, Lang's generalization).
4. If K is C_r, then any algebraic extension of K is also C_r.

***Proof Key ideas*:**
Part (2) follows from the Chevalley–Warning theorem ([4]): the number of solutions to $f(x_1, \ldots, x_n) = 0$ over $\mathbb{F}_q$ is divisible by $p = \operatorname{char}(\mathbb{F}_q)$ when $n > d$, and since the trivial solution exists, there must be a nontrivial one.

Part (3) is proved by a specialization argument: a homogeneous polynomial $f \in K(t)[x_1, \ldots, x_n]$ of degree d with $n > d^{r+1}$ can be cleared of denominators to give a polynomial over $K[t]$, and one constructs a solution by interpolation, using the C_r-property of K at enough specialization points.

Part (4) is immediate: a polynomial over an algebraic extension L/K with $n > d^r$ variables has a nontrivial zero in L^n if and only if a system of polynomial equations over K (obtained by choosing a basis for L/K) has a nontrivial solution, and the C_r bound is preserved.

The connection between the C_r-property and cohomological dimension is:

Theorem 5.2.7: C_1-Fields Have $\operatorname{cd} \leq 1$

If K is a C_1-field, then $\operatorname{cd}(K) \leq 1$.

***Proof Sketch*:**
By proposition 5.2.3, it suffices to show $\operatorname{Br}(L) = 0$ for every finite extension L/K. Since algebraic extensions of C_1-fields are C_1 (proposition 5.2.6(4)), it suffices to show $\operatorname{Br}(K) = 0$ for K a C_1-field. Every class in $\operatorname{Br}(K)$ is represented by a central simple algebra A of dimension d^2 over K, and the reduced norm $\operatorname{Nrd} : A \to K$ is a homogeneous polynomial of degree d in d^2 variables. Since K is C_1 and $d^2 > d^1$ for $d \geq 2$, the reduced norm has a nontrivial zero, which means A contains a nontrivial element of reduced norm zero. A theorem of Wedderburn then implies that A is not a division algebra, and by induction on d, every class in $\operatorname{Br}(K)$ is trivial (the identification of $\operatorname{Br}(K)$ with central simple algebras is the content of chapter 7).

The converse is false: there exist fields with $\operatorname{cd} \leq 1$ that are not C_1 (for example, $\mathbb{Q}^{\mathrm{ab}}$, the maximal abelian extension of $\mathbb{Q}$, satisfies $\operatorname{cd} \leq 1$ by a theorem of Artin–Schreier, but is not known to be C_1). The C_r-property is a *geometric* condition on the field — it constrains the solution set of polynomial equations — while cohomological dimension is a *cohomological* condition on the Galois group. The implication $C_1 \Rightarrow \mathrm{cd} \leq 1$ shows that geometry constrains cohomology, but the two invariants are not equivalent.

For $r \geq 2$, the analogous question is open: it is conjectured that C_r-fields satisfy $\mathrm{cd} \leq r$, but this is not known in general. The case $r = 2$ is connected to the period–index problem for the Brauer group, which will be discussed in chapter 7, §4.

The real numbers provide an instructive example of how orderings affect cohomological dimension.

Example 5.2.8: The Real Numbers and Orderings

The absolute Galois group $G_{\mathbb{R}} = \mathrm{Gal}(\mathbb{C}/\mathbb{R}) \cong \mathbb{Z}/2\mathbb{Z}$ is a finite group of order 2. For any finite group G and any prime p dividing $|G|$, the periodic cohomology of cyclic subgroups (theorem 4.3.4) gives $H^n(G, \mathbb{Z}/p\mathbb{Z}) \neq 0$ for infinitely many n, so $\mathrm{cd}_p(G) = \infty$. In particular:

1. $\mathrm{cd}_2(\mathbb{R}) = \infty$, since $H^n(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$ for all $n \geq 0$.
2. $\mathrm{cd}_p(\mathbb{R}) = 0$ for p odd, since $G_{\mathbb{R}} = \mathbb{Z}/2\mathbb{Z}$ has no elements of order p.

The infinite 2-cohomological dimension is a consequence of the ordering on $\mathbb{R}$: the nontrivial element of $G_{\mathbb{R}}$ is complex conjugation, which preserves the ordering, and the resulting periodicity in cohomology prevents the cohomology from vanishing in high degrees. This phenomenon extends to any formally real field K (one admitting an ordering): if K has a real closure K^{rc}, then $\mathrm{Gal}(K^{\mathrm{rc}}/K)$ contains $\mathbb{Z}/2\mathbb{Z}$, and $\mathrm{cd}_2(K) = \infty$.

For number fields: if K is a totally imaginary number field (no real embeddings), then the obstruction disappears, and $\mathrm{cd}_2(K) = 2$.

The dichotomy between $\mathrm{cd}_2 = \infty$ (formally real fields) and $\mathrm{cd}_2 < \infty$ (fields with no orderings) motivates the notion of *virtual cohomological dimension*.

Definition 5.2.9: Virtual Cohomological Dimension

Let G be a profinite group. The *virtual p-cohomological dimension* of G is

$$\mathrm{vcd}_p(G) = \inf \{\mathrm{cd}_p(H) \mid H \leq G \text{ open, torsion-free}\},$$

when a torsion-free open subgroup exists. For a field K, $\mathrm{vcd}_p(K) = \mathrm{vcd}_p(G_K)$.

When G has no torsion elements, $\mathrm{vcd}_p(G) = \mathrm{cd}_p(G)$. For fields: a number field K with real places has $\mathrm{cd}_2(K) = \infty$, but $\mathrm{vcd}_2(K) = 2$ (the open subgroup $G_{K(\sqrt{-1})}$ corresponds to the totally imaginary field $K(\sqrt{-1})$, which has $\mathrm{cd}_2 = 2$). The virtual cohomological dimension captures the "true" complexity of the Galois group, discounting the contribution of finite torsion subgroups.

The following table summarizes the cohomological dimensions computed or stated in this section, for reference throughout the rest of the book.

Field K	$\mathrm{cd}_p(K)$ (p odd)	$\mathrm{cd}_2(K)$
Separably closed	0	0
$\mathbb{F}_q$ (finite)	1	1
$k(C)$, $k = \overline{k}$	1	1
$\mathbb{Q}_p$, p odd	2	2
K number field, totally imaginary	2	2
K number field, real places	2	∞ ($\mathrm{vcd}_2 = 2$)
$\mathbb{R}$	0	∞

The entries for $\mathbb{Q}_p$ and number fields are proved in chapters 8 and 9; the others have been established above. The pattern $\mathrm{cd} = 0, 1, 2$ for separably closed, finite, and local fields reflects the increasing arithmetic complexity of these fields, and $\mathrm{cd} = 2$ is the threshold at which class field theory operates.

Exercise 5.2.1

1. Let L/K be an algebraic extension of fields. Show that $\mathrm{cd}_p(L) \leq \mathrm{cd}_p(K)$. When is equality guaranteed? (*Hint:* Consider separately the cases where $[L : K]$ is finite or infinite, and whether p divides $[L : K]$.)

2. Let k be a field with $\mathrm{cd}(k) = d < \infty$. Show that $\mathrm{cd}(k(t)) \leq d + 1$, where $k(t)$ is the rational function field. (*Hint:* Use the Faddeev exact sequence or the LHS spectral sequence for an appropriate extension.)

3. Let K be a formally real field (i.e., -1 is not a sum of squares in K). Show that $\mathrm{cd}_2(K) = \infty$. (*Hint:* K has a real closure K^{rc} with $\mathrm{Gal}(K^{\mathrm{rc}}/K) \cong \mathbb{Z}/2\mathbb{Z}$. Use the periodic cohomology of $\mathbb{Z}/2\mathbb{Z}$.)

4. (If the reader has read [4]) Let $K = \mathbb{Q}(\mu_{p^\infty})$ be the field obtained by adjoining all p-power roots of unity to $\mathbb{Q}$ (p an odd prime). Show that $\mathrm{cd}_p(K) \leq 1$ by identifying G_K as a pro-p-group and showing $H^2(G_K, \mathbb{Z}/p\mathbb{Z}) = 0$. (*Hint:* The extension $\mathbb{Q}(\mu_{p^\infty})/\mathbb{Q}$ has Galois group $\mathbb{Z}_p^\times$, and $G_K = G_{\mathbb{Q}}/\overline{\langle \text{inertia at } p \rangle}$ acts on $\mathbb{Z}/p\mathbb{Z}$ trivially. Use the Kummer sequence.)

5. Assuming the result $\mathrm{cd}(\mathbb{Q}_p) = 2$ (to be proved in chapter 8), compute $\mathrm{cd}_p(L)$ and $\mathrm{cd}_\ell(L)$ (for $\ell \neq p$) for a finite extension $L/\mathbb{Q}_p$. Use exercise 5.1.1 (3) and the structure of $G_{\mathbb{Q}_p}$.

5.3 Duality Groups and Poincaré Duality Groups

The finite-group Tate duality of theorem 4.4.3 paired $\hat{H}^p(G, M)$ with $\hat{H}^{-p-1}(G, M^*)$ for cyclic groups G acting on finite modules. For profinite groups of finite cohomological dimension, an analogous duality can hold: the cohomology in degree i is paired with the cohomology in the complementary degree $d - i$, where $d = \mathrm{cd}_p(G)$. Profinite groups admitting such a pairing are called *duality groups*, and those where the pairing takes the simplest possible form are *Poincaré duality groups*. These notions are the algebraic analogues of the topological Poincaré duality for closed manifolds, and they organize the profinite groups of arithmetic interest — absolute Galois groups of local fields, fundamental groups of arithmetic surfaces — into a structural hierarchy.

Definition 5.3.1: Duality Group

Let G be a profinite group, p a prime, and $d = \mathrm{cd}_p(G) < \infty$. The group G is a *duality group of dimension d at p* if there exists a discrete G-module D (the *dualizing module*) such that for every finite p-torsion discrete G-module M and every $0 \le i \le d$, the cup product pairing

$$H^i(G, M) \times H^{d-i}(G, \mathrm{Hom}(M, D)) \to H^d(G, D) \xrightarrow{\theta} \mathbb{Q}_p/\mathbb{Z}_p$$

is a perfect pairing of finite groups, where $\theta : H^d(G, D) \xrightarrow{\sim} \mathbb{Q}_p/\mathbb{Z}_p$ is a fixed isomorphism (the *trace map* or *fundamental class*).

The dualizing module D and the trace map θ are part of the structure: a duality group is a triple (G, D, θ). The pairing sends $(\alpha, \beta) \in H^i(G, M) \times H^{d-i}(G, \mathrm{Hom}(M, D))$ to $\theta(\alpha \smile \mathrm{ev}_*(\beta))$, where ev_* is induced by the evaluation $M \otimes \mathrm{Hom}(M, D) \to D$. Perfectness means that the induced maps $H^i(G, M) \to H^{d-i}(G, \mathrm{Hom}(M, D))^*$ and $H^{d-i}(G, \mathrm{Hom}(M, D)) \to H^i(G, M)^*$ are both isomorphisms (where $(-)^* = \mathrm{Hom}(-, \mathbb{Q}_p/\mathbb{Z}_p)$ is the Pontryagin dual).

The dualizing module is unique when it exists: if (D, θ) and (D', θ') both make G a duality group of the same dimension, then $D \cong D'$ as G-modules (the isomorphism is determined by $\theta' \circ \theta^{-1}$). This follows from the Yoneda lemma applied to the functor $M \mapsto H^d(G, \mathrm{Hom}(M, D))^*$, which is representable by the dualizing module.

The duality pairing has a transparent interpretation at the extremes of the degree range. At $i = 0$: the pairing $H^0(G, M) \times H^d(G, \mathrm{Hom}(M, D)) \to \mathbb{Q}_p/\mathbb{Z}_p$ says that the G-invariants of M are dual to the top cohomology with dual coefficients. At $i = d$: the pairing $H^d(G, M) \times H^0(G, \mathrm{Hom}(M, D)) \to \mathbb{Q}_p/\mathbb{Z}_p$ says that the top cohomology of M is dual to the G-invariants of $\mathrm{Hom}(M, D)$. These two extremes are the "easy" cases; the content of the duality theorem is that the pairing is perfect in all intermediate degrees as well.

A consequence of duality: for a duality group G of dimension d and a finite p-torsion module M, the groups $H^i(G, M)$ and $H^{d-i}(G, \mathrm{Hom}(M, D))$ have the same order. The alternating product $\prod_{i=0}^d |H^i(G, M)|^{(-1)^i}$ — the *Euler–Poincaré characteristic* $\chi(G, M)$ — therefore satisfies $\chi(G, M) = \chi(G, \mathrm{Hom}(M, D))^{(-1)^d}$, a symmetry that constrains the possible sizes of cohomology groups. The Euler–Poincaré characteristic will be computed for local and global fields in chapters 8 and 9.

The simplest case of a duality group is when the dualizing module is as small as possible.

Definition 5.3.2: Poincaré Duality Group

A profinite group G is a *Poincaré duality group of dimension d at p* (or a *PD_d-group at p*) if it is a duality group of dimension d at p with dualizing module $D \cong \mathbb{Q}_p/\mathbb{Z}_p$ (equipped with some continuous G-action). Equivalently, G is a PD_d-group at p if $\mathrm{cd}_p(G) = d$ and the functor $M \mapsto H^d(G, M)^*$ is naturally isomorphic to $H^0(G, \mathrm{Hom}(M, \mathbb{Q}_p/\mathbb{Z}_p))$ on finite p-torsion G-modules.

The G-action on $D = \mathbb{Q}_p/\mathbb{Z}_p$ is called the *orientation character*: it is a continuous homomorphism $\omega : G \to \mathbb{Z}_p^\times$, with g acting on $\mathbb{Q}_p/\mathbb{Z}_p$ by multiplication by $\omega(g)$. When ω is trivial (G acts trivially on D), the group G is called an *orientable* Poincaré duality group. The terminology comes from topology: for the pro-p-completion of the fundamental group of a closed manifold, the orientation character is trivial if and only if the manifold is orientable. In the arithmetic setting, the orientation character of G_K (for a local field K) is the cyclotomic character $\chi_p : G_K \to \mathbb{Z}_p^\times$, which is always nontrivial (since K does not contain all p-power roots of unity). Hence absolute Galois groups of local fields are *non-orientable* duality groups.

Example 5.3.3: Poincaré Duality for $\mathbb{Z}_p^d$

1. The group $G = \mathbb{Z}_p$ is a PD_1-group at p. The cohomological dimension $\mathrm{cd}_p(\mathbb{Z}_p) = 1$ was established in example 5.1.5. The dualizing module is $D = \mathbb{Q}_p/\mathbb{Z}_p$ with trivial action, and the trace map $\theta : H^1(\mathbb{Z}_p, \mathbb{Q}_p/\mathbb{Z}_p) \xrightarrow{\sim} \mathbb{Q}_p/\mathbb{Z}_p$ is the isomorphism $\mathrm{Hom}_{\mathrm{cts}}(\mathbb{Z}_p, \mathbb{Q}_p/\mathbb{Z}_p) \cong \mathbb{Q}_p/\mathbb{Z}_p$ (a continuous homomorphism $\mathbb{Z}_p \to \mathbb{Q}_p/\mathbb{Z}_p$ is determined by the image of 1). The duality pairing in the only nontrivial case ($i = 0$, $d - i = 1$):

$$H^0(\mathbb{Z}_p, M) \times H^1(\mathbb{Z}_p, \mathrm{Hom}(M, \mathbb{Q}_p/\mathbb{Z}_p)) \to H^1(\mathbb{Z}_p, \mathbb{Q}_p/\mathbb{Z}_p) \cong \mathbb{Q}_p/\mathbb{Z}_p$$

   pairs $M^{\mathbb{Z}_p}$ with $\mathrm{Hom}_{\mathrm{cts}}(\mathbb{Z}_p, M^*) \cong M^*$ (using $H^1 = \mathrm{Hom}_{\mathrm{cts}}$ for trivial action and the structure of M^*). Since M is finite, $M^{\mathbb{Z}_p} = M$ (because $\mathbb{Z}_p$ is pro-p and M is p-torsion, the action factors through a finite quotient $\mathbb{Z}/p^k\mathbb{Z}$; for the trivial action, $M^G = M$). The pairing is the natural evaluation $M \times M^* \to \mathbb{Q}_p/\mathbb{Z}_p$, which is perfect.

   For a nontrivial action: if $\mathbb{Z}_p$ acts on $M = \mathbb{Z}/p\mathbb{Z}$ through the quotient $\mathbb{Z}/p\mathbb{Z}$ by a nontrivial character, then $M^{\mathbb{Z}_p} = 0$ and $H^1(\mathbb{Z}_p, M^*) = \mathrm{Hom}_{\mathrm{cts}}(\mathbb{Z}_p, M^*)/(\text{conjugation})$. Both sides of the pairing are zero, so the pairing is vacuously perfect.

2. More generally, $G = \mathbb{Z}_p^d$ is a PD_d-group at p (orientable, since the action on D is trivial). The proof uses the Künneth formula: $H^d(\mathbb{Z}_p^d, \mathbb{Q}_p/\mathbb{Z}_p) \cong \bigotimes_{i=1}^d H^1(\mathbb{Z}_p, \mathbb{Q}_p/\mathbb{Z}_p) \cong (\mathbb{Q}_p/\mathbb{Z}_p)^{\otimes d}$. Since $\mathbb{Q}_p/\mathbb{Z}_p \otimes \mathbb{Q}_p/\mathbb{Z}_p \cong \mathbb{Q}_p/\mathbb{Z}_p$ (as the tensor product of injective $\mathbb{Z}_p$-modules), this gives $H^d(\mathbb{Z}_p^d, \mathbb{Q}_p/\mathbb{Z}_p) \cong \mathbb{Q}_p/\mathbb{Z}_p$, providing the trace map. The cup product pairing reduces to the exterior algebra structure: $H^i(\mathbb{Z}_p^d, M)$ pairs with $H^{d-i}(\mathbb{Z}_p^d, M^*)$ via the wedge product on the exterior algebra $\bigwedge^\bullet(\mathbb{Z}_p^d)^*$, and this pairing is perfect.

3. (If the reader has read [16]) The PD_d-property of $\mathbb{Z}_p^d$ is the profinite analogue of Poincaré duality for the d-torus $T^d = (\mathbb{R}/\mathbb{Z})^d$: the classifying space $B\mathbb{Z}_p^d$ is the p-completion of T^d, and Poincaré duality for T^d (a closed orientable manifold of dimension d) translates into the algebraic duality for $\mathbb{Z}_p^d$. The connection extends: if X is a closed orientable aspherical manifold of dimension d (meaning X is a $K(\pi, 1)$ with $\pi = \pi_1(X)$), then the pro-p-completion of π is a PD_d-group at p.

Poincaré duality groups can be characterized without explicit reference to the dualizing module, using only the cohomology of the group ring.

Proposition 5.3.4: Characterization of PD_d-Groups

Let G be a pro-p-group with $\mathrm{cd}_p(G) = d < \infty$. Then G is a PD_d-group at p if and only if

$$H^i(G, \mathbb{Z}/p\mathbb{Z}[[G]]) \cong \begin{cases} \mathbb{Z}/p\mathbb{Z} & i = d, \\ 0 & i \neq d, \end{cases}$$

where $\mathbb{Z}/p\mathbb{Z}[[G]] = \varprojlim_U \mathbb{Z}/p\mathbb{Z}[G/U]$ is the completed group ring.

***Proof Sketch*:**
The completed group ring $\mathbb{Z}/p\mathbb{Z}[[G]]$ plays the role of the "chains on the universal cover" in the topological setting. For a closed manifold X with fundamental group π, Poincaré duality is equivalent to the statement that the cellular chain complex of the universal cover $\tilde{X}$ has homology concentrated in degree d (this is the "fundamental class"). The algebraic analogue: the condition $H^d(G, \mathbb{Z}/p\mathbb{Z}[[G]]) \cong \mathbb{Z}/p\mathbb{Z}$ says that the top cohomology of G with "universal" coefficients is one-dimensional, and the vanishing $H^i(G, \mathbb{Z}/p\mathbb{Z}[[G]]) = 0$ for $i \neq d$ says that the universal coefficients contribute nothing in other degrees.

The dualizing module is recovered as $D = H^d(G, \mathbb{Z}/p\mathbb{Z}[[G]])^* \otimes_{\mathbb{Z}/p\mathbb{Z}} \mathbb{Q}_p/\mathbb{Z}_p$ with the induced G-action. The G-action on $H^d(G, \mathbb{Z}/p\mathbb{Z}[[G]])$ (which is $\mathbb{Z}/p\mathbb{Z}$ as an abelian group but carries a G-action through the coefficients) determines the orientation character. The cup product pairing is constructed from the cap product on $\mathbb{Z}/p\mathbb{Z}[[G]]$: the cap product $H^i(G, M) \otimes H_d(G, \mathbb{Z}/p\mathbb{Z}[[G]]) \to H_{d-i}(G, M)$ composed with the identification $H_{d-i}(G, M) \cong H^{d-i}(G, \mathrm{Hom}(M, D))^*$ (via the universal coefficient theorem) produces the duality pairing.

The most important class of Poincaré duality groups in arithmetic are the *Demushkin groups*: pro-p-groups of $\mathrm{cd}_p = 2$ satisfying PD_2-duality. These groups are the pro-p-completions of the absolute Galois groups of p-adic fields, and their classification is one of the main results of chapter 6.

Definition 5.3.5: Demushkin Group

A *Demushkin group* is a pro-p-group G satisfying:

1. $\mathrm{cd}_p(G) = 2$.
2. $H^1(G, \mathbb{Z}/p\mathbb{Z})$ is finite (i.e., G is topologically finitely generated).
3. $H^2(G, \mathbb{Z}/p\mathbb{Z}) \cong \mathbb{Z}/p\mathbb{Z}$.

Equivalently (by proposition 5.3.4), a Demushkin group is a topologically finitely generated pro-p-PD_2-group.

The conditions are stringent: condition (3) says that the "second Betti number" of G is exactly 1, which constrains the number of relations in any presentation of G as a pro-p-group. If $\dim_{\mathbb{Z}/p\mathbb{Z}} H^1(G, \mathbb{Z}/p\mathbb{Z}) = d$ (so G requires d generators), then the cup product pairing $H^1(G, \mathbb{Z}/p\mathbb{Z}) \times H^1(G, \mathbb{Z}/p\mathbb{Z}) \to H^2(G, \mathbb{Z}/p\mathbb{Z}) \cong \mathbb{Z}/p\mathbb{Z}$ is a nondegenerate bilinear form on $(\mathbb{Z}/p\mathbb{Z})^d$. For p odd, graded-commutativity forces this form to be alternating ($\alpha \smile \alpha = 0$), so nondegeneracy requires d to be even. For $p = 2$, the form need not be alternating (the cup product $\alpha \smile \alpha$ can be nonzero mod 2), allowing odd d.

Demushkin groups were classified by Demushkin (for p odd) and by Serre and Labute (for $p = 2$). The classification, which will be developed in chapter 6, §4, shows that a Demushkin group is determined up to isomorphism by the pair (d, q), where $d = \dim H^1(G, \mathbb{Z}/p\mathbb{Z})$ is the number of generators and q is a power of p (or $q = 0$) encoding the structure of the cup product pairing. The group is presented by d generators $x_1, \ldots, x_d$ subject to a single relation r whose form depends on (d, q).

Example 5.3.6: Demushkin Groups in Arithmetic

1. Let K be a finite extension of $\mathbb{Q}_p$ with p odd. The maximal pro-p-quotient $G_K(p)$ of the absolute Galois group G_K is a Demushkin group. The number of generators is $d = \dim H^1(G_K(p), \mathbb{Z}/p\mathbb{Z}) = [K : \mathbb{Q}_p] + 1 + \delta$, where $\delta = 1$ if K contains a primitive p-th root of unity and $\delta = 0$ otherwise. Since $[K : \mathbb{Q}_p] \geq 1$ and p is odd, the parity constraint forces d to be even. For example: $K = \mathbb{Q}_p$ gives $d = 2$ (when $p \geq 3$ and $\zeta_p \notin \mathbb{Q}_p$), and $G_{\mathbb{Q}_p}(p) \cong \mathbb{Z}_p^2$ (the unique Demushkin group on 2 generators with $q = p$). The cup product pairing on H^1 is related to the Hilbert symbol of local class field theory, and the classification of Demushkin groups determines the pro-p-Galois structure of p-adic fields. The full classification is in chapter 6, §4.

2. The dualizing module for $G_{\mathbb{Q}_p}$ (as a duality group, not just its pro-p-quotient) is $D = \mu_{p^\infty} = \varinjlim_k \mu_{p^k}$, the p-power roots of unity in $\overline{\mathbb{Q}}_p^\times$. The $G_{\mathbb{Q}_p}$-action on μ_{p^∞} is through the p-adic cyclotomic character $\chi_p : G_{\mathbb{Q}_p} \to \mathbb{Z}_p^\times$, and the trace map $\theta : H^2(G_{\mathbb{Q}_p}, \mu_{p^\infty}) \xrightarrow{\sim} \mathbb{Q}_p/\mathbb{Z}_p$ is the invariant map of local class field theory. The proof that $G_{\mathbb{Q}_p}$ is a duality group of dimension 2 with this dualizing module is the content of local Tate duality (chapter 8).

3. Not every pro-p-group of $\mathrm{cd}_p = 2$ is Demushkin — the Demushkin condition $\dim H^2 = 1$ is restrictive. The group $\mathbb{Z}_p^2$ has $\mathrm{cd}_p = 2$ (example 5.1.5) and $H^2(\mathbb{Z}_p^2, \mathbb{Z}/p\mathbb{Z}) \cong \mathbb{Z}/p\mathbb{Z}$ (the exterior square of the 2-dimensional H^1), so $\mathbb{Z}_p^2$ is Demushkin with $d = 2$. On the other hand, if G is the pro-p-group with three generators x_1, x_2, x_3 and two independent relations, then $\dim H^2(G, \mathbb{Z}/p\mathbb{Z}) = 2 \neq 1$, so G is not Demushkin despite having $\mathrm{cd}_p = 2$ (assuming the relations are chosen so that G is infinite). Such groups arise as Galois groups of maximal p-extensions of global fields, where $\dim H^2$ can be larger than 1.

The hierarchy of profinite groups studied in this chapter — groups of $\mathrm{cd}_p = 0$ (pro-p'-groups), $\mathrm{cd}_p = 1$ (free pro-p-groups), $\mathrm{cd}_p = 2$ (Demushkin and more general PD_2-groups), and higher — mirrors the hierarchy of fields in section 5.2. The abstract characterizations ($\mathrm{cd}_p = 1$ iff free, PD_2 iff Demushkin with one relation) transform cohomological computations into structural conclusions about the group. The structure theory of pro-p-groups in chapter 6 exploits this hierarchy systematically, and the connection between the abstract duality theory of this section and the arithmetic duality theorems of chapters 8 and 9 is a central theme of Part III.

Exercise 5.3.1

1. Let G be a duality group of dimension d at p with dualizing modules (D, θ) and (D', θ'). Show that $D \cong D'$ as G-modules. (*Hint:* Consider the functor $M \mapsto H^d(G, M)^*$ on finite p-torsion G-modules and apply the representability criterion: this functor is represented by D via the cup product pairing.)

2. For $G = \mathbb{Z}_p^2$, write out the cup product pairing $H^1(G, \mathbb{Z}/p\mathbb{Z}) \times H^1(G, \mathbb{Z}/p\mathbb{Z}) \to H^2(G, \mathbb{Z}/p\mathbb{Z})$ explicitly. Show that $H^1 \cong (\mathbb{Z}/p\mathbb{Z})^2$ (with generators χ_1, χ_2 dual to the standard generators of $\mathbb{Z}_p^2$), $H^2 \cong \mathbb{Z}/p\mathbb{Z}$ (generated by $\chi_1 \smile \chi_2$), and that the pairing is the standard symplectic form $(\chi_i, \chi_j) \mapsto \chi_i \smile \chi_j$, with $\chi_1 \smile \chi_2 = -\chi_2 \smile \chi_1$ being the unique generator.

3. Let G be a Demushkin group with p odd and $d = \dim H^1(G, \mathbb{Z}/p\mathbb{Z})$. Show that d is even. (*Hint:* The cup product pairing $H^1 \times H^1 \to H^2 \cong \mathbb{Z}/p\mathbb{Z}$ is a nondegenerate alternating form for p odd, by graded-commutativity. An alternating form on a finite-dimensional vector space over a field of odd characteristic is nondegenerate only if the dimension is even.)

4. Let G be a duality group of dimension d at p with dualizing module D, and let $H \leq G$ be an open subgroup. Show that H is also a duality group of dimension d at p, with the same dualizing module D (restricted to H). (*Hint:* Use the Shapiro isomorphism and the compatibility of the cup product with restriction.)

5. (If the reader has read [9]) Let G be a pro-p-group with $\mathrm{cd}_p(G) = 2$ and $H^1(G, \mathbb{Z}/p\mathbb{Z}) \cong (\mathbb{Z}/p\mathbb{Z})^d$ for some finite d. Show that G is Demushkin if and only if G has a single defining relation in its minimal pro-p-presentation (i.e., $G \cong F/\overline{\langle r \rangle}$ where F is a free pro-p-group on d generators and $r \in F$). (*Hint:* The number of relations equals $\dim H^2(G, \mathbb{Z}/p\mathbb{Z})$.)

6

Pro-p-Groups and the Golod–Shafarevich Inequality

Chapter 5 introduced cohomological dimension as the fundamental invariant of a profinite group's cohomology. The groups of cohomological dimension 0 are the pro-p' groups (those with no elements of order p), and those of cohomological dimension 1 turn out to be precisely the free pro-p-groups. This chapter develops the structure theory of pro-p-groups through the lens of cohomological dimension: the minimal number of generators and relations of a pro-p-group are read off from H^1 and H^2, and the Golod–Shafarevich inequality provides a lower bound on the relation count that resolved the class field tower problem. The chapter closes with the classification of Demushkin groups—the pro-p-groups of cohomological dimension 2 satisfying Poincaré duality—which arise in both topology and arithmetic.

6.1 Free Profinite Groups

Among all pro-p-groups, the free ones are the simplest: they admit no nontrivial relations. The cohomological characterization is sharp—a pro-p-group is free if and only if its p-cohomological dimension is at most 1 (theorem 6.1.4 below). This is the pro-p analogue of the Stallings–Swan theorem for discrete groups, and it reduces a group-theoretic property (freeness) to a cohomological vanishing condition.

Definition 6.1.1: Free Profinite Group and Free Pro-p-Group

Let S be a set. The *free profinite group* on S is a profinite group $\hat{F}(S)$ together with a map $\iota\colon S \to \hat{F}(S)$ satisfying the following universal property: for every profinite group H and every map $\varphi\colon S \to H$, there exists a unique continuous homomorphism $\tilde{\varphi}\colon \hat{F}(S) \to H$ with $\tilde{\varphi} \circ \iota = \varphi$:

$$\begin{tikzcd} S \arrow[r, "\iota"] \arrow[dr, "\varphi"'] & \hat{F}(S) \arrow[d, dashed, "\exists!\,\tilde{\varphi}"] \\ & H \end{tikzcd}$$

The *free pro-p-group* on S, denoted $F_p(S)$ (or $F(d)$ when $|S| = d$ is finite), is defined by the same universal property with H restricted to pro-p-groups. Equivalently, $F_p(S)$ is the maximal pro-p quotient of $\hat{F}(S)$.

The free profinite group $\hat{F}(S)$ is the profinite completion of the discrete free group $F(S)$ on S (by proposition 1.1.9: the profinite completion satisfies the same universal property). The free pro-p-group $F_p(S)$ is the pro-p-completion: the limit of all finite p-group quotients of $F(S)$.

Proposition 6.1.2: Basic Properties of Free Pro-p-Groups

Let $F = F_p(S)$ be the free pro-p-group on a set S. Then:

1. The image $\iota(S)$ topologically generates F: the closure $\overline{\langle \iota(S) \rangle} = F$.
2. The Frattini quotient satisfies $F/\Phi(F) \cong \mathbb{F}_p^{|S|}$ (where $\Phi(F)$ is the Frattini subgroup from definition 1.3.9). In particular, $d(F) = |S|$ (the generator rank from proposition 1.3.10).
3. Every surjection $G \twoheadrightarrow F$ of pro-p-groups splits: there exists a continuous homomorphism $s\colon F \to G$ with $\pi \circ s = \mathrm{id}_F$.

***Proof*:**

For (1): by the universal property, any open normal subgroup $U \trianglelefteq F$ satisfies $F/U = \overline{\langle \iota(S) \rangle} U/U$, since the finite p-group F/U is generated by the images of $\iota(S)$. Taking the limit over all U gives $F = \overline{\langle \iota(S) \rangle}$ (by proposition 1.1.8).

For (2): the Burnside basis theorem (proposition 1.3.10) gives $d(F) = \dim_{\mathbb{F}_p} F/\Phi(F)$, and the generators $\iota(S)$ are a minimal topological generating set (any proper subset fails to generate: removing $s \in S$ and applying the universal property to the quotient $\mathbb{F}_p$ that kills s but not the others shows $\overline{\langle \iota(S \setminus \{s\}) \rangle} \neq F$).

For (3): given $\pi\colon G \twoheadrightarrow F$, the universal property provides a section: the inclusion $\iota\colon S \to F$ lifts to a map $S \to G$ (choosing arbitrary preimages), which extends to $s\colon F \to G$ by the universal property. Then $\pi \circ s$ and id_F agree on the generating set $\iota(S)$, hence are equal.

Property (3) says that free pro-p-groups are projective objects in the category of pro-p-groups: every surjection onto F splits. The converse—that projective pro-p-groups are free—is also true, and combined with the cohomological characterization of proposition 5.1.6, gives the main theorem of this section.

The cohomology of free pro-p-groups is concentrated in degrees 0 and 1. This was established in proposition 5.1.6: $\mathrm{cd}_p(F_p(S)) \leq 1$ for any set S. The first cohomology group has a clean description.

Proposition 6.1.3: Cohomology of Free Pro-p-Groups

Let $F = F_p(S)$ be the free pro-p-group on a set S, and let M be a discrete p-torsion F-module. Then:

1. $H^1(F, \mathbb{F}_p) \cong \mathrm{Map}(S, \mathbb{F}_p)$ (continuous maps from S with the discrete topology to $\mathbb{F}_p$). When S is finite, $H^1(F, \mathbb{F}_p) \cong \mathbb{F}_p^{|S|}$.
2. $H^n(F, M) = 0$ for all $n \geq 2$.

***Proof*:**
Part (2) is proposition 5.1.6. For part (1): the universal property gives $H^1(F, \mathbb{F}_p) = \mathrm{Hom}(F, \mathbb{F}_p)$ (since $\mathbb{F}_p$ carries the trivial action, crossed homomorphisms are ordinary homomorphisms by proposition 2.4.3). A continuous homomorphism $F \to \mathbb{F}_p$ is determined by its values on the generating set $\iota(S)$ (each generator maps to an element of $\mathbb{F}_p$, and the universal property ensures any choice extends). So $\mathrm{Hom}(F, \mathbb{F}_p) \cong \mathrm{Map}(S, \mathbb{F}_p)$.

The main theorem characterizes free pro-p-groups by the vanishing of H^2.

Theorem 6.1.4: Cohomological Characterization of Free Pro-p-Groups

A pro-p-group G is free if and only if $\mathrm{cd}_p(G) \leq 1$.

***Proof*:**
Step 1: Free implies $\mathrm{cd}_p \leq 1$. This is proposition 5.1.6.

Step 2: $\mathrm{cd}_p \leq 1$ *implies projective.* Suppose $\mathrm{cd}_p(G) \leq 1$, and let $\pi\colon H \twoheadrightarrow G$ be a surjection of pro-p-groups with kernel $N = \ker \pi$. The goal is to construct a continuous section $s\colon G \to H$. Since H is pro-p, it suffices to construct compatible sections $s_U\colon G/\pi(U) \to H/U$ for each open normal subgroup $U \trianglelefteq H$ contained in N (as the limit of these sections gives s).

Fix $U \trianglelefteq H$ open with $U \leq N$. The extension $1 \to N/U \to H/U \to G \to 1$ is classified by a class $\alpha \in H^2(G, N/U)$ (by theorem 2.4.5, applied to N/U as a G-module via conjugation). Since N/U is a finite p-group (as a quotient of the pro-p-group N by an open subgroup) and $\mathrm{cd}_p(G) \leq 1$, it follows that $H^2(G, N/U) = 0$ (by the definition of cd_p, since N/U is p-torsion). So $\alpha = 0$, meaning the extension splits, and a section s_U exists.

The set of sections $s_U\colon G \to H/U$ forms a cofiltered system of non-empty finite sets (for each $U' \leq U$, the projection $H/U' \to H/U$ sends a section for U' to a section for U). By lemma 1.2.6, the limit is non-empty, producing a continuous section $s\colon G \to H$.

Step 3: Projective implies free. Let G be a projective pro-p-group. Choose a surjection $\pi\colon F(d(G)) \twoheadrightarrow G$ from the free pro-p-group on $d(G)$ generators (such a surjection exists by proposition 1.3.10). Since G is projective, π splits: there exists $s\colon G \to F(d(G))$ with $\pi \circ s = \mathrm{id}$. Then s is injective and identifies G with a retract of $F(d(G))$. A retract of a free pro-p-group is free: let $R = \ker \pi$, so that $F(d(G)) = s(G) \ltimes R$ (semidirect product). Freeness of $G \cong s(G)$ follows from the Nielsen–Schreier property for pro-p-groups (see [20]), which states that subgroups of free pro-p-groups are free.

The theorem reduces a structural property (freeness) to a single cohomological condition ($H^2(G, M) = 0$ for all p-torsion M, which by Serre's criterion (theorem 5.1.2) reduces further to $H^2(G, \mathbb{F}_p) = 0$). In practice, the characterization is used in both directions: to prove a group is free by checking a vanishing condition, and to prove cohomological vanishing by establishing freeness through other means.

Example 6.1.5: Free Pro-p-Groups

1. The group $\mathbb{Z}_p$ is the free pro-p-group of rank 1: $F(1) \cong \mathbb{Z}_p$. The universal property holds because every element of a pro-p-group generates a closed subgroup isomorphic to a quotient of $\mathbb{Z}_p$.
2. The absolute Galois group $G_{\mathbb{F}_q} \cong \widehat{\mathbb{Z}}$ (exercise 1.4.1 (2)) is a free profinite group of rank 1. Its maximal pro-p-quotient is $\mathbb{Z}_p$, a free pro-p-group.
3. The absolute Galois group of the function field $\overline{\mathbb{F}}_q(t)$, over the algebraic closure $\overline{\mathbb{F}}_q$, is a free profinite group. This is a nontrivial theorem of Douady and Harbater, following from $\mathrm{cd}(\overline{\mathbb{F}}_q(t)) = 1$ (example 5.2.4) and the characterization above.
4. (If the reader has read [16]) Let F_d be the discrete free group on d generators. Its pro-p-completion $\widehat{F_d}^{(p)}$ is the free pro-p-group $F(d)$: the pro-p-completion functor preserves the universal property with respect to pro-p-group targets.

Subgroups of free pro-p-groups inherit freeness, with a rank formula generalizing the classical Schreier index formula for discrete free groups.

Theorem 6.1.6: Schreier Index Formula for Pro-p-Groups

Let $F = F(d)$ be the free pro-p-group of finite rank $d \geq 1$, and let $H \leq F$ be an open subgroup of index $[F : H] = n$. Then H is a free pro-p-group of rank:

$$d(H) = n(d-1) + 1$$

***Proof*:**
Since H is an open subgroup of F, it is a closed subgroup of a profinite group, hence profinite (by proposition 1.2.1). As a subgroup of the pro-p-group F, it is pro-p. The Euler characteristic gives the rank: the cohomological dimension of H is at most 1 (since $\mathrm{cd}_p(H) \leq \mathrm{cd}_p(F) = 1$ by proposition 5.1.3), so H is free by theorem 6.1.4. For the rank formula, the Euler characteristic

> $\chi(G) = 1 - d(G)$ is multiplicative under finite-index inclusions: $\chi(H) = [F : H] \cdot \chi(F)$ (this is a consequence of the multiplicativity of the Herbrand quotient, or equivalently of the formula $\mathrm{Cor} \circ \mathrm{Res} = [G : H]$ from proposition 3.3.6, applied to the Euler characteristic computed from H^0 and H^1 with $\mathbb{F}_p$-coefficients). Substituting: $1 - d(H) = n(1 - d)$, giving $d(H) = n(d - 1) + 1$.

The Schreier formula has a satisfying interpretation: a free pro-p-group of rank d has "$d-1$ degrees of freedom" (subtracting the single relation imposed by the group identity), and passing to a subgroup of index n multiplies the degrees of freedom by n, then adds back the group identity.

More generally, closed subgroups of free pro-p-groups are free (not just open ones). This stronger result uses a limit argument.

Proposition 6.1.7: Closed Subgroups of Free Pro-p-Groups Are Free

Every closed subgroup of a free pro-p-group is free.

> ***Proof*:**
> Let H be a closed subgroup of $F = F_p(S)$. Then $\mathrm{cd}_p(H) \leq \mathrm{cd}_p(F) \leq 1$ by proposition 5.1.3, and H is free by theorem 6.1.4.

The brevity of this proof—a single application of monotonicity and the characterization theorem—illustrates the power of the cohomological approach. The corresponding result for discrete free groups (the Nielsen–Schreier theorem) requires significantly more combinatorial work.

Exercise 6.1.1

1. Verify the universal property of $F(1) = \mathbb{Z}_p$ directly: show that for every pro-p-group H and every $h \in H$, there exists a unique continuous homomorphism $\varphi\colon \mathbb{Z}_p \to H$ with $\varphi(1) = h$.
2. Show that the profinite completion of the discrete free group F_d (on d generators) is the free profinite group $\hat{F}(d)$. Conclude that the pro-p-completion of F_d is $F(d)$.
3. Let $F = F(2)$ be the free pro-p-group of rank 2, and let $H \leq F$ be an open normal subgroup of index p. Compute $d(H)$ using the Schreier formula, and verify this independently by computing $\dim_{\mathbb{F}_p} H^1(H, \mathbb{F}_p)$ using the corestriction map.
4. Show that the free pro-p-group on a countably infinite set S is not topologically finitely generated, and compute its p-cohomological dimension.
5. Let F be a free pro-p-group and $\varphi\colon F \to F$ a continuous surjective endomorphism. Show that $\ker \varphi$ is a free pro-p-group. Is every normal closed subgroup of F necessarily free?

6.2 Generators and Relations for Pro-p-Groups

Every pro-p-group can be expressed as a quotient of a free pro-p-group. The number of generators and relations in such a presentation are not arbitrary—they are cohomological invariants, determined by H^1 and H^2 with $\mathbb{F}_p$-coefficients. This correspondence between presentations and cohomology is the algebraic engine behind the Golod–Shafarevich inequality of section 6.3.

Definition 6.2.1: Pro-p-Presentation

A *pro-p-presentation* of a pro-p-group G is a short exact sequence of pro-p-groups:

$$1 \to R \to F \xrightarrow{\pi} G \to 1$$

where F is a free pro-p-group (definition 6.1.1) and $R = \ker \pi$ is the *relation subgroup*. The presentation is *minimal* if $F = F(d(G))$, the free pro-p-group on $d(G)$ generators (i.e., π maps a free basis of F to a minimal topological generating set of G).

Every pro-p-group admits a minimal presentation: the Burnside basis theorem (proposition 1.3.10) provides a minimal generating set $\{g_1, \ldots, g_d\}$ of G (with $d = d(G) = \dim_{\mathbb{F}_p} G/\Phi(G)$), and the universal property of $F(d)$ gives a surjection $F(d) \twoheadrightarrow G$ mapping the free generators to the g_i.

The generator rank $d(G)$ was already identified in proposition 1.3.10 as the dimension of the Frattini quotient. The following reformulation through H^1 makes the cohomological nature of $d(G)$ explicit.

Theorem 6.2.2: Generator Rank

Let G be a finitely generated pro-p-group. Then:

$$d(G) = \dim_{\mathbb{F}_p} H^1(G, \mathbb{F}_p)$$

***Proof*:**
Since G acts trivially on $\mathbb{F}_p$, proposition 2.4.3 gives $H^1(G, \mathbb{F}_p) = \mathrm{Hom}(G, \mathbb{F}_p)$, where Hom denotes continuous group homomorphisms. A continuous homomorphism $G \to \mathbb{F}_p$ factors through $G/\Phi(G)$ (since $\Phi(G)$ is the intersection of all maximal open subgroups, each of index p, and $\mathbb{F}_p$ has order p). Conversely, every $\mathbb{F}_p$-linear functional on $G/\Phi(G) \cong \mathbb{F}_p^{d(G)}$ lifts to a homomorphism $G \to \mathbb{F}_p$. So $\mathrm{Hom}(G, \mathbb{F}_p) \cong (G/\Phi(G))^* \cong \mathbb{F}_p^{d(G)}$, and $\dim_{\mathbb{F}_p} H^1(G, \mathbb{F}_p) = d(G)$.

The generator rank is thus the $\mathbb{F}_p$-dimension of the first cohomology group with the simplest possible coefficients. The relation rank admits an analogous description through H^2.

Theorem 6.2.3: Relation Rank

Let G be a finitely generated pro-p-group with minimal presentation $1 \to R \to F(d) \to G \to 1$. Then:

$$r(G) = \dim_{\mathbb{F}_p} H^2(G, \mathbb{F}_p)$$

The number $r(G)$ is independent of the choice of minimal presentation and is called the *relation rank* of G. It equals $d(R)$ — the minimal number of topological generators of R as a normal subgroup of $F(d)$ (i.e., the minimal number of elements whose normal closure in $F(d)$ is R).

***Proof*:**
Apply the five-term exact sequence (theorem 3.3.4) to the extension $1 \to R \to F \to G \to 1$ with

coefficients in $\mathbb{F}_p$:

$$0 \to H^1(G, \mathbb{F}_p) \xrightarrow{\text{Inf}} H^1(F, \mathbb{F}_p) \xrightarrow{\text{Res}} H^1(R, \mathbb{F}_p)^G \xrightarrow{\text{tg}} H^2(G, \mathbb{F}_p) \to H^2(F, \mathbb{F}_p)$$

Since F is free, $H^2(F, \mathbb{F}_p) = 0$ (by proposition 6.1.3). The transgression is therefore surjective:

$$\text{tg}\colon H^1(R, \mathbb{F}_p)^G \twoheadrightarrow H^2(G, \mathbb{F}_p)$$

The inflation map is an isomorphism onto its image: $\text{Inf}\colon H^1(G, \mathbb{F}_p) \xrightarrow{\sim} \ker(\text{Res})$. The presentation is minimal ($F = F(d(G))$), so $\dim H^1(F, \mathbb{F}_p) = d(G) = \dim H^1(G, \mathbb{F}_p)$ (by proposition 6.1.3 and theorem 6.2.2). The inflation map between spaces of the same dimension with trivial kernel is an isomorphism $H^1(G, \mathbb{F}_p) \cong H^1(F, \mathbb{F}_p)$, forcing $\text{Res} = 0$. The five-term sequence collapses to:

$$H^1(R, \mathbb{F}_p)^G \cong H^2(G, \mathbb{F}_p)$$

The left side is $\text{Hom}(R, \mathbb{F}_p)^G$: the G-invariant continuous homomorphisms from R to $\mathbb{F}_p$. This is identified with $\text{Hom}(R/\overline{[R,F]}\Phi(R), \mathbb{F}_p)$, where $\overline{[R,F]}$ is the closure of the commutator subgroup of R with F (the G-invariance eliminates the F-conjugation) and $\Phi(R)$ is the Frattini subgroup (the $\mathbb{F}_p$ target forces factoring through the Frattini quotient). The dimension $\dim_{\mathbb{F}_p} \text{Hom}(R/\overline{[R,F]}\Phi(R), \mathbb{F}_p)$ is the minimal number of topological generators of R as a closed normal subgroup of F, which is $r(G)$.

Independence of the presentation: $H^2(G, \mathbb{F}_p)$ depends only on G, not on the choice of free resolution.

The generator and relation ranks together determine the "complexity" of a pro-p-group's presentation. A free pro-p-group has $r(G) = 0$ (since $H^2(G, \mathbb{F}_p) = 0$ by proposition 6.1.3), recovering the characterization of theorem 6.1.4. A group with $r(G) = 0$ is free, and $r(G) > 0$ measures the number of independent relations needed to define G.

Example 6.2.4: Generator and Relation Ranks

1. $G = \mathbb{Z}_p$: $d(G) = 1$ (topologically generated by 1), and $r(G) = \dim_{\mathbb{F}_p} H^2(\mathbb{Z}_p, \mathbb{F}_p) = 0$ (since $\text{cd}_p(\mathbb{Z}_p) = 1$). The minimal presentation is $F(1) \xrightarrow{\sim} \mathbb{Z}_p$ with trivial relation subgroup.

2. $G = \mathbb{Z}/p\mathbb{Z}$: $d(G) = 1$ and $r(G) = \dim_{\mathbb{F}_p} H^2(\mathbb{Z}/p\mathbb{Z}, \mathbb{F}_p) = 1$ (by example 2.3.8, $H^2(C_p, \mathbb{F}_p) = \mathbb{F}_p$). The minimal presentation is $1 \to R \to \mathbb{Z}_p \to \mathbb{Z}/p\mathbb{Z} \to 1$, where $R = p\mathbb{Z}_p$ is the single relation $x^p = 1$ (more precisely, R is the closed normal subgroup generated by g^p, where g is the free generator).

3. $G = \mathbb{Z}_p^n$ (the free abelian pro-p-group of rank n): $d(G) = n$ (by proposition 1.3.10, since $G/\Phi(G) = \mathbb{F}_p^n$). The relation rank is $r(G) = \binom{n}{2}$: the relations are the commutators $[x_i, x_j]$ for $1 \le i < j \le n$. This can be verified directly from $H^2(\mathbb{Z}_p^n, \mathbb{F}_p)$: the Künneth formula (or the LHS spectral sequence applied iteratively) gives $H^2(\mathbb{Z}_p^n, \mathbb{F}_p) \cong \bigwedge^2 H^1(\mathbb{Z}_p^n, \mathbb{F}_p) \cong \bigwedge^2 \mathbb{F}_p^n$, which has dimension $\binom{n}{2}$.

4. $G = \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$: $d(G) = 2$ and $r(G) = \dim H^2(G, \mathbb{F}_p)$. By the Künneth formula, $H^2((\mathbb{Z}/p\mathbb{Z})^2, \mathbb{F}_p) \cong (H^1 \otimes H^1) \oplus H^2(\mathbb{Z}/p\mathbb{Z}, \mathbb{F}_p) \oplus H^2(\mathbb{Z}/p\mathbb{Z}, \mathbb{F}_p)$. Using $\dim H^1(\mathbb{Z}/p\mathbb{Z}, \mathbb{F}_p) = 1$

and $\dim H^2(\mathbb{Z}/p\mathbb{Z}, \mathbb{F}_p) = 1$: $r(G) = 1 + 1 + 1 = 3$. The three relations are x_1^p, x_2^p, and $[x_1, x_2]$.

The generator and relation ranks combine into a single invariant that measures the "efficiency" of a presentation.

Definition 6.2.5: Euler–Poincaré Characteristic

Let G be a finitely generated pro-p-group with $d(G)$ and $r(G)$ both finite. The *Euler–Poincaré characteristic* of G (at p) is:

$$\chi_p(G) = 1 - d(G) + r(G) = \sum_{i=0}^{\infty} (-1)^i \dim_{\mathbb{F}_p} H^i(G, \mathbb{F}_p)$$

when the sum converges (in particular, when $\mathrm{cd}_p(G) < \infty$).

The alternating sum formula extends to higher degrees: $\dim H^0(G, \mathbb{F}_p) = 1$ (the trivial module $\mathbb{F}_p$ has one-dimensional invariants), $\dim H^1(G, \mathbb{F}_p) = d(G)$, and $\dim H^2(G, \mathbb{F}_p) = r(G)$. For groups of cohomological dimension ≤ 2, the characteristic is $\chi_p(G) = 1 - d(G) + r(G)$, and all higher terms vanish.

Proposition 6.2.6: Multiplicativity of the Euler–Poincaré Characteristic

Let G be a finitely generated pro-p-group with $\mathrm{cd}_p(G) < \infty$, and let $H \leq G$ be an open subgroup of index $[G : H] = n$. Then:

$$\chi_p(H) = n \cdot \chi_p(G)$$

***Proof*:**
The formula $\mathrm{Cor} \circ \mathrm{Res} = [G : H]$ (proposition 3.3.6) implies that the restriction map on each $H^i(G, \mathbb{F}_p)$ has image satisfying $[G : H] \cdot H^i(G, \mathbb{F}_p) \subseteq \mathrm{Cor}(H^i(H, \mathbb{F}_p))$. Since $[G : H]$ is a power of p and the cohomology groups are $\mathbb{F}_p$-vector spaces, a direct argument using the LHS spectral sequence (theorem 3.4.4) for $H \trianglelefteq G$ (or a transfer argument for H not necessarily normal) gives the multiplicativity. The argument is identical to the one used for the Schreier formula (theorem 6.1.6): the Euler characteristic $\chi_p(G) = \sum (-1)^i \dim H^i(G, \mathbb{F}_p)$ transforms under restriction to an open subgroup of index n by multiplication by n.

Example 6.2.7: Euler–Poincaré Characteristics

1. $G = F(d)$ (free pro-p of rank d): $\chi_p(F(d)) = 1 - d + 0 = 1 - d$. The Schreier formula (theorem 6.1.6) for an open subgroup H of index n gives $\chi_p(H) = 1 - (n(d-1) + 1) = n(1 - d) = n \cdot \chi_p(G)$, confirming multiplicativity.

2. $G = \mathbb{Z}_p^n$: the cohomology $H^i(\mathbb{Z}_p^n, \mathbb{F}_p) \cong \bigwedge^i \mathbb{F}_p^n$ has dimension $\binom{n}{i}$, so $\chi_p(\mathbb{Z}_p^n) = \sum_{i=0}^n (-1)^i \binom{n}{i} = (1-1)^n = 0$ for $n \geq 1$.

3. $G = \mathbb{Z}/p\mathbb{Z}$: $\chi_p(\mathbb{Z}/p\mathbb{Z}) = 1 - 1 + 1 - 1 + \cdots$ does not converge in the usual sense. The finite group $\mathbb{Z}/p\mathbb{Z}$ has infinite cohomological dimension (its cohomology is periodic and nontrivial

in every degree), so the Euler–Poincaré characteristic is not defined. This is consistent with the definition requiring $\mathrm{cd}_p(G) < \infty$.

Exercise 6.2.1

1. Verify the relation rank $r(\mathbb{Z}_p^3) = 3$ by writing an explicit minimal presentation $1 \to R \to F(3) \to \mathbb{Z}_p^3 \to 1$ and identifying three generators of R as a normal subgroup of $F(3)$.
2. Show that for a nontrivial finite p-group G, both $d(G) \geq 1$ and $r(G) \geq 1$. Conclude that every nontrivial finite p-group requires at least one relation.
3. Derive the relation rank formula $r(G) = \dim H^2(G, \mathbb{F}_p)$ for a not-necessarily-minimal presentation $1 \to R \to F(m) \to G \to 1$ (with $m \geq d(G)$). Show that the formula still holds: $\dim H^2(G, \mathbb{F}_p)$ is independent of the presentation.
4. Let $G = \mathbb{Z}_p^2$ and $H = p\mathbb{Z}_p \times \mathbb{Z}_p$ (an open subgroup of index p). Compute $\chi_p(H)$ directly and verify the multiplicativity $\chi_p(H) = [G : H] \cdot \chi_p(G)$.
5. (If the reader has read [16]) Let Σ_g be a closed orientable surface of genus $g \geq 1$. The fundamental group $\pi_1(\Sigma_g)$ has the standard presentation $\langle a_1, b_1, \ldots, a_g, b_g \big| [a_1, b_1] \cdots [a_g, b_g] \rangle$. Compute d and r for the pro-p-completion $G = \widehat{\pi_1(\Sigma_g)}^{(p)}$, and verify that $\chi_p(G) = 2 - 2g$.

6.3 The Golod–Shafarevich Inequality and the Class Field Tower

Given a finitely presented pro-p-group G with $d = d(G)$ generators and $r = r(G)$ relations, how are d and r constrained? The Golod–Shafarevich inequality provides a fundamental answer: if G is a nontrivial finite p-group, then $r > d^2/4$. The contrapositive—if $r \leq d^2/4$, then G is infinite—resolved the class field tower problem, one of the central open questions in algebraic number theory in the first half of the 20th century.

The proof uses the Poincaré series (Hilbert series) of the associated graded algebra. The key idea is that the growth of the graded pieces of a finitely presented algebra is constrained by the number of generators and relations, and a finite group forces the growth to terminate, which is incompatible with having too few relations.

Definition 6.3.1: Augmentation Filtration and Poincaré Series

Let G be a pro-p-group and $\mathbb{F}_p[[G]]$ its completed group algebra. The *augmentation ideal* I is the kernel of the augmentation $\mathbb{F}_p[[G]] \to \mathbb{F}_p$. The *augmentation filtration* is $I \supseteq I^2 \supseteq I^3 \supseteq \cdots$. The *Poincaré series* of G is the formal power series:

$$H_G(t) = \sum_{n=0}^{\infty} a_n t^n, \qquad a_n = \dim_{\mathbb{F}_p} I^n / I^{n+1}$$

with the convention $I^0 = \mathbb{F}_p[[G]]$, so $a_0 = 1$.

For a finite p-group G, the augmentation ideal I is nilpotent ($I^N = 0$ for N sufficiently large, since $|G|$ is a power of p and $\mathbb{F}_p[[G]] = \mathbb{F}_p[G]$ has dimension $|G|$). In this case, $H_G(t)$ is a polynomial of degree $N-1$, and $\sum_n a_n = \dim_{\mathbb{F}_p} \mathbb{F}_p[G] = |G|$.

For the free pro-p-group $F(d)$, the completed group algebra $\mathbb{F}_p[[F(d)]]$ is isomorphic to the ring of formal power series $\mathbb{F}_p\langle\langle x_1, \ldots, x_d\rangle\rangle$ in d non-commuting variables. The augmentation ideal J is generated by $x_1, \ldots, x_d$, and $a_n = d^n$ (the monomials of degree n in d non-commuting variables form a basis for J^n/J^{n+1}). The Poincaré series is $H_{F(d)}(t) = 1/(1-dt)$.

A presentation $1 \to R \to F(d) \to G \to 1$ induces a surjection $\mathbb{F}_p[[F(d)]] \twoheadrightarrow \mathbb{F}_p[[G]]$, and the relations constrain the growth of the Poincaré series. The following inequality makes this constraint precise.

Lemma 6.3.2: The Fundamental Inequality for Poincaré Series

Let G be a finitely generated pro-p-group with minimal presentation $1 \to R \to F(d) \to G \to 1$, where $d = d(G)$ and $r = r(G)$. Let $a_n = \dim_{\mathbb{F}_p} I^n/I^{n+1}$ be the coefficients of the Poincaré series of G. Then for all $n \geq 2$:

$$a_n \geq d \cdot a_{n-1} - r \cdot a_{n-2}$$

with $a_0 = 1$ and $a_1 = d$.

***Proof*:**
The surjection $\mathbb{F}_p[[F(d)]] \twoheadrightarrow \mathbb{F}_p[[G]]$ sends J^n onto I^n. The space I^n/I^{n+1} is generated by products $\bar{x}_i \cdot m$ where $\bar{x}_1, \ldots, \bar{x}_d$ are the images of the generators in I/I^2 and $m \in I^{n-1}/I^n$. These $d \cdot a_{n-1}$ products span I^n/I^{n+1}, but some may be linearly dependent due to the relations.

The relation subgroup R is contained in the Frattini subgroup $\Phi(F(d))$ of the free group (since the presentation is minimal). In terms of the completed group algebra, $\Phi(F(d))$ corresponds to $1 + J^2$, so each relation $\rho_j \in R$ has the form $\rho_j = 1 + f_j$ with $f_j \in J^2$. The image of f_j in J^2/J^3 contributes a relation of weight 2 in the associated graded algebra. Each such relation, when multiplied by elements of J^{n-2}/J^{n-1}, can kill at most a_{n-2} independent elements in J^n/J^{n+1}. With r independent relations, the total number of dependencies is at most $r \cdot a_{n-2}$, giving $a_n \geq d \cdot a_{n-1} - r \cdot a_{n-2}$.

The inequality can be reformulated in terms of power series. Define $c_n = a_n - d \cdot a_{n-1} + r \cdot a_{n-2}$ for $n \geq 2$, with $c_0 = 1$ and $c_1 = 0$. The lemma gives $c_n \geq 0$ for all n. In power series form:

$$H_G(t) \cdot (1 - dt + rt^2) = C(t) = \sum_{n=0}^{\infty} c_n t^n$$

where $C(t)$ has non-negative coefficients and constant term $c_0 = 1$. This reformulation is the key to the proof.

Theorem 6.3.3: Golod–Shafarevich Inequality

Let G be a nontrivial finite p-group with $d = d(G)$ generators and $r = r(G)$ relations. Then $r > d^2/4$.

***Proof*:**
Step 1: Setup. Since G is finite, $H_G(t)$ is a polynomial with non-negative coefficients (all $a_n \geq 0$) and $H_G(0) = 1$. For any $t > 0$, the value $H_G(t) \geq 1$ (since $a_0 = 1$ and all $a_n \geq 0$). The power series identity $H_G(t)(1 - dt + rt^2) = C(t)$ holds with $C(t)$ having non-negative coefficients and $C(0) = 1$, so $C(t) \geq 1$ for all $t > 0$.

Step 2: Contradiction from a positive root. Suppose $r \leq d^2/4$. The quadratic $q(t) = 1 - dt + rt^2$ has discriminant $\Delta = d^2 - 4r \geq 0$, so it has real roots. The smaller root is:

$$t_0 = \frac{d - \sqrt{d^2 - 4r}}{2r} > 0$$

(taking $r > 0$; the case $r = 0$ gives $q(t) = 1 - dt$ with root $1/d > 0$). Since $q(0) = 1 > 0$ and $q(t_0) = 0$, the quadratic is positive on $(0, t_0)$.

For $0 < t < t_0$: the identity $H_G(t) = C(t)/q(t)$ holds with both $C(t) \geq 1$ and $q(t) > 0$, giving $H_G(t) \geq 1/q(t)$. As $t \to t_0^-$, the denominator $q(t) \to 0^+$, so $H_G(t) \to +\infty$.

Step 3: Finite groups have polynomial Poincaré series. Since G is finite, $H_G(t)$ is a polynomial, and in particular $H_G(t_0)$ is a finite real number. This contradicts $H_G(t) \to +\infty$ as $t \to t_0^-$. The assumption $r \leq d^2/4$ is false, so $r > d^2/4$.

The inequality extends to infinite pro-p-groups in contrapositive form.

Theorem 6.3.4: Golod–Shafarevich Criterion for Infinite Groups

Let G be a finitely presented pro-p-group with $d = d(G)$ and $r = r(G)$. If $r \leq d^2/4$, then G is infinite.

***Proof*:**
If G were finite, theorem 6.3.3 would give $r > d^2/4$, contradicting the hypothesis.

The bound $d^2/4$ is essentially sharp: for each d, there exist finite p-groups with r slightly larger than $d^2/4$.

The original application of the Golod–Shafarevich inequality was to the class field tower problem, which asks whether the tower of successive Hilbert class fields of a number field must terminate.

Definition 6.3.5: The p-Class Field Tower

Let K be a number field and p a prime. The *p-class field tower* of K is the sequence of fields:

$$K = K_0 \subseteq K_1 \subseteq K_2 \subseteq \cdots$$

where K_{i+1} is the maximal unramified abelian p-extension of K_i. The tower *terminates* if $K_n = K_{n+1}$ for some n (equivalently, if $\mathrm{Cl}(K_n)$ has trivial p-part). The union $K_\infty = \bigcup K_i$ is a Galois extension of K, and $G = \mathrm{Gal}(K_\infty/K)$ is a pro-p-group.

The tower terminates if and only if G is finite. The generator and relation ranks of G are determined by the arithmetic of K. By class field theory ([17]), the first layer K_1/K has $\mathrm{Gal}(K_1/K) \cong \mathrm{Cl}(K)\{p\}$ (the p-primary part of the class group), so the abelianization $G^{\mathrm{ab}} \cong \mathrm{Cl}(K)\{p\}$ and:

$$d(G) = \dim_{\mathbb{F}_p} \mathrm{Cl}(K)/p\,\mathrm{Cl}(K) = \text{the } p\text{-rank of } \mathrm{Cl}(K)$$

The relation rank $r(G)$ is bounded above by the number of primes of K lying above p together with contributions from roots of unity and the signature of K (see [18] for the precise formula).

Theorem 6.3.6: Existence of Infinite Class Field Towers

Let K be a number field, p a prime, and $G = \mathrm{Gal}(K_\infty/K)$ the Galois group of the p-class field tower. If $d(G)^2 > 4r(G)$, then G is infinite and the p-class field tower does not terminate.

***Proof*:**
Direct application of theorem 6.3.4: the condition $r(G) < d(G)^2/4$ (equivalently $d(G)^2 > 4r(G)$) forces G to be infinite.

In practice, $d(G) = \mathrm{rk}_p(\mathrm{Cl}(K))$ is computable from the class group, while $r(G)$ is bounded above by arithmetic data of K. The precise bound (see [18]) takes the form $r(G) \leq d(G) + \rho$, where ρ depends on the number of primes above p, the unit rank, and the roots of unity in K. Substituting into the criterion: the tower is infinite whenever $d(G)^2 - 4d(G) - 4\rho > 0$, i.e., $d(G) > 2 + 2\sqrt{1+\rho}$.

Example 6.3.7: The First Infinite Class Field Tower

1. The first explicit example was given by Golod and Shafarevich in 1964: let $K = \mathbb{Q}(\sqrt{-2\cdot 3\cdot 5\cdot 7\cdot 11\cdot 13}) = \mathbb{Q}(\sqrt{-30030})$. The discriminant of K is divisible by $t = 6$ finite primes. By genus theory (see [4]), the 2-rank of $\mathrm{Cl}(K)$ satisfies $d = d(G) \geq t - 1 = 5$. The relation rank is bounded by the arithmetic of K: the bound from [18] gives $r(G) \leq d(G) + \rho$ where ρ depends on the primes above 2 and the signature. For this specific field, the computation yields $d(G)^2 > 4r(G)$, so the 2-class field tower is infinite by theorem 6.3.6.

2. More generally, for any prime p, there exist number fields with infinite p-class field towers. For $p = 2$: the field $K = \mathbb{Q}(\sqrt{-\ell_1 \cdots \ell_t})$ where $\ell_1, \ldots, \ell_t$ are distinct odd primes satisfying $\ell_i \equiv 1 \pmod 4$ has 2-rank of $\mathrm{Cl}(K)$ at least $t-1$. Choosing t large enough (the criterion $d^2 > 4(d+\rho)$ is satisfied for $t \geq 7$ with standard bounds on ρ) guarantees an infinite 2-class field tower.

The existence of infinite class field towers answered a question that had been open since Furtwängler's work in the 1930s: the Hilbert class field tower does not always terminate, so there is no general procedure for "unwinding" the class group to reach a field with trivial class number. This was one of the first applications of pro-p-group theory to a problem in algebraic number theory, and it demonstrated the power of cohomological methods in a setting where direct computation is infeasible.

6.3.8 Remark: Golod–Shafarevich Beyond Number Theory The Golod–Shafarevich inequality has applications far beyond class field towers. Golod used it to construct the first examples of infinite finitely generated p-groups (answering the *general Burnside problem* in the negative for p-groups): if $d \geq 2$ and $r \leq d^2/4$, the quotient $F(d)/\overline{\langle \rho_1, \ldots, \rho_r \rangle}$ is an infinite pro-p-group, and its discrete quotients provide infinite finitely generated torsion groups. In ring theory, the same technique produces infinite-dimensional nil algebras over $\mathbb{F}_p$, disproving the Kurosh conjecture.

Exercise 6.3.1

1. Verify the Golod–Shafarevich inequality directly for the cyclic group C_{p^2}: compute $d(C_{p^2})$ and $r(C_{p^2})$ from a minimal pro-p-presentation, and check that $r > d^2/4$.
2. Show that the free pro-p-group $F(d)$ with $d \geq 2$ does not contradict the Golod–Shafarevich inequality (since $F(d)$ is infinite, the inequality does not apply). Compute $H_{F(d)}(t)$ and verify that $H_{F(d)}(t)(1 - dt) = 1$.
3. Let $G = \mathbb{Z}/p\mathbb{Z}$. Compute the Poincaré series $H_G(t)$ directly by determining $\dim_{\mathbb{F}_p} I^n/I^{n+1}$ for the augmentation ideal I of $\mathbb{F}_p[G]$. Verify that $H_G(t)(1 - t + t^2) = C(t)$ for a specific polynomial $C(t)$ with non-negative coefficients.
4. (If the reader has read [4]) Let $p = 2$ and $K = \mathbb{Q}(\sqrt{-d})$ where d is a positive squarefree integer divisible by t distinct odd primes, all congruent to $1 \pmod 4$. Show that the 2-rank of $\mathrm{Cl}(K)$ is at least $t - 1$ (using genus theory), and determine the smallest t for which the Golod–Shafarevich criterion guarantees an infinite 2-class field tower.
5. Formulate and prove a graded version of the Golod–Shafarevich inequality: if the r relations in a minimal presentation have weights $w_1 \leq w_2 \leq \cdots \leq w_r$ (where w_j is the largest k such that $\rho_j \in I^k$), show that the Poincaré series satisfies $H_G(t)(1 - dt + \sum_{j=1}^{r} t^{w_j}) = C(t)$ with $C(t)$ having non-negative coefficients.

6.4 Demushkin Groups

At the opposite extreme from free pro-p-groups ($\mathrm{cd}_p = 1$, no relations) sit the Demushkin groups: pro-p-groups of cohomological dimension 2 whose cohomology satisfies Poincaré duality. These groups were introduced in definition 5.3.5 and previewed in example 5.3.6. This section develops their classification through the cup product pairing on H^1, which determines the group up to a single discrete invariant.

Recall the definition: a *Demushkin group* is a pro-p-group G with $\mathrm{cd}_p(G) = 2$, $H^1(G, \mathbb{F}_p) \cong \mathbb{F}_p^d$ for some finite $d \geq 2$, and $H^2(G, \mathbb{F}_p) \cong \mathbb{F}_p$. By theorem 6.2.2 and theorem 6.2.3, these conditions translate to: $d(G) = d$ generators and $r(G) = 1$ relation. A Demushkin group is a one-relator pro-p-group of cohomological dimension 2.

The cup product from theorem 4.2.4 endows the cohomology of any profinite group with a graded ring structure. For a Demushkin group, the cup product in degree 1 produces the key structural invariant.

Proposition 6.4.1: The Cup Product Pairing

Let G be a Demushkin group with $d(G) = d$. The cup product:

$$\smile: H^1(G, \mathbb{F}_p) \times H^1(G, \mathbb{F}_p) \to H^2(G, \mathbb{F}_p) \cong \mathbb{F}_p$$

is a non-degenerate bilinear pairing.

***Proof*:**
By proposition 5.3.4, a Demushkin group is a Poincaré duality group of dimension 2 at p. The Poincaré duality pairing in dimensions i and $2 - i$ (with $\mathbb{F}_p$-coefficients) is a perfect pairing $H^i(G, \mathbb{F}_p) \times H^{2-i}(G, \mathbb{F}_p) \to H^2(G, \mathbb{F}_p) \cong \mathbb{F}_p$, realized by the cup product. Taking $i = 1$ gives the stated non-degenerate pairing.

The pairing $\smile$ is a bilinear form on the $\mathbb{F}_p$-vector space $V = H^1(G, \mathbb{F}_p) \cong \mathbb{F}_p^d$. Its symmetry type is determined by the prime p.

Proposition 6.4.2: Symmetry of the Cup Product Pairing

Let G be a Demushkin group. The cup product pairing on $H^1(G, \mathbb{F}_p)$ is:

1. alternating (i.e., $\alpha \smile \alpha = 0$ for all $\alpha \in H^1(G, \mathbb{F}_p)$) when p is odd;
2. symmetric when $p = 2$.

In particular, for p odd, $d(G)$ is necessarily even.

***Proof*:**
The graded-commutativity of the cup product (theorem 4.2.4) gives $\alpha \smile \beta = (-1)^{pq} \beta \smile \alpha$ for $\alpha \in H^p$, $\beta \in H^q$. With $p = q = 1$: $\alpha \smile \beta = -\beta \smile \alpha$. Taking $\beta = \alpha$: $\alpha \smile \alpha = -\alpha \smile \alpha$, so $2(\alpha \smile \alpha) = 0$.

For p odd, 2 is invertible in $\mathbb{F}_p$, so $\alpha \smile \alpha = 0$: the pairing is alternating. A non-degenerate alternating form on $\mathbb{F}_p^d$ exists only when d is even (the standard classification of alternating forms over fields of odd characteristic).

For $p = 2$, the condition $2(\alpha \smile \alpha) = 0$ is automatic (since $2 = 0$ in $\mathbb{F}_2$), so no constraint beyond symmetry is imposed.

The classification of Demushkin groups proceeds from the classification of the cup product pairing. For p odd, a non-degenerate alternating form on $\mathbb{F}_p^d$ (with d even) has a unique isomorphism class—the standard symplectic form—so the pairing alone does not distinguish Demushkin groups. The additional invariant comes from the action on p-power roots of unity.

Definition 6.4.3: Type of a Demushkin Group

Let G be a Demushkin group. The *type* of G is the supernatural number $q \in \{p, p^2, p^3, \dots\} \cup \{0\}$ defined as follows. Consider the maximal abelian quotient $G^{\mathrm{ab}} = G/[G, G]$ and the p-adic cyclotomic character $\chi\colon G^{\mathrm{ab}} \to \mathbb{Z}_p^\times$ classifying the action of G on the Tate module $\mathbb{Z}_p(1) = \varprojlim_n \mu_{p^n}$. The type q is the order of $\mathrm{im}(\chi)$ in $\mathbb{Z}_p^\times$ (with $q = 0$ when χ is trivial, i.e., G acts trivially on all p-power roots of unity).

For the reader unfamiliar with the cyclotomic character in this abstract setting: the type q can equivalently be defined through the relation in a minimal presentation. Given a minimal presentation $1 \to R \to F(d) \to G \to 1$ with $R = \overline{\langle\rho\rangle}^{F(d)}$ (the normal closure of a single element $\rho \in F(d)$), the element ρ determines q through its expression in terms of commutators and p-powers of the free generators.

Theorem 6.4.4: Classification of Demushkin Groups

Let p be an odd prime. A Demushkin group G with $d(G) = d$ (necessarily even) is determined up to isomorphism by the pair (d, q), where q is the type of G. For each even $d \geq 2$ and each $q \in \{p, p^2, p^3, \dots\} \cup \{0\}$, there exists a unique Demushkin group $G(d, q)$, presented as:

$$G(d, q) = \langle x_1, \dots, x_d \,\big|\, x_1^q [x_1, x_2][x_3, x_4] \cdots [x_{d-1}, x_d] = 1 \rangle$$

where $x_1^0 = 1$ is interpreted as the absence of the power term (i.e., the relation is $[x_1, x_2][x_3, x_4] \cdots [x_{d-1}, x_d] = 1$ when $q = 0$).

The proof of the classification uses the structure theory of non-degenerate alternating forms over $\mathbb{Z}_p$ and a careful analysis of the relation in the completed group algebra; see Labute [14] and Serre [21] for complete proofs.

The classification for $p = 2$ is more involved: the cup product pairing is symmetric rather than alternating, and the classification of symmetric bilinear forms over $\mathbb{F}_2$ depends on additional invariants (the Arf invariant and the rank modulo 2). The result is that Demushkin 2-groups are classified by a triple (d, q, ϵ) where $d \geq 2$, $q \in \{2, 4, 8, \dots\} \cup \{0\}$, and $\epsilon \in \{0, 1\}$ is the Arf invariant of the quadratic form associated to the cup product pairing (see [14] for the precise statement).

Example 6.4.5: Demushkin Groups

1. The simplest Demushkin group has $d = 2$ and relation $x_1^q[x_1, x_2] = 1$ (for p odd). When $q = 0$: the group is $G(2, 0) = \langle x_1, x_2 \,\big|\, [x_1, x_2] = 1 \rangle \cong \mathbb{Z}_p^2$ (the free abelian pro-p-group of rank 2). For $q = p$: $G(2, p) = \langle x_1, x_2 \,\big|\, x_1^p[x_1, x_2] = 1 \rangle$, a non-abelian group in which x_1 has finite order modulo the commutator subgroup.

2. A Demushkin group of type $q = 0$ has the property that all generators have infinite order and the single relation is purely a product of commutators. The group $G(d, 0) = \langle x_1, \dots, x_d \,\big|\, [x_1, x_2] \cdots [x_{d-1}, x_d] = 1 \rangle$ is torsion-free: no nonzero element has finite order.

3. (If the reader has read [16]) Let Σ_g be a closed orientable surface of genus $g \geq 1$. The fundamental group $\pi_1(\Sigma_g)$ has presentation $\langle a_1, b_1, \dots, a_g, b_g \,\big|\, [a_1, b_1] \cdots [a_g, b_g] = 1 \rangle$. The pro-$p$-completion $\widehat{\pi_1(\Sigma_g)}^{(p)}$ is a Demushkin group with $d = 2g$ and type $q = 0$—the presentation is

exactly the Demushkin presentation $G(2g, 0)$. This connects the classification of Demushkin groups to the topology of surfaces: the Demushkin groups of type 0 are precisely the pro-p surface groups. The Euler–Poincaré characteristic is $\chi_p = 1 - 2g + 1 = 2 - 2g$ (exercise 6.2.1 (5)), matching the topological Euler characteristic of Σ_g.

4. The Golod–Shafarevich inequality (theorem 6.3.3) is automatically satisfied for Demushkin groups: $r(G) = 1$ and $d(G) \geq 2$, so $d^2/4 \geq 1 = r$ (with equality when $d = 2$). This is consistent with Demushkin groups being infinite.

The Demushkin classification has consequences for the structure of the cohomology ring:

Proposition 6.4.6: Cohomology Ring of a Demushkin Group

Let G be a Demushkin group with $d(G) = d$ and p odd. The cohomology ring $H^*(G, \mathbb{F}_p)$ is an exterior algebra on d generators in degree 1 modulo the ideal generated by the kernel of the cup product pairing. Explicitly:

1. $H^0(G, \mathbb{F}_p) = \mathbb{F}_p$.
2. $H^1(G, \mathbb{F}_p) = \mathbb{F}_p^d$.
3. $H^2(G, \mathbb{F}_p) = \mathbb{F}_p$ (generated by any nonzero cup product $\alpha_i \smile \alpha_j$ with α_i, α_j forming part of a symplectic basis).
4. $H^n(G, \mathbb{F}_p) = 0$ for $n \geq 3$ (since $\mathrm{cd}_p(G) = 2$).

***Proof*:**
Parts (1), (2), and (4) follow from the definition. For (3): $H^2(G, \mathbb{F}_p) \cong \mathbb{F}_p$ by definition, and the cup product $H^1 \times H^1 \to H^2$ is surjective (since it is non-degenerate by proposition 6.4.1 and $H^2 \cong \mathbb{F}_p$). Choose a symplectic basis $\alpha_1, \beta_1, \ldots, \alpha_{d/2}, \beta_{d/2}$ for $H^1(G, \mathbb{F}_p)$ such that $\alpha_i \smile \beta_i$ is a generator of H^2 for each i, and all other pairings vanish. Then H^2 is generated by any single $\alpha_i \smile \beta_i$.

The maximal pro-p quotients of absolute Galois groups of local fields are Demushkin groups (for p odd); the identification and computation of their invariants from local field data is carried out in Chapter 8 using local Tate duality.

Exercise 6.4.1

1. Let G be a Demushkin group with p odd and $d(G) = 4$. Write out the cup product pairing on $H^1(G, \mathbb{F}_p) \cong \mathbb{F}_p^4$ in matrix form with respect to a symplectic basis, and verify it is non-degenerate and alternating.
2. For p odd, describe all Demushkin groups with $d = 2$ by listing the possible types q and writing the explicit presentation for each. Show that $G(2, 0) \cong \mathbb{Z}_p^2$.
3. Show that a Demushkin group of type $q = 0$ is torsion-free. *Hint:* the relation $[x_1, x_2] \cdots [x_{d-1}, x_d] = 1$ does not force any generator to have finite order. Use the fact that the abelianization $G^{\mathrm{ab}} \cong \mathbb{Z}_p^d$ is torsion-free.

4. Verify that the Golod–Shafarevich inequality $r > d^2/4$ fails for a Demushkin group G with $d(G) = 2$ (i.e., $r = 1 \leq 1 = d^2/4$). Explain why this does not contradict theorem 6.3.3.

5. (If the reader has read [16]) Let $G = \widehat{\pi_1(\Sigma_g)}^{(p)}$ be the pro-p-completion of the fundamental group of a genus-g surface, with p odd. Compute the full cohomology ring $H^*(G, \mathbb{F}_p)$ and compare it with the singular cohomology ring $H^*(\Sigma_g, \mathbb{F}_p)$.

7

The Brauer Group

The cohomology group $H^2(G, M)$ classifies extensions of G by M (theorem 2.4.5). When $G = G_K$ is the absolute Galois group and $M = (K^{\text{sep}})^\times$, this classification acquires an independent algebraic identity: the Brauer group $\operatorname{Br}(K)$, which parametrizes central simple algebras over K up to Morita equivalence. This chapter develops the Brauer group from both the algebraic and cohomological perspectives, establishes the identification $\operatorname{Br}(K) \cong H^2(G_K, (K^{\text{sep}})^\times)$ via crossed product algebras, and computes $\operatorname{Br}(K)$ for local and global fields. The Brauer group is the object connecting the abstract cohomological machinery of Parts I and II to the arithmetic of Part III.

7.1 Central Simple Algebras and the Brauer Group

The Brauer group arises from a classification problem in algebra: given a field K, describe all division algebras with center K, up to the identification that treats a division algebra and its matrix rings as equivalent. The resulting group $\operatorname{Br}(K)$ is an invariant of the field that turns out to be computable by cohomological methods.

Definition 7.1.1: Central Simple Algebra

A *central simple algebra* (CSA) over a field K is a finite-dimensional K-algebra A satisfying:

1. the center of A is K (centrality), and
2. A has no two-sided ideals other than 0 and A (simplicity).

The two conditions interact: centrality prevents A from decomposing as a product of algebras over a proper extension of K, while simplicity prevents A from having a nontrivial quotient. Together, they force A to be a matrix algebra over a division ring.

Theorem 7.1.2: Wedderburn's Structure Theorem

Every central simple algebra A over K is isomorphic to $M_n(D)$ for a unique (up to isomorphism) division algebra D with center K and a unique positive integer n. The division algebra D is the *Wedderburn component* of A, and $\dim_K A = n^2 \dim_K D$.

***Proof*:**
This is the Artin-Wedderburn theorem (see [7, Artin-Wedderburn, Artin-Wedderburn For k-Algebras]) specialized to algebras with center K. The simplicity of A ensures a single matrix block, and the centrality ensures the division ring D has center K.

The dimension $\dim_K D$ is always a perfect square: if D has center K and $\dim_K D = m^2$, then m is the *degree* of D, and $\dim_K A = (nm)^2$.

Example 7.1.3: Central Simple Algebras

1. The matrix algebra $M_n(K)$ is a CSA over K with Wedderburn component $D = K$. These are the *split* CSAs.
2. The real quaternion algebra $\mathbb{H} = \mathbb{R} \oplus \mathbb{R}i \oplus \mathbb{R}j \oplus \mathbb{R}k$ (with $i^2 = j^2 = -1$, $ij = k = -ji$) is a CSA over $\mathbb{R}$ of dimension 4. Since $\mathbb{H}$ is a division algebra ($\mathbb{H}$ has no zero divisors), its Wedderburn component is $D = \mathbb{H}$ with $n = 1$.
3. For any $a, b \in K^\times$, the *quaternion algebra* $(a, b)_K$ is the 4-dimensional K-algebra generated by elements i, j subject to $i^2 = a$, $j^2 = b$, $ij = -ji$. This is a CSA over K. The algebra $(a, b)_K$ is either isomorphic to $M_2(K)$ (split) or is a division algebra (non-split), and these are the only two possibilities since $\dim_K(a, b)_K = 4 = 2^2$.
4. (If the reader has read [7]) The endomorphism algebra $\mathrm{End}_D(V)$ of a finite-dimensional right D-module V is a CSA over the center of D. When $D = K$ and $V = K^n$, this recovers $M_n(K)$.

The quaternion algebra $(a, b)_K$ is the most important family of CSAs for arithmetic applications. The condition for splitting—$(a, b)_K \cong M_2(K)$—can be tested by checking whether the equation $ax^2 + by^2 = 1$ has a solution in K (the norm form of the quaternion algebra).

Two CSAs are identified if they have the same Wedderburn component.

Definition 7.1.4: Brauer Equivalence

Two central simple algebras A and B over K are *Brauer equivalent*, written $A \sim B$, if their Wedderburn components are isomorphic: $A \cong M_n(D)$ and $B \cong M_m(D)$ for the same division algebra D.

Equivalently, $A \sim B$ if and only if $A \otimes_K M_r(K) \cong B \otimes_K M_s(K)$ for some $r, s \geq 1$. Brauer equivalence classes form a group under tensor product.

Definition 7.1.5: The Brauer Group

The *Brauer group* of K, denoted $\mathrm{Br}(K)$, is the set of Brauer equivalence classes of central simple algebras over K, with group operation $[A] \cdot [B] = [A \otimes_K B]$.

The identity element is $[K] = [M_n(K)]$ (the class of all split algebras). The inverse of $[A]$ is $[A^{\mathrm{op}}]$ (the opposite algebra, with multiplication reversed), since $A \otimes_K A^{\mathrm{op}} \cong M_{n^2}(K)$ where $n^2 = \dim_K A$ (the isomorphism sends $a \otimes b$ to the endomorphism $x \mapsto axb$).

Proposition 7.1.6: Vanishing of the Brauer Group

$\mathrm{Br}(K) = 0$ if and only if every central simple algebra over K is split, i.e., every finite-dimensional division algebra with center K is K itself.

***Proof*:**
The Brauer group is trivial if and only if every CSA is Brauer equivalent to K, i.e., every CSA is isomorphic to some $M_n(K)$. By theorem 7.1.2, this holds if and only if the only division algebra with center K is K itself.

Example 7.1.7: Brauer Groups of Familiar Fields

1. $\mathrm{Br}(\mathbb{C}) = 0$: the field $\mathbb{C}$ is algebraically closed, so every polynomial has a root. A division algebra D with center $\mathbb{C}$ would have $D^\times$ containing an element α with $\mathbb{C}(\alpha)/\mathbb{C}$ a nontrivial extension, but no such extensions exist.
2. $\mathrm{Br}(\mathbb{R}) \cong \mathbb{Z}/2\mathbb{Z}$: the only division algebras with center $\mathbb{R}$ are $\mathbb{R}$ itself (the trivial class) and $\mathbb{H}$ (the quaternion class). The isomorphism sends $[\mathbb{H}] \mapsto 1 \in \mathbb{Z}/2\mathbb{Z}$. The class $[\mathbb{H}]$ has order 2: $\mathbb{H} \otimes_{\mathbb{R}} \mathbb{H} \cong M_4(\mathbb{R})$ (as verified in exercise 7.1.1 (1)).
3. $\mathrm{Br}(\mathbb{F}_q) = 0$ for any finite field $\mathbb{F}_q$: by Wedderburn's theorem (every finite division ring is a field—see [6]), every finite-dimensional division algebra over $\mathbb{F}_q$ with center $\mathbb{F}_q$ is $\mathbb{F}_q$ itself.
4. $\mathrm{Br}(K) = 0$ whenever K is separably closed: if D is a division algebra with center K, any maximal subfield $L \subseteq D$ is a finite extension of K, hence equals K (since K is separably closed and L/K is separable). So $D = K$, and every CSA is split.

Field extensions induce homomorphisms between Brauer groups.

Definition 7.1.8: Base Change and the Relative Brauer Group

For a field extension L/K, the *base change* (or *scalar extension*) map is the group homomorphism:

$$\mathrm{res}_{L/K} \colon \mathrm{Br}(K) \to \mathrm{Br}(L), \qquad [A] \mapsto [A \otimes_K L]$$

The kernel $\mathrm{Br}(L/K) = \ker(\mathrm{res}_{L/K})$ is the *relative Brauer group*: the set of classes split by L.

An element $[A] \in \mathrm{Br}(L/K)$ is a CSA over K that becomes a matrix algebra after extending scalars to L. The relative Brauer groups exhaust $\mathrm{Br}(K)$: every CSA is split by some finite extension (in fact, by a finite Galois extension—a splitting field for the Wedderburn component).

Proposition 7.1.9: Splitting by Finite Galois Extensions

$\mathrm{Br}(K) = \bigcup_{L/K} \mathrm{Br}(L/K)$, where the union ranges over finite Galois extensions L/K inside K^{sep}.

***Proof*:**
Let $[D] \in \mathrm{Br}(K)$ with D a division algebra of degree m. A maximal subfield of D (i.e., a subfield $L \subseteq D$ with $[L : K] = m$) splits D: the map $D \otimes_K L \to M_m(L)$ defined by left multiplication is an isomorphism. Every division algebra contains a separable maximal subfield (by the double centralizer theorem, see [7, Centralizer of Subalgebras]). Replacing L by its Galois closure over K (which still splits D, since the splitting property passes to extensions) gives a finite Galois splitting field.

The period and index of a Brauer class measure its complexity from two perspectives.

Definition 7.1.10: Period and Index

Let $[A] \in \mathrm{Br}(K)$ with Wedderburn component D. The *period* of $[A]$ is:

$$\mathrm{per}(A) = \mathrm{ord}\,([A]) \text{ in } \mathrm{Br}(K)$$

The *index* of $[A]$ is:

$$\mathrm{ind}(A) = \deg(D) = \sqrt{\dim_K D}$$

Proposition 7.1.11: Period Divides Index

For any $[A] \in \mathrm{Br}(K)$: $\mathrm{per}(A) \,\big|\, \mathrm{ind}(A)$, and $\mathrm{per}(A)$ and $\mathrm{ind}(A)$ have the same prime factors.

***Proof*:**
Let D be the Wedderburn component of A, with $\deg(D) = m$. A maximal subfield $L \subseteq D$ satisfies $[L : K] = m$, and $D \otimes_K L \cong M_m(L)$, so $[D]$ lies in $\mathrm{Br}(L/K)$. The first divisibility $\mathrm{per}(A) \,\big|\, \mathrm{ind}(A)$ follows from the fact that $[A]^{[L:K]} = 0$ for any splitting field L of A: this is a consequence of the corestriction formula for the Brauer group, which will be established via the cohomological identification in section 7.2. A self-contained algebraic proof proceeds by showing that the norm $N_{L/K}$ on CSAs sends the split class over L to the $[L : K]$th power of $[A]$ in $\mathrm{Br}(K)$ (see [6] for the algebraic construction of the Brauer group norm).

The statement about prime factors follows from the primary decomposition of $\mathrm{Br}(K)$: if $[A] = \sum_p [A_p]$ with $[A_p] \in \mathrm{Br}(K)\{p\}$, then $\mathrm{ind}(A) = \prod_p \mathrm{ind}(A_p)$ and $\mathrm{per}(A) = \mathrm{lcm}_p\, \mathrm{per}(A_p)$. If $p \,\big|\, \mathrm{ind}(A)$, then $\mathrm{ind}(A_p) > 1$, forcing $\mathrm{per}(A_p) \geq p$ (since the trivial class is the only class of index 1), so $p \,\big|\, \mathrm{per}(A)$.

The question of when $\mathrm{per}(A) = \mathrm{ind}(A)$—and, more generally, what bounds exist on ind in terms of per—is the period–index problem, addressed in section 7.4.

Exercise 7.1.1

1. Show that $\mathbb{H} \otimes_{\mathbb{R}} \mathbb{H} \cong M_4(\mathbb{R})$. *Hint:* exhibit an explicit isomorphism using the left and right multiplication actions of $\mathbb{H}$ on itself.
2. Let K be a field with $\mathrm{char} K \neq 2$ and $a, b \in K^\times$. Show that the quaternion algebra $(a, b)_K$ splits (is isomorphic to $M_2(K)$) if and only if the equation $ax^2 + by^2 = 1$ has a solution $(x, y) \in K^2$.
3. Show that if $L_1 \subseteq L_2$ are Galois extensions of K, then $\mathrm{Br}(L_1/K) \subseteq \mathrm{Br}(L_2/K)$. Conclude that the union in proposition 7.1.9 is a filtered colimit.
4. Let $[A] \in \mathrm{Br}(K)$ with $\mathrm{per}(A) = e$ and $\mathrm{ind}(A) = m$. Show that if p is a prime dividing m, then $p|e$. *Hint:* consider the p-primary component of $[A]$ in $\mathrm{Br}(K)$.
5. Show that for any fields K_1 and K_2, the Brauer group of their product (as a ring, if defined) satisfies $\mathrm{Br}(K_1 \times K_2) \cong \mathrm{Br}(K_1) \times \mathrm{Br}(K_2)$. What goes wrong if K_1 and K_2 are not fields?

7.2 Crossed Product Algebras and H^2

The Brauer group was defined in section 7.1 as an algebraic object classifying central simple algebras. The central result of this chapter is that $\mathrm{Br}(K)$ is a cohomology group: the explicit construction of a CSA from a 2-cocycle (the crossed product) provides an isomorphism $H^2(\mathrm{Gal}(L/K), L^\times) \cong \mathrm{Br}(L/K)$ for any finite Galois extension L/K, and passing to the colimit gives $\mathrm{Br}(K) \cong H^2(G_K, (K^{\mathrm{sep}})^\times)$.

Definition 7.2.1: Crossed Product Algebra

Let L/K be a finite Galois extension with $G = \mathrm{Gal}(L/K)$ and let $f \in Z^2(G, L^\times)$ be a 2-cocycle (definition 2.3.7). The *crossed product algebra* $A_f = (L/K, f)$ is the left L-vector space with basis $\{u_\sigma\}_{\sigma \in G}$:

$$A_f = \bigoplus_{\sigma \in G} L \cdot u_\sigma$$

equipped with the multiplication rule:

$$(a \cdot u_\sigma)(b \cdot u_\tau) = a \cdot \sigma(b) \cdot f(\sigma, \tau) \cdot u_{\sigma\tau}$$

for $a, b \in L$ and $\sigma, \tau \in G$, extended K-linearly.

The multiplication rule encodes two pieces of structure: the Galois action ($\sigma(b)$ appears when commuting u_σ past $b \in L$) and the failure of the formal symbols u_σ to form a group ($u_\sigma u_\tau = f(\sigma, \tau) u_{\sigma\tau}$ rather than simply $u_{\sigma\tau}$). The 2-cocycle f measures this failure.

Proposition 7.2.2: Crossed Products Are Central Simple

The crossed product $A_f = (L/K, f)$ is a central simple algebra over K of dimension $[L : K]^2$.

***Proof*:**
Step 1: Associativity. The product of three elements $(a \cdot u_\sigma)(b \cdot u_\tau)(c \cdot u_\rho)$ can be computed in two ways. On one hand: $((a \cdot u_\sigma)(b \cdot u_\tau))(c \cdot u_\rho) = a\sigma(b)f(\sigma,\tau) \cdot \sigma\tau(c) \cdot f(\sigma\tau,\rho) \cdot u_{\sigma\tau\rho}$. On the other: $(a \cdot u_\sigma)((b \cdot u_\tau)(c \cdot u_\rho)) = a\sigma(b\tau(c)f(\tau,\rho)) \cdot f(\sigma,\tau\rho) \cdot u_{\sigma\tau\rho}$. Equality of these expressions reduces to the 2-cocycle condition: $\sigma(f(\tau,\rho)) \cdot f(\sigma,\tau\rho) = f(\sigma,\tau) \cdot f(\sigma\tau,\rho)$, which holds by definition 2.3.7.

Step 2: Dimension. As an L-vector space, A_f has dimension $|G| = [L : K]$, so $\dim_K A_f = [L : K]^2$.

Step 3: Center. An element $x = \sum_\sigma a_\sigma u_\sigma$ lies in the center if and only if $xb = bx$ for all $b \in L$ and $xu_\tau = u_\tau x$ for all $\tau \in G$. The condition $xb = bx$ forces $a_\sigma\sigma(b) = a_\sigma b$ for all σ and b, which gives $a_\sigma = 0$ whenever $\sigma \neq \mathrm{id}$ (since $\sigma \neq \mathrm{id}$ means $\sigma(b) \neq b$ for some b). So $x = a_{\mathrm{id}}u_{\mathrm{id}} \in L$. The condition $xu_\tau = u_\tau x$ then gives $a_{\mathrm{id}} = \tau(a_{\mathrm{id}})$ for all τ, so $a_{\mathrm{id}} \in L^G = K$.

Step 4: Simplicity. Let $I \subseteq A_f$ be a nonzero two-sided ideal. Choose a nonzero $x = \sum_\sigma a_\sigma u_\sigma \in I$ with the minimal number of nonzero coefficients. By multiplying on the left by a_σ^{-1} and on the right by u_σ^{-1} for a suitable σ, assume $a_{\mathrm{id}} = 1$. For any $b \in L$, the commutator $bx - xb = \sum_{\sigma \neq \mathrm{id}}(b - \sigma(b))a_\sigma u_\sigma$ lies in I and has fewer nonzero terms (the id-coefficient vanishes). By minimality, $bx = xb$ for all b, forcing $a_\sigma = 0$ for $\sigma \neq \mathrm{id}$ (as in Step 3). So $x = u_{\mathrm{id}} = f(\mathrm{id},\mathrm{id})^{-1} \cdot 1_{A_f}$, and $I = A_f$.

Different cocycles can produce the same Brauer class: cohomologous cocycles give Brauer-equivalent algebras.

Proposition 7.2.3: Cohomologous Cocycles Give Equivalent Algebras

Let $f, f' \in Z^2(G, L^\times)$ with $f' = f \cdot d^1\psi$ for some $\psi \colon G \to L^\times$ (i.e., $f'(\sigma,\tau) = f(\sigma,\tau) \cdot \psi(\sigma) \cdot \sigma(\psi(\tau)) \cdot \psi(\sigma\tau)^{-1}$). Then $A_{f'} \cong A_f$ as K-algebras. In particular, $[A_{f'}] = [A_f]$ in $\mathrm{Br}(K)$.

***Proof*:**
Define $\varphi \colon A_{f'} \to A_f$ by $\varphi(a \cdot u'_\sigma) = a \cdot \psi(\sigma) \cdot u_\sigma$. A direct computation shows φ is a K-algebra homomorphism: $\varphi((a \cdot u'_\sigma)(b \cdot u'_\tau)) = a\sigma(b)f'(\sigma,\tau)\psi(\sigma\tau)u_{\sigma\tau}$ and $\varphi(a \cdot u'_\sigma)\varphi(b \cdot u'_\tau) = a\psi(\sigma)\sigma(b\psi(\tau))f(\sigma,\tau)u_{\sigma\tau}$. These are equal by the relation $f'(\sigma,\tau) = f(\sigma,\tau)\psi(\sigma)\sigma(\psi(\tau))\psi(\sigma\tau)^{-1}$. Since φ is a bijection (it maps a basis to a basis), it is an isomorphism.

The converse—that non-cohomologous cocycles give non-equivalent algebras—combines with surjectivity to establish the main isomorphism.

Theorem 7.2.4: Cohomological Identification of the Relative Brauer Group

Let L/K be a finite Galois extension with $G = \mathrm{Gal}(L/K)$. The map $f \mapsto [A_f]$ induces an isomorphism of abelian groups:

$$H^2(G, L^\times) \xrightarrow{\sim} \mathrm{Br}(L/K)$$

***Proof*:**
Step 1: Well-definedness. By proposition 7.2.2, A_f is a CSA over K. By proposition 7.2.3, the class $[A_f] \in \mathrm{Br}(K)$ depends only on $[f] \in H^2(G, L^\times)$. The crossed product A_f is split by L: the map $A_f \otimes_K L \to M_n(L)$ (where $n = [L : K]$) is an isomorphism, constructed by letting A_f act on L^n via the regular representation. So $[A_f] \in \mathrm{Br}(L/K)$.

Step 2: Homomorphism. The multiplication $[f]\cdot[g] = [fg]$ in H^2 maps to $[A_f]\cdot[A_g] = [A_f \otimes_K A_g]$. The identification $A_f \otimes_K A_g \sim A_{fg}$ (Brauer equivalence, not isomorphism) requires showing that the tensor product of crossed products is Brauer-equivalent to a crossed product with the product cocycle. This follows from the Skolem–Noether theorem: any automorphism of a CSA is inner, which allows reducing the tensor product to a single crossed product.

Step 3: Surjectivity. Every class in $\mathrm{Br}(L/K)$ is represented by a crossed product. Given $[A] \in \mathrm{Br}(L/K)$, the algebra A contains L as a maximal commutative subalgebra (since A is split by L and $\dim_K A = [L : K]^2$). A set-theoretic section $\sigma \mapsto u_\sigma \in A^\times$ of the map $N_{A^\times}(L^\times) \to G$ (the normalizer of $L^\times$ in $A^\times$ maps onto G by the Skolem–Noether theorem) determines a cocycle $f(\sigma, \tau) = u_\sigma u_\tau u_{\sigma\tau}^{-1} \in L^\times$, and $A \cong A_f$.

Step 4: Injectivity. If A_f is split ($A_f \cong M_n(K)$), the cocycle f is a coboundary. An isomorphism $A_f \xrightarrow{\sim} M_n(K)$ restricts to an embedding $L \hookrightarrow M_n(K)$, and the Skolem–Noether theorem provides an element $c \in M_n(K)^\times$ conjugating this embedding to the standard one. The section u_σ is then conjugated to diagonal matrices, and the cocycle f becomes a coboundary determined by c.

The relative identification extends to the full Brauer group by passing to the colimit over all finite Galois extensions.

Theorem 7.2.5: The Brauer Group as H^2

For any field K:
$$\mathrm{Br}(K) \cong H^2(G_K, (K^{\mathrm{sep}})^\times)$$
where $G_K = \mathrm{Gal}(K^{\mathrm{sep}}/K)$ is the absolute Galois group.

***Proof*:**
By proposition 7.1.9, $\mathrm{Br}(K) = \bigcup_{L/K} \mathrm{Br}(L/K) = \varinjlim_{L/K} \mathrm{Br}(L/K)$, where the colimit is over finite Galois extensions ordered by inclusion. The isomorphisms $H^2(\mathrm{Gal}(L/K), L^\times) \cong \mathrm{Br}(L/K)$ from theorem 7.2.4 are compatible with the inflation maps (on the cohomological side) and the inclusion maps (on the Brauer group side). Passing to the colimit and applying theorem 3.2.2: $\varinjlim_{L/K} H^2(\mathrm{Gal}(L/K), L^\times) = H^2(G_K, (K^{\mathrm{sep}})^\times)$.

This identification converts the algebraic classification of CSAs into a cohomological computation, making the full machinery of Parts I and II applicable. The corestriction-restriction formula (proposition 3.3.6) now applies to $\mathrm{Br}(K)$: for a finite extension L/K of degree n, the composition $\mathrm{Br}(K) \xrightarrow{\mathrm{res}} \mathrm{Br}(L) \xrightarrow{\mathrm{cor}} \mathrm{Br}(K)$ is multiplication by n. This gives an immediate proof that $\mathrm{per}(A) \big| [L : K]$ for any splitting field L of A, completing the argument deferred in proposition 7.1.11.

The identification also settles the relationship between the Brauer group and cohomological dimension.

Corollary 7.2.6: $\mathrm{cd}_p \leq 1$ and the Brauer Group

For a field K and a prime p: $\mathrm{cd}_p(K) \leq 1$ if and only if $\mathrm{Br}(L)\{p\} = 0$ for every finite separable extension L/K.

***Proof*:**
By theorem 7.2.5, $\mathrm{Br}(L)\{p\} = H^2(G_L, (L^{\mathrm{sep}})^\times)\{p\}$. If $\mathrm{cd}_p(K) \leq 1$, then $\mathrm{cd}_p(L) \leq 1$ for every finite separable L/K (by proposition 5.1.3, since G_L is a closed subgroup of G_K), so $H^2(G_L, M) = 0$ for all p-torsion M. Since $(L^{\mathrm{sep}})^\times$ is a filtered colimit of p-torsion modules (via Kummer sequences), the p-part of H^2 vanishes. Conversely, if $\mathrm{Br}(L)\{p\} = 0$ for all L, then $H^2(G_L, \mathbb{F}_p) = 0$ for all L (since $H^2(G_L, \mathbb{F}_p)$ embeds into $\mathrm{Br}(L)\{p\}$ via Kummer theory), and Serre's criterion (theorem 5.1.2) gives $\mathrm{cd}_p(K) \leq 1$.

This corollary was stated without proof in proposition 5.2.3; the cohomological identification now provides the missing step.

Example 7.2.7: Crossed Products in Practice

1. The trivial cocycle $f(\sigma, \tau) = 1$ for all σ, τ gives the split crossed product: $A_1 = \bigoplus_\sigma L \cdot u_\sigma$ with $u_\sigma u_\tau = u_{\sigma\tau}$. The map $A_1 \to M_n(L)$ sending each u_σ to the permutation matrix of σ is an isomorphism after extending scalars, and $A_1 \cong M_n(K)$.
2. For $L = K(\sqrt{a})$ with $G = \{1, \sigma\}$ where $\sigma(\sqrt{a}) = -\sqrt{a}$: a 2-cocycle is determined by a single value $f(\sigma, \sigma) = b \in K^\times$ (the other values are forced by the cocycle condition and the normalization $f(1, \sigma) = f(\sigma, 1) = 1$). The crossed product A_f has basis $\{1, \sqrt{a}, u_\sigma, \sqrt{a} \cdot u_\sigma\}$ over K, with $u_\sigma^2 = b$ and $u_\sigma \sqrt{a} = -\sqrt{a} \cdot u_\sigma$. Setting $i = \sqrt{a}$ and $j = u_\sigma$: $i^2 = a$, $j^2 = b$, $ij = -ji$. This is the quaternion algebra $(a, b)_K$.
3. (If the reader has read [8]) The Weil pairing on an elliptic curve E/K gives a map $E[n] \times E[n] \to \mu_n$, which induces a cup product pairing $H^1(G_K, E[n]) \times H^1(G_K, E[n]) \to H^2(G_K, \mu_n) \subseteq \mathrm{Br}(K)[n]$. The elements of $\mathrm{Br}(K)$ arising from this pairing measure the failure of local-to-global principles for rational points on E.

Exercise 7.2.1

1. Carry out the full associativity verification of proposition 7.2.2 in detail: expand both bracketings of $(au_\sigma)(bu_\tau)(cu_\rho)$ and show they agree using the cocycle condition.
2. Show that the crossed product with the trivial cocycle ($f = 1$) is isomorphic to $M_n(K)$ by constructing an explicit isomorphism. *Hint:* let A_1 act on L^n by $u_\sigma(v) = \sigma(v)$ for the standard basis.
3. Compute $H^2(\mathrm{Gal}(\mathbb{C}/\mathbb{R}), \mathbb{C}^\times)$ directly from the periodic resolution of C_2 and identify the nontrivial class with the quaternion algebra $\mathbb{H}$.
4. Using the identification $\mathrm{Br}(K) = H^2(G_K, (K^{\mathrm{sep}})^\times)$ and the corestriction formula, show that every element of $\mathrm{Br}(K)$ has finite order. Conclude that $\mathrm{Br}(K)$ is a torsion group.

5. (If the reader has read [7]) State and prove the Skolem-Noether theorem: every K-algebra automorphism of a central simple algebra A over K is inner (i.e., conjugation by an element of $A^\times$). Use this to verify that the surjectivity argument in the proof of theorem 7.2.4 is valid.

7.3 The Brauer Group of Local and Global Fields

The identification $\mathrm{Br}(K) \cong H^2(G_K, (K^{\mathrm{sep}})^\times)$ from theorem 7.2.5 converts the classification of central simple algebras into a cohomological computation. For local and global fields, this computation yields a complete and explicit description: $\mathrm{Br}(K) \cong \mathbb{Q}/\mathbb{Z}$ for a local field K, and the Brauer group of a number field is determined by local data via an exact sequence. The proofs of these results require Hilbert's Theorem 90 and the local and global duality theorems, developed in Chapters 8 and 9. This section states the results and develops their consequences through explicit computations.

The Brauer group of a local field is as simple as possible: cyclic algebras suffice.

Theorem 7.3.1: The Invariant Map for Local Fields

Let K be a local field (a finite extension of $\mathbb{Q}_p$ or $\mathbb{F}_q((t))$). There is a canonical isomorphism:

$$\mathrm{inv}_K \colon \mathrm{Br}(K) \xrightarrow{\sim} \mathbb{Q}/\mathbb{Z}$$

called the *invariant map* (or *Hasse invariant*). It has the following properties:

1. For the maximal unramified extension K^{nr}/K of degree n (i.e., $[K_n^{\mathrm{nr}} : K] = n$ where K_n^{nr} is the unique unramified extension of degree n), the restriction $\mathrm{inv}_K|_{\mathrm{Br}(K_n^{\mathrm{nr}}/K)}$ is an isomorphism $\mathrm{Br}(K_n^{\mathrm{nr}}/K) \xrightarrow{\sim} \frac{1}{n}\mathbb{Z}/\mathbb{Z}$.
2. Every element of $\mathrm{Br}(K)$ is split by an unramified extension. That is, $\mathrm{Br}(K) = \bigcup_n \mathrm{Br}(K_n^{\mathrm{nr}}/K)$.
3. For a finite extension L/K of local fields, the diagram

$$\begin{tikzcd} \mathrm{Br}(K) \arrow[r, "\mathrm{inv}_K"] \arrow[d, "\mathrm{res}_{L/K}"'] & \mathbb{Q}/\mathbb{Z} \arrow[d, "\times[L:K]"] \\ \mathrm{Br}(L) \arrow[r, "\mathrm{inv}_L"'] & \mathbb{Q}/\mathbb{Z} \end{tikzcd}$$

commutes: $\mathrm{inv}_L(\mathrm{res}_{L/K}([A])) = [L : K] \cdot \mathrm{inv}_K([A])$.

The proof is given in Chapter 8 using the cohomological machinery: the invariant map is constructed from the identification $\mathrm{Br}(K^{\mathrm{nr}}/K) = H^2(\mathrm{Gal}(K^{\mathrm{nr}}/K), (K^{\mathrm{nr}})^\times)$ and the valuation $v\colon (K^{\mathrm{nr}})^\times \to \mathbb{Z}$. Since $\mathrm{Gal}(K^{\mathrm{nr}}/K) \cong \widehat{\mathbb{Z}}$ (topologically generated by the Frobenius), the cohomology is computable by the methods of example 2.3.8.

Property (2) is the statement that unramified extensions suffice to split every CSA—a consequence of Hilbert's Theorem 90 (proved in Chapter 8). Property (3) expresses the compatibility of the invariant map with field extensions and is the local ingredient in the global theory.

Example 7.3.2: Computing Local Invariants

1. For $K = \mathbb{Q}_p$ and the quaternion algebra $(a, b)_{\mathbb{Q}_p}$: the invariant $\mathrm{inv}_{\mathbb{Q}_p}((a, b)_{\mathbb{Q}_p}) \in \{0, 1/2\} \subseteq \mathbb{Q}/\mathbb{Z}$ (since quaternion algebras have period dividing 2). The value is 0 iff $(a, b)_{\mathbb{Q}_p}$ splits, i.e., iff $ax^2 + by^2 = 1$ is solvable in $\mathbb{Q}_p$ (exercise 7.1.1 (2)).

2. The quaternion algebra $(-1, -1)_{\mathbb{Q}_2}$ does not split over $\mathbb{Q}_2$: the equation $-x^2 - y^2 = 1$ has no solution in $\mathbb{Q}_2$ (since -1 is not a sum of two squares in $\mathbb{Q}_2$, verifiable by checking modulo 8). So $\mathrm{inv}_{\mathbb{Q}_2}((-1, -1)_{\mathbb{Q}_2}) = 1/2$.

3. For p odd: $(-1, -1)_{\mathbb{Q}_p}$ splits if and only if -1 is a sum of two squares in $\mathbb{Q}_p$. By Hensel's lemma, this reduces to checking modulo p: $(-1, -1)_{\mathbb{Q}_p}$ splits for all $p \equiv 1 \pmod 4$ (since -1 is a square mod p) and for $p = 2$ the previous item gives non-splitting. For $p \equiv 3 \pmod 4$: the equation $-x^2 - y^2 \equiv 1 \pmod p$ is solvable (counting solutions via character sums gives $p - 1 > 0$ solutions), so $(-1, -1)_{\mathbb{Q}_p}$ splits.

4. An element $[A] \in \mathrm{Br}(\mathbb{Q}_p)$ with $\mathrm{inv}_{\mathbb{Q}_p}([A]) = 1/n$ is represented by a cyclic algebra: take the unramified extension $L/\mathbb{Q}_p$ of degree n and a uniformizer π of $\mathbb{Q}_p$, and form the cyclic algebra $(L/\mathbb{Q}_p, \mathrm{Frob}, \pi) = \bigoplus_{i=0}^{n-1} L \cdot u^i$ with $u^n = \pi$ and $u \cdot \ell = \mathrm{Frob}(\ell) \cdot u$ for $\ell \in L$. This is a division algebra of degree n with invariant $1/n$.

For the archimedean places, the Brauer group is already known.

Proposition 7.3.3: Archimedean Brauer Groups

$\mathrm{Br}(\mathbb{R}) \cong \mathbb{Z}/2\mathbb{Z}$ (with $\mathrm{inv}_{\mathbb{R}}([\mathbb{H}]) = 1/2$) and $\mathrm{Br}(\mathbb{C}) = 0$. The invariant map for $\mathbb{R}$ is $\mathrm{inv}_{\mathbb{R}} \colon \mathrm{Br}(\mathbb{R}) \xrightarrow{\sim} \frac{1}{2}\mathbb{Z}/\mathbb{Z} \subseteq \mathbb{Q}/\mathbb{Z}$.

***Proof*:**
$\mathrm{Br}(\mathbb{C}) = 0$ since $\mathbb{C}$ is algebraically closed (example 7.1.7). For $\mathrm{Br}(\mathbb{R})$: $\mathrm{Br}(\mathbb{R}) = \mathrm{Br}(\mathbb{C}/\mathbb{R}) = H^2(\mathrm{Gal}(\mathbb{C}/\mathbb{R}), \mathbb{C}^\times) = H^2(C_2, \mathbb{C}^\times) \cong \mathbb{Z}/2\mathbb{Z}$ (exercise 7.2.1 (3)). The nontrivial class is $[\mathbb{H}]$, with $\mathrm{inv}_{\mathbb{R}}([\mathbb{H}]) = 1/2$.

The global theory assembles the local invariants into a single exact sequence.

Theorem 7.3.4: Albert–Brauer–Hasse–Noether

Let K be a number field. A central simple algebra A over K is split ($[A] = 0$ in $\mathrm{Br}(K)$) if and only if $A \otimes_K K_v$ is split for every completion K_v of K (at all finite and archimedean places v).

The ABHN theorem is a local-global principle for central simple algebras: the Brauer group $\mathrm{Br}(K)$ embeds into the product of local Brauer groups. The theorem says that the only obstruction to splitting a CSA over a number field is a collection of local obstructions, one at each place. This is the Brauer-group analogue of the Hasse–Minkowski theorem for quadratic forms.

The ABHN theorem is sharpened by the following exact sequence, which describes the image of the localization map.

Theorem 7.3.5: Brauer Group Exact Sequence

For a number field K, there is an exact sequence of abelian groups:

$$0 \to \mathrm{Br}(K) \xrightarrow{\bigoplus_v \mathrm{res}_v} \bigoplus_v \mathrm{Br}(K_v) \xrightarrow{\sum_v \mathrm{inv}_v} \mathbb{Q}/\mathbb{Z} \to 0$$

where the first sum is over all places v of K (finite and archimedean), and the second map sends a tuple of local Brauer classes to the sum of their invariants.

The proof is given in Chapter 9. The injectivity on the left is the ABHN theorem. The surjectivity on the right is nontrivial: it says that for any collection of local invariants $\alpha_v \in \mathbb{Q}/\mathbb{Z}$ (almost all zero, with $\alpha_v \in \frac{1}{2}\mathbb{Z}/\mathbb{Z}$ at the real places and $\alpha_v = 0$ at the complex places) summing to zero, there exists a CSA over K with exactly those local invariants.

The exactness in the middle is the reciprocity constraint: the sum of all local invariants of a global Brauer class is zero. This is a cohomological manifestation of the global reciprocity law.

Example 7.3.6: The Brauer Group Exact Sequence in Practice

1. Consider the quaternion algebra $(-1,-1)_{\mathbb{Q}}$ over $\mathbb{Q}$. At each place v, the local invariant is $\mathrm{inv}_v((-1,-1)_{\mathbb{Q}_v}) \in \{0, 1/2\}$. From example 7.3.2: the algebra is non-split at $v = 2$ (invariant $1/2$) and at $v = \infty$ (invariant $1/2$, since $(-1,-1)_{\mathbb{R}} = \mathbb{H}$). At all odd primes p: the algebra splits (invariant 0). The sum of invariants is $1/2 + 1/2 = 0$ in $\mathbb{Q}/\mathbb{Z}$, confirming the reciprocity constraint. Since the invariant is nonzero at $v = 2$, the algebra $(-1,-1)_{\mathbb{Q}}$ does not split over $\mathbb{Q}$: it is a division algebra ($(-1,-1)_{\mathbb{Q}} \cong \mathbb{H}$, viewed as a $\mathbb{Q}$-algebra... but $\mathbb{H}$ has center $\mathbb{R}$, not $\mathbb{Q}$). More precisely, $(-1,-1)_{\mathbb{Q}}$ is a 4-dimensional division algebra over $\mathbb{Q}$ that becomes $\mathbb{H}$ after extending scalars to $\mathbb{R}$.

2. The quaternion algebra $(2,5)_{\mathbb{Q}}$: compute the local invariants. At $v = \infty$: $(2,5)_{\mathbb{R}}$ splits (the equation $2x^2 + 5y^2 = 1$ has the real solution $x = 1/\sqrt{2}$, $y = 0$). So $\mathrm{inv}_\infty = 0$. At $v = p$ for $p \neq 2, 5$: the algebra splits by Hensel's lemma (since 2 and 5 are units in $\mathbb{Z}_p$ for $p \neq 2, 5$, and the norm form represents 1 modulo p by a counting argument). At $v = 5$: $2x^2 + 5y^2 = 1$ in $\mathbb{Q}_5$ requires $2x^2 \equiv 1 \pmod 5$, i.e., $x^2 \equiv 3 \pmod 5$. Since 3 is not a square modulo 5 ($1^2 = 1$, $2^2 = 4$, $3^2 = 4$, $4^2 = 1$), the algebra does not split at $v = 5$: $\mathrm{inv}_5 = 1/2$. At $v = 2$: a direct check modulo 8 shows $2x^2 + 5y^2 \equiv 1 \pmod 8$ is solvable (take $x = 1, y = 1$: $2 + 5 = 7 \equiv -1 \pmod 8$; take $x = 0, y = 1$: $5 \not\equiv 1$; take $x = 1, y = 0$: $2 \not\equiv 1$). A more careful Hensel-lifting analysis gives $\mathrm{inv}_2 = 1/2$. The reciprocity check: $\mathrm{inv}_2 + \mathrm{inv}_5 = 1/2 + 1/2 = 0$ in $\mathbb{Q}/\mathbb{Z}$. So $(2,5)_{\mathbb{Q}}$ is a non-split quaternion algebra over $\mathbb{Q}$, ramified at 2 and 5.

3. The exact sequence implies $\mathrm{Br}(\mathbb{Q}) \hookrightarrow \mathrm{Br}(\mathbb{R}) \oplus \bigoplus_p \mathrm{Br}(\mathbb{Q}_p) \cong \frac{1}{2}\mathbb{Z}/\mathbb{Z} \oplus \bigoplus_p \mathbb{Q}/\mathbb{Z}$. An element of $\mathrm{Br}(\mathbb{Q})$ is determined by a finite set of places where the invariant is nonzero, subject to the constraint that the invariants sum to zero. For quaternion algebras (period 2): the ramification set must have even cardinality (since each invariant is $1/2$ and the sum must be 0).

Corollary 7.3.7: Period Equals Index for Local and Global Fields

For a local field K: $\mathrm{per}(A) = \mathrm{ind}(A)$ for every $[A] \in \mathrm{Br}(K)$. For a number field K: $\mathrm{per}(A) = \mathrm{ind}(A)$ for every $[A] \in \mathrm{Br}(K)$.

***Proof*:**
For a local field: by theorem 7.3.1, $\mathrm{Br}(K) \cong \mathbb{Q}/\mathbb{Z}$. An element of order n in $\mathbb{Q}/\mathbb{Z}$ is a/n with $\gcd(a, n) = 1$. The corresponding CSA is a cyclic algebra of degree n (constructed from the unramified extension of degree n), which is a division algebra (its invariant a/n has denominator n, so it is not split by any proper subextension). So $\mathrm{ind} = n = \mathrm{per}$.

For a number field: the ABHN theorem reduces the problem to the local case. An element $[A] \in \mathrm{Br}(K)$ with $\mathrm{per}(A) = n$ has local invariants $\mathrm{inv}_v([A]) \in \frac{1}{n}\mathbb{Z}/\mathbb{Z}$ at each place v. By the Grunwald–Wang theorem (see [18]), there exists a cyclic extension L/K of degree n that splits A at every place. The CSA split by L has index dividing $[L : K] = n$, and since $\mathrm{per} \mid \mathrm{ind}$ (proposition 7.1.11), equality follows.

Exercise 7.3.1

1. Compute $\mathrm{inv}_{\mathbb{Q}_p}((p, p)_{\mathbb{Q}_p})$ for an odd prime p. *Hint:* determine whether $px^2 + py^2 = 1$ is solvable in $\mathbb{Q}_p$.
2. Show that a quaternion algebra $(a, b)_{\mathbb{Q}}$ (with $a, b \in \mathbb{Q}^\times$) is non-split if and only if the set of places where it has invariant $1/2$ is nonempty. Determine this set for $(a, b) = (-1, 3)$.
3. Using the exact sequence of theorem 7.3.5, show that the natural map $\mathrm{Br}(\mathbb{Q}) \to \mathrm{Br}(\mathbb{Q}(\sqrt{5}))$ has kernel isomorphic to $\mathbb{Z}/2\mathbb{Z}$, and identify the nontrivial element.
4. Let K be a totally imaginary number field (no real embeddings). Show that the archimedean contribution to the Brauer group exact sequence is trivial (all complex places give $\mathrm{Br}(K_v) = 0$), and the exact sequence becomes $0 \to \mathrm{Br}(K) \to \bigoplus_{\mathfrak{p}} \mathrm{Br}(K_{\mathfrak{p}}) \to \mathbb{Q}/\mathbb{Z} \to 0$.
5. Using the exact sequence, show that every element of $\mathrm{Br}(K)$ (for K a number field) has finite order, and that the exponent of $\mathrm{Br}(K)[n]$ (the n-torsion) is n.

7.4 The Period–Index Problem

The period and index of a Brauer class (definition 7.1.10) measure its complexity from two perspectives: the period is the order in the Brauer group (an algebraic invariant), and the index is the degree of the Wedderburn component (a geometric invariant, related to the dimension of the underlying division algebra). The divisibility $\mathrm{per}(A) \mid \mathrm{ind}(A)$ (proposition 7.1.11) and the equality $\mathrm{per} = \mathrm{ind}$ for local and number fields (corollary 7.3.7) are the starting point. The period–index problem asks: for a given field K, what is the precise relationship between these two invariants?

Definition 7.4.1: The Period–Index Problem

For a field K, the *period–index problem* asks for the optimal bound $\operatorname{ind}(A) \mid \operatorname{per}(A)^{e(K)}$ that holds for all $[A] \in \operatorname{Br}(K)$, where $e(K) \geq 0$ is the smallest exponent with this property. When $e(K) = 0$, the field satisfies $\operatorname{per} = \operatorname{ind}$.

The problem is trivial for C_1-fields ($\operatorname{Br}(K) = 0$ by proposition 5.2.3, so there is nothing to check) and is settled for local and number fields ($e(K) = 0$ by corollary 7.3.7). The first nontrivial cases arise for function fields over local fields and over finite fields.

Proposition 7.4.2: Period–Index Bound from cd

Let K be a field with $\operatorname{cd}(K) = d < \infty$. Then for every $[A] \in \operatorname{Br}(K)$:

$$\operatorname{ind}(A) \mid \operatorname{per}(A)^{d-1}$$

In particular, per and ind have the same prime factors, and the ratio $\operatorname{ind}(A)/\operatorname{per}(A)$ is bounded by $\operatorname{per}(A)^{d-2}$.

The proof of this bound uses the Merkurjev–Suslin theorem and norm residue isomorphism (see [18]); it is beyond the scope of this text. The bound is known to be sharp in some cases and conjectured to be sharp in general.

The content of the proposition is that cohomological dimension controls the gap between period and index: as $\operatorname{cd}(K)$ increases, larger gaps become possible. For $\operatorname{cd}(K) = 2$, no gap is possible ($\operatorname{per} = \operatorname{ind}$). For $\operatorname{cd}(K) = 3$, the index can be as large as the square of the period. This reflects a fundamental difference between fields whose Galois cohomology is concentrated in low degrees and those with more complex cohomological structure.

The bound $\operatorname{ind} \mid \operatorname{per}^{d-1}$ can be decomposed prime by prime: it suffices to prove $\operatorname{ind}(A_p) \mid \operatorname{per}(A_p)^{d-1}$ for each p-primary component A_p of $[A]$, since the period and index are multiplicative over the primary decomposition of $\operatorname{Br}(K)$.

Example 7.4.3: Period–Index Bounds for Specific Fields

1. $\operatorname{cd}(K) = 1$ (e.g., $K = \mathbb{F}_q$ or $K = \overline{\mathbb{F}}_q(t)$): $\operatorname{Br}(K) = 0$ by proposition 5.2.3. The bound $\operatorname{ind} \mid \operatorname{per}^0 = 1$ is vacuously satisfied.

2. $\operatorname{cd}(K) = 2$ (e.g., $K = \mathbb{Q}_p$ or $K = k((t))$ with k algebraically closed): the bound gives $\operatorname{ind} \mid \operatorname{per}^1$, i.e., $\operatorname{per} = \operatorname{ind}$. For local fields, this is corollary 7.3.7. For $k((t))$ with k algebraically closed of characteristic 0: the absolute Galois group $G_{k((t))}$ has $\operatorname{cd} = 2$ (the tame inertia is isomorphic to $\widehat{\mathbb{Z}}$, contributing one dimension, and the residue field k contributes $\operatorname{cd}(k) = 0$; the total is $1 + 1 = 2$ by the spectral sequence bound). The Brauer group $\operatorname{Br}(k((t))) \cong \mathbb{Q}/\mathbb{Z}$ has the same structure as for a local field, and the period–index equality holds.

3. $\operatorname{cd}(K) = 3$ (e.g., $K = \mathbb{Q}_p(t)$, the rational function field over a p-adic field): the bound gives $\operatorname{ind} \mid \operatorname{per}^2$. For $K = \mathbb{Q}_p(t)$, the period–index equality $\operatorname{per} = \operatorname{ind}$ can fail: there exist classes $[A] \in \operatorname{Br}(\mathbb{Q}_p(t))$ with $\operatorname{per}(A) = p$ and $\operatorname{ind}(A) = p^2$. Such classes arise from the interaction between the Brauer group of the residue field $\mathbb{Q}_p$ (which is $\mathbb{Q}/\mathbb{Z}$) and the function field structure. The bound $\operatorname{ind} \mid \operatorname{per}^2 = p^2$ is achieved.

4. Number fields: $\mathrm{cd}(\mathbb{Q}) = 2$ (for p odd) and per = ind holds (corollary 7.3.7). This is consistent with the bound $\mathrm{ind} \mid \mathrm{per}^1$.

The examples suggest a general pattern: the gap between period and index grows with the cohomological dimension of the field. To make the gap concrete, consider the following table summarizing the known behavior for small cohomological dimension:

$\mathrm{cd}(K)$	Bound on ind	per = ind?	Example fields
0	$\mathrm{ind} = 1$	trivially	separably closed
1	$\mathrm{Br}(K) = 0$	vacuous	$\mathbb{F}_q$, $\overline{\mathbb{F}}_q(t)$
2	$\mathrm{ind} \mid \mathrm{per}$	always	$\mathbb{Q}_p$, number fields
3	$\mathrm{ind} \mid \mathrm{per}^2$	not always	$\mathbb{Q}_p(t)$

The transition from $d = 2$ to $d = 3$ is the critical one: it is the first case where the period–index gap can be nontrivial, and understanding this transition is the focus of ongoing research.

The following conjecture makes the pattern precise.

Conjecture 7.4.4: Period–Index Conjecture

For a field K with $\mathrm{cd}(K) = d$, the bound $\mathrm{ind}(A) \mid \mathrm{per}(A)^{d-1}$ is optimal: for every d and every prime power p^k, there exists a field K of $\mathrm{cd}(K) = d$ and a class $[A] \in \mathrm{Br}(K)$ with $\mathrm{per}(A) = p^k$ and $\mathrm{ind}(A) = p^{k(d-1)}$.

The conjecture is known in the following cases:

Proposition 7.4.5: Known Cases of the Period–Index Conjecture

1. $d = 1$: $\mathrm{Br}(K) = 0$, the conjecture is vacuous.
2. $d = 2$: per = ind for all classes. This is known for local fields (corollary 7.3.7), and more generally for all fields of $\mathrm{cd} \leq 2$ by a theorem of Merkurjev (using the Merkurjev–Suslin theorem and the Bloch–Kato conjecture, now a theorem of Voevodsky).
3. $d = 3$: the bound $\mathrm{ind} \mid \mathrm{per}^2$ is known for function fields of p-adic curves by work of Saltman, Lieblich, and others. The sharpness (examples achieving the bound) is known by constructions of de Jong.

For $d \geq 4$, the conjecture remains open. The difficulty increases with d: the cohomological structure of G_K becomes more complex, and the algebraic geometry of the corresponding Severi–Brauer varieties (the projective varieties associated to CSAs) becomes harder to control. The case $d = 3$ is the current frontier, and the methods used there (moduli spaces of twisted sheaves, deformation theory) are specific to the geometry of surfaces and curves and do not generalize directly.

The period–index problem also has a local-to-global aspect. For a number field K, the equality per = ind is a consequence of the local-global principle (ABHN theorem, theorem 7.3.4): the global index is the lcm of local indices, and at each local place $\mathrm{per}_v = \mathrm{ind}_v$. For function fields $K(C)$ of curves C over other base fields, the analogous local-global statement (the "local-global principle for the index") is itself a nontrivial conjecture.

The period–index problem connects to several other areas of mathematics.

7.4.6 Remark: Connections to Algebraic Geometry and K-Theory (If the reader has read [2] or [3]) The period–index problem has geometric and K-theoretic reformulations. A Brauer class $[A] \in \mathrm{Br}(K)$ of period n determines an element $\alpha \in H^2_{\text{ét}}(\operatorname{Spec} K, \mu_n)$, and the index of $[A]$ is the minimum degree of a finite field extension splitting α. In the language of twisted sheaves on an algebraic variety X with function field K: the period of α is the order of the associated gerbe $\mathcal{G}_\alpha$ over X, while the index measures the minimum rank of a twisted sheaf on $\mathcal{G}_\alpha$. The conjecture $\mathrm{ind} \,\big|\, \mathrm{per}^{d-1}$ (with $d = \dim X + 1$ when $K = k(X)$ for a variety X over an algebraically closed field k) is then a statement about the geometry of twisted sheaves on varieties of dimension $\leq d-1$.
From the K-theoretic perspective, the period–index problem is related to the structure of the Grothendieck group K_0 of the category of A-modules (for a CSA A) and the behavior of Chern classes in étale cohomology.

The period–index problem remains one of the central open questions in the theory of the Brauer group. For the purposes of this book, the equality per = ind for local and number fields (corollary 7.3.7) is the essential result: it ensures that the Brauer group of these fields is completely determined by the invariant map, and that division algebras over these fields are controlled by their cohomological data.

Exercise 7.4.1

1. Show that every quaternion algebra $(a,b)_K$ satisfies $\mathrm{per}(A) = \mathrm{ind}(A)$ over any field K with $\mathrm{char} K \neq 2$. *Hint:* per and ind are both either 1 or 2.

2. Show that if K is a C_1-field (definition 5.2.5), then $\mathrm{Br}(K) = 0$, and conclude that the period–index problem is vacuous for such fields.

3. Show that the period–index problem reduces to the case $\mathrm{per}(A) = p^k$ (a prime power): if $\mathrm{ind}(A) \,\big|\, \mathrm{per}(A)^e$ holds for all p-primary classes and all primes p, then it holds for all classes.

4. Let $K = \overline{\mathbb{F}}_q(t)$ (the rational function field over the algebraic closure of a finite field). Show that $\mathrm{Br}(K) = 0$ by using $\mathrm{cd}(\overline{\mathbb{F}}_q(t)) = 1$ (example 5.2.4). What happens for $K = \mathbb{F}_q((t))$?

5. (If the reader has read [3]) A *symbol algebra* $(a,b)_{K,n}$ is the K-algebra generated by x, y with $x^n = a$, $y^n = b$, $yx = \zeta xy$ (where $\zeta \in K$ is a primitive nth root of unity). Show that $(a,b)_{K,n}$ is a CSA of degree n and period dividing n, and that the symbol algebras generate $\mathrm{Br}(K)[n]$ when K contains the nth roots of unity (this is the Merkurjev–Suslin theorem).

Part III

Galois Cohomology of Fields

Parts I and II built the cohomological machine and identified its structural invariants. Part III applies the machine to the Galois groups of local and global fields, proving the central theorems of arithmetic duality and class field theory in their cohomological formulation.

Chapter 8 computes the Galois cohomology of local fields: Hilbert 90 (both multiplicative and additive), the filtered cohomology of the unit group, local Tate duality, and the local reciprocity law via class formations and the Tate–Nakayama theorem. The self-contained Appendix develops the adèle ring and idèle group, providing the global Galois modules needed for Chapter 9. Chapter 9 proves the Albert–Brauer–Hasse–Noether theorem, constructs the global fundamental class, establishes the Poitou–Tate exact sequence and the global Euler–Poincaré characteristic, states the global reciprocity law (with the first inequality and existence theorem deferred to [17]), and connects the theory to Selmer groups and the Shafarevich–Tate group. Chapter 10 moves beyond the abelian setting: Kummer theory, Galois descent, non-abelian H^1 and H^2, and the connections to étale cohomology that situate the entire theory within modern arithmetic geometry.

8

Local Fields

The abstract cohomological machine of Parts I and II now meets the arithmetic of local fields. A local field K — a finite extension of $\mathbb{Q}_p$ or $\mathbb{F}_q((t))$ — has absolute Galois group G_K whose structure was described in example 1.4.4: a filtration by inertia and wild inertia, with the unramified quotient topologically generated by the Frobenius. This chapter computes $H^n(G_K, M)$ for the coefficient modules that arise in number theory, proves local Tate duality, and establishes local class field theory via a purely cohomological argument. The results fulfill the promises of Chapter 7: the invariant map, the structure of $\mathrm{Br}(K)$, and the period–index equality are all consequences of the computations developed here.

8.1 Cohomology of Local Fields: Unramified and Tamely Ramified Extensions

The first step in understanding the Galois cohomology of local fields is the vanishing theorem of Hilbert: the first cohomology of the multiplicative group is trivial. This result, together with the explicit structure of unramified extensions, determines the Brauer group of a local field and constructs the invariant map.

Throughout this section, K denotes a local field with ring of integers $\mathcal{O}_K$, maximal ideal $\mathfrak{m}_K$, residue field $\mathbb{F}_K = \mathcal{O}_K/\mathfrak{m}_K$ of cardinality q, and uniformizer π_K. For a finite Galois extension L/K, write $G = \mathrm{Gal}(L/K)$ and use the analogous notation $\mathcal{O}_L, \mathfrak{m}_L, \mathbb{F}_L, \pi_L$ for L.

Theorem 8.1.1: Hilbert's Theorem 90 (Multiplicative Form)

For any finite Galois extension L/K: $H^1(\mathrm{Gal}(L/K), L^\times) = 0$.

***Proof*:**
A 1-cocycle $a\colon G \to L^\times$ satisfies $a_{\sigma\tau} = a_\sigma \cdot \sigma(a_\tau)$ for all $\sigma, \tau \in G$ (definition 2.3.5). By the linear independence of characters (see [6, Artin's Lemma, linear independence of Automorphisms (Dedekind's Lemma)]), the elements $\{\sigma\}_{\sigma\in G}$, viewed as L-linear maps $L \to L$, are L-linearly independent. The element $b = \sum_{\sigma\in G} a_\sigma \cdot \sigma(c) \in L$ is nonzero for a suitable choice of $c \in L^\times$ (since the sum $\sum a_\sigma \sigma$ is a nonzero L-linear combination of distinct characters, it cannot vanish identically). For such c, the cocycle condition gives:

$$\tau(b) = \sum_\sigma \tau(a_\sigma) \cdot \tau\sigma(c) = \sum_\sigma a_\tau^{-1} a_{\tau\sigma} \cdot \tau\sigma(c) = a_\tau^{-1} \sum_\rho a_\rho \cdot \rho(c) = a_\tau^{-1} \cdot b$$

where the second equality uses the cocycle condition in the form $\tau(a_\sigma) = a_\tau^{-1} a_{\tau\sigma}$, and the third substitutes $\rho = \tau\sigma$. Rearranging: $a_\tau = b/\tau(b)$ for all $\tau \in G$. This is a 1-coboundary (definition 2.3.6), so $[a] = 0$ in $H^1(G, L^\times)$.

Hilbert's Theorem 90 says that the norm-1 elements in $L^\times$ are exactly the elements of the form $b/\sigma(b)$: the equation $N_{L/K}(x) = 1$ is solvable by $x = b/\sigma(b)$ whenever L/K is cyclic (using $\hat{H}^{-1}(G, L^\times) \cong H^1(G, L^\times) = 0$ by proposition 4.1.5 for cyclic G).

Theorem 8.1.2: Hilbert's Theorem 90 (Additive Form)

For any finite Galois extension L/K: $H^n(\mathrm{Gal}(L/K), L) = 0$ for all $n \geq 1$.

***Proof*:**
The normal basis theorem [6, Normal Basis Theorem] states that $L \cong K[G]$ as G-modules: there exists $\theta \in L$ such that $\{\sigma(\theta)\}_{\sigma\in G}$ is a K-basis for L. Thus L is a free $K[G]$-module of rank 1, and in particular an induced module: $L \cong \mathrm{Ind}_{\{1\}}^G(K)$. Induced modules are acyclic (theorem 2.2.6 for ordinary cohomology, or proposition 4.1.7 for Tate cohomology), giving $H^n(G, L) = 0$ for all $n \geq 1$.

With Hilbert 90 established, the cohomology of unramified extensions can be computed explicitly. The key input is the decomposition of $L^\times$ by the valuation.

Let L/K be an unramified extension of degree n, so that $G = \mathrm{Gal}(L/K)$ is cyclic of order n, generated by the Frobenius Frob (example 1.4.4). The valuation gives a short exact sequence of G-modules:

$$1 \to \mathcal{O}_L^\times \to L^\times \xrightarrow{v_L} \mathbb{Z} \to 0$$

where G acts trivially on $\mathbb{Z}$ (since $v_L(\sigma(x)) = v_L(x)$ for all $\sigma \in G$ in an unramified extension).

Theorem 8.1.3: Cohomology of Unramified Extensions

Let L/K be an unramified extension of local fields with $G = \mathrm{Gal}(L/K)$ cyclic of order n. Then:

1. $H^1(G, \mathcal{O}_L^\times) = 0$.
2. $H^2(G, L^\times) \cong \mathbb{Z}/n\mathbb{Z}$.
3. $\hat{H}^0(G, L^\times) = K^\times/N_{L/K}(L^\times) \cong \mathbb{Z}/n\mathbb{Z}$.

***Proof*:**
Step 1: The Herbrand quotient of $L^\times$. Since G is cyclic, $h(G, L^\times)$ is defined (definition 4.1.9). The long exact sequence from $1 \to \mathcal{O}_L^\times \to L^\times \to \mathbb{Z} \to 0$ and the multiplicativity of the Herbrand quotient (theorem 4.1.10) give:

$$h(G, L^\times) = h(G, \mathcal{O}_L^\times) \cdot h(G, \mathbb{Z})$$

The trivial G-module $\mathbb{Z}$ has $h(G, \mathbb{Z}) = n$ (since $\hat{H}^0(G, \mathbb{Z}) = \mathbb{Z}/n\mathbb{Z}$ and $\hat{H}^{-1}(G, \mathbb{Z}) = 0$ by example 4.1.8).

Step 2: The Herbrand quotient of $\mathcal{O}_L^\times$. The group $\mathcal{O}_L^\times$ has a filtration by the higher unit groups $U_L^{(i)} = 1 + \mathfrak{m}_L^i$, with graded pieces $\mathcal{O}_L^\times/U_L^{(1)} \cong \mathbb{F}_L^\times$ and $U_L^{(i)}/U_L^{(i+1)} \cong \mathbb{F}_L^+$ for $i \geq 1$. Each graded piece is a finite G-module. By the multiplicativity of h and the fact that $h(G, M) = 1$ for finite M (proposition 4.1.11): $h(G, \mathcal{O}_L^\times/U_L^{(i)}) = 1$ for all i. Since $\mathcal{O}_L^\times = \varprojlim_i \mathcal{O}_L^\times/U_L^{(i)}$ and the Herbrand quotient is compatible with limits of this form, $h(G, \mathcal{O}_L^\times) = 1$.

Step 3: Computing H^1 and H^2. From Steps 1 and 2: $h(G, L^\times) = 1 \cdot n = n$. The long exact sequence gives:

$$L^{G,\times} \xrightarrow{v_K} \mathbb{Z} \xrightarrow{\delta} H^1(G, \mathcal{O}_L^\times) \to \underbrace{H^1(G, L^\times)}_{=0} \to H^1(G, \mathbb{Z}) \to \cdots$$

Since $v_K \colon K^\times \to \mathbb{Z}$ is surjective, $\delta = 0$, so $H^1(G, \mathcal{O}_L^\times)$ injects into $H^1(G, L^\times) = 0$. This gives $H^1(G, \mathcal{O}_L^\times) = 0$, proving (1). Since $h(G, \mathcal{O}_L^\times) = 1$ and $H^1 = 0$: $\hat{H}^0(G, \mathcal{O}_L^\times) = 0$ as well (both numerator and denominator of the Herbrand quotient agree). The norm map $N_{L/K} \colon \mathcal{O}_L^\times \to \mathcal{O}_K^\times$ is therefore surjective.

For (2) and (3): the long exact sequence in Tate cohomology gives:

$$\hat{H}^0(G, \mathcal{O}_L^\times) \to \hat{H}^0(G, L^\times) \to \hat{H}^0(G, \mathbb{Z}) \xrightarrow{\delta} H^1(G, \mathcal{O}_L^\times)$$

Since $\hat{H}^0(G, \mathcal{O}_L^\times) = 0$ and $H^1(G, \mathcal{O}_L^\times) = 0$: $\hat{H}^0(G, L^\times) \cong \hat{H}^0(G, \mathbb{Z}) = \mathbb{Z}/n\mathbb{Z}$, proving (3). For (2): by periodicity (theorem 4.3.4), $H^2(G, L^\times) \cong \hat{H}^0(G, L^\times) = \mathbb{Z}/n\mathbb{Z}$.

Part (3) identifies $K^\times/N_{L/K}(L^\times) \cong \mathbb{Z}/n\mathbb{Z}$ for an unramified extension of degree n. The norm map $N_{L/K}$ surjects onto $\mathcal{O}_K^\times$ (from the vanishing of $\hat{H}^0(G, \mathcal{O}_L^\times)$), and the cokernel of $N_{L/K}$ on $K^\times$ is generated by the class of any uniformizer π_K (since $v_K(\pi_K) = 1$ generates $\mathbb{Z}/n\mathbb{Z}$ under the isomorphism $K^\times/N_{L/K}(L^\times) \cong \mathbb{Z}/n\mathbb{Z}$).

The construction of the invariant map follows.

Corollary 8.1.4: Construction of the Invariant Map

Let K be a local field and K_n^{nr}/K the unramified extension of degree n. The isomorphism $H^2(\mathrm{Gal}(K_n^{\mathrm{nr}}/K), (K_n^{\mathrm{nr}})^\times) \cong \mathbb{Z}/n\mathbb{Z}$ from theorem 8.1.3(2) defines:

$$\mathrm{inv}_K \colon \mathrm{Br}(K_n^{\mathrm{nr}}/K) \xrightarrow{\sim} \frac{1}{n}\mathbb{Z}/\mathbb{Z}$$

These isomorphisms are compatible as n varies (under inflation), and passing to the colimit gives $\mathrm{inv}_K \colon \mathrm{Br}(K^{\mathrm{nr}}/K) \xrightarrow{\sim} \mathbb{Q}/\mathbb{Z}$.

***Proof*:**
By theorem 7.2.4, $\mathrm{Br}(K_n^{\mathrm{nr}}/K) \cong H^2(\mathrm{Gal}(K_n^{\mathrm{nr}}/K), (K_n^{\mathrm{nr}})^\times) \cong \mathbb{Z}/n\mathbb{Z}$. The identification with $\frac{1}{n}\mathbb{Z}/\mathbb{Z}$ sends the generator of $\mathbb{Z}/n\mathbb{Z}$ to $1/n$. Compatibility with inflation: for $m \big| n$, the inflation $H^2(\mathrm{Gal}(K_m^{\mathrm{nr}}/K), -) \to H^2(\mathrm{Gal}(K_n^{\mathrm{nr}}/K), -)$ corresponds to the inclusion $\frac{1}{m}\mathbb{Z}/\mathbb{Z} \hookrightarrow \frac{1}{n}\mathbb{Z}/\mathbb{Z}$ under the invariant maps. The colimit is $\mathbb{Q}/\mathbb{Z}$.

To complete the proof of theorem 7.3.1, it remains to show that every element of $\mathrm{Br}(K)$ is split by an unramified extension.

Theorem 8.1.5: Every CSA Is Split by an Unramified Extension

$\mathrm{Br}(K) = \mathrm{Br}(K^{\mathrm{nr}}/K)$. In particular, $\mathrm{inv}_K \colon \mathrm{Br}(K) \xrightarrow{\sim} \mathbb{Q}/\mathbb{Z}$.

***Proof*:**
It suffices to show $\mathrm{Br}(L/K) \subseteq \mathrm{Br}(K^{\mathrm{nr}}/K)$ for every finite Galois L/K. The inflation-restriction sequence (proposition 3.3.3) for the normal subgroup $I = \mathrm{Gal}(L/L \cap K^{\mathrm{nr}})$ (the inertia subgroup of $\mathrm{Gal}(L/K)$) gives:

$$0 \to H^2(\mathrm{Gal}(L \cap K^{\mathrm{nr}}/K), (L \cap K^{\mathrm{nr}})^\times) \xrightarrow{\mathrm{Inf}} H^2(\mathrm{Gal}(L/K), L^\times)$$

The inflation map lands in $\mathrm{Br}(K^{\mathrm{nr}}/K)$ (its image consists of classes split by the maximal unramified subextension). The cokernel is controlled by $H^1(I, L^\times)$ via the five-term exact sequence (theorem 3.3.4), and since $L^\times$ is replaced by $\mathcal{O}_L^\times$ (with $L^\times/\mathcal{O}_L^\times \cong \mathbb{Z}$, on which I acts trivially), the relevant group $H^1(I, \mathcal{O}_L^\times)$ is analyzed through the unit filtration. For the unramified quotient G/I acting on $H^1(I, \mathcal{O}_L^\times)$: the vanishing of H^1 for units in unramified extensions (theorem 8.1.3(1)) propagates through the spectral sequence to give the full result. The details require the LHS spectral sequence (theorem 3.4.4) applied to the extension $1 \to I \to G \to G/I \to 1$ with coefficients in $L^\times$.

Example 8.1.6: Explicit Computations for $\mathbb{Q}_p$

1. For $K = \mathbb{Q}_p$ and $L = \mathbb{Q}_{p^2}$ (the unramified quadratic extension): $G = \langle \mathrm{Frob} \rangle \cong C_2$. The norm $N_{L/K} : L^\times \to K^\times$ satisfies $K^\times / N(L^\times) \cong \mathbb{Z}/2\mathbb{Z}$, generated by the class of p (any uniformizer). The units $\mathbb{Z}_p^\times$ are norms (since the norm surjects onto $\mathcal{O}_K^\times$ by theorem 8.1.3). The Brauer class with invariant $1/2$ is represented by the crossed product $(L/K, \mathrm{Frob}, p)$: the cyclic algebra $L \oplus L \cdot u$ with $u^2 = p$ and $u \cdot \ell = \mathrm{Frob}(\ell) \cdot u$ for $\ell \in L$.

2. The fundamental class $u_{\mathbb{Q}_{p^n}/\mathbb{Q}_p} \in H^2(C_n, \mathbb{Q}_{p^n}^\times)$ is the cocycle $f(\mathrm{Frob}^i, \mathrm{Frob}^j) = \begin{cases} 1 & i + j < n \\ p & i + j \geq n \end{cases}$ (for $0 \leq i, j < n$). Its invariant is $1/n \in \mathbb{Q}/\mathbb{Z}$. The crossed product with this cocycle is the unique division algebra over $\mathbb{Q}_p$ of degree n.

3. (If the reader has read [10]) The Hilbert symbol $(a, b)_p = \mathrm{inv}_{\mathbb{Q}_p}((a, b)_{\mathbb{Q}_p}) \in \{0, 1/2\}$ for $a, b \in \mathbb{Q}_p^\times$ computes the invariant of the quaternion algebra. The values computed in example 7.3.2 are recovered: $(a, b)_p = 0$ iff $ax^2 + by^2 = 1$ is solvable in $\mathbb{Q}_p$.

Exercise 8.1.1

1. Let $L = K(\sqrt{a})$ be a quadratic extension of a local field K (with $\mathrm{char} K \neq 2$). Verify Hilbert 90 directly: show that if $x \in L^\times$ satisfies $N_{L/K}(x) = 1$, then $x = b/\sigma(b)$ for some $b \in L^\times$, where σ is the nontrivial automorphism.

2. Let L/K be the unramified extension of degree n. Show that the norm map $N_{L/K} : \mathcal{O}_L^\times \to \mathcal{O}_K^\times$ is surjective by reducing to the surjectivity of the norm $N_{\mathbb{F}_L/\mathbb{F}_K} : \mathbb{F}_L^\times \to \mathbb{F}_K^\times$ on residue fields and applying Hensel's lemma.

3. Compute the invariant of the cyclic algebra $(\mathbb{Q}_{p^3}/\mathbb{Q}_p, \mathrm{Frob}, p^2)$ (the crossed product with $u^3 = p^2$). Show that $\mathrm{inv}_{\mathbb{Q}_p} = 2/3 \in \mathbb{Q}/\mathbb{Z}$.

4. Let L/K be a totally ramified cyclic extension of degree p (with $\mathrm{char} K = 0$). Show that $H^1(\mathrm{Gal}(L/K), \mathcal{O}_L^\times) \neq 0$ in general. *Hint:* use the exact sequence $1 \to U_L^{(1)} \to \mathcal{O}_L^\times \to \mathbb{F}_L^\times \to 1$ and the fact that G acts trivially on $\mathbb{F}_L^\times$ (since L/K is totally ramified).

5. Give a direct proof that the quaternion algebra $(-1, p)_{\mathbb{Q}_p}$ (for p odd) is split by the unramified quadratic extension $\mathbb{Q}_{p^2}/\mathbb{Q}_p$ by exhibiting an explicit embedding $\mathbb{Q}_{p^2} \hookrightarrow M_2(\mathbb{Q}_p)$ that commutes with the quaternion algebra structure.

8.2 Filtered Cohomology: Higher Unit Groups

For unramified extensions, the cohomology of local units vanishes (theorem 8.1.3(1)). For ramified extensions, the situation is more complex: the unit group $\mathcal{O}_L^\times$ has a natural filtration by the higher unit groups $U_L^{(i)}$ that reflects the ramification filtration of the Galois group, and the cohomology of this filtration connects the abstract theory to the arithmetic of ramification.

Definition 8.2.1: Higher Unit Groups

Let L be a local field with ring of integers $\mathcal{O}_L$ and maximal ideal $\mathfrak{m}_L$. The *higher unit groups* are:

$$U_L^{(0)} = \mathcal{O}_L^\times, \qquad U_L^{(i)} = 1 + \mathfrak{m}_L^i \quad \text{for } i \geq 1$$

These form a descending filtration $\mathcal{O}_L^\times = U_L^{(0)} \supset U_L^{(1)} \supset U_L^{(2)} \supset \cdots$ with $\bigcap_i U_L^{(i)} = \{1\}$.

The graded pieces of this filtration have a clean algebraic structure.

Proposition 8.2.2: Graded Pieces of the Unit Filtration

There are isomorphisms of abelian groups:

1. $U_L^{(0)}/U_L^{(1)} \cong \mathbb{F}_L^\times$ (the residue field multiplicative group), via $u \mapsto u \bmod \mathfrak{m}_L$.
2. $U_L^{(i)}/U_L^{(i+1)} \cong \mathbb{F}_L^+$ (the residue field additive group) for $i \geq 1$, via $1 + a\pi_L^i \mapsto a \bmod \mathfrak{m}_L$, where π_L is a uniformizer.

If L/K is Galois with $G = \mathrm{Gal}(L/K)$, each $U_L^{(i)}$ is a G-submodule of $\mathcal{O}_L^\times$, and the isomorphisms are G-equivariant (where G acts on $\mathbb{F}_L$ via the natural action on the residue field).

***Proof*:**
For (1): the map $\mathcal{O}_L^\times \to \mathbb{F}_L^\times$ sending $u \mapsto \bar{u}$ is a surjective group homomorphism with kernel $U_L^{(1)} = 1 + \mathfrak{m}_L$. For (2): the map $U_L^{(i)} \to \mathbb{F}_L^+$ sending $1 + a\pi_L^i \mapsto \bar{a}$ is a surjective group homomorphism (using $(1 + a\pi_L^i)(1 + b\pi_L^i) = 1 + (a+b)\pi_L^i + ab\pi_L^{2i} \equiv 1 + (a+b)\pi_L^i \pmod{\mathfrak{m}_L^{i+1}}$) with kernel $U_L^{(i+1)}$. The G-equivariance follows from $\sigma(1 + a\pi_L^i) = 1 + \sigma(a)\sigma(\pi_L)^i$ and the fact that $\sigma(\pi_L)/\pi_L \in U_L^{(0)}$ (so the leading term modulo $\mathfrak{m}_L^{i+1}$ is $\sigma(a) \cdot (\sigma(\pi_L)/\pi_L)^i$, which reduces to $\bar{\sigma}(\bar{a})$ in $\mathbb{F}_L$).

The filtration interacts with the ramification filtration of the Galois group in a precise way.

Proposition 8.2.3: Ramification Groups and Unit Groups

Let L/K be a finite Galois extension of local fields with $G = \mathrm{Gal}(L/K)$. The ramification subgroups G_i (in lower numbering, see [10, Ramification Groups in the Lower Numbering]) satisfy:

1. G_0 (the inertia group) acts trivially on $\mathbb{F}_L^\times$: $\sigma(\bar{u}) = \bar{u}$ for all $\sigma \in G_0$ and $u \in \mathcal{O}_L^\times$.
2. G_i acts trivially on $U_L^{(0)}/U_L^{(i)}$ for each $i \geq 1$.
3. G_1 (the wild inertia) acts trivially on $U_L^{(0)}/U_L^{(1)} \cong \mathbb{F}_L^\times$.

***Proof*:**
For (1): $G_0 = \mathrm{Gal}(L/L \cap K^{\mathrm{nr}})$ fixes the maximal unramified subextension, hence acts trivially on the residue field $\mathbb{F}_L$ (which is determined by the unramified part). For (2): by definition,

$\sigma \in G_i$ satisfies $v_L(\sigma(\pi_L) - \pi_L) \geq i+1$ (see [10, Ramification Groups in the Lower Numbering]), so $\sigma(1 + a\pi_L^j) = 1 + \sigma(a)\sigma(\pi_L)^j \equiv 1 + a\pi_L^j \pmod{\mathfrak{m}_L^{j+i}}$ for $j < i$, giving trivial action on $U_L^{(j)}/U_L^{(j+i)}$. In particular, G_i acts trivially on each quotient $U_L^{(j)}/U_L^{(j+1)}$ for $j < i$, hence on $U_L^{(0)}/U_L^{(i)}$. Part (3) is the special case $i = 1$.

The interaction between ramification and the unit filtration governs the cohomology. The tamely ramified case (where the wild inertia G_1 is trivial) is significantly simpler than the wildly ramified case.

Theorem 8.2.4: Cohomology of Tamely Ramified Extensions

Let L/K be a finite tamely ramified Galois extension (i.e., $G_1 = \{1\}$, so the ramification is concentrated in the tame inertia $G_0/G_1 = G_0$, which has order coprime to the residue characteristic p). Then:

1. $H^1(G, U_L^{(1)}) = 0$.
2. The long exact sequence for $1 \to U_L^{(1)} \to \mathcal{O}_L^\times \to \mathbb{F}_L^\times \to 1$ gives $H^1(G, \mathcal{O}_L^\times) \cong \ker(H^1(G_0, \mathbb{F}_L^\times) \to H^2(G, U_L^{(1)}))$.

***Proof*:**

Step 1: Vanishing of $H^1(G, U_L^{(1)})$. Each graded piece $U_L^{(i)}/U_L^{(i+1)} \cong \mathbb{F}_L^+$ (for $i \geq 1$) is a finite G-module. The additive form of Hilbert 90 (theorem 8.1.2) applied to the residue field extension $\mathbb{F}_L/\mathbb{F}_K$ gives $H^1(G_0/G_1, \mathbb{F}_L^+) = 0$ (since $G_0 = G_0/G_1$ acts on $\mathbb{F}_L$ and the normal basis theorem applies to $\mathbb{F}_L/\mathbb{F}_K$). The vanishing propagates up the filtration: the long exact sequence for $1 \to U_L^{(i+1)} \to U_L^{(1)} \to U_L^{(1)}/U_L^{(i+1)} \to 1$ and induction on i give $H^1(G, U_L^{(1)}/U_L^{(i)}) = 0$ for all i. Passing to the limit (using $U_L^{(1)} = \varprojlim_i U_L^{(1)}/U_L^{(i)}$ and the compatibility of cohomology with limits of discrete modules) gives $H^1(G, U_L^{(1)}) = 0$.

Part (2) follows from the long exact sequence applied to the short exact sequence $1 \to U_L^{(1)} \to \mathcal{O}_L^\times \to \mathbb{F}_L^\times \to 1$: the vanishing $H^1(G, U_L^{(1)}) = 0$ gives $H^1(G, \mathcal{O}_L^\times) \hookrightarrow H^1(G, \mathbb{F}_L^\times)$, and the connecting map $\mathbb{F}_K^\times \to H^1(G, U_L^{(1)}) = 0$ shows the inflation $H^1(G/G_0, (\mathbb{F}_L^\times)^{G_0}) \to H^1(G, \mathbb{F}_L^\times)$ captures the full H^1.

For wildly ramified extensions ($G_1 \neq \{1\}$), the cohomology of $U_L^{(1)}$ is nontrivial. The wild part of the cohomology is a p-group and is controlled by the higher ramification groups.

Proposition 8.2.5: Wild Cohomology

Let L/K be a finite Galois extension of local fields with $G = \mathrm{Gal}(L/K)$ and $G_1 \neq \{1\}$ (wild ramification). Then $H^1(G_1, U_L^{(1)})$ is a p-group, and its order is related to the discriminant of L/K via the formula:

$$|H^1(G_1, U_L^{(1)})| = |G_1|/e_{\text{wild}}$$

where $e_{\text{wild}} = |G_1|$ is the wild part of the ramification index (the precise relationship involves the higher ramification breaks; see [10] and [21]).

The detailed structure of $H^1(G_1, U_L^{(i)})$ for each layer of the filtration depends on the ramification breaks of L/K. A full treatment requires the Hasse–Arf theorem and the upper numbering of ramification groups; the essential point for the applications in this chapter is the Herbrand quotient computation, which holds uniformly regardless of ramification type.

Theorem 8.2.6: Herbrand Quotient for Local Fields

Let L/K be a cyclic extension of local fields of degree n with $G = \mathrm{Gal}(L/K) \cong C_n$. Then:

1. $h(G, L^\times) = n$.
2. $h(G, \mathcal{O}_L^\times) = 1$.
3. $|\hat{H}^0(G, L^\times)| = |K^\times/N_{L/K}(L^\times)| = n$.

***Proof*:**

Step 1: The Herbrand quotient of $\mathbb{Z}$. The valuation exact sequence $1 \to \mathcal{O}_L^\times \to L^\times \xrightarrow{v_L} \mathbb{Z} \to 0$ has G acting on $\mathbb{Z}$ via multiplication by $e = e(L/K)$ (the ramification index): $\sigma \cdot 1 = 1$ when L/K is unramified, and more generally $v_L(\sigma(\pi_L)) = v_L(\pi_L) = 1$ so G acts trivially on $\mathbb{Z}$ via v_L. However, the image of v_L restricted to $K^\times$ is $e\mathbb{Z}$ (since $v_L(\pi_K) = e$). The induced sequence on invariants gives the index $[v_L(K^\times) : v_L(N(L^\times))]$.

Step 2: Multiplicativity. By theorem 4.1.10: $h(G, L^\times) = h(G, \mathcal{O}_L^\times) \cdot h(G, \mathbb{Z})$. Since G acts trivially on $\mathbb{Z}$: $h(G, \mathbb{Z}) = |\hat{H}^0(G, \mathbb{Z})|/|\hat{H}^{-1}(G, \mathbb{Z})| = n/1 = n$ (by example 4.1.8). So $h(G, L^\times) = n \cdot h(G, \mathcal{O}_L^\times)$.

Step 3: The Herbrand quotient of $\mathcal{O}_L^\times$. By the same filtration argument as in theorem 8.1.3: the higher unit groups $U_L^{(i)}/U_L^{(i+1)}$ are finite G-modules, so $h(G, U_L^{(i)}/U_L^{(i+1)}) = 1$ (proposition 4.1.11). Multiplicativity along the filtration gives $h(G, \mathcal{O}_L^\times/U_L^{(i)}) = 1$ for all i, and passing to the limit: $h(G, \mathcal{O}_L^\times) = 1$. This gives $h(G, L^\times) = n$.

Step 4: Computing $\hat{H}^0$. Since $h(G, L^\times) = n = |\hat{H}^0(G, L^\times)|/|\hat{H}^{-1}(G, L^\times)|$ and $\hat{H}^{-1}(G, L^\times) = \hat{H}^1(G, L^\times) = H^1(G, L^\times) = 0$ (Hilbert 90, theorem 8.1.1), the denominator is 1, giving $|\hat{H}^0(G, L^\times)| = n$.

Part (3) determines the norm index: for a cyclic extension of local fields of degree n, the group

$K^\times/N_{L/K}(L^\times)$ has order exactly n. This is the "local norm index equality," and it is the local input for the reciprocity law of section 8.4.

Example 8.2.7: The Unit Filtration for $\mathbb{Q}_p$

1. For $K = \mathbb{Q}_p$ with p odd: $\mathcal{O}_K^\times = \mathbb{Z}_p^\times$ has the filtration $\mathbb{Z}_p^\times \supset 1 + p\mathbb{Z}_p \supset 1 + p^2\mathbb{Z}_p \supset \cdots$ with graded pieces $\mathbb{F}_p^\times$ (of order $p-1$) and $\mathbb{F}_p^+$ (of order p) in each subsequent layer. The logarithm map $\log\colon 1 + p\mathbb{Z}_p \xrightarrow{\sim} p\mathbb{Z}_p$ (converging for p odd) gives an isomorphism $U_{\mathbb{Q}_p}^{(1)} \cong \mathbb{Z}_p$ of topological groups.
2. For a totally ramified cyclic extension $L/\mathbb{Q}_p$ of degree p (e.g., $L = \mathbb{Q}_p(\zeta_p)$ when p is odd): $G = C_p$ acts nontrivially on $U_L^{(1)}$, and $H^1(G, U_L^{(1)}) \neq 0$ (by exercise 8.1.1 (4)). The Herbrand quotient $h(G, \mathcal{O}_L^\times) = 1$ means that $|\hat{H}^0(G, \mathcal{O}_L^\times)| = |H^1(G, \mathcal{O}_L^\times)|$: the "cohomological" and "homological" sides have the same order, even though neither vanishes.
3. For $K = \mathbb{Q}_p$ and the unramified extension $L = \mathbb{Q}_{p^n}$: the unit filtration is preserved by $G = \langle \mathrm{Frob} \rangle$, and Frob acts on each graded piece $\mathbb{F}_{p^n}^+$ by the Frobenius $x \mapsto x^p$. The vanishing $H^1(G, U_L^{(1)}) = 0$ (theorem 8.2.4, since unramified implies tamely ramified) confirms theorem 8.1.3(1) through the filtration.

Exercise 8.2.1

1. Let L/K be an unramified extension of degree n. Compute $H^1(G, \mathbb{F}_L^\times)$ directly (where G acts on $\mathbb{F}_L^\times$ via the Frobenius), and verify that it vanishes.
2. Let L/K be a totally ramified cyclic extension of degree p (with $\mathrm{char} K = 0$). Verify $h(G, L^\times) = p$ by computing $|\hat{H}^0(G, L^\times)| = |K^\times/N(L^\times)|$ and $|\hat{H}^{-1}(G, L^\times)| = |H^1(G, L^\times)| = 1$ (Hilbert 90) independently.
3. Let $K = \mathbb{Q}_p$ (with p odd) and $L = K(\sqrt{u})$ where $u \in \mathbb{Z}_p^\times$ is a non-square. This is an unramified quadratic extension. Determine $K^\times/N_{L/K}(L^\times)$ explicitly: show that a uniformizer p is not a norm, while every unit is a norm.
4. Let $L = \mathbb{Q}_p(\zeta_p)$ for p an odd prime. This is a totally ramified cyclic extension of $\mathbb{Q}_p$ of degree $p-1$. Compute $h(G, \mathcal{O}_L^\times)$ and determine $|H^1(G, \mathcal{O}_L^\times)|$. *Hint:* first compute $|\hat{H}^0(G, \mathcal{O}_L^\times)|$ using the exact sequence $1 \to \mathcal{O}_L^\times \to L^\times \to \mathbb{Z} \to 0$.
5. For $K = \mathbb{Q}_2$ and $L = \mathbb{Q}_2(\sqrt{-1})$ (a totally ramified quadratic extension): write out the unit filtration of $\mathcal{O}_L^\times$, identify the G-action on each graded piece, and compute $H^1(G, U_L^{(i)}/U_L^{(i+1)})$ for $i = 0, 1, 2$.

8.3 Local Tate Duality

The cohomology of a local field satisfies a duality that pairs H^r with H^{2-r}: a local analogue of Poincaré duality for manifolds. This duality — the local Tate duality theorem — is the most important structural result about the cohomology of local Galois groups. It determines the cohomological dimension of local fields, identifies the maximal pro-p quotients as Demushkin groups, and provides the Euler–Poincaré characteristic that controls the size of cohomology groups.

Definition 8.3.1: Cartier Dual

Let K be a local field and M a finite discrete G_K-module. The *Cartier dual* of M is the finite G_K-module:

$$M^* = \mathrm{Hom}_{\mathbb{Z}}(M, (K^{\mathrm{sep}})^\times)$$

with G_K acting by $(g \cdot \varphi)(m) = g(\varphi(g^{-1}m))$ for $g \in G_K$, $\varphi \in M^*$, $m \in M$.

The Cartier dual is a finite G_K-module of the same order as M. For $M = \mathbb{Z}/n\mathbb{Z}$ with trivial action: $M^* = \mathrm{Hom}(\mathbb{Z}/n\mathbb{Z}, (K^{\mathrm{sep}})^\times) \cong \mu_n$ (the nth roots of unity, with the natural Galois action). For $M = \mu_n$: $M^* \cong \mathbb{Z}/n\mathbb{Z}$ (with trivial action when $\mu_n \subseteq K$). The double dual satisfies $M^{**} \cong M$ canonically.

Theorem 8.3.2: Local Tate Duality

Let K be a local field and M a finite discrete G_K-module. For $r = 0, 1, 2$, the cup product followed by the invariant map:

$$H^r(G_K, M) \times H^{2-r}(G_K, M^*) \xrightarrow{\smile} H^2(G_K, (K^{\mathrm{sep}})^\times) \xrightarrow{\mathrm{inv}_K} \mathbb{Q}/\mathbb{Z}$$

induces a perfect pairing of finite groups. That is, the induced map $H^r(G_K, M) \to \mathrm{Hom}(H^{2-r}(G_K, M^*), \mathbb{Q}/\mathbb{Z})$ is an isomorphism.

***Proof*:**
The proof proceeds in four steps: the $r = 2$ case by explicit construction, the $r = 0$ case by duality, the Euler–Poincaré characteristic by the Herbrand quotient, and the $r = 1$ case by a counting argument and Kummer theory.

Step 1: The case $r = 2$. The proof reduces by dévissage to the case of a simple G_K-module. For a short exact sequence $0 \to M' \to M \to M'' \to 0$: the long exact sequence in cohomology and the exactness of Pontryagin duality on finite abelian groups give a commutative diagram with exact rows:

$$\begin{array}{ccccccc} H^2(G_K, M') & \longrightarrow & H^2(G_K, M) & \longrightarrow & H^2(G_K, M'') & \longrightarrow & 0 \\ \downarrow & & \downarrow & & \downarrow & & \\ (M')^{G_K,*,\vee} & \longrightarrow & M^{G_K,*,\vee} & \longrightarrow & (M'')^{G_K,*,\vee} & \longrightarrow & 0 \end{array}$$

where $(-)^\vee = \mathrm{Hom}(-, \mathbb{Q}/\mathbb{Z})$. The top row is exact on the right because $\mathrm{cd}(K) \leq 2$ (so $H^3 = 0$, proved in corollary 8.3.3 below using only the $r = 2$ result for simple modules). By the five lemma, it suffices to prove the result for simple G_K-modules M.

For $M = \mathbb{F}_p$ with trivial G_K-action (the essential case when p is odd; the case $p = 2$ is analogous): $M^* = \mu_p$, and $H^0(G_K, M^*) = \mu_p(K)$. Each $\zeta \in \mu_p(K)$ determines a G_K-equivariant map $\varphi_\zeta \colon \mathbb{F}_p \to (K^{\mathrm{sep}})^\times$ by $\varphi_\zeta(a) = \zeta^a$, inducing $(\varphi_\zeta)_* \colon H^2(G_K, \mathbb{F}_p) \to H^2(G_K, (K^{\mathrm{sep}})^\times) = \mathrm{Br}(K)$.

The Kummer sequence $1 \to \mu_p \to (K^{\mathrm{sep}})^\times \xrightarrow{(-)^p} (K^{\mathrm{sep}})^\times \to 1$ gives an exact sequence:

$$H^1(G_K, (K^{\mathrm{sep}})^\times) \to H^2(G_K, \mu_p) \to \mathrm{Br}(K) \xrightarrow{\times p} \mathrm{Br}(K)$$

Hilbert 90 (theorem 8.1.1) kills the left term, so $H^2(G_K, \mu_p) \cong \mathrm{Br}(K)[p] \cong \mathbb{Z}/p\mathbb{Z}$. The evaluation map $\mathbb{F}_p \otimes \mu_p \to \mu_p$ (sending $a \otimes \zeta \mapsto \zeta^a$) is an isomorphism of G_K-modules, so $H^2(G_K, \mathbb{F}_p \otimes \mu_p) \cong H^2(G_K, \mu_p) \cong \mathbb{Z}/p\mathbb{Z}$. When $\mu_p \subseteq K$: fixing a generator $\zeta \in \mu_p(K)$, the map $(\varphi_\zeta)_*$ is an isomorphism $H^2(G_K, \mathbb{F}_p) \xrightarrow{\sim} \mathrm{Br}(K)[p]$. The pairing $H^2(G_K, \mathbb{F}_p) \times \mu_p(K) \to \mathbb{Q}/\mathbb{Z}$ is then non-degenerate (both sides have order p, and $(\varphi_\zeta)_*$ is nonzero for $\zeta \neq 1$). When $\mu_p \not\subseteq K$: $H^0(G_K, \mu_p) = \{1\}$, and $H^2(G_K, \mathbb{F}_p) = 0$ (since $\mathbb{F}_p \otimes \mu_p \cong \mu_p$ as G_K-modules, and $H^2(G_K, \mathbb{F}_p)$ maps injectively into $H^2(G_K, \mu_p) \cong \mathbb{Z}/p\mathbb{Z}$ via the evaluation, but the G_K-action on μ_p is non-trivial, forcing the invariant part to vanish). The pairing is vacuously perfect (both sides are zero).

Step 2: The case $r = 0$. Apply Step 1 with M replaced by M^*: the pairing $H^2(G_K, M^*) \times H^0(G_K, M^{**}) \to \mathbb{Q}/\mathbb{Z}$ is perfect. Since $M^{**} \cong M$ canonically, this gives a perfect pairing $H^0(G_K, M) \times H^2(G_K, M^*) \to \mathbb{Q}/\mathbb{Z}$ (transposing factors; the sign from the graded-commutativity of the cup product (theorem 4.2.4) does not affect non-degeneracy).

Step 3: The Euler–Poincaré characteristic. The formula $\chi(G_K, M) = |H^0| \cdot |H^2|/|H^1| = |M|^{-[K:\mathbb{Q}_p]}$ is established in theorem 8.3.4 below. Granting this: Steps 1 and 2 give $|H^0(G_K, M)| = |H^2(G_K, M^*)|$ and $|H^2(G_K, M)| = |H^0(G_K, M^*)|$. The Euler–Poincaré formula applied to M gives:

$$\frac{|H^0(G_K, M)| \cdot |H^2(G_K, M)|}{|H^1(G_K, M)|} = |M|^{-[K:\mathbb{Q}_p]}$$

Applied to M^* (using $|M^*| = |M|$):

$$\frac{|H^0(G_K, M^*)| \cdot |H^2(G_K, M^*)|}{|H^1(G_K, M^*)|} = |M|^{-[K:\mathbb{Q}_p]}$$

Substituting $|H^0(G_K, M^*)| = |H^2(G_K, M)|$ and $|H^2(G_K, M^*)| = |H^0(G_K, M)|$ into the second equation reproduces the first, giving $|H^1(G_K, M)| = |H^1(G_K, M^*)|$. The $r = 1$ pairing maps $H^1(G_K, M)$ into $\mathrm{Hom}(H^1(G_K, M^*), \mathbb{Q}/\mathbb{Z})$, and these groups have the same finite order.

Step 4: The case $r = 1$. By Step 3, it suffices to prove injectivity: if $\alpha \in H^1(G_K, M)$ is nonzero, there exists $\beta \in H^1(G_K, M^*)$ with $\mathrm{inv}_K(\alpha \smile \beta) \neq 0$. By dévissage (filtering M by simple submodules and using the long exact sequence), it suffices to treat the case $M = \mathbb{F}_p$ with trivial action, where $M^* = \mu_p$.

The pairing is $H^1(G_K, \mathbb{F}_p) \times H^1(G_K, \mu_p) \to H^2(G_K, \mu_p) \xrightarrow{\mathrm{inv}_K} \frac{1}{p}\mathbb{Z}/\mathbb{Z}$. The group $H^1(G_K, \mathbb{F}_p) = \mathrm{Hom}(G_K, \mathbb{F}_p)$ classifies cyclic p-extensions of K: a nonzero α determines a surjection $G_K \twoheadrightarrow \mathbb{Z}/p\mathbb{Z}$ with kernel $\mathrm{Gal}(K^{\mathrm{sep}}/L)$ for a cyclic extension L/K of degree p. The group $H^1(G_K, \mu_p)$ is identified with $K^\times/(K^\times)^p$ by Kummer theory (the connecting homomorphism in the Kummer sequence). The cup product $\alpha \smile [a]$ for $[a] \in K^\times/(K^\times)^p$ computes the norm residue symbol: its invariant is $\mathrm{inv}_K(\alpha \smile [a]) = 0$ if and only if $a \in N_{L/K}(L^\times)$.

The norm index computation (theorem 8.2.6(3)) gives $|K^\times/N_{L/K}(L^\times)| = p$. So $N_{L/K}(L^\times)$ is a proper subgroup of $K^\times$ containing $(K^\times)^p$ (since norms of pth powers are pth powers). Choosing

$a \in K^\times \setminus N_{L/K}(L^\times)$ gives $[a] \in K^\times/(K^\times)^p$ with $\mathrm{inv}_K(\alpha \smile [a]) \neq 0$. This is the required β, completing the injectivity argument.

Local Tate duality has immediate consequences for the cohomological dimension and the structure of local Galois groups.

Corollary 8.3.3: Cohomological Dimension of Local Fields

For a local field K: $\mathrm{cd}(K) = 2$.

***Proof*:**
The inequality $\mathrm{cd}(K) \geq 2$ follows from $\mathrm{cd}_2(K) = 2$: by Tate duality, $H^2(G_K, \mathbb{Z}/2\mathbb{Z})$ is dual to $H^0(G_K, \mu_2)$, and $\mu_2(K) = \{\pm 1\} \neq \{1\}$ for every local field of characteristic $\neq 2$, giving $H^2(G_K, \mathbb{Z}/2\mathbb{Z}) \neq 0$. By Serre's criterion (theorem 5.1.2), $\mathrm{cd}_2(K) \geq 2$.

For $\mathrm{cd}(K) \leq 2$: it suffices to show $H^3(G_K, \mathbb{Z}/\ell\mathbb{Z}) = 0$ for every prime ℓ (by Serre's criterion). The vanishing follows from the dimension-shifting argument: embed $\mathbb{Z}/\ell\mathbb{Z} \hookrightarrow I$ with I coinduced (hence acyclic), and the long exact sequence reduces $H^3(G_K, \mathbb{Z}/\ell\mathbb{Z})$ to $H^2(G_K, I/(\mathbb{Z}/\ell\mathbb{Z}))$, which is controlled by the structure of $\mathrm{Br}(K) \cong \mathbb{Q}/\mathbb{Z}$ (the only H^2 with values in $(K^{\mathrm{sep}})^\times$ is the Brauer group, which has no elements requiring H^3 to be nontrivial). The full argument uses the spectral sequence for the decomposition $G_K \to G_K/I_K \cong \widehat{\mathbb{Z}}$ (theorem 3.4.4) and the fact that $\mathrm{cd}(\widehat{\mathbb{Z}}) = 1$.

The local Euler–Poincaré characteristic, used in the proof of Tate duality, is a powerful computational tool in its own right.

Theorem 8.3.4: Local Euler–Poincaré Characteristic

Let K be a finite extension of $\mathbb{Q}_p$ and M a finite G_K-module. Then:

$$\frac{|H^0(G_K, M)| \cdot |H^2(G_K, M)|}{|H^1(G_K, M)|} = \frac{1}{|M|^{[K:\mathbb{Q}_p]}}$$

***Proof*:**
Step 1: Multiplicativity. Define $\chi(G_K, M) = |H^0| \cdot |H^2|/|H^1|$. For a short exact sequence $0 \to M' \to M \to M'' \to 0$: the long exact sequence in cohomology (using $H^3 = 0$ since $\mathrm{cd}(K) \leq 2$) gives an exact sequence of finite groups $0 \to H^0(M') \to \cdots \to H^2(M'') \to 0$. The alternating product of orders in an exact sequence of finite groups equals 1, giving $\chi(G_K, M) = \chi(G_K, M') \cdot \chi(G_K, M'')$.

Step 2: Reduction to the case $M = \mathbb{F}_p$ with trivial action. By multiplicativity and dévissage (filtering M by a composition series of simple G_K-modules), it suffices to verify the formula for each simple module. A simple finite G_K-module M is an $\mathbb{F}_p$-vector space on which G_K acts through a finite quotient $G = \mathrm{Gal}(L/K)$. By Shapiro's lemma (theorem 3.5.5), $H^i(G_K, M)$ can be expressed in terms of the cohomology of G_L with simpler coefficients. The general reduction (see [21]) shows $\chi(G_K, M) = \chi(G_K, \mathbb{F}_p)^{\dim M} \cdot \chi(G_L, \mathbb{F}_p)^{-\dim M + \dim M^G} \cdots$, but the cleanest approach is to verify the formula directly for $M = \mathbb{F}_p$ (trivial action) and extend by multiplicativity.

Step 3: The case $M = \mathbb{F}_p$. The colimit formula for profinite cohomology (theorem 3.2.2) gives $H^i(G_K, \mathbb{F}_p) = \varinjlim_L H^i(\mathrm{Gal}(L/K), \mathbb{F}_p)$ over finite Galois L/K. For a cyclic quotient $\mathrm{Gal}(L'/K)$ of order p^k: the Herbrand quotient $h(\mathrm{Gal}(L'/K), \mathbb{F}_p) = |\hat{H}^0|/|\hat{H}^{-1}|$ is computable from the Tate cohomology of cyclic groups (example 4.1.8). Passing to a sufficiently large L: $H^i(G_K, \mathbb{F}_p)$ stabilizes, and the orders are determined by the pro-p-structure of G_K.

For $\mathbb{F}_p$ with trivial G_K-action: $|H^0(G_K, \mathbb{F}_p)| = p$ (the invariants of a trivial module). For $|H^2|$: by the $r = 2$ duality (Step 1 of the Tate duality proof), $|H^2(G_K, \mathbb{F}_p)| = |H^0(G_K, \mu_p)| = |\mu_p(K)|$. Let $\delta = 1$ if $\mu_p \subseteq K$ and $\delta = 0$ otherwise, so $|H^2| = p^\delta$. For $|H^1|$: the group $H^1(G_K, \mathbb{F}_p) = \mathrm{Hom}(G_K, \mathbb{F}_p)$ classifies continuous homomorphisms $G_K \to \mathbb{F}_p$. The number of independent such homomorphisms is the $\mathbb{F}_p$-dimension of $G_K^{\mathrm{ab}}/(G_K^{\mathrm{ab}})^p \otimes \mathbb{F}_p$. By explicit computation of the abelianization of G_K using the local unit structure (the maximal abelian pro-p extension of K is controlled by $K^\times/(K^\times)^p$ via Kummer theory when $\mu_p \subseteq K$, and by Artin–Schreier theory otherwise): $|H^1| = p^{[K:\mathbb{Q}_p]+1+\delta}$.

The formula gives $\chi = p \cdot p^\delta / p^{[K:\mathbb{Q}_p]+1+\delta} = p^{-[K:\mathbb{Q}_p]} = |\mathbb{F}_p|^{-[K:\mathbb{Q}_p]}$, as required.

The identification of local Galois groups as Demushkin groups, foreshadowed in definition 5.3.5 and example 5.3.6, is now a consequence of Tate duality.

Theorem 8.3.5: Structure of the Maximal Pro-p Quotient

Let K be a finite extension of $\mathbb{Q}_p$ with p odd, and let $G_K^{(p)}$ denote the maximal pro-p quotient of G_K. Then:

1. If $\mu_p \not\subseteq K$: the group $G_K^{(p)}$ is a free pro-p-group (theorem 6.1.4) of rank $d(G_K^{(p)}) = [K : \mathbb{Q}_p] + 1$.
2. If $\mu_p \subseteq K$: the group $G_K^{(p)}$ is a Demushkin group (definition 5.3.5) with $d(G_K^{(p)}) = [K : \mathbb{Q}_p] + 2$, $H^2(G_K^{(p)}, \mathbb{F}_p) \cong \mathbb{F}_p$, and type $q = |\mu_{p^\infty}(K)|$ (definition 6.4.3).

***Proof*:**

Step 1: Generator rank. By theorem 6.2.2, $d(G_K^{(p)}) = \dim_{\mathbb{F}_p} H^1(G_K, \mathbb{F}_p)$ (since H^1 with $\mathbb{F}_p$-coefficients factors through the maximal pro-p quotient). The Euler–Poincaré characteristic (theorem 8.3.4) with $M = \mathbb{F}_p$ (trivial action) gives:

$$\frac{p \cdot |H^2(G_K, \mathbb{F}_p)|}{|H^1(G_K, \mathbb{F}_p)|} = p^{-[K:\mathbb{Q}_p]}$$

By Tate duality (theorem 8.3.2), $|H^2(G_K, \mathbb{F}_p)| = |H^0(G_K, \mu_p)| = |\mu_p(K)|$. If $\mu_p \not\subseteq K$: $|\mu_p(K)| = 1$, giving $|H^1| = p^{[K:\mathbb{Q}_p]+1}$, so $d = [K : \mathbb{Q}_p] + 1$. If $\mu_p \subseteq K$: $|\mu_p(K)| = p$, giving $|H^1| = p^{[K:\mathbb{Q}_p]+2}$, so $d = [K : \mathbb{Q}_p] + 2$.

Step 2: The case $\mu_p \not\subseteq K$. Tate duality gives $|H^2(G_K, \mathbb{F}_p)| = |\mu_p(K)| = 1$, so $H^2(G_K^{(p)}, \mathbb{F}_p) = 0$. By Serre's criterion (theorem 5.1.2), $\mathrm{cd}_p(G_K^{(p)}) \leq 1$. Since $G_K^{(p)}$ is nontrivial (it has nontrivial

H^1), the characterization of theorem 6.1.4 gives $G_K^{(p)} \cong F([K : \mathbb{Q}_p] + 1)$, a free pro-$p$-group.

Step 3: The case $\mu_p \subseteq K$. Tate duality gives $|H^2(G_K, \mathbb{F}_p)| = |\mu_p(K)| = p$, so $H^2(G_K^{(p)}, \mathbb{F}_p) \cong \mathbb{F}_p$. Serre's criterion gives $\mathrm{cd}_p(G_K^{(p)}) = 2$ (since $H^2 \neq 0$ and $H^3 = 0$ by corollary 8.3.3). The conditions for a Demushkin group (definition 5.3.5) are satisfied: $\mathrm{cd}_p = 2$, $H^1 \cong \mathbb{F}_p^d$ with d finite, and $H^2 \cong \mathbb{F}_p$.

Step 4: The type. When $\mu_p \subseteq K$, the type q is determined by the action of $G_K^{(p)}$ on the p-power roots of unity: $q = |\mu_{p^\infty}(K)| = p^k$ where k is the largest integer with $\mu_{p^k} \subseteq K$. This identifies $G_K^{(p)}$ with the Demushkin group $G(d, q)$ from theorem 6.4.4.

The dichotomy in the theorem reflects a fundamental arithmetic distinction: when K contains the pth roots of unity, Kummer theory provides a nontrivial H^2 (the Brauer group has p-torsion classes that are not split by p-extensions), and the resulting constraint on the pro-p Galois group forces the Demushkin structure. When $\mu_p \not\subseteq K$, no such constraint exists, and the group is as free as its generator count allows.

Example 8.3.6: Tate Duality in Practice

1. For $M = \mu_p$ with p odd and $K \supseteq \mu_p$: the Cartier dual $M^* = \mathrm{Hom}(\mu_p, (K^{\mathrm{sep}})^\times) \cong \mathbb{Z}/p\mathbb{Z}$ (with trivial action, since K contains μ_p). Tate duality gives perfect pairings: $H^0(G_K, \mu_p) \times H^2(G_K, \mathbb{Z}/p\mathbb{Z}) \to \mathbb{Q}/\mathbb{Z}$ (pairing $\mu_p(K) \cong \mathbb{F}_p$ with $H^2(G_K, \mathbb{Z}/p\mathbb{Z}) \cong \mathbb{F}_p$) and $H^1(G_K, \mu_p) \times H^1(G_K, \mathbb{Z}/p\mathbb{Z}) \to \mathbb{Q}/\mathbb{Z}$. The group $H^1(G_K, \mu_p) \cong K^\times/(K^\times)^p$ (by Kummer theory) has dimension $[K : \mathbb{Q}_p]+2$ over $\mathbb{F}_p$, matching $|H^1(G_K, \mathbb{Z}/p\mathbb{Z})| = p^{[K:\mathbb{Q}_p]+2}$. The pairing identifies cyclic p-extensions of K (classified by $H^1(G_K, \mathbb{Z}/p\mathbb{Z}) = \mathrm{Hom}(G_K, \mathbb{Z}/p\mathbb{Z})$) with elements of $K^\times/(K^\times)^p$ — a precursor to the local reciprocity map of section 8.4.
2. For $M = \mathbb{Z}/n\mathbb{Z}$ with trivial action and $\gcd(n, \mathrm{char}\mathbb{F}_K) = 1$: the dual is $M^* = \mu_n$, and Tate duality gives $H^1(G_K, \mathbb{Z}/n\mathbb{Z}) \times H^1(G_K, \mu_n) \to \frac{1}{n}\mathbb{Z}/\mathbb{Z}$. This pairing sends a character $\chi\colon G_K \to \mathbb{Z}/n\mathbb{Z}$ (classifying a cyclic extension L/K of degree dividing n) and a Kummer class $a \in K^\times/(K^\times)^n$ to $\mathrm{inv}_K((a, L/K))$, where $(a, L/K)$ is the norm residue symbol.
3. (If the reader has read [8]) For an elliptic curve E/K and $M = E[p]$ (the p-torsion): the Weil pairing gives $E[p]^* \cong E[p]$ (self-duality). Tate duality gives a perfect pairing $H^1(G_K, E[p]) \times H^1(G_K, E[p]) \to \mathbb{Q}/\mathbb{Z}$, and the image of the local Kummer map $E(K)/pE(K) \hookrightarrow H^1(G_K, E[p])$ is its own orthogonal complement under this pairing. This is the local condition defining the Selmer group (developed further in section 9.5).

Exercise 8.3.1

1. Let $K = \mathbb{Q}_p$ with p odd and $M = \mathbb{Z}/p\mathbb{Z}$ (trivial action). Compute $|H^r(G_{\mathbb{Q}_p}, M)|$ for $r = 0, 1, 2$ using the Euler–Poincaré characteristic and Tate duality. Determine which case ($\mu_p \subseteq \mathbb{Q}_p$ or not) applies.
2. A finite G_K-module M is *self-dual* if $M^* \cong M$ as G_K-modules. Show that Tate duality gives a perfect pairing $H^1(G_K, M) \times H^1(G_K, M) \to \mathbb{Q}/\mathbb{Z}$ for self-dual M.
3. Verify $\mathrm{cd}(\mathbb{Q}_p) = 2$ directly (for p odd) by showing $H^2(G_{\mathbb{Q}_p}, \mathbb{Z}/2\mathbb{Z}) \neq 0$ (using Tate duality and

$\mu_2(\mathbb{Q}_p) = \{\pm 1\}$) and $H^3(G_{\mathbb{Q}_p}, \mathbb{Z}/\ell\mathbb{Z}) = 0$ for all primes ℓ (using the LHS spectral sequence for $1 \to I_p \to G_{\mathbb{Q}_p} \to \widehat{\mathbb{Z}} \to 1$).

4. Determine the structure of $G_K^{(p)}$ (free or Demushkin, with invariants d and q) for: (a) $p \geq 5$ and $K = \mathbb{Q}_p$; (b) $p \geq 3$ and $K = \mathbb{Q}_p(\zeta_p)$.

5. Using the Euler–Poincaré characteristic, compute $|H^1(G_K, \mu_n)|$ for $K = \mathbb{Q}_p$ and $n = p$. Verify the result agrees with the Kummer theory computation $H^1(G_K, \mu_p) \cong K^\times/(K^\times)^p$.

8.4 The Cohomological Reciprocity Isomorphism

The invariant map $\mathrm{inv}_K \colon \mathrm{Br}(K) \xrightarrow{\sim} \mathbb{Q}/\mathbb{Z}$ (corollary 8.1.4, theorem 8.1.5) and the cup product pairing combine to produce the local reciprocity map: a canonical homomorphism $\mathrm{rec}_K \colon K^\times \to G_K^{\mathrm{ab}}$ that describes the abelian extensions of K in terms of the multiplicative group. The proof is purely cohomological, using the Tate–Nakayama theorem from Chapter 4. The abstract structure underlying the proof is that of a *class formation*: an axiomatic framework isolating the properties of $(G_K, (K^{\mathrm{sep}})^\times)$ that make reciprocity work.

What this section establishes is the reciprocity map as an isomorphism *of the cohomological formalism*: rec_K is the degree -2 image of the fundamental class under Tate–Nakayama, and its construction is complete within the cohomology developed here. Its *development* into class field theory belongs to *Class Field Theory* [17]: the identification of rec_K with the norm-residue symbol, the norm-group and conductor description of the abelian extensions of K, the local existence theorem, and the explicit reciprocity laws. That book takes the isomorphism proved here as its starting point.

Definition 8.4.1: Class Formation

A *class formation* is a pair (G, C) consisting of a profinite group G and a discrete G-module C satisfying two axioms:

1. *(Hilbert 90)* For every open subgroup $H \leq G$: $H^1(H, C) = 0$.

2. *(Brauer group structure)* For every pair of open subgroups $H \trianglelefteq H' \leq G$ with H'/H finite, there is a canonical isomorphism:

$$\mathrm{inv}_{H'/H} \colon H^2(H'/H, C^H) \xrightarrow{\sim} \frac{1}{|H'/H|}\mathbb{Z}/\mathbb{Z}$$

compatible with inflation: for $H \trianglelefteq H' \trianglelefteq H''$, the diagram

$$\begin{tikzcd}
H^2(H''/H, C^H) \arrow[r, "\mathrm{inv}"] & \frac{1}{|H''/H|}\mathbb{Z}/\mathbb{Z} \\
H^2(H''/H', C^{H'}) \arrow[u, leftarrow, "\mathrm{Inf}"'] \arrow[r, "\mathrm{inv}"'] & \frac{1}{|H''/H'|}\mathbb{Z}/\mathbb{Z} \arrow[u, hookleftarrow]
\end{tikzcd}$$

commutes (where the right vertical map is the natural inclusion).

Axiom (1) abstracts Hilbert's Theorem 90. Axiom (2) abstracts the invariant map on the Brauer group: it says that H^2 of the formation module, for any finite quotient of G, is cyclic of the expected order, with compatible identifications across the tower of quotients.

Theorem 8.4.2: The Local Class Formation

For a local field K, the pair $(G_K, (K^{\mathrm{sep}})^\times)$ is a class formation.

***Proof*:**
Axiom (1): for any open subgroup $H = \mathrm{Gal}(K^{\mathrm{sep}}/L)$ (corresponding to a finite extension L/K), $H^1(H, (K^{\mathrm{sep}})^\times) = \varinjlim_{L'/L} H^1(\mathrm{Gal}(L'/L), (L')^\times) = 0$ by Hilbert 90 (theorem 8.1.1).

Axiom (2): for $H = \mathrm{Gal}(K^{\mathrm{sep}}/L)$ and $H' = \mathrm{Gal}(K^{\mathrm{sep}}/K)$, the quotient $H'/H = \mathrm{Gal}(L/K)$ and $C^H = (K^{\mathrm{sep}})^{\times,\mathrm{Gal}(K^{\mathrm{sep}}/L)} = L^\times$. The invariant map $\mathrm{inv}_{\mathrm{Gal}(L/K)} \colon H^2(\mathrm{Gal}(L/K), L^\times) \xrightarrow{\sim} \frac{1}{[L:K]}\mathbb{Z}/\mathbb{Z}$ is the composite of the identification $H^2(\mathrm{Gal}(L/K), L^\times) \cong \mathrm{Br}(L/K)$ (theorem 7.2.4) and the invariant map $\mathrm{inv}_K|_{\mathrm{Br}(L/K)}$ (corollary 8.1.4). Compatibility with inflation follows from the compatibility of the invariant map (theorem 7.3.1(3)).

The class formation axioms are the input for the Tate–Nakayama theorem, which produces the reciprocity map. The argument proceeds through two key identifications from earlier chapters. The first, from the Tate cohomology of finite groups (proposition 4.1.5, theorem 2.5.2): the group $\hat{H}^{-2}(G, \mathbb{Z}) \cong H_1(G, \mathbb{Z}) \cong G^{\mathrm{ab}}$ for any finite group G, where $G^{\mathrm{ab}} = G/[G, G]$ is the abelianization. The second, from the periodicity theorem (theorem 4.3.4): for a cyclic group G, the cup product with the fundamental class $u \in \hat{H}^2(G, \mathbb{Z})$ gives an isomorphism $\hat{H}^r(G, M) \xrightarrow{\sim} \hat{H}^{r+2}(G, M)$ for all $r \in \mathbb{Z}$ and all G-modules M. The Tate–Nakayama theorem generalizes this: the cup product with the fundamental class $u_{L/K} \in H^2(\mathrm{Gal}(L/K), L^\times)$ of a class formation gives the same periodicity, even when $\mathrm{Gal}(L/K)$ is not cyclic.

Theorem 8.4.3: Tate–Nakayama for the Local Class Formation

Let L/K be a finite Galois extension of local fields with $G = \mathrm{Gal}(L/K)$. The *fundamental class* $u_{L/K} \in H^2(G, L^\times)$ is the unique element with $\mathrm{inv}_K(u_{L/K}) = 1/[L : K] \in \mathbb{Q}/\mathbb{Z}$. The cup product with $u_{L/K}$ gives an isomorphism:

$$\hat{H}^r(G, \mathbb{Z}) \xrightarrow{- \smile u_{L/K}} \hat{H}^{r+2}(G, L^\times)$$

for all $r \in \mathbb{Z}$.

***Proof*:**
Step 1: Reduction to cyclic groups. The Tate–Nakayama theorem for class formations (see [18]) states: if (G, C) is a class formation and $u \in H^2(G, C)$ is a generator (in the sense that $\mathrm{inv}(u)$ generates $\frac{1}{|G|}\mathbb{Z}/\mathbb{Z}$), then the cup product with u is an isomorphism $\hat{H}^r(G, \mathbb{Z}) \xrightarrow{\sim} \hat{H}^{r+2}(G, C)$ for all r, provided the isomorphism is verified for all cyclic subquotients of G. The reduction to cyclic subquotients uses the Nakayama criterion (theorem 4.4.4 from Chapter 4): a G-module M is cohomologically trivial if $\hat{H}^0(H, M) = \hat{H}^{-1}(H, M) = 0$ for all subgroups $H \leq G$. The cup product with u defines a map of δ-functors, and its cone is cohomologically trivial if the two key degrees vanish for all cyclic subgroups.

Step 2: Verification for cyclic extensions. For G cyclic: the class formation axioms give $|H^2(G, L^\times)| = [L : K]$ and $H^1(G, L^\times) = 0$. By the Herbrand quotient (theorem 8.2.6): $|\hat{H}^0(G, L^\times)| = [L : K]$ and $|\hat{H}^{-1}(G, L^\times)| = 1$. The periodicity theorem for cyclic groups (theorem 4.3.4) gives $\hat{H}^r(G, \mathbb{Z}) \xrightarrow{\sim} \hat{H}^{r+2}(G, \mathbb{Z})$ via the cup product with the canonical generator of $\hat{H}^2(G, \mathbb{Z}) \cong \mathbb{Z}/[L : K]\mathbb{Z}$. The fundamental class $u_{L/K}$ is the image of this generator under the isomorphism $\hat{H}^2(G, \mathbb{Z}) \cong H^2(G, L^\times)$ (matching the invariant $1/[L : K]$). The cup product with $u_{L/K}$ is therefore an isomorphism for G cyclic.

Step 3: Extension to general G. The Nakayama criterion reduces the general case to verifying $\hat{H}^0(H, \mathbb{Z}) \xrightarrow{\sim} \hat{H}^2(H, L^{\times, H'})$ for all cyclic subgroups H of quotients of G, where H' is the fixed field. This is handled by Step 2 applied to each cyclic subextension, using the compatibility of the fundamental class with restriction (proposition 4.2.5).

The local reciprocity law is the case $r = -2$ of the Tate–Nakayama isomorphism.

Theorem 8.4.4: Local Reciprocity

Let L/K be a finite Galois extension of local fields with $G = \mathrm{Gal}(L/K)$. There is a canonical isomorphism:

$$\mathrm{rec}_{L/K} \colon K^\times / N_{L/K}(L^\times) \xrightarrow{\sim} \mathrm{Gal}(L/K)^{\mathrm{ab}}$$

Passing to the limit over all finite Galois extensions L/K inside K^{sep} gives the *local Artin map*:

$$\mathrm{rec}_K \colon K^\times \to G_K^{\mathrm{ab}}$$

a continuous homomorphism with dense image, whose kernel is the intersection of all norm groups $\bigcap_L N_{L/K}(L^\times) = \{1\}$.

***Proof*:**
The Tate–Nakayama isomorphism (theorem 8.4.3) at $r = -2$ gives:

$$\hat{H}^{-2}(G, \mathbb{Z}) \xrightarrow{\sim} \hat{H}^0(G, L^\times)$$

The left side is $\hat{H}^{-2}(G, \mathbb{Z}) \cong H_1(G, \mathbb{Z}) \cong G^{\mathrm{ab}}$ (proposition 4.1.5, theorem 2.5.2). The right side is $\hat{H}^0(G, L^\times) = (L^\times)^G / N_G(L^\times) = K^\times / N_{L/K}(L^\times)$ (definition 4.1.2). The isomorphism $G^{\mathrm{ab}} \xrightarrow{\sim} K^\times / N_{L/K}(L^\times)$ is inverted to give $\mathrm{rec}_{L/K} \colon K^\times / N_{L/K}(L^\times) \xrightarrow{\sim} G^{\mathrm{ab}}$.

For the limit: the reciprocity maps $\mathrm{rec}_{L/K}$ are compatible as L varies (by the compatibility of the fundamental classes with inflation). Passing to the projective limit $G_K^{\mathrm{ab}} = \varprojlim_L \mathrm{Gal}(L/K)^{\mathrm{ab}}$ and the inductive limit on the norm groups gives a continuous map $\mathrm{rec}_K \colon K^\times \to G_K^{\mathrm{ab}}$. The density of the image follows from the surjectivity of each $\mathrm{rec}_{L/K}$. The triviality of the kernel $\bigcap_L N_{L/K}(L^\times) = \{1\}$ is a consequence of the existence theorem of local class field theory (every open subgroup of finite index in $K^\times$ is a norm group; see [17]).

The local Artin map has explicit descriptions on specific elements of $K^\times$.

Proposition 8.4.5: Properties of the Local Artin Map

Let K be a local field with residue field $\mathbb{F}_q$. The local Artin map $\mathrm{rec}_K : K^\times \to G_K^{\mathrm{ab}}$ satisfies:

1. A uniformizer π_K maps to a lift of the arithmetic Frobenius $\mathrm{Frob}_K \in \mathrm{Gal}(K^{\mathrm{nr}}/K)$ (the automorphism acting as $x \mapsto x^q$ on the residue field of K^{nr}).
2. The unit group $\mathcal{O}_K^\times$ maps onto the inertia subgroup $I_K^{\mathrm{ab}} \subseteq G_K^{\mathrm{ab}}$: $\mathrm{rec}_K(\mathcal{O}_K^\times) = I_K^{\mathrm{ab}}$.
3. For a finite abelian extension L/K: the kernel of rec_K composed with $G_K^{\mathrm{ab}} \twoheadrightarrow \mathrm{Gal}(L/K)$ is $N_{L/K}(L^\times)$.

***Proof*:**
For (1): by the construction of the invariant map (corollary 8.1.4), the fundamental class $u_{K_n^{\mathrm{nr}}/K}$ is determined by the Frobenius and the uniformizer. The Tate–Nakayama isomorphism at $r = -2$ sends the generator of $\hat{H}^{-2}(G, \mathbb{Z})$ (corresponding to $\mathrm{Frob}_K \in G^{\mathrm{ab}}$) to the class of π_K in $\hat{H}^0(G, (K_n^{\mathrm{nr}})^\times) = K^\times/N((K_n^{\mathrm{nr}})^\times)$. Inverting: $\mathrm{rec}_K(\pi_K) = \mathrm{Frob}_K$.

For (2): the exact sequence $1 \to \mathcal{O}_K^\times \to K^\times \xrightarrow{v_K} \mathbb{Z} \to 0$ and the decomposition $G_K^{\mathrm{ab}} \twoheadrightarrow \mathrm{Gal}(K^{\mathrm{nr}}/K) \cong \widehat{\mathbb{Z}}$ (with kernel I_K^{ab}) are compatible with rec_K: the composition $K^\times \xrightarrow{\mathrm{rec}_K} G_K^{\mathrm{ab}} \to \mathrm{Gal}(K^{\mathrm{nr}}/K)$ sends $\pi_K \mapsto \mathrm{Frob}_K$ (by (1)) and $\mathcal{O}_K^\times \mapsto 1$ (since units are norms in unramified extensions by theorem 8.1.3(1)). So $\mathrm{rec}_K(\mathcal{O}_K^\times) \subseteq I_K^{\mathrm{ab}}$. Surjectivity follows from the surjectivity of $\mathrm{rec}_{L/K}$ for totally ramified abelian L/K.

Part (3) is the definition of the reciprocity map: $\mathrm{rec}_{L/K}$ is an isomorphism $K^\times/N_{L/K}(L^\times) \xrightarrow{\sim} \mathrm{Gal}(L/K)$.

Example 8.4.6: The Local Artin Map

1. For $K = \mathbb{Q}_p$ (with p odd): $K^\times \cong p^{\mathbb{Z}} \times \mathbb{Z}_p^\times \cong \mathbb{Z} \times \mathbb{Z}/(p-1)\mathbb{Z} \times \mathbb{Z}_p$. The Artin map sends $p \mapsto \mathrm{Frob}_p^{-1}$ (the geometric Frobenius, by convention) and $\mathbb{Z}_p^\times \to I_p^{\mathrm{ab}}$. The inertia acts on the tame extensions (generated by p-power roots of p and roots of unity), and the unit part of the Artin map encodes the Galois action on these roots.
2. For $K = \mathbb{R}$: $K^\times = \mathbb{R}^\times$, $G_K = \mathrm{Gal}(\mathbb{C}/\mathbb{R}) \cong C_2$, and $N_{\mathbb{C}/\mathbb{R}}(z) = |z|^2$. So $\mathbb{R}^\times/N(\mathbb{C}^\times) = \mathbb{R}^\times/\mathbb{R}_{>0} \cong \{+1, -1\} \cong C_2$. The Artin map sends $-1 \mapsto \sigma$ (complex conjugation) and $\mathbb{R}_{>0} \mapsto 1$.
3. For $K = \mathbb{Q}_p$ and the unramified extension $L = \mathbb{Q}_{p^n}$: the norm group $N(L^\times)$ contains all units $\mathbb{Z}_p^\times$ (by theorem 8.1.3(1)) and $p^n\mathbb{Z} \subseteq N(p^{\mathbb{Z}})$ (since $N(p) = p^n$). So $\mathbb{Q}_p^\times/N(L^\times) \cong \mathbb{Z}/n\mathbb{Z}$, generated by the class of p. The Artin map sends $p \mapsto \mathrm{Frob}_p$, confirming $\mathrm{rec}_{L/K}(p) = \mathrm{Frob}_p$.

The local reciprocity law established in this section is used in two directions. The book [17] cites it as the cohomological construction of the local Artin map, and Chapter 9 uses the local reciprocity maps as ingredients in the global theory.

Exercise 8.4.1

1. Verify axiom (2) of the class formation definition for the pair $(G_{\mathbb{Q}_p}, \bar{\mathbb{Q}}_p^\times)$ in the case of the unramified extension $\mathbb{Q}_{p^2}/\mathbb{Q}_p$: show that $H^2(\mathrm{Gal}(\mathbb{Q}_{p^2}/\mathbb{Q}_p), \mathbb{Q}_{p^2}^\times) \cong \mathbb{Z}/2\mathbb{Z}$ and identify the invariant map.

2. Let $K = \mathbb{Q}_p$ (with p odd) and $L = K(\sqrt[p]{\pi})$ where $\pi = p$ is a uniformizer. This is a totally ramified cyclic extension of degree p. Determine the norm group $N_{L/K}(L^\times) \subseteq \mathbb{Q}_p^\times$ and describe $\mathrm{rec}_{L/K}$ explicitly.

3. Show that the restriction of the Artin map to $\mathcal{O}_K^\times$ is surjective onto I_K^{ab} for $K = \mathbb{Q}_p$ by showing that every finite totally ramified abelian extension L/K satisfies $N_{L/K}(\mathcal{O}_L^\times) \subsetneq \mathcal{O}_K^\times$.

4. Let $K \subseteq L \subseteq M$ be a tower of finite abelian extensions of local fields. Show that the Artin maps are compatible: $\mathrm{rec}_{M/K} = \mathrm{Inf} \circ \mathrm{rec}_{L/K}$ on $K^\times/N_{M/K}(M^\times)$, where $\mathrm{Inf}\colon \mathrm{Gal}(L/K) \hookrightarrow \mathrm{Gal}(M/K)$ is the natural map.

5. (If the reader has read [17]) The *existence theorem* of local class field theory states that every open subgroup of finite index in $K^\times$ is the norm group $N_{L/K}(L^\times)$ for a unique finite abelian extension L/K. Using this, show that the local Artin map $\mathrm{rec}_K\colon K^\times \to G_K^{\mathrm{ab}}$ induces a bijection between open subgroups of finite index in $K^\times$ and open subgroups of G_K^{ab}.

9

Global Fields

The local theory of Chapter 8 computed the cohomology of G_K for a local field K, established local Tate duality, and proved the local reciprocity law via the Tate–Nakayama theorem. The global theory faces a fundamentally different challenge: the absolute Galois group of a number field acts on modules that carry information at every place simultaneously, and the relationship between local and global cohomology is governed by nontrivial exact sequences rather than by a single computation.

Early cohomological approaches to global class field theory (Artin–Tate, 1950s) attempted to work directly with the multiplicative group $L^\times$ and the ideal class group, mirroring the ideal-theoretic formulation of Takagi and Artin. The difficulty is that $H^n(G, L^\times)$ captures global information (the Brauer group, Hilbert 90) but offers no access to the individual local completions: there is no natural map from $H^n(G, L^\times)$ to $H^n(G_w, L_w^\times)$ that accounts for all places simultaneously. The idèlic reformulation resolves this. The appendix assembled local data into the idèle group $\mathbb{I}_L$ and established the Shapiro decomposition $H^n(G, \mathbb{I}_L) \cong \bigoplus_v H^n(G_w, L_w^\times)$ (theorem A.3.5), making the local cohomology groups—computed in Chapter 8—directly visible as summands of a global object. The fundamental exact sequence $1 \to L^\times \to \mathbb{I}_L \to C_L \to 1$ (proposition A.3.6) then provides the exact relationship between the global cohomology $H^n(G, L^\times)$ and the local decomposition $H^n(G, \mathbb{I}_L)$, with the idèle class group C_L measuring the discrepancy. This chapter exploits this structure, culminating in the Poitou–Tate exact sequence, the global Euler–Poincaré characteristic, and the global reciprocity law. The chapter closes with Selmer groups and the Shafarevich–Tate group, connecting the cohomological machinery to the arithmetic of elliptic curves and abelian varieties.

9.1 Idèle Cohomology and the Global Fundamental Class

The fundamental exact sequence $1 \to L^\times \to \mathbb{I}_L \to C_L \to 1$ of G-modules (proposition A.3.6), combined with the Shapiro decomposition of $H^n(G, \mathbb{I}_L)$ (theorem A.3.5), reduces the study of $H^n(G, C_L)$ to understanding the relationship between the global cohomology $H^n(G, L^\times)$ (the Brauer group and its relatives) and the local cohomology groups $H^n(G_w, L_w^\times)$. This section carries out this analysis in low degrees, establishes the vanishing of $H^1(G, C_L)$ (via the Albert–Brauer–Hasse–Noether theorem), and constructs the global fundamental class $u_{L/K} \in H^2(G, C_L)$ that will drive the global reciprocity law in section 9.4.

Throughout this section, L/K denotes a finite Galois extension of number fields with $G = \mathrm{Gal}(L/K)$. For each place v of K, a place w of L above v is fixed, with decomposition group $G_w = \mathrm{Gal}(L_w/K_v) \leq G$.

The long exact sequence of the fundamental short exact sequence begins:

$$1 \to K^\times \to \mathbb{I}_K \to C_K \xrightarrow{\delta^0} H^1(G, L^\times) \xrightarrow{\alpha_1} H^1(G, \mathbb{I}_L) \xrightarrow{\beta_1} H^1(G, C_L) \xrightarrow{\delta^1} H^2(G, L^\times) \xrightarrow{\alpha_2} H^2(G, \mathbb{I}_L)$$

The first three terms are the definition of C_K (the fixed points $\mathbb{I}_L^G = \mathbb{I}_K$ by theorem A.3.2 and $C_L^G = C_K$). Hilbert 90 (theorem 8.1.1) gives $H^1(G, L^\times) = 0$, simplifying the sequence to:

$$0 \to H^1(G, \mathbb{I}_L) \to H^1(G, C_L) \xrightarrow{\delta^1} H^2(G, L^\times) \xrightarrow{\alpha_2} H^2(G, \mathbb{I}_L) \to \cdots$$

The Shapiro decomposition (theorem A.3.5) and local Hilbert 90 (theorem 8.1.1) give $H^1(G, \mathbb{I}_L) \cong \bigoplus_v H^1(G_w, L_w^\times) = 0$. The sequence reduces further:

Proposition 9.1.1: H^1 of the Idèle Class Group

$H^1(G, C_L) \cong \ker\big(H^2(G, L^\times) \xrightarrow{\alpha_2} H^2(G, \mathbb{I}_L)\big)$.

***Proof*:**

The vanishing of $H^1(G, L^\times)$ and $H^1(G, \mathbb{I}_L)$ reduces the long exact sequence to $0 \to H^1(G, C_L) \xrightarrow{\delta^1} H^2(G, L^\times) \xrightarrow{\alpha_2} H^2(G, \mathbb{I}_L)$, and δ^1 identifies $H^1(G, C_L)$ with $\ker \alpha_2$.

The map $\alpha_2\colon H^2(G, L^\times) \to H^2(G, \mathbb{I}_L)$ sends a global Brauer class to its collection of local components. Identifying $H^2(G, L^\times) \cong \mathrm{Br}(L/K)$ (theorem 7.2.4) and $H^2(G, \mathbb{I}_L) \cong \bigoplus_v H^2(G_w, L_w^\times) \cong \bigoplus_v \mathrm{Br}(L_w/K_v)$ (theorem A.3.5), the map α_2 is the localization map sending a global central simple algebra to its local invariants. The group $H^1(G, C_L)$ therefore measures the failure of the Hasse principle for the Brauer group: it consists of global Brauer classes that are split at every place.

The following theorem, whose proof was deferred from Chapter **??**, states that this failure does not occur.

Theorem 9.1.2: Albert–Brauer–Hasse–Noether

The localization sequence:

$$0 \to \mathrm{Br}(L/K) \xrightarrow{\text{loc}} \bigoplus_v \mathrm{Br}(L_w/K_v) \xrightarrow{\sum \mathrm{inv}_v} \mathbb{Q}/\mathbb{Z}$$

is exact. In particular:

1. A central simple algebra over K split by L is trivial if and only if it is locally trivial at every place (Hasse principle for the Brauer group).
2. $H^1(G, C_L) = 0$.

***Proof*:**
Step 1: The sequence is a complex. For a global class $[\alpha] \in \mathrm{Br}(L/K)$ with local invariants $\mathrm{inv}_v(\alpha_v) \in \mathbb{Q}/\mathbb{Z}$ at each place v: $\sum_v \mathrm{inv}_v(\alpha_v) = 0$ in $\mathbb{Q}/\mathbb{Z}$. This follows from the compatibility of the global and local invariant maps with the product formula. Given the fundamental class $u_{L_w/K_v} \in H^2(G_w, L_w^\times)$ at each place and the global class $[\alpha] \in H^2(G, L^\times)$: the restriction $\mathrm{res}_w([\alpha])$ to G_w maps to $H^2(G_w, L_w^\times) \cong \mathrm{Br}(L_w/K_v)$, and the local invariant is $\mathrm{inv}_v(\mathrm{res}_w([\alpha]))$. The sum $\sum_v \mathrm{inv}_v = 0$ follows from the global reciprocity law applied to the norm residue symbol (or equivalently, from the product formula for the Hilbert symbol - proved for $\mathbb{Q}$ and quadratic symbols in [10, The Product Formula]).

Step 2: Injectivity (Hasse principle) for cyclic extensions. Suppose G is cyclic of order n. The argument requires two inputs.

The *second inequality* (algebraic): $|\hat{H}^0(G, C_L)| \le [L : K]$. The Tate cohomology long exact sequence (using $\hat{H}^1(G, L^\times) = 0$ by Hilbert 90) gives a surjection $\hat{H}^0(G, \mathbb{I}_L) \twoheadrightarrow \hat{H}^0(G, C_L)$, i.e., the norm index $[C_K : N(C_L)]$ is bounded by the index $[\mathbb{I}_K : K^\times \cdot N(\mathbb{I}_L)]$. The latter is at most n: the subgroup $K^\times \cdot N(\mathbb{I}_L)$ contains $K^\times$ and the local norm images $N(L_w^\times)$ at each place, and the local norm index $[K_v^\times : N(L_w^\times)] \le [L_w : K_v] \le n$ at each v.

The *first inequality* (analytic): $|\hat{H}^0(G, C_L)| \ge [L : K]$. This requires the non-vanishing of Hecke L-functions at $s = 1$; the proof is deferred to [17].

Combining: $|\hat{H}^0(G, C_L)| = n$. By the periodicity of Tate cohomology for cyclic groups (theorem 4.3.4), $|H^2(G, C_L)| = n$. The Herbrand quotient $h(G, C_L) = |\hat{H}^0(G, C_L)|/|H^1(G, C_L)| = n/|H^1(G, C_L)|$ can also be computed independently as $h(G, C_L) = n$ (via the multiplicativity of the Herbrand quotient along the fundamental exact sequence, using the compactness of C_L^0; see [18] for the detailed computation). This gives $|H^1(G, C_L)| = 1$, and proposition 9.1.1 then gives $\ker(\alpha_2) = 0$.

The "first inequality" $|H^2(G, C_L)| \ge [L : K]$ (equivalently, $|\hat{H}^0(G, C_L)| \ge [L : K]$, i.e., $[C_K : N(C_L)] \ge [L : K]$) requires input beyond the cohomological formalism. For cyclic extensions, it can be proved using the theory of L-functions (the non-vanishing of $L(1, \chi)$ for nontrivial Dirichlet characters); see [17] for the complete argument. Combining both inequalities: $|H^2(G, C_L)| = [L : K]$, $h(G, C_L) = 1$, and therefore $|H^1(G, C_L)| = 1$, i.e., $H^1(G, C_L) = 0$.

Step 3: Reduction to cyclic. For general G: the vanishing of $H^1(G, C_L)$ for all cyclic subextensions L'/K' within L/K implies the vanishing for G itself, by the following argument. Let $[\alpha] \in \ker(\alpha_2)$, so $[\alpha]$ is locally trivial. For any cyclic subgroup $H \leq G$ with fixed field $K' = L^H$: $\mathrm{res}_{K'}([\alpha]) \in H^2(\mathrm{Gal}(L/K'), L^\times) = \mathrm{Br}(L/K')$ is still locally trivial (restriction commutes with localization), and by the cyclic case $\mathrm{res}_{K'}([\alpha]) = 0$. The restriction-corestriction formula $\mathrm{cor} \circ \mathrm{res} = [G : H]$ (**??** from Chapter 3) shows that $[G : H] \cdot [\alpha] = 0$ for every cyclic subgroup H. The gcd of the indices $[G : H]$ over all cyclic subgroups is 1 (a finite group is generated by its cyclic subgroups), so $[\alpha] = 0$.

Part (2) follows from (1) and proposition 9.1.1: $H^1(G, C_L) \cong \ker(\alpha_2) = 0$.

The ABHN theorem passes to the limit over all finite Galois extensions L/K to give the global Brauer group exact sequence.

Corollary 9.1.3: Global Brauer Group Sequence

The sequence:

$$0 \to \mathrm{Br}(K) \xrightarrow{\mathrm{loc}} \bigoplus_v \mathrm{Br}(K_v) \xrightarrow{\sum \mathrm{inv}_v} \mathbb{Q}/\mathbb{Z} \to 0$$

is exact.

***Proof*:**
Passing to the colimit $\mathrm{Br}(K) = \varinjlim_L \mathrm{Br}(L/K)$ and $\bigoplus_v \mathrm{Br}(K_v) = \varinjlim_L \bigoplus_v \mathrm{Br}(L_w/K_v)$: the injectivity of loc follows from theorem 9.1.2(1), and exactness at the middle term follows from the complex property (Step 1) and the surjectivity of each local invariant map $\mathrm{inv}_v \colon \mathrm{Br}(K_v) \xrightarrow{\sim} \mathbb{Q}/\mathbb{Z}$ (corollary 8.1.4). Surjectivity at $\mathbb{Q}/\mathbb{Z}$: for any $a/n \in \mathbb{Q}/\mathbb{Z}$, the local class at a single finite place v with $\mathrm{inv}_v = a/n$ and all other invariants zero lifts to a global class (choose L/K containing the local extension of degree n at v).

This exact sequence is the definitive statement about the Brauer group of a number field: a central simple algebra over K is determined up to Brauer equivalence by its local invariants, which are rational numbers summing to zero. This fulfills the promise made in section 7.3 of Chapter **??**.

With $H^1(G, C_L) = 0$ established, the construction of the global fundamental class proceeds as in the local case.

Definition 9.1.4: Global Invariant Map

The *global invariant map* $\mathrm{inv}_{L/K}\colon H^2(G, C_L) \to \mathbb{Q}/\mathbb{Z}$ is the composition:

$$H^2(G, C_L) \xrightarrow{\delta^{-1}} \mathrm{coker}(\alpha_2) \hookrightarrow \mathbb{Q}/\mathbb{Z}$$

where the first map is the connecting homomorphism from the long exact sequence (using the surjectivity of α_2 onto $\bigoplus_v \mathrm{Br}(L_w/K_v)$ within the image of $\mathrm{Br}(L/K)$, and the cokernel is identified via $\sum \mathrm{inv}_v$). More directly: the long exact sequence and $H^1(G, C_L) = 0$ give an exact sequence:

$$H^2(G, L^\times) \xrightarrow{\alpha_2} H^2(G, \mathbb{I}_L) \to H^2(G, C_L) \xrightarrow{\delta^2} H^3(G, L^\times)$$

The global invariant map is defined by the property that for each place v, the composition $H^2(G_w, L_w^\times) \to H^2(G, \mathbb{I}_L) \to H^2(G, C_L) \xrightarrow{\mathrm{inv}} \mathbb{Q}/\mathbb{Z}$ equals the local invariant inv_v.

Theorem 9.1.5: The Global Fundamental Class

The global invariant map $\mathrm{inv}_{L/K}\colon H^2(G, C_L) \to \mathbb{Q}/\mathbb{Z}$ is an isomorphism onto $\frac{1}{[L:K]}\mathbb{Z}/\mathbb{Z}$, compatible with inflation. The *global fundamental class* is the unique element $u_{L/K} \in H^2(G, C_L)$ with:

$$\mathrm{inv}_{L/K}(u_{L/K}) = \frac{1}{[L:K]}$$

Its image in $H^2(G, \mathbb{I}_L) \cong \bigoplus_v H^2(G_w, L_w^\times)$ has v-component equal to the local fundamental class $u_{L_w/K_v} \in H^2(G_w, L_w^\times)$ (theorem 8.4.3).

***Proof*:**
Step 1: The cyclic case — second inequality. For G cyclic of order n: the Tate cohomology long exact sequence (with $\hat{H}^1(G, L^\times) = H^1(G, L^\times) = 0$ by Hilbert 90) gives a surjection $\hat{H}^0(G, \mathbb{I}_L) \twoheadrightarrow \hat{H}^0(G, C_L)$, i.e., $\mathbb{I}_K/N(\mathbb{I}_L) \twoheadrightarrow C_K/N(C_L)$. The norm index satisfies $|C_K/N(C_L)| \leq [L : K]$: the subgroup $K^\times \cdot N(\mathbb{I}_L)$ has index at most n in $\mathbb{I}_K$, since $N(\mathbb{I}_L)$ contains $\mathcal{O}_v^\times$ at all unramified places (theorem 8.1.3(1)) and the approximation theorem (see [6, Approximation Theorem]) implies that $K^\times$ and the local norms together generate a subgroup of bounded index. Combined with $H^1(G, C_L) = 0$ (theorem 9.1.2(2)) and the periodicity of Tate cohomology for cyclic groups (theorem 4.3.4): $|H^2(G, C_L)| = |\hat{H}^0(G, C_L)| \leq [L : K]$.

Step 2: The first inequality. The reverse bound $|H^2(G, C_L)| \geq [L : K]$ (equivalently, $[C_K : N(C_L)] \geq [L : K]$) requires input beyond the cohomological formalism: for cyclic extensions, it is proved using the non-vanishing of Hecke L-functions at $s = 1$ (see [17] for the complete argument). Combining both inequalities: $|H^2(G, C_L)| = [L : K]$.

Step 3: The general case. For arbitrary G: the restriction-corestriction argument of theorem 9.1.2(Step 3) reduces to the cyclic case. The compatibility of the global invariant map with restriction and the known structure for cyclic subextensions determine $\mathrm{inv}_{L/K}$ as an isomorphism

$H^2(G, C_L) \xrightarrow{\sim} \frac{1}{[L:K]}\mathbb{Z}/\mathbb{Z}$. Compatibility with inflation follows from the compatibility of the local fundamental classes with inflation (proposition 4.2.5 from Chapter 4).

The global fundamental class is the central input for the global reciprocity law in section 9.4. Its local-global compatibility—the v-component is the local fundamental class—is the mechanism by which the global Artin map restricts to the local Artin maps at each place.

Example 9.1.6: The Global Fundamental Class for Quadratic Extensions

Let $L = K(\sqrt{d})$ for some non-square $d \in K^\times$, so $G \cong C_2$. The global invariant map gives $H^2(G, C_L) \cong \mathbb{Z}/2\mathbb{Z}$, generated by $u_{L/K}$ with $\mathrm{inv}(u_{L/K}) = 1/2$. The norm index $[C_K : N_{L/K}(C_L)] = 2$ (from the first and second inequalities). The local components: $\mathrm{inv}_v(u_{L_w/K_v}) = 1/2$ if v is ramified or v is a real place where $d < 0$, and $\mathrm{inv}_v = 0$ otherwise. The constraint $\sum_v \mathrm{inv}_v = 0 \in \mathbb{Q}/\mathbb{Z}$ recovers the fact that the number of ramified places (including archimedean) is even.

Exercise 9.1.1

1. Let L/K be a cyclic extension of prime degree p. Show the second inequality $[C_K : N_{L/K}(C_L)] \leq p$ directly: for each place v, the local norm index $[K_v^\times : N(L_w^\times)]$ divides p, and the global norm index is at most the product of the local norm indices at the finitely many non-split places.
2. Show that the ABHN sequence is a complex: for any $[\alpha] \in \mathrm{Br}(K)$, $\sum_v \mathrm{inv}_v(\alpha_v) = 0$ in $\mathbb{Q}/\mathbb{Z}$. (*Hint:* use the Hilbert symbol and the product formula - proved for $\mathbb{Q}$ and quadratic symbols in [10, The Product Formula], and in general a form of the reciprocity law.)
3. Compute $H^2(G, C_L)$ for $L = \mathbb{Q}(i)$, $K = \mathbb{Q}$, by identifying the ramified and archimedean places and verifying that the local invariants sum to zero.
4. Show that the ABHN theorem implies the Hasse norm theorem for cyclic extensions: an element $a \in K^\times$ is a norm from $L^\times$ if and only if $a \in N_{L_w/K_v}(L_w^\times)$ for every place v. (*Hint:* use $\hat{H}^0(G, L^\times) = K^\times/N(L^\times)$ and the Tate cohomology long exact sequence.)
5. Using corollary 9.1.3, compute $\mathrm{Br}(\mathbb{Q})$. Show that every element of $\mathrm{Br}(\mathbb{Q})$ is represented by a quaternion algebra $(a, b)_{\mathbb{Q}}$ for suitable $a, b \in \mathbb{Q}^\times$, and describe the local invariants of $(a, b)_{\mathbb{Q}}$ in terms of the Hilbert symbol.

9.2 The Poitou–Tate Exact Sequence

Local Tate duality (theorem 8.3.2 from Chapter 8) provides a perfect pairing $H^r(G_v, M) \times H^{2-r}(G_v, M^*) \to \mathbb{Q}/\mathbb{Z}$ for finite G_{K_v}-modules M at each place v of a number field K. The Poitou–Tate exact sequence is the global counterpart: it assembles these local pairings into a nine-term exact sequence that controls the relationship between global and local cohomology of finite Galois modules. The sequence is the master result of global Galois cohomology—every arithmetic duality theorem for number fields is either a special case of it or a consequence of its structure.

Throughout this section, K is a number field, M is a finite G_K-module (a finite abelian group with a continuous action of G_K), and M^* denotes its Cartier dual.

Definition 9.2.1: Cartier Dual

Let M be a finite G_K-module. The *Cartier dual* of M is the finite G_K-module:

$$M^* = \mathrm{Hom}_{\mathbb{Z}}(M, (K^{\mathrm{sep}})^\times)$$

where G_K acts by $(\sigma \cdot f)(m) = \sigma(f(\sigma^{-1}m))$ for $\sigma \in G_K$, $f \in M^*$, $m \in M$. Since M is finite of exponent n, every homomorphism $M \to (K^{\mathrm{sep}})^\times$ lands in $\mu_n \subseteq (K^{\mathrm{sep}})^\times$, so $M^* = \mathrm{Hom}_{\mathbb{Z}}(M, \mu_n)$ as a G_K-module.

The Cartier dual satisfies $(M^*)^* \cong M$ (canonically), and for $M = \mathbb{Z}/n\mathbb{Z}$ (with trivial action), $M^* \cong \mu_n$ (with the cyclotomic action). The duality exchanges "constant" and "multiplicative" modules.

Definition 9.2.2: S-Ramified Galois Group

Let S be a finite set of places of K containing all archimedean places and all places where M is ramified (i.e., where the inertia group I_v acts nontrivially on M). The *maximal S-ramified extension* K_S is the maximal algebraic extension of K unramified at all $v \notin S$, and:

$$G_{K,S} = \mathrm{Gal}(K_S/K)$$

The action of G_K on M factors through $G_{K,S}$ (since I_v acts trivially for $v \notin S$). A critical distinction: while the action factors through $G_{K,S}$, cocycles $G_K \to M$ need not factor through $G_{K,S}$, so $H^n(G_K, M) \neq H^n(G_{K,S}, M)$ in general. For instance, $H^1(G_K, \mathbb{Z}/p\mathbb{Z}) = \mathrm{Hom}(G_K, \mathbb{Z}/p\mathbb{Z})$ classifies *all* cyclic p-extensions of K and is infinite, while $H^1(G_{K,S}, \mathbb{Z}/p\mathbb{Z})$ classifies only those unramified outside S and is finite. The Poitou–Tate sequence requires finite cohomology groups and is formulated for $G_{K,S}$.

For each place v of K, the choice of an embedding $\overline{K} \hookrightarrow \overline{K}_v$ determines a continuous homomorphism $G_v = G_{K_v} \to G_K$, well-defined up to conjugacy. The restriction along this map gives the *localization maps.*

Definition 9.2.3: Localization Maps

For each $r \geq 0$ and each place v of K, the *localization map* is the restriction:

$$\mathrm{loc}_v \colon H^r(G_{K,S}, M) \to H^r(G_v, M)$$

These assemble into a single map:

$$\mathrm{loc} \colon H^r(G_{K,S}, M) \to \bigoplus_{v \in S} H^r(G_v, M)$$

The restriction to $v \in S$ is essential: at $v \notin S$, the unramified cohomology $H^1_{\mathrm{nr}}(G_v, M) = H^1(G_v/I_v, M)$ can be nontrivial (e.g., $H^1_{\mathrm{nr}}(G_v, \mathbb{Z}/p\mathbb{Z}) = \mathbb{Z}/p\mathbb{Z}$ for every v), so the sum over all v would be infinite. The localization maps for $v \in S$ factor through $G_{K,S}$ via the decomposition groups at places in S.

The local Tate duality pairings allow the local cohomology groups to be identified with duals of cohomology groups of M^*. The *dual localization map* is the composition:

$$\bigoplus_{v\in S} H^r(G_v, M) \xrightarrow{\bigoplus_v \mathrm{inv}_v} \bigoplus_{v\in S} H^{2-r}(G_v, M^*)^* \xleftarrow{\mathrm{loc}^*} H^{2-r}(G_{K,S}, M^*)^*$$

where inv_v is the local Tate duality isomorphism (theorem 8.3.2) and loc^* is the Pontryagin dual of loc for M^*.

Definition 9.2.4: Shafarevich–Tate Groups

For a finite G_K-module M and $r \geq 0$, the *rth Shafarevich–Tate group* is:

$$\text{Ш}^r(K, M) = \ker\Big(H^r(G_{K,S}, M) \xrightarrow{\mathrm{loc}} \bigoplus_{v\in S} H^r(G_v, M)\Big)$$

the subgroup of global cohomology classes that are locally trivial at every place of S. For S large enough (containing all archimedean places, all places where M is ramified, and all places dividing $|M|$), the group $\text{Ш}^r(K, M)$ is independent of S: at $v \notin S$, every class in $H^r(G_{K,S}, M)$ is automatically unramified, and the unramified local cohomology $H^r_{\mathrm{nr}}(G_v, M)$ vanishes for $r \geq 1$ and $v \nmid |M|$.

The groups $\text{Ш}^r(K, M)$ measure the failure of the local-global principle at the level of cohomology: a class in Ш^r is "globally nontrivial but locally invisible." For $r = 0$: $\text{Ш}^0(K, M) = \ker(M^{G_K} \to \prod_v M^{G_v})$, which is trivial (an element fixed by G_K is fixed by every decomposition group). For $r = 1$ and $r = 2$, these groups are nontrivial in general and are the central objects of the Poitou–Tate sequence.

Theorem 9.2.5: Poitou–Tate Exact Sequence

Let M be a finite G_K-module with Cartier dual M^*. There is a nine-term exact sequence of finite groups:

$$\begin{aligned} 0 \to \text{Ш}^1(K, M) \to H^1(G_{K,S}, M) &\xrightarrow{\mathrm{loc}} \bigoplus_{v\in S} H^1(G_v, M) \\ \to H^1(G_{K,S}, M^*)^* \to H^2(G_{K,S}, M) &\xrightarrow{\mathrm{loc}} \bigoplus_{v\in S} H^2(G_v, M) \\ &\to H^0(G_{K,S}, M^*)^* \to 0 \end{aligned} \tag{9.1}$$

The maps from local cohomology to the duals of global cohomology are induced by the local Tate duality pairings and the sum of local invariants $\sum_v \mathrm{inv}_v \colon \bigoplus_v \mathrm{Br}(K_v) \to \mathbb{Q}/\mathbb{Z}$.

***Proof*:**
The proof assembles three ingredients: the global Brauer group exact sequence (corollary 9.1.3), local Tate duality (theorem 8.3.2), and the cup product pairing $H^r(G_{K,S}, M) \times H^{2-r}(G_{K,S}, M^*) \xrightarrow{\smile} H^2(G_{K,S}, \mu) \xrightarrow{\sum \mathrm{inv}_v} \mathbb{Q}/\mathbb{Z}$.

Step 1: The sequence is a complex. The composition of consecutive maps must vanish. The critical composition is:

$$H^1(G_{K,S}, M) \xrightarrow{\text{loc}} \bigoplus_{v \in S} H^1(G_v, M) \to H^1(G_{K,S}, M^*)^*$$

For a global class $\alpha \in H^1(G_{K,S}, M)$ and a global class $\beta \in H^1(G_{K,S}, M^*)$: the image of $\text{loc}(\alpha)$ under the dual map evaluates to $\sum_v \text{inv}_v(\text{loc}_v(\alpha) \cup \text{loc}_v(\beta))$. By the compatibility of the global and local cup products with the localization maps: $\text{loc}_v(\alpha) \cup \text{loc}_v(\beta) = \text{loc}_v(\alpha \cup \beta)$. So the sum becomes $\sum_v \text{inv}_v(\text{loc}_v(\alpha \cup \beta)) = 0$ by the global Brauer group exact sequence (corollary 9.1.3: the sum of local invariants of a global Brauer class is zero). The argument for the other compositions is analogous.

Step 2: Exactness at $H^1(G_{K,S}, M)$. The kernel of loc is $Ш^1(K, M)$ by definition, so exactness at the first H^1 term is tautological. The content is the exactness at $\bigoplus_{v \in S} H^1(G_v, M)$: a local family $(\alpha_v)_v$ lies in the kernel of the map to $H^1(G_{K,S}, M^*)^*$ if and only if $\sum_v \text{inv}_v(\alpha_v \cup \beta_v) = 0$ for every $\beta \in H^1(G_{K,S}, M^*)$ (where $\beta_v = \text{loc}_v(\beta)$). The proof that this kernel equals im(loc) uses a duality argument: the orthogonal complement of $\text{loc}(H^1(G_{K,S}, M^*))$ under the sum of local pairings equals $\text{loc}(H^1(G_{K,S}, M))$, by the non-degeneracy of the local pairings and the ABHN exact sequence applied to the cup product $H^1(G_{K,S}, M) \times H^1(G_{K,S}, M^*) \to \text{Br}(K)$.

Step 3: Exactness at $H^1(G_{K,S}, M^*)^*$. The map $\bigoplus_{v \in S} H^1(G_v, M) \to H^1(G_{K,S}, M^*)^*$ is the Pontryagin dual of $H^1(G_{K,S}, M^*) \xrightarrow{\text{loc}} \bigoplus_{v \in S} H^1(G_v, M^*)$, composed with the local duality isomorphisms. Its cokernel maps injectively into $H^2(G_{K,S}, M)$ via the connecting map from the global-to-local spectral sequence. The exactness at $H^1(G_{K,S}, M^*)^*$ follows from the snake lemma applied to the commutative diagram relating the global and local long exact sequences.

Step 4: Exactness at the H^2 terms. The argument mirrors Steps 2–3 with $r = 2$: the exactness at $\bigoplus_{v \in S} H^2(G_v, M)$ uses the global Brauer group sequence and the duality between $Ш^2(K, M)$ and $Ш^1(K, M^*)$. The surjectivity onto $H^0(G_{K,S}, M^*)^*$ (the final nonzero term) follows from the finiteness of all groups involved and a counting argument using the global Euler–Poincaré characteristic.

Step 5: Finiteness. All groups in the sequence are finite. The global cohomology groups $H^r(G_{K,S}, M)$ are finite for finite M (a consequence of the finiteness of the class group and the Dirichlet unit theorem; see [18]). The local groups $H^r(G_v, M)$ are finite by local Tate duality (theorem 8.3.2). The Sha groups, as kernels of maps between finite groups, are finite. The Pontryagin duals of finite groups are finite.

The proof of the Poitou–Tate sequence is among the most technical in algebraic number theory. The full details—particularly the precise identification of the connecting maps and the verification of exactness at $H^1(G_{K,S}, M^*)^*$—require careful diagram chases using the machinery of derived categories or the Artin–Verdier duality formalism. The complete proof is given in [18]; the argument above establishes the structural framework and the key inputs from which the exactness follows.

The most striking consequence of the Poitou–Tate sequence is the duality between the Shafarevich–Tate groups.

Theorem 9.2.6: Poitou–Tate Duality

For a finite G_K-module M:

$$Ш^1(K, M) \cong Ш^2(K, M^*)^*$$

as finite abelian groups, where $(-)^*$ on the right denotes the Pontryagin dual.

***Proof*:**
The nine-term exact sequence (theorem 9.2.5) applied to M^* gives:

$$0 \to Ш^1(K, M^*) \to H^1(G_{K,S}, M^*) \xrightarrow{\text{loc}} \bigoplus_{v \in S} H^1(G_v, M^*) \to H^1(G_{K,S}, M)^* \to \cdots$$

The exactness at $\bigoplus_{v\in S} H^1(G_v, M^*)$ identifies $\operatorname{coker}\big(\text{loc}\colon H^1(G_{K,S}, M^*) \to \bigoplus_{v\in S} H^1(G_v, M^*)\big)$ with a subgroup of $H^1(G_{K,S}, M)^*$. Dualizing: the Pontryagin dual of $Ш^1(K, M^*)$ appears as the cokernel of the transposed localization map in the sequence for M, which is $Ш^2(K, M)$. The identification $Ш^1(K, M) \cong Ш^2(K, M^*)^*$ follows by applying the same argument with M and M^* interchanged.

Poitou–Tate duality swaps the degree and the module: $Ш^1$ for M is dual to $Ш^2$ for the Cartier dual M^*. Since the Cartier dual of $\mathbb{Z}/n\mathbb{Z}$ is μ_n and vice versa, the duality interchanges "additive" and "multiplicative" local-global obstructions.

Example 9.2.7: Poitou–Tate for $\mathbb{Z}/n\mathbb{Z}$ and μ_n

1. For $M = \mathbb{Z}/n\mathbb{Z}$ (constant module, trivial G_K-action): $M^* = \mu_n$. The group $H^1(G_{K,S}, \mathbb{Z}/n\mathbb{Z}) \cong \operatorname{Hom}(G_{K,S}, \mathbb{Z}/n\mathbb{Z})$ classifies cyclic extensions of K of degree dividing n, and $Ш^1(K, \mathbb{Z}/n\mathbb{Z})$ consists of cyclic extensions that split at every place — i.e., everywhere locally trivial abelian extensions. By the Poitou–Tate sequence, $Ш^1(K, \mathbb{Z}/n\mathbb{Z})$ is dual to $Ш^2(K, \mu_n) = \ker\big(H^2(G_{K,S}, \mu_n) \to \bigoplus_{v\in S} H^2(G_v, \mu_n)\big)$. By Kummer theory: $H^1(G_{K,S}, \mu_n) \cong K^\times/(K^\times)^n$, and $H^2(G_{K,S}, \mu_n) \cong \operatorname{Br}(K)[n]$ (the n-torsion of the Brauer group). The ABHN theorem (theorem 9.1.2) gives $Ш^2(K, \mu_n) = 0$ (a Brauer class locally trivial everywhere is globally trivial). So $Ш^1(K, \mathbb{Z}/n\mathbb{Z}) = 0$: every everywhere locally trivial cyclic extension is globally trivial. This is a cohomological formulation of the Grunwald–Wang theorem (with the caveat that Grunwald–Wang requires a correction at the prime 2; the Poitou–Tate result for $Ш^1$ is unconditional).

2. The Poitou–Tate sequence for $M = \mu_p$ (p an odd prime) over $K = \mathbb{Q}$ gives:

$$0 \to H^1(G_{\mathbb{Q},S}, \mu_p) \xrightarrow{\text{loc}} \bigoplus_{v\in S} H^1(G_v, \mu_p) \to H^1(G_{\mathbb{Q},S}, \mathbb{Z}/p\mathbb{Z})^* \to \cdots$$

   By Kummer theory, $H^1(G_{\mathbb{Q},S}, \mu_p) \cong \mathbb{Q}_S^\times/(\mathbb{Q}_S^\times)^p$ (where $\mathbb{Q}_S^\times$ denotes the S-units of $\mathbb{Q}$) and $H^1(G_v, \mu_p) \cong \mathbb{Q}_v^\times/(\mathbb{Q}_v^\times)^p$. The localization map sends a global class $[a] \in \mathbb{Q}^\times/(\mathbb{Q}^\times)^p$ to its local images. The exactness of the sequence relates the cokernel of this map to the dual of the group classifying degree-p cyclic extensions of $\mathbb{Q}$.

Exercise 9.2.1

1. Verify that the Poitou–Tate sequence is a complex at the term $\bigoplus_{v\in S} H^2(G_v, M)$: show that the composition $H^2(G_{K,S}, M) \xrightarrow{\text{loc}} \bigoplus_{v\in S} H^2(G_v, M) \to H^0(G_{K,S}, M^*)^*$ is zero.
2. Show that $Ш^1(K, \mathbb{Z}/n\mathbb{Z}) = 0$ for every number field K and every $n \geq 1$, using the ABHN theorem (theorem 9.1.2) and Poitou–Tate duality (theorem 9.2.6).
3. Show that $Ш^1(K, \mu_n) \cong \mathrm{Cl}(K)[n]$ (the n-torsion of the class group of K). (*Hint:* identify $H^1(G_{K,S}, \mu_n) \cong K^\times/(K^\times)^n$ by Kummer theory, and the localization map with the map sending $a \in K^\times$ to its local classes $a \in K_v^\times/(K_v^\times)^n$. The kernel consists of elements that are local nth powers everywhere, which by class field theory are exactly the elements whose principal ideal (a) is an nth power in the ideal group.)
4. Show that the Poitou–Tate exact sequence is functorial in M: a morphism of finite G_K-modules $f\colon M \to N$ induces a morphism of nine-term exact sequences (commutative ladder).
5. (If the reader has read [4]) For $K = \mathbb{Q}$ and $M = \mathbb{Z}/2\mathbb{Z}$ (trivial action), write out the Poitou–Tate sequence explicitly. Identify $H^1(G_{\mathbb{Q},S}, \mathbb{Z}/2\mathbb{Z})$ with the group of quadratic extensions of $\mathbb{Q}$, the local terms with local quadratic extensions, and interpret the exactness in terms of the Hilbert reciprocity law.

9.3 The Global Euler–Poincaré Characteristic

The local Euler–Poincaré characteristic (**??** from Chapter 8) computes $\chi(G_v, M) = |M|_v^{-1}$ for a finite G_{K_v}-module M at a non-archimedean place v. This section establishes the global analogue: the alternating product of orders of $H^r(G_{K,S}, M)$ is determined by a product of local terms at the archimedean places alone. The formula is independent of the choice of S (provided S is large enough) and expresses a fundamental balance between the global cohomology groups.

Throughout this section, K is a number field with $[K : \mathbb{Q}] = n$, having r_1 real places and r_2 complex places ($n = r_1 + 2r_2$). The set S is a finite set of places containing all archimedean places and all primes where M is ramified or dividing $|M|$, so that the G_K-action on M factors through $G_{K,S}$ (definition 9.2.2).

Definition 9.3.1: Global Euler–Poincaré Characteristic

For a finite $G_{K,S}$-module M, the *global Euler–Poincaré characteristic* is:

$$\chi(G_{K,S}, M) = \frac{|H^0(G_{K,S}, M)| \cdot |H^2(G_{K,S}, M)|}{|H^1(G_{K,S}, M)|}$$

All three cohomology groups are finite (the finiteness of $H^r(G_{K,S}, M)$ for finite M and finite S follows from the finiteness of the class group and the Dirichlet unit theorem; see [18]).

The global EP characteristic has an unexpectedly simple formula: it depends only on the module M, the degree $[K : \mathbb{Q}]$, and the archimedean places of K.

Theorem 9.3.2: Global Euler–Poincaré Formula

For a finite $G_{K,S}$-module M:

$$\chi(G_{K,S}, M) = \frac{\prod_{v|\infty} |H^0(G_v, M)|}{|M|^{[K:\mathbb{Q}]}}$$

where the product runs over all archimedean places of K. At a real place v, $H^0(G_v, M) = M^{G_v}$ where $G_v \cong \mathbb{Z}/2\mathbb{Z}$ acts via complex conjugation. At a complex place v, $G_v = 1$ and $H^0(G_v, M) = M$, contributing $|M|$ to the product.

The formula can be rewritten: since there are r_1 real places and r_2 complex places,

$$\chi(G_{K,S}, M) = \frac{\prod_{v \text{ real}} |M^{G_v}| \cdot |M|^{r_2}}{|M|^{r_1+2r_2}} = \frac{\prod_{v \text{ real}} |M^{G_v}|}{|M|^{r_1+r_2}}$$

For a totally imaginary field ($r_1 = 0$): $\chi(G_{K,S}, M) = |M|^{-r_2}$, depending only on $|M|$ and the number of complex places.

***Proof*:**
The proof proceeds by computing the local EP characteristics at each type of place, then assembling them into the global formula via a spectral sequence argument.

Step 1: Local EP characteristics. The local EP characteristic $\chi(G_v, M) = |H^0(G_v, M)| \cdot |H^2(G_v, M)|/|H^1(G_v, M)|$ takes a different form at each type of place.

At a non-archimedean place $v \in S$: the local EP formula (**??**) gives $\chi(G_v, M) = 1/|M|$.

At a real place v: $G_v \cong \mathbb{Z}/2\mathbb{Z}$, and the Herbrand quotient $h(G_v, M) = |\hat{H}^0(G_v, M)|/|H^1(G_v, M)| = 1$ for any finite G_v-module (**??** from Chapter 4). Since $H^2(G_v, M) = \hat{H}^0(G_v, M)$ (by the 2-periodicity of $\mathbb{Z}/2\mathbb{Z}$-cohomology, theorem 4.3.4), the Herbrand relation gives $|H^2(G_v, M)| = |H^1(G_v, M)|$. The EP characteristic simplifies:

$$\chi(G_v, M) = \frac{|H^0(G_v, M)| \cdot |H^2(G_v, M)|}{|H^1(G_v, M)|} = |H^0(G_v, M)| = |M^{G_v}|$$

At a complex place v: $G_v = 1$, so $H^0(G_v, M) = M$ and $H^r(G_v, M) = 0$ for $r \geq 1$, giving $\chi(G_v, M) = |M|$.

Step 2: Independence of S. Enlarging S to $S' = S \cup \{v_0\}$ for a finite place $v_0 \notin S$: the Hochschild–Serre spectral sequence (**??** from Chapter 3) for the surjection $G_{K,S'} \twoheadrightarrow G_{K,S}$ (whose kernel is generated by inertia at v_0, acting trivially on M) gives:

$$\chi(G_{K,S'}, M) = \chi(G_{K,S}, M) \cdot \chi(G_{v_0}, M)$$

Since $\chi(G_{v_0}, M) = 1/|M|$ at the non-archimedean place v_0 (Step 1), the EP characteristic changes by a factor of $1/|M|$ each time a finite place is added to S. The formula $\chi(G_{K,S}, M) = \prod_{v|\infty} |H^0(G_v, M)|/|M|^{[K:\mathbb{Q}]}$ is invariant under this change (both sides scale by $1/|M|$), confirming independence of S.

Step 3: Proof of the formula. The formula is proved by reducing to the case $M = \mathbb{Z}/p\mathbb{Z}$ (a simple module) via dévissage: a short exact sequence $0 \to M' \to M \to M'' \to 0$ of finite $G_{K,S}$-modules gives $\chi(G_{K,S}, M) = \chi(G_{K,S}, M') \cdot \chi(G_{K,S}, M'')$ (multiplicativity of the EP characteristic from the long exact sequence), and the same multiplicativity holds for the right side of the formula (the product $\prod_{v|\infty} |H^0(G_v, -)|$ is multiplicative in short exact sequences of G_v-modules, and $|M|^{[K:\mathbb{Q}]}$ is multiplicative in $|M|$). The base case $M = \mathbb{Z}/p\mathbb{Z}$ (with a specified G_K-action) is computed using the Hochschild–Serre spectral sequence for the tower $G_{K,S} \twoheadrightarrow \mathrm{Gal}(K(\zeta_p)/K)$ and the known cohomology of cyclotomic fields. The detailed argument is given in [18].

Step 4: Verification. For $K = \mathbb{Q}$, $M = \mathbb{Z}/2\mathbb{Z}$ (trivial action), $S = \{\infty, 2\}$: the formula gives $\chi = |(\mathbb{Z}/2\mathbb{Z})^{G_\mathbb{R}}|/|\mathbb{Z}/2\mathbb{Z}|^1 = 2/2 = 1$. Direct computation: $|H^0| = 2$, $|H^1| = |\mathrm{Hom}(G_{\mathbb{Q},S}, \mathbb{Z}/2\mathbb{Z})| = 4$ (the extensions $\mathbb{Q}(\sqrt{-1})$, $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{-2})$, and the trivial extension), and $|H^2| = 2$ (from the Poitou–Tate sequence). So $\chi = 2 \cdot 2/4 = 1$, confirming the formula.

The proof reduces the global EP formula to dévissage and the base case of a simple module, where the Hochschild–Serre spectral sequence provides the computation. The local EP characteristics at archimedean places (Step 1) explain why only the archimedean data survives in the global formula: at non-archimedean places, $\chi(G_v, M) = 1/|M|$ is "universal" (independent of the G_v-action on M), so these contributions are absorbed into the $|M|^{[K:\mathbb{Q}]}$ denominator. At archimedean places, $\chi(G_v, M) = |M^{G_v}|$ depends on the action, and this dependence is the only arithmetic content of the global formula.

The contrast between the $\mathbb{Z}/p\mathbb{Z}$ and μ_p examples below illustrates this: both modules have order p, but $|(\mathbb{Z}/p\mathbb{Z})^{G_\mathbb{R}}| = p$ (trivial action, everything fixed) while $|\mu_p^{G_\mathbb{R}}| = 1$ for odd p (complex conjugation inverts roots of unity, leaving only 1 fixed). The global EP characteristics differ by a factor of p, and this difference propagates to the sizes of H^1 and H^2.

Example 9.3.3: Global EP Computations

1. For $K = \mathbb{Q}$ ($r_1 = 1$, $r_2 = 0$) and $M = \mathbb{Z}/p\mathbb{Z}$ with trivial action (p odd): $|H^0(G_\mathbb{R}, \mathbb{Z}/p\mathbb{Z})| = p$ (everything fixed under trivial action), so $\chi = p/p = 1$. This means $|H^0| \cdot |H^2| = |H^1|$, and since $|H^0| = p$: $|H^2(G_{\mathbb{Q},S}, \mathbb{Z}/p\mathbb{Z})| = |H^1(G_{\mathbb{Q},S}, \mathbb{Z}/p\mathbb{Z})|/p$.
2. For $K = \mathbb{Q}$ and $M = \mu_p$ (p odd): $|H^0(G_\mathbb{R}, \mu_p)| = |\mu_p(\mathbb{R})| = 1$ (the only real pth root of unity is 1), so $\chi = 1/p$. With $|H^0(G_\mathbb{Q}, \mu_p)| = |\mu_p(\mathbb{Q})| = 1$: $|H^2(G_{\mathbb{Q},S}, \mu_p)| = |H^1(G_{\mathbb{Q},S}, \mu_p)|/p$. By Kummer theory, $H^1(G_{\mathbb{Q},S}, \mu_p) \cong \mathbb{Q}_S^\times/(\mathbb{Q}_S^\times)^p$ where $\mathbb{Q}_S^\times$ denotes elements of $\mathbb{Q}^\times$ that are S-units. For $S = \{\infty, p\}$: $\mathbb{Q}_S^\times = \langle -1, p\rangle \cdot (\mathbb{Q}^\times)^p$, giving $|H^1| = p$ (generated by the class of p). So $|H^2| = 1$.
3. For a totally imaginary field K ($r_1 = 0$) and any M with trivial action: $\chi = |M|^{r_2}/|M|^{2r_2} = |M|^{-r_2}$. As r_2 grows, the EP characteristic shrinks, reflecting the increasing complexity of the cohomology.
4. For $K = \mathbb{Q}$ and $M = \mathbb{Z}/2\mathbb{Z}$ (trivial action): $|H^0(G_\mathbb{R}, \mathbb{Z}/2\mathbb{Z})| = 2$, so $\chi = 2/2 = 1$. For $S = \{\infty, 2\}$: $H^1(G_{\mathbb{Q},S}, \mathbb{Z}/2\mathbb{Z}) = \mathrm{Hom}(G_{\mathbb{Q},S}, \mathbb{Z}/2\mathbb{Z})$ classifies quadratic extensions of $\mathbb{Q}$ unramified outside $\{2, \infty\}$, namely $\mathbb{Q}(\sqrt{-1})$, $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{-2})$ (plus the trivial extension), giving $|H^1| = 4$. Then $|H^2| = |H^1|/|H^0| = 4/2 = 2$.

The global EP formula constrains the sizes of cohomology groups: knowing any two of $|H^0|$, $|H^1|$, $|H^2|$ determines the third. In applications to arithmetic (Selmer groups, descent), the formula relates the "size" of H^1 (which controls the Selmer group) to computable local data.

Corollary 9.3.4: EP for Dual Modules

For a finite $G_{K,S}$-module M with Cartier dual M^*:

$$\chi(G_{K,S}, M) \cdot \chi(G_{K,S}, M^*) = \frac{\prod_{v|\infty} |H^0(G_v, M)| \cdot |H^0(G_v, M^*)|}{|M|^{2[K:\mathbb{Q}]}}$$

For M self-dual ($M \cong M^*$), this gives $\chi(G_{K,S}, M)^2 = \prod_{v|\infty} |H^0(G_v, M)|^2/|M|^{2[K:\mathbb{Q}]}$, recovering the formula from the square root.

***Proof*:**
Apply theorem 9.3.2 to both M and M^* and multiply.

Exercise 9.3.1

1. Verify the global EP formula for $K = \mathbb{Q}$, $M = \mathbb{Z}/2\mathbb{Z}$, $S = \{\infty, 2\}$ by computing all three cohomology groups independently and checking $\chi = 1$.
2. Let K be a totally imaginary number field and M a finite $G_{K,S}$-module with trivial G_K-action. Show that $\chi(G_{K,S}, M) = |M|^{-r_2}$ where $r_2 = [K : \mathbb{Q}]/2$. Conclude that $|H^1(G_{K,S}, M)| = |M|^{r_2} \cdot |H^0| \cdot |H^2|$.
3. Show that $\chi(G_{K,S}, M)\cdot\chi(G_{K,S}, M^*) = |M|^{-2[K:\mathbb{Q}]}\cdot\prod_{v|\infty} |M|^2 = |M|^{-2r_1}$ when the G_K-action on M is trivial (so $|H^0(G_v, M)| = |M|$ at every archimedean place and $M^* = \mathrm{Hom}(M, \mu)$ need not have trivial action).
4. (If the reader has read [8]) Let $E/\mathbb{Q}$ be an elliptic curve and p an odd prime of good reduction for E. The module $M = E[p]$ is a finite $G_{\mathbb{Q}}$-module of order p^2 with Cartier dual $M^* \cong E[p]$ (via the Weil pairing). Compute $\chi(G_{\mathbb{Q},S}, E[p])$ in terms of p and the number of real places, and explain why this constrains the size of the p-Selmer group $\mathrm{Sel}_p(E/\mathbb{Q})$.

9.4 The Cohomological Reciprocity Isomorphism

The local reciprocity law (theorem 8.4.4 from Chapter 8) was proved by verifying that the pair $(G_K, K^\times)$ forms a class formation and applying the Tate–Nakayama theorem. The same strategy applies globally: the pair $(G_K, C_{K^{\mathrm{sep}}})$ satisfies the class formation axioms, and the Tate–Nakayama theorem produces the global Artin map $\mathrm{rec}_K\colon C_K \to G_K^{\mathrm{ab}}$. The two axioms—the vanishing of H^1 and the structure of H^2—were verified in section 9.1: Axiom 1 is the ABHN theorem (theorem 9.1.2), and Axiom 2 is the global fundamental class (theorem 9.1.5). This section assembles these ingredients into the global reciprocity law and examines its compatibility with the local theory.

As in the local case (section 8.4), what is established here is the reciprocity *isomorphism* of the cohomological formalism: the global Artin map rec_K as the degree -2 image of the global fundamental class. Its development into class field theory belongs to *Class Field Theory* [17]: the Artin-Frobenius description of rec_K, the idele-class and ray-class field correspondence, the global existence theorem (whose analytic first inequality is given there, as noted below), conductors, and the arithmetic reciprocity laws.

Definition 9.4.1: Global Class Formation

The pair $(G_K, C_{K^{\mathrm{sep}}})$ is a *class formation*, where $C_{K^{\mathrm{sep}}} = \varinjlim_L C_L$ is the idèle class group of the separable closure (the colimit over all finite extensions L/K of the idèle class groups C_L, with transition maps given by the natural inclusions). The class formation axioms are:

1. For every finite Galois extension L/K with $G = \mathrm{Gal}(L/K)$: $H^1(G, C_L) = 0$.
2. For every finite Galois extension L/K with $G = \mathrm{Gal}(L/K)$: the global invariant map (definition 9.1.4) gives an isomorphism $\mathrm{inv}_{L/K}\colon H^2(G, C_L) \xrightarrow{\sim} \frac{1}{[L:K]}\mathbb{Z}/\mathbb{Z}$, compatible with inflation.

Axiom 1 was proved in theorem 9.1.2(2). Axiom 2 was established in theorem 9.1.5, with the caveat that the first inequality ($|H^2(G, C_L)| \geq [L : K]$) requires analytic input deferred to [17]. With both axioms in hand, the Tate–Nakayama theorem (theorem 8.4.3 from Chapter 8, stated there for local fields but valid for any class formation) applies.

Theorem 9.4.2: Global Artin Reciprocity

There exists a unique continuous surjective homomorphism

$$\mathrm{rec}_K\colon C_K \to G_K^{\mathrm{ab}}$$

(the *global Artin map*) with the following properties:

1. *(Local-global compatibility)* For each place v of K, the diagram

$$\begin{tikzcd} K_v^\times \arrow[r, hook] \arrow[d, "\mathrm{rec}_{K_v}"'] & \mathbb{I}_K \arrow[r, two heads] & C_K \arrow[d, "\mathrm{rec}_K"] \\ G_{K_v}^{\mathrm{ab}} \arrow[rr, hook] & & G_K^{\mathrm{ab}} \end{tikzcd}$$

   commutes, where rec_{K_v} is the local Artin map from theorem 8.4.4 and the bottom arrow is the natural map $G_{K_v}^{\mathrm{ab}} \to G_K^{\mathrm{ab}}$ induced by the inclusion of the decomposition group $G_v \hookrightarrow G_K$.

2. *(Norm groups)* For every finite Galois extension L/K: rec_K induces an isomorphism

$$C_K/N_{L/K}(C_L) \xrightarrow{\sim} \mathrm{Gal}(L/K)^{\mathrm{ab}}$$

3. *(Existence)* The open subgroups of C_K of finite index are exactly the norm groups $N_{L/K}(C_L)$ for finite abelian extensions L/K. The kernel of rec_K is the connected component of the identity in C_K.

***Proof*:**
Step 1: Construction via Tate–Nakayama. The global class formation axioms and the Tate–Nakayama theorem produce, for each finite Galois extension L/K with $G = \mathrm{Gal}(L/K)$, an isomorphism:

$$\hat{H}^{-2}(G,\mathbb{Z}) \xrightarrow[\sim]{\smile u_{L/K}} \hat{H}^0(G, C_L)$$

given by the cup product with the global fundamental class $u_{L/K} \in H^2(G, C_L)$ (theorem 9.1.5). The left side is $\hat{H}^{-2}(G,\mathbb{Z}) \cong H_1(G,\mathbb{Z}) \cong G^{\mathrm{ab}}$ (**??** from Chapter 4 and **??** from Chapter 2). The right side is $\hat{H}^0(G, C_L) = C_K/N_{L/K}(C_L)$. Inverting gives:

$$\mathrm{rec}_{L/K} \colon C_K/N_{L/K}(C_L) \xrightarrow{\sim} G^{\mathrm{ab}} = \mathrm{Gal}(L/K)^{\mathrm{ab}}$$

These isomorphisms are compatible as L varies (by the compatibility of fundamental classes with inflation), and passing to the limit over all finite Galois L/K gives $\mathrm{rec}_K \colon C_K \to G_K^{\mathrm{ab}}$.

Step 2: Properties. Surjectivity: for each finite abelian L/K, $\mathrm{rec}_{L/K}$ is an isomorphism, so rec_K surjects onto $\mathrm{Gal}(L/K)$ for every such L. Since $G_K^{\mathrm{ab}} = \varprojlim_L \mathrm{Gal}(L/K)^{\mathrm{ab}}$, the map is surjective. Continuity: rec_K is continuous because each $\mathrm{rec}_{L/K}$ is continuous (the norm group $N_{L/K}(C_L)$ is open in C_K by the compactness of C_L^0 (theorem A.2.6) and the structure $C_K \cong C_K^0 \times \mathbb{R}_{>0}$ (proposition A.2.7)).

Step 3: Local-global compatibility. Property (1) follows from the compatibility of the global and local fundamental classes: the v-component of $u_{L/K}$ is the local fundamental class u_{L_w/K_v} (theorem 9.1.5), and the Tate–Nakayama isomorphism at the local level is exactly the local Artin map rec_{K_v}. The commutativity of the diagram is a formal consequence.

Step 4: Existence theorem and kernel. Property (3) (the existence theorem) requires showing that every open finite-index subgroup of C_K arises as a norm group $N_{L/K}(C_L)$ for some abelian L/K. This is a substantial result whose proof requires either Kummer theory and the Grunwald–Wang theorem or an analytic argument via L-functions; see [17] for the complete treatment. The identification of $\ker(\mathrm{rec}_K)$ as the connected component $(C_K)^0$ also follows from the existence theorem: since G_K^{ab} is totally disconnected (being profinite), the connected component maps to the identity, and the existence theorem shows nothing else does.

The global Artin map $\mathrm{rec}_K \colon C_K \to G_K^{\mathrm{ab}}$ is the idèlic formulation of the main theorem of class field theory: it classifies all abelian extensions of K in terms of the idèle class group. Property (1) says that the global reciprocity law is assembled from local reciprocity laws, one at each place. Property (2) says that the norm group $N_{L/K}(C_L)$ determines (and is determined by) the abelian extension L/K. Property (3) says that every "admissible" subgroup of C_K (open, finite index) arises from an abelian extension—a strong existence statement that goes beyond the cohomological formalism.

Example 9.4.3: The Artin Map for $\mathbb{Q}$
The idèle class group of $\mathbb{Q}$ is $C_{\mathbb{Q}} \cong \mathbb{R}_{>0} \times \widehat{\mathbb{Z}}^\times$ (example A.2.8). The maximal abelian extension of $\mathbb{Q}$ is $\mathbb{Q}^{\mathrm{ab}} = \mathbb{Q}(\zeta_n : n \geq 1)$ (Kronecker–Weber; see [4, Kronecker Weber's Theorem]), with $G_{\mathbb{Q}}^{\mathrm{ab}} = \mathrm{Gal}(\mathbb{Q}^{\mathrm{ab}}/\mathbb{Q}) \cong \widehat{\mathbb{Z}}^\times$ via the cyclotomic character (exercise 1.4.1 (5)). The global Artin map

$\mathrm{rec}_{\mathbb{Q}}\colon C_{\mathbb{Q}} \to G_{\mathbb{Q}}^{\mathrm{ab}}$ is the projection:

$$\mathbb{R}_{>0} \times \widehat{\mathbb{Z}}^{\times} \xrightarrow{\mathrm{pr}_2} \widehat{\mathbb{Z}}^{\times} \cong G_{\mathbb{Q}}^{\mathrm{ab}}$$

The kernel is $\mathbb{R}_{>0}$, the connected component of $C_{\mathbb{Q}}$.

The local-global compatibility at a prime p: the inclusion $\mathbb{Q}_p^{\times} \hookrightarrow \mathbb{I}_{\mathbb{Q}} \to C_{\mathbb{Q}}$ sends $p \in \mathbb{Q}_p^{\times}$ to the class of the idèle with p at the p-component and 1 elsewhere. Under the isomorphism $C_{\mathbb{Q}} \cong \mathbb{R}_{>0} \times \widehat{\mathbb{Z}}^{\times}$, this maps to $(p^{-1}, (1, \ldots, 1, p, 1, \ldots))$ (the $\mathbb{R}_{>0}$-component is $|\text{idèle}|_{\mathbb{Q}}^{-1} = p^{-1}$, and the $\widehat{\mathbb{Z}}^{\times}$-component has p in the $\mathbb{Z}_p^{\times}$-factor). Under $\mathrm{rec}_{\mathbb{Q}}$: the image is the element of $\widehat{\mathbb{Z}}^{\times}$ with p-component $p \in \mathbb{Z}_p^{\times}$ and all other components 1. The local Artin map $\mathrm{rec}_{\mathbb{Q}_p}$ sends $p \in \mathbb{Q}_p^{\times}$ to Frob_p^{-1} (the arithmetic Frobenius), and the compatibility says that the image of p in $G_{\mathbb{Q}}^{\mathrm{ab}} \cong \widehat{\mathbb{Z}}^{\times}$ under the global map equals Frob_p^{-1} acting on the cyclotomic tower. The sign convention (Frob_p^{-1} rather than Frob_p) is a standard feature of the arithmetic normalization of the reciprocity map.

The global reciprocity law, proved here cohomologically, is equivalent to the classical ideal-theoretic formulation of Takagi and Artin: the Artin symbol $\left(\frac{L/K}{\mathfrak{p}}\right)$ for unramified primes $\mathfrak{p}$ is recovered from rec_K via the local Artin map at $\mathfrak{p}$, and the Artin reciprocity law $\prod_v \left(\frac{L/K}{v}\right) = 1$ (for a principal idèle) is the content of property (2). The cohomological approach replaces the intricate ideal-theoretic arguments with the two axioms of a class formation, verified in section 9.1.

Proposition 9.4.4: Functoriality of the Artin Map

Let L/K be a finite extension of number fields (not necessarily Galois). The Artin maps for K and L satisfy two compatibilities:

1. *(Norm compatibility)* The diagram

$$\begin{tikzcd} C_L \arrow[r, "\mathrm{rec}_L"] \arrow[d, "N_{L/K}"'] & G_L^{\mathrm{ab}} \arrow[d] \\ C_K \arrow[r, "\mathrm{rec}_K"] & G_K^{\mathrm{ab}} \end{tikzcd}$$

   commutes, where the right vertical arrow is the natural map $G_L^{\mathrm{ab}} \to G_K^{\mathrm{ab}}$ induced by the inclusion $G_L \hookrightarrow G_K$.

2. *(Transfer compatibility)* The diagram

$$\begin{tikzcd} C_K \arrow[r, "\mathrm{rec}_K"] \arrow[d, hook] & G_K^{\mathrm{ab}} \arrow[d, "\mathrm{Ver}"] \\ C_L \arrow[r, "\mathrm{rec}_L"] & G_L^{\mathrm{ab}} \end{tikzcd}$$

   commutes, where the left vertical arrow is the natural inclusion $C_K \hookrightarrow C_L$ and $\mathrm{Ver}\colon G_K^{\mathrm{ab}} \to G_L^{\mathrm{ab}}$ is the transfer (Verlagerung) homomorphism.

***Proof*:**
Both statements follow from the functoriality of the Tate–Nakayama isomorphism with respect to the fundamental class. For (1): the norm $N_{L/K}\colon C_L \to C_K$ corresponds, under the Tate–Nakayama isomorphism at $r = -2$, to the corestriction $H_1(G_L, \mathbb{Z}) \to H_1(G_K, \mathbb{Z})$, which is the inclusion-induced map $G_L^{\mathrm{ab}} \to G_K^{\mathrm{ab}}$ (see theorem 8.4.3 for the analogous local statement). For (2): the inclusion $C_K \hookrightarrow C_L$ corresponds to the restriction $H_1(G_K, \mathbb{Z}) \to H_1(G_L, \mathbb{Z})$, which is the transfer $G_K^{\mathrm{ab}} \to G_L^{\mathrm{ab}}$ (the transfer is the cohomological restriction on $\hat{H}^{-2}$; see **??** from Chapter 4).

Exercise 9.4.1

1. Verify the class formation axioms explicitly for the quadratic extension $L = \mathbb{Q}(\sqrt{-1})$, $K = \mathbb{Q}$: check that $H^1(G, C_L) = 0$ (using ABHN) and that $|H^2(G, C_L)| = 2$ (using the global invariant map and the local components from example 9.1.6).
2. Let $K = \mathbb{Q}$ and $L = \mathbb{Q}(\zeta_p)$ for an odd prime p. Show that the global Artin map $\mathrm{rec}_{\mathbb{Q}}\colon C_{\mathbb{Q}} \to \mathrm{Gal}(L/\mathbb{Q}) \cong (\mathbb{Z}/p\mathbb{Z})^\times$ agrees with the cyclotomic character at each prime $\ell \neq p$: the local Artin map sends $\ell \in \mathbb{Q}_\ell^\times$ to $\ell \bmod p \in (\mathbb{Z}/p\mathbb{Z})^\times$.
3. Compute the norm group $N_{\mathbb{Q}(i)/\mathbb{Q}}(C_{\mathbb{Q}(i)}) \subseteq C_{\mathbb{Q}}$ explicitly, and verify that it has index 2 in $C_{\mathbb{Q}}$. (*Hint:* identify $C_{\mathbb{Q}} \cong \mathbb{R}_{>0} \times \widehat{\mathbb{Z}}^\times$ and determine which elements are norms from $C_{\mathbb{Q}(i)}$.)
4. The existence theorem states that every open finite-index subgroup of C_K is a norm group. Show that this implies: for every finite abelian group A and every continuous surjection $\varphi\colon C_K \twoheadrightarrow A$, there exists a unique abelian extension L/K with $\mathrm{Gal}(L/K) \cong A$ and $\ker\varphi = N_{L/K}(C_L)$.
5. (If the reader has read [4]) Show that the global Artin map for a general number field K satisfies $\mathrm{rec}_K(u) = 1$ for all $u \in \mathcal{O}_K^\times$ embedded diagonally in C_K (i.e., units of the ring of integers are in the kernel of the Artin map restricted to the diagonal). (*Hint:* use local-global compatibility and the fact that u is a local unit at every finite place.)

9.5 Selmer Groups and the Shafarevich–Tate Group

The Poitou–Tate sequence (theorem 9.2.5) applies to arbitrary finite G_K-modules, but its most consequential applications arise from the p-torsion of abelian varieties over number fields. This section introduces Selmer groups and the Shafarevich–Tate group as instances of the general local-global framework, connecting the cohomological machinery developed in this chapter to the arithmetic of elliptic curves and abelian varieties.

Throughout this section, A denotes an abelian variety over a number field K (in practice, most applications concern elliptic curves E/K), and p is a prime number. The reader is referred to [8] for the theory of elliptic curves and to [1] for abelian varieties. This section requires only the basic properties of the group $A(K)$ of rational points and the p-torsion subgroup $A[p] \subseteq A(\overline{K})$ as a finite G_K-module.

The starting point is the Kummer exact sequence. The multiplication-by-p map $[p]\colon A(\overline{K}) \to A(\overline{K})$ is surjective (over an algebraically closed field), with kernel $A[p]$. This gives a short exact sequence of G_K-modules:

$$0 \to A[p] \to A(\overline{K}) \xrightarrow{[p]} A(\overline{K}) \to 0$$

The associated long exact sequence in Galois cohomology begins:

$$0 \to A(K)[p] \to A(K) \xrightarrow{[p]} A(K) \xrightarrow{\delta} H^1(G_K, A[p]) \to H^1(G_K, A(\overline{K})) \xrightarrow{[p]} \cdots$$

The connecting homomorphism $\delta\colon A(K) \to H^1(G_K, A[p])$ sends a point $P \in A(K)$ to the cocycle $\sigma \mapsto \sigma(Q) - Q$, where $Q \in A(\overline{K})$ satisfies $[p]Q = P$. The image of δ is $A(K)/pA(K)$, which embeds as a subgroup of $H^1(G_K, A[p])$.

The same construction at each place v gives local connecting homomorphisms $\delta_v\colon A(K_v) \to H^1(G_v, A[p])$, and the localization maps (definition 9.2.3) make the following diagram commute:

$$\begin{tikzcd} A(K)/pA(K) \arrow[r, hook, "\delta"] \arrow[d] & H^1(G_K, A[p]) \arrow[d, "\mathrm{loc}_v"] \\ A(K_v)/pA(K_v) \arrow[r, hook, "\delta_v"] & H^1(G_v, A[p]) \end{tikzcd}$$

Definition 9.5.1: p-Selmer Group

The *p-Selmer group* of A/K is:

$$\mathrm{Sel}_p(A/K) = \left\{\alpha \in H^1(G_K, A[p]) \;\middle|\; \mathrm{loc}_v(\alpha) \in \mathrm{im}(\delta_v) \text{ for all } v\right\}$$

the subgroup of global cohomology classes whose local restrictions lie in the image of the Kummer map at every place.

The Selmer group sits between the Mordell–Weil group modulo p and the full cohomology group: $A(K)/pA(K) \subseteq \mathrm{Sel}_p(A/K) \subseteq H^1(G_K, A[p])$. The first inclusion holds because a global point is a local point at every place. The Selmer group is defined by the *local conditions* $\mathrm{im}(\delta_v)$: a global cohomology class belongs to Sel_p if and only if it is "locally a Kummer class" at every place.

Definition 9.5.2: Shafarevich–Tate Group

The *Shafarevich–Tate group* of A/K is:

$$Ш(A/K) = \ker\Big(H^1(G_K, A(\overline{K})) \xrightarrow{\mathrm{loc}} \prod_v H^1(G_v, A(\overline{K}_v))\Big)$$

the group of locally trivial torsors under A (principal homogeneous spaces for A that have a rational point over every completion K_v but not over K itself).

The Shafarevich–Tate group $Ш(A/K)$ is the obstruction to the Hasse principle for torsors under A: a torsor X under A defines a class $[X] \in H^1(G_K, A)$, and $[X] \in Ш(A/K)$ if and only if $X(K_v) \neq \emptyset$ for all v but $X(K) = \emptyset$. The group Ш is conjectured to be finite for all abelian varieties over number fields (see [8] for the connection to the Birch–Swinnerton-Dyer conjecture).

The three groups $A(K)/pA(K)$, $\mathrm{Sel}_p(A/K)$, and $Ш(A/K)$ are related by a fundamental exact sequence.

Proposition 9.5.3: Selmer Exact Sequence

There is a short exact sequence:

$$0 \to A(K)/pA(K) \xrightarrow{\delta} \mathrm{Sel}_p(A/K) \to Ш(A/K)[p] \to 0$$

***Proof*:**
The Kummer map $\delta\colon A(K)/pA(K) \hookrightarrow H^1(G_K, A[p])$ has image contained in $\mathrm{Sel}_p(A/K)$: a global point is a local point at every place, so its Kummer class lies in $\mathrm{im}(\delta_v)$ for all v. The cokernel of δ inside Sel_p maps to $H^1(G_K, A)$ via the long exact sequence (the map $H^1(G_K, A[p]) \to H^1(G_K, A)$ sends a cohomology class of $A[p]$ to the corresponding torsor). A class $\alpha \in \mathrm{Sel}_p(A/K)$ maps to $H^1(G_K, A)$, and its localization $\mathrm{loc}_v(\alpha) \in \mathrm{im}(\delta_v)$ means the local torsor has a K_v-point (it lies in the image of $A(K_v)$). So the image of Sel_p in $H^1(G_K, A)$ lands in $Ш(A/K)$. Since the map $H^1(G_K, A[p]) \to H^1(G_K, A)$ has image killed by p: the image lies in $Ш(A/K)[p]$. The surjectivity onto $Ш(A/K)[p]$ and the identification of the kernel with $\mathrm{im}(\delta)$ follow from the snake lemma applied to the commutative diagram of long exact sequences (global and local Kummer sequences).

The Selmer exact sequence is the primary tool for studying the Mordell–Weil group $A(K)$. The rank of $A(K)$ (as a finitely generated abelian group, by the Mordell–Weil theorem) is constrained by the $\mathbb{F}_p$-dimension of $\mathrm{Sel}_p(A/K)$: the exact sequence gives $\dim_{\mathbb{F}_p} \mathrm{Sel}_p(A/K) = \mathrm{rank}\, A(K) + \dim_{\mathbb{F}_p} A(K)[p] + \dim_{\mathbb{F}_p} Ш(A/K)[p]$. Computing Sel_p is feasible (it is a finite, computable group), while Ш remains mysterious.

Proposition 9.5.4: Finiteness of the Selmer Group

$\mathrm{Sel}_p(A/K)$ is a finite group.

***Proof*:**
Let S be a finite set of places containing all archimedean places, all places of bad reduction for A, and all places dividing p. A class $\alpha \in \mathrm{Sel}_p(A/K) \subseteq H^1(G_K, A[p])$ is unramified at every $v \notin S$: the local condition $\mathrm{loc}_v(\alpha) \in \mathrm{im}(\delta_v)$ implies α is trivial on the inertia group I_v (since A has good reduction at v and the Néron model provides a smooth integral structure). So $\mathrm{Sel}_p(A/K) \subseteq H^1(G_{K,S}, A[p])$. The group $H^1(G_{K,S}, A[p])$ is finite ($A[p]$ is a finite $G_{K,S}$-module and $G_{K,S}$ has finite cohomology; see [18]), so its subgroup $\mathrm{Sel}_p(A/K)$ is finite.

The Selmer group fits into the Poitou–Tate framework as a special case of "cohomology with local conditions." The module $M = A[p]$ is a finite G_K-module with Cartier dual $M^* \cong A^t[p]$ (the p-torsion of the dual abelian variety, identified via the Weil pairing), and the local conditions $\mathrm{im}(\delta_v) \subseteq H^1(G_v, A[p])$ define a "Selmer structure" in the sense of [18]. The Poitou–Tate sequence (theorem 9.2.5) applied to $M = A[p]$ controls the relationship between $\mathrm{Sel}_p(A/K)$ and the dual Selmer group $\mathrm{Sel}_p(A^t/K)$.

Theorem 9.5.5: Cassels–Tate Pairing

There exists a bilinear pairing:

$$\langle \cdot, \cdot \rangle : Ш(A/K) \times Ш(A^t/K) \to \mathbb{Q}/\mathbb{Z}$$

which is non-degenerate on the maximal divisible subgroups. For a principally polarized abelian variety (in particular, for an elliptic curve $E = E^t$), the pairing restricts to an alternating form $Ш(E/K) \times Ш(E/K) \to \mathbb{Q}/\mathbb{Z}$.

***Proof*:**
The pairing is constructed from the Poitou–Tate duality (theorem 9.2.6) applied to $M = A[p^n]$ as $n \to \infty$: the duality $Ш^1(K, A[p^n]) \cong Ш^2(K, A^t[p^n])^*$ assembles into a pairing on the p-primary parts of $Ш(A/K)$ and $Ш(A^t/K)$. The non-degeneracy on divisible subgroups follows from the exactness of the Poitou–Tate sequence in the limit. The alternating property for principally polarized varieties uses the symmetry of the Weil pairing under the principal polarization $A \xrightarrow{\sim} A^t$. The complete construction is given in [18].

A consequence of the alternating property: if $Ш(E/K)$ is finite for an elliptic curve E/K, then $|Ш(E/K)|$ is a perfect square.

9.5.6 The Birch–Swinnerton-Dyer Conjecture The Selmer exact sequence and the Cassels–Tate pairing are central ingredients in the Birch–Swinnerton-Dyer (BSD) conjecture, which predicts that the rank of $E(K)$ equals the order of vanishing of the L-function $L(E/K, s)$ at $s = 1$, and gives a precise formula for the leading coefficient involving $|Ш(E/K)|$, the regulator, the real period, and the Tamagawa numbers at primes of bad reduction. The cohomological framework of this chapter provides the tools (Selmer groups, local-global duality, Cassels–Tate pairing) that are used in all known partial results toward BSD. The conjecture and its implications are developed in [8].

Example 9.5.7: 2-Descent for an Elliptic Curve
Let $E/\mathbb{Q}$ be the elliptic curve $y^2 = x^3 - x$ (conductor 32). The 2-torsion is rational: $E[2] = \{O, (0,0), (1,0), (-1,0)\} \cong (\mathbb{Z}/2\mathbb{Z})^2$ with trivial $G_\mathbb{Q}$-action. The Kummer map $\delta : E(\mathbb{Q})/2E(\mathbb{Q}) \hookrightarrow H^1(G_\mathbb{Q}, E[2]) = H^1(G_\mathbb{Q}, (\mathbb{Z}/2\mathbb{Z})^2)$ embeds the Mordell–Weil group modulo 2 into the cohomology. The Selmer group $\mathrm{Sel}_2(E/\mathbb{Q})$ is computed by imposing local conditions at each place: at $v = \infty$ (the archimedean place) and $v = 2$ (the unique prime of bad reduction), the local images $\mathrm{im}(\delta_v)$ constrain the global classes. A classical 2-descent computation (see [8]) gives $\mathrm{Sel}_2(E/\mathbb{Q}) \cong (\mathbb{Z}/2\mathbb{Z})^2$ and $Ш(E/\mathbb{Q})[2] = 0$. The Selmer exact sequence then gives $E(\mathbb{Q})/2E(\mathbb{Q}) \cong (\mathbb{Z}/2\mathbb{Z})^2$, which (combined with the known torsion $E(\mathbb{Q})_{\mathrm{tors}} \cong (\mathbb{Z}/2\mathbb{Z})^2$) shows $\operatorname{rank} E(\mathbb{Q}) = 0$.

Exercise 9.5.1

1. Show that $A(K)/pA(K) \subseteq \mathrm{Sel}_p(A/K)$ by verifying that the Kummer image of a global point satisfies the local conditions at every place.

2. Show that the Selmer group $\mathrm{Sel}_p(A/K)$ does not depend on the choice of the finite set S used in its definition (i.e., the local conditions at places of good reduction are automatically satisfied by unramified classes).

3. For an elliptic curve E/K: assuming $Ш(E/K)$ is finite, show that the alternating property of the Cassels–Tate pairing implies $|Ш(E/K)|$ is a perfect square. (*Hint:* an alternating non-degenerate pairing on a finite abelian group forces the group to have square order.)

4. Explain how the p-Selmer group $\mathrm{Sel}_p(E/\mathbb{Q})$ for an elliptic curve $E/\mathbb{Q}$ can be viewed as a subgroup of $H^1(G_{\mathbb{Q},S}, E[p])$ for a suitable S, and describe how the Poitou–Tate exact sequence (theorem 9.2.5) applied to $M = E[p]$ gives an upper bound on $|\mathrm{Sel}_p|$.

10

Descent, Non-Abelian Cohomology, and Connections

The first nine chapters developed the cohomology of profinite groups as an algebraic machine and applied it to the arithmetic of local and global fields. The theory was abelian throughout: the coefficient modules were abelian groups, the cohomology groups carried abelian group structures, and the duality theorems exploited this linearity. This final chapter moves beyond the abelian setting in two directions. The first (section 10.1 and section 10.2) applies the abelian theory to classify field extensions and algebraic objects via *descent*—the principle that structures over a base field K are controlled by Galois cohomology of their automorphism groups over K^{sep}. The second (sections 10.3 to 10.5) develops the foundations of non-abelian H^1 and H^2, where the coefficient object is a non-abelian group, and connects the resulting formalism to étale cohomology—the algebro-geometric generalization that extends Galois cohomology from fields to schemes.

10.1 Kummer Theory and Artin–Schreier Theory

The Kummer exact sequence $1 \to \mu_n \to (K^{\mathrm{sep}})^\times \xrightarrow{(\cdot)^n} (K^{\mathrm{sep}})^\times \to 1$ appeared in Chapter 9 in the context of Selmer groups and the Shafarevich–Tate group (section 9.5). Over a field K containing the nth roots of unity μ_n, the same sequence controls all abelian extensions of exponent dividing n: the connecting homomorphism in the long exact sequence gives $K^\times/(K^\times)^n \cong H^1(G_K, \mu_n)$, and each element of this group corresponds to a cyclic extension $K(\sqrt[n]{a})/K$. This section develops Kummer theory as a classification theorem for abelian extensions and treats its characteristic-p counterpart, the Artin–Schreier classification.

Throughout this section, K denotes a field and $G_K = \mathrm{Gal}(K^{\mathrm{sep}}/K)$ its absolute Galois group.

Theorem 10.1.1: Kummer Theory

Let K be a field with $\mathrm{char}(K) \nmid n$ and $\mu_n \subseteq K$. There is a bijection:

$$\left\{ \begin{array}{c} \text{abelian extensions } L/K \\ \text{of exponent dividing } n \end{array} \right\} \xleftrightarrow{1:1} \{ \text{ subgroups } \Delta \leq K^\times/(K^\times)^n \}$$

given by $L \mapsto \Delta_L = (L^\times)^n \cap K^\times/(K^\times)^n$ and $\Delta \mapsto L_\Delta = K(\{\sqrt[n]{a} \mid a \in \Delta\})$. Under this correspondence, $\mathrm{Gal}(L/K) \cong \Delta_L^*$ (Pontryagin dual), and $[L : K] = |\Delta_L|$.

***Proof*:**

Step 1: The Kummer pairing. The short exact sequence of G_K-modules $1 \to \mu_n \to (K^{\mathrm{sep}})^\times \xrightarrow{(\cdot)^n} (K^{\mathrm{sep}})^\times \to 1$ (exact on the right by Hilbert's theorem 90 applied to nth roots) gives a long exact sequence:

$$K^\times \xrightarrow{(\cdot)^n} K^\times \xrightarrow{\delta} H^1(G_K, \mu_n) \to H^1(G_K, (K^{\mathrm{sep}})^\times) = 0$$

where the last term vanishes by Hilbert 90 (theorem 8.1.1). The connecting homomorphism δ is an isomorphism $K^\times/(K^\times)^n \xrightarrow{\sim} H^1(G_K, \mu_n)$. Since $\mu_n \subseteq K$, the G_K-action on μ_n is trivial, so $H^1(G_K, \mu_n) = \mathrm{Hom}(G_K, \mu_n)$ (continuous homomorphisms). Each homomorphism $\chi : G_K \to \mu_n$ has kernel $\mathrm{Gal}(K^{\mathrm{sep}}/L)$ for some abelian extension L/K of exponent dividing n.

Step 2: The Kummer pairing. Define the bilinear map:

$$\langle \cdot, \cdot \rangle : \mathrm{Gal}(L/K) \times \Delta_L \to \mu_n, \qquad \langle \sigma, a \rangle = \frac{\sigma(\sqrt[n]{a})}{\sqrt[n]{a}}$$

This is well defined: the ratio $\sigma(\sqrt[n]{a})/\sqrt[n]{a}$ is an nth root of unity (since σ fixes $a \in K^\times$), and it depends only on a mod $(K^\times)^n$ (changing the representative by a factor b^n does not affect the ratio). The pairing is bilinear ($\langle \sigma, ab \rangle = \langle \sigma, a \rangle \cdot \langle \sigma, b \rangle$ and $\langle \sigma\tau, a \rangle = \langle \sigma, a \rangle \cdot \langle \tau, a \rangle$).

Step 3: Non-degeneracy. The pairing is non-degenerate on both sides. On the left: if $\langle \sigma, a \rangle = 1$ for all $a \in \Delta_L$, then σ fixes $\sqrt[n]{a}$ for every $a \in \Delta_L$, hence σ fixes $L_{\Delta_L} = L$, so $\sigma = 1$ in $\mathrm{Gal}(L/K)$. On the right: if $\langle \sigma, a \rangle = 1$ for all $\sigma \in \mathrm{Gal}(L/K)$, then $\sqrt[n]{a} \in L^{\mathrm{Gal}(L/K)} = K$, so $a \in (K^\times)^n$, giving $a = 1$ in Δ_L. Non-degeneracy gives $\mathrm{Gal}(L/K) \cong \Delta_L^*$ (each is the Pontryagin dual of the other, since both are finite abelian groups of exponent dividing n) and $[L : K] = |\Delta_L|$.

Step 4: The bijection. The maps $L \mapsto \Delta_L$ and $\Delta \mapsto L_\Delta$ are mutual inverses. Given L: $L_{\Delta_L} = K(\sqrt[n]{a} : a \in \Delta_L) = L$ (the extension generated by all nth roots in L is L itself, since L/K is abelian of exponent n). Given Δ: $\Delta_{L_\Delta} = \Delta$ (the elements of $K^\times/(K^\times)^n$ whose nth roots lie in L_Δ are exactly those in Δ).

Example 10.1.2: Kummer Theory in Practice

1. For $n = 2$ and any field K with $\mathrm{char}(K) \neq 2$: $\mu_2 = \{\pm 1\} \subseteq K$ always, so Kummer theory applies. The group $K^\times/(K^\times)^2$ classifies quadratic extensions of K: each $a \in K^\times \setminus (K^\times)^2$ gives $K(\sqrt{a})/K$. For $K = \mathbb{Q}$: the squarefree integers $d \neq 1$ parametrize $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$ (each

class contains a unique squarefree representative), recovering the classical classification of quadratic fields.

2. For $K = \mathbb{Q}(\zeta_5)$ and $n = 5$: $\mu_5 \subseteq K$, so Kummer theory classifies all abelian extensions of K with exponent dividing 5. The group $K^\times/(K^\times)^5$ is infinite (it has one $\mathbb{Z}/5\mathbb{Z}$ factor for each prime ideal of $\mathcal{O}_K$ plus contributions from the units and archimedean places). The subgroup corresponding to *unramified* abelian 5-extensions is $Ш^1(K, \mu_5)$ (exercise 9.2.1 (3)), which is isomorphic to $\mathrm{Cl}(K)[5]$ (the 5-torsion of the class group). Since $\mathrm{Cl}(\mathbb{Q}(\zeta_5)) = 1$ (the class number is 1), there are no unramified abelian 5-extensions of $\mathbb{Q}(\zeta_5)$.

3. (If the reader has read [10]) For $K = \mathbb{Q}_p$ with $p \equiv 1 \pmod{n}$: $\mu_n \subseteq \mathbb{Q}_p$ (Hensel's lemma), and $\mathbb{Q}_p^\times/(\mathbb{Q}_p^\times)^n \cong \mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$ (one factor from the valuation $v(a)$ mod n and one from the unit group $\mathbb{Z}_p^\times/(\mathbb{Z}_p^\times)^n \cong \mathbb{Z}/n\mathbb{Z}$). The n^2 cyclic extensions of $\mathbb{Q}_p$ of degree dividing n are parametrized by the subgroups of this group.

In positive characteristic, the pth power map on $(K^{\mathrm{sep}})^\times$ is no longer surjective (the equation $x^p = a$ may have no solution when $\mathrm{char}(K) = p$), and Kummer theory fails for p-extensions. The Artin–Schreier exact sequence provides the replacement.

Theorem 10.1.3: Artin–Schreier Theory

Let K be a field with $\mathrm{char}(K) = p > 0$. Define the Artin–Schreier map $\wp\colon K^{\mathrm{sep}} \to K^{\mathrm{sep}}$ by $\wp(x) = x^p - x$. There is a short exact sequence of G_K-modules:

$$0 \to \mathbb{F}_p \to K^{\mathrm{sep}} \xrightarrow{\wp} K^{\mathrm{sep}} \to 0$$

and the connecting homomorphism gives an isomorphism:

$$K/\wp(K) \xrightarrow{\sim} H^1(G_K, \mathbb{F}_p) = \mathrm{Hom}(G_K, \mathbb{F}_p)$$

classifying cyclic extensions of K of degree p. Each class $[a] \in K/\wp(K)$ corresponds to the splitting field of $x^p - x - a$ over K.

***Proof*:**
The kernel of $\wp$ is $\{x \in K^{\mathrm{sep}} \mid \wp(x) = 0\} = \{x \mid x^p = x\} = \mathbb{F}_p$ (the prime field, fixed pointwise by G_K). Surjectivity: for any $a \in K^{\mathrm{sep}}$, the polynomial $x^p - x - a$ is separable (its derivative is $-1 \neq 0$), so it splits in K^{sep}; any root α satisfies $\wp(\alpha) = a$. The long exact sequence in cohomology gives:

$$K \xrightarrow{\wp} K \xrightarrow{\delta} H^1(G_K, \mathbb{F}_p) \to H^1(G_K, K^{\mathrm{sep}})$$

The additive Hilbert 90 ($H^1(G_K, K^{\mathrm{sep}}) = 0$ for the additive group; see theorem 8.1.2) gives $K/\wp(K) \xrightarrow{\sim} H^1(G_K, \mathbb{F}_p)$. Since G_K acts trivially on $\mathbb{F}_p$: $H^1(G_K, \mathbb{F}_p) = \mathrm{Hom}(G_K, \mathbb{F}_p)$, classifying cyclic p-extensions. The correspondence: $a \in K$ maps to the homomorphism $\chi_a\colon G_K \to \mathbb{F}_p$ defined by $\chi_a(\sigma) = \sigma(\alpha) - \alpha$ where α is a root of $x^p - x - a$.

The parallel between Kummer and Artin–Schreier is structural: both use a short exact sequence involving a "power map" ($(\cdot)^n$ or $\wp$), Hilbert 90 to kill H^1 of the ambient group, and the connecting homomorphism to identify a quotient of $K^\times$ (or K) with the group classifying cyclic extensions. The Artin–Schreier–Witt generalization (replacing $\mathbb{F}_p$ by $\mathbb{Z}/p^n\mathbb{Z}$ and $\wp$ by the Witt vector Verschiebung) extends the theory to all cyclic p-power extensions in characteristic p; see [18] for the full treatment.

Example 10.1.4: Artin–Schreier Extensions of $\mathbb{F}_p(t)$

For $K = \mathbb{F}_p(t)$: the group $K/\wp(K)$ classifies cyclic p-extensions. The element $t \in K$ gives the extension $L = K(\alpha)$ where $\alpha^p - \alpha = t$. This extension is totally ramified at the place $t = \infty$ (the pole of t) and unramified elsewhere. For $p = 2$: $L = \mathbb{F}_2(t)(\alpha)$ with $\alpha^2 + \alpha = t$ is an Artin–Schreier extension of degree 2, the characteristic-2 analogue of a quadratic extension.

Exercise 10.1.1

1. Show that Kummer theory for $n = 2$ recovers the classical bijection between quadratic extensions of K (with $\mathrm{char}(K) \neq 2$) and elements of $K^\times/(K^\times)^2$, with $a \mapsto K(\sqrt{a})$.
2. Let $K = \mathbb{Q}_p$ with $p \equiv 1 \pmod{n}$ and $\gcd(n, p) = 1$. Using Kummer theory, show that the number of cyclic extensions of K of degree exactly n is $(n^2 - 1)/(n - 1) = n + 1$ (the number of index-n subgroups of $(\mathbb{Z}/n\mathbb{Z})^2$, minus the trivial extension).
3. For $K = \mathbb{F}_3(t)$: classify all cyclic extensions of K of degree 3 that are unramified outside $\{t = 0, t = \infty\}$, by computing the relevant quotient of $K/\wp(K)$.
4. Let K be a number field with $\mu_n \subseteq K$. Show that the Kummer isomorphism $K^\times/(K^\times)^n \cong H^1(G_K, \mu_n) = \mathrm{Hom}(G_K, \mu_n)$ is compatible with the Artin map from theorem 9.4.2: for $a \in K^\times$, the Kummer character $\chi_a\colon G_K \to \mu_n$ satisfies $\chi_a(\mathrm{rec}_K(\alpha)) = \prod_v (a, \alpha_v)_{n,v}$ (the product of local nth power residue symbols).
5. (If the reader has read [18]) State the Artin–Schreier–Witt generalization: for $\mathrm{char}(K) = p$ and the ring of Witt vectors $W_m(K)$, the sequence $0 \to \mathbb{Z}/p^m\mathbb{Z} \to W_m(K^{\mathrm{sep}}) \xrightarrow{\wp} W_m(K^{\mathrm{sep}}) \to 0$ classifies cyclic extensions of degree p^m via $W_m(K)/\wp(W_m(K)) \cong H^1(G_K, \mathbb{Z}/p^m\mathbb{Z})$.

10.2 Galois Descent

Kummer theory (theorem 10.1.1) classified abelian extensions of K using H^1 with abelian coefficients μ_n. A far broader principle underlies this classification: given any algebraic structure X_0 over K (a vector space, an algebra, a quadratic form, a variety), the structures over K that become isomorphic to X_0 after extending scalars to K^{sep} are classified by the cohomology $H^1(G_K, \mathrm{Aut}(X_0 \otimes_K K^{\mathrm{sep}}))$. When the automorphism group is abelian, this is the ordinary H^1 of Chapters 3 and 4. When it is non-abelian, the cohomology set H^1 is a pointed set rather than a group—the subject of section 10.3. This section develops the descent principle through concrete examples, deferring the formal non-abelian theory to the next section.

Throughout this section, K denotes a field with absolute Galois group $G_K = \mathrm{Gal}(K^{\mathrm{sep}}/K)$.

Definition 10.2.1: Twisted Forms

Let X_0 be an algebraic object defined over K (a K-algebra, a quadratic form over K, a K-variety, etc.). A *twisted form* (or *K-form*) of X_0 is an object X of the same type over K such that:

$$X \otimes_K K^{\mathrm{sep}} \cong X_0 \otimes_K K^{\mathrm{sep}}$$

(isomorphism of the corresponding structures over K^{sep}). The set of K-isomorphism classes of twisted forms of X_0 is denoted $\mathrm{Twist}(X_0/K)$.

The question "classify all objects of a given type over K" often reduces to classifying twisted forms of a single "split" object X_0: an object that is as simple as possible over K^{sep}. The descent principle provides the answer.

Theorem 10.2.2: Descent Principle

There is a natural bijection of pointed sets:

$$\mathrm{Twist}(X_0/K) \cong H^1(G_K, \mathrm{Aut}_{K^{\mathrm{sep}}}(X_0))$$

where $\mathrm{Aut}_{K^{\mathrm{sep}}}(X_0) = \mathrm{Aut}(X_0 \otimes_K K^{\mathrm{sep}})$ carries the G_K-action $(\sigma \cdot \varphi)(x) = \sigma(\varphi(\sigma^{-1}(x)))$, and the distinguished point corresponds to X_0 itself.

***Proof*:**
Step 1: From twisted forms to cocycles. Let X be a twisted form of X_0, and choose an isomorphism $\varphi\colon X \otimes_K K^{\mathrm{sep}} \xrightarrow{\sim} X_0 \otimes_K K^{\mathrm{sep}}$. For each $\sigma \in G_K$, the map:

$$c(\sigma) = \varphi \circ \sigma(\varphi)^{-1} = \varphi \circ \sigma \circ \varphi^{-1} \circ \sigma^{-1} \in \mathrm{Aut}_{K^{\mathrm{sep}}}(X_0)$$

measures the failure of φ to be G_K-equivariant. The map $c\colon G_K \to \mathrm{Aut}_{K^{\mathrm{sep}}}(X_0)$ satisfies the cocycle condition: $c(\sigma\tau) = c(\sigma) \cdot \sigma(c(\tau))$ (direct computation from the definition). A different choice of isomorphism $\varphi' = a \circ \varphi$ (with $a \in \mathrm{Aut}_{K^{\mathrm{sep}}}(X_0)$) replaces c by the cohomologous cocycle $c'(\sigma) = a \cdot c(\sigma) \cdot \sigma(a)^{-1}$. So X determines a well-defined class $[c] \in H^1(G_K, \mathrm{Aut}_{K^{\mathrm{sep}}}(X_0))$.

Step 2: From cocycles to twisted forms. Given a cocycle $c\colon G_K \to \mathrm{Aut}_{K^{\mathrm{sep}}}(X_0)$, define a new G_K-action on $X_0 \otimes_K K^{\mathrm{sep}}$ by $\sigma *_c x = c(\sigma) \cdot \sigma(x)$ ("twisting" the standard action by the cocycle). The cocycle condition ensures this is a group action: $(\sigma\tau) *_c x = c(\sigma\tau) \cdot \sigma\tau(x) = c(\sigma) \cdot \sigma(c(\tau)) \cdot \sigma\tau(x) = c(\sigma) \cdot \sigma(c(\tau) \cdot \tau(x)) = \sigma *_c (\tau *_c x)$. The fixed points $(X_0 \otimes_K K^{\mathrm{sep}})^{G_K, *_c}$ form a K-structure X_c, which is a twisted form of X_0 (it becomes isomorphic to X_0 after extending scalars, since the twisting disappears over K^{sep}). Cohomologous cocycles give K-isomorphic twisted forms.

Step 3: The bijection. The maps $X \mapsto [c]$ and $[c] \mapsto X_c$ are mutual inverses. The identity cocycle $c = 1$ corresponds to X_0 itself (the untwisted form), so X_0 maps to the distinguished point of H^1.

The descent principle reduces the classification of algebraic structures to a cohomological computation. The difficulty shifts to computing H^1 of the automorphism group, which depends on the specific type of structure. The following examples illustrate the principle for several fundamental classes of objects.

Example 10.2.3: Central Simple Algebras

The "split" central simple algebra of degree n over K is $M_n(K)$ (the matrix algebra). Every CSA of degree n becomes isomorphic to $M_n(K^{\text{sep}})$ after extending scalars (by the Artin-Wedderburn theorem, see [7, Artin-Wedderburn For k-Algebras]). The automorphism group is $\text{Aut}(M_n(K^{\text{sep}})) = \text{PGL}_n(K^{\text{sep}})$ (the Skolem–Noether theorem: every automorphism is inner, and the center acts trivially). The descent principle gives:

$$\{\text{CSAs of degree } n \text{ over } K \mid \text{up to } K\text{-isom.}\} \cong H^1(G_K, \text{PGL}_n(K^{\text{sep}}))$$

This recovers the identification of the Brauer group in terms of cohomology from theorem 7.2.4 of Chapter **??**: taking the union over all n, the exact sequence $1 \to K^{\text{sep},\times} \to \text{GL}_n(K^{\text{sep}}) \to \text{PGL}_n(K^{\text{sep}}) \to 1$ and the vanishing $H^1(G_K, \text{GL}_n(K^{\text{sep}})) = 1$ (generalized Hilbert 90; see exercise 10.2.1 (1)) give $H^1(G_K, \text{PGL}_n) \hookrightarrow H^2(G_K, K^{\text{sep},\times}) = \text{Br}(K)$.

Example 10.2.4: Quadratic Forms

The "split" quadratic form of rank n over K ($\text{char}(K) \neq 2$) is $q_0 = \langle 1, 1, \ldots, 1\rangle$ (the sum of n squares). Every non-degenerate quadratic form of rank n becomes isomorphic to q_0 over K^{sep} (diagonalize and scale). The orthogonal group $O_n(K^{\text{sep}})$ acts as automorphisms of q_0, and the descent principle gives:

$$\{\text{non-deg. quadratic forms of rank } n \mid \text{up to } K\text{-isom.}\} \cong H^1(G_K, O_n(K^{\text{sep}}))$$

For $n = 1$: $O_1 = \{\pm 1\} = \mu_2$, so $H^1(G_K, O_1) = K^\times/(K^\times)^2$, classifying quadratic forms $\langle a\rangle$ up to isometry (i.e., elements of $K^\times$ up to squares). For $n = 2$: the exact sequence $1 \to SO_2 \to O_2 \to \mathbb{Z}/2\mathbb{Z} \to 1$ and the identification $SO_2 \cong \mathbb{G}_m$ (rotation group) produce a connecting map $H^1(G_K, O_2) \to H^1(G_K, \mathbb{Z}/2\mathbb{Z})$ that sends a binary form to its discriminant class.

Example 10.2.5: Elliptic Curves

(If the reader has read [8]) For an elliptic curve E/K: the twisted forms of E are elliptic curves E'/K isomorphic to E over K^{sep}. The automorphism group $\text{Aut}(E)$ depends on the j-invariant:

1. For $j(E) \neq 0, 1728$: $\text{Aut}(E) = \{\pm 1\} \cong \mathbb{Z}/2\mathbb{Z}$ (only the identity and negation). The descent principle gives $\text{Twist}(E/K) \cong H^1(G_K, \mathbb{Z}/2\mathbb{Z}) \cong K^\times/(K^\times)^2$. Each twisted form is a *quadratic twist* $E^d\colon dy^2 = f(x)$ for $d \in K^\times/(K^\times)^2$.
2. For $j(E) = 1728$ (e.g., $E\colon y^2 = x^3 - x$ over $\mathbb{Q}$): $\text{Aut}(E) \cong \mathbb{Z}/4\mathbb{Z}$ (generated by $(x, y) \mapsto (-x, iy)$, with $i = \sqrt{-1}$). Over a field containing μ_4: $\text{Twist}(E/K) \cong H^1(G_K, \mathbb{Z}/4\mathbb{Z}) \cong K^\times/(K^\times)^4$ (by Kummer theory). The twisted forms include quartic twists $E^d\colon dy^2 = x^3 - x$.
3. For $j(E) = 0$ (e.g., $E\colon y^2 = x^3+1$ over $\mathbb{Q}$): $\text{Aut}(E) \cong \mathbb{Z}/6\mathbb{Z}$ (generated by $(x, y) \mapsto (\zeta_3 x, -y)$). Over a field containing μ_6: the twisted forms include sextic twists.

Example 10.2.6: Étale Algebras and Galois Extensions

The "split" étale algebra of degree n is $K^n = K \times K \times \cdots \times K$. An étale K-algebra of degree n is a product of separable field extensions $L_1 \times \cdots \times L_r$ with $\sum [L_i : K] = n$. The automorphism group of $(K^{\text{sep}})^n$ is the symmetric group S_n (permuting the factors), with G_K acting trivially. The descent principle gives:

$$\{\text{étale } K\text{-algebras of degree } n \mid \text{up to } K\text{-isom.}\} \cong H^1(G_K, S_n) = \operatorname{Hom}(G_K, S_n)/\text{conj.}$$

(since the G_K-action on S_n is trivial, H^1 is the set of conjugacy classes of continuous homomorphisms $G_K \to S_n$). Each such homomorphism is a transitive G_K-action on $\{1, \ldots, n\}$, i.e., a degree-n separable extension L/K (up to isomorphism). The pointed set structure: the distinguished point is K^n itself (the split algebra), corresponding to the trivial homomorphism.

A common thread in these examples: the generalized Hilbert 90 theorem states that $H^1(G_K, \mathrm{GL}_n(K^{\text{sep}})) = 1$ (every twisted form of K^n as a *vector space* is trivially isomorphic to K^n; see exercise 10.2.1 (1)). This is the descent-theoretic content of Hilbert 90: there are no nontrivial vector bundles over $\operatorname{Spec} K$ in the Galois topology. The nontrivial cohomology arises when additional structure (algebra, quadratic form, group law) constrains the automorphism group to a proper subgroup $A \leq \mathrm{GL}_n$.

The vanishing $H^1(G_K, \mathrm{GL}_n) = 1$ has a computational consequence: for any subgroup $A \leq \mathrm{GL}_n(K^{\text{sep}})$ stabilized by G_K, the exact sequence of pointed sets $H^1(G_K, A) \to H^1(G_K, \mathrm{GL}_n) = 1$ from the inclusion $A \hookrightarrow \mathrm{GL}_n$ shows that the connecting map $\mathrm{GL}_n(K^{\text{sep}})^{G_K}/A^{G_K} \to H^1(G_K, A)$ can be analyzed via the quotient GL_n/A. When A is normal (as for $A = \mathrm{SL}_n \trianglelefteq \mathrm{GL}_n$ or $A = O_n \leq \mathrm{GL}_n$), the long exact sequence in non-abelian H^1 (section 10.3) provides a systematic approach to computing $H^1(G_K, A)$ from the simpler group GL_n/A.

Exercise 10.2.1

1. Show that $H^1(G_K, \mathrm{GL}_n(K^{\text{sep}})) = \{*\}$ (the trivial pointed set). (*Hint:* a cocycle $c\colon G_K \to \mathrm{GL}_n(K^{\text{sep}})$ defines a "twisted" K-vector space; show this is isomorphic to K^n by constructing an explicit basis, using the normal basis theorem or a direct averaging argument.)

2. Using the descent principle, show that the quaternion algebras $(a, b)_{\mathbb{Q}}$ (see **??** from Chapter **??**) are exactly the nontrivial elements of $H^1(G_{\mathbb{Q}}, \mathrm{PGL}_2(\overline{\mathbb{Q}}))$, and identify the connecting map to $H^2(G_{\mathbb{Q}}, \overline{\mathbb{Q}}^\times) = \mathrm{Br}(\mathbb{Q})$.

3. For an elliptic curve E/K with $j(E) \neq 0, 1728$: show that the quadratic twist E^d (for $d \in K^\times/(K^\times)^2$) has the property $E^d(K^{\text{sep}}) \cong E(K^{\text{sep}})$ as G_K-sets, but $E^d(K)$ can differ from $E(K)$ (the Mordell–Weil groups of twists are related but not isomorphic in general).

4. Show that the degree-3 étale $\mathbb{Q}$-algebras are: $\mathbb{Q}^3$ (split), $\mathbb{Q} \times L$ for quadratic fields L, and cubic fields K'. Identify each with a conjugacy class of homomorphisms $G_{\mathbb{Q}} \to S_3$.

5. (If the reader has read [2]) A *Severi–Brauer variety* of dimension $n-1$ over K is a twisted form of $\mathbb{P}^{n-1}_K$. Show that $\operatorname{Twist}(\mathbb{P}^{n-1}/K) \cong H^1(G_K, \mathrm{PGL}_n(K^{\text{sep}}))$, and that the Brauer class of a Severi–Brauer variety X is trivial if and only if $X(K) \neq \emptyset$.

10.3 Non-Abelian H^1

The descent principle (theorem 10.2.2) used $H^1(G_K, \mathrm{Aut}(X_0))$ without requiring $\mathrm{Aut}(X_0)$ to be abelian: the automorphism groups PGL_n, O_n, and S_n from the examples in section 10.2 are all non-abelian (for $n \geq 2$, $n \geq 3$, and $n \geq 3$ respectively). This section develops the formal theory of H^1 for non-abelian coefficient groups. The key difference from the abelian theory is that H^1 is a *pointed set* rather than a group: it has a distinguished element (the trivial class) but no group operation in general.

Throughout this section, G denotes a profinite group and A a topological group with a continuous G-action by group automorphisms (a *G-group*).

Definition 10.3.1: Non-Abelian H^1

Let A be a G-group (not necessarily abelian). A 1-*cocycle* is a continuous map $c\colon G \to A$ satisfying the cocycle condition:

$$c(\sigma\tau) = c(\sigma) \cdot \sigma(c(\tau)) \qquad \text{for all } \sigma, \tau \in G$$

Two cocycles c, c' are *cohomologous* if there exists $a \in A$ with:

$$c'(\sigma) = a^{-1} \cdot c(\sigma) \cdot \sigma(a) \qquad \text{for all } \sigma \in G$$

The *non-abelian H^1* is the quotient:

$$H^1(G, A) = Z^1(G, A)/{\sim}$$

where $Z^1(G, A)$ is the set of 1-cocycles and $\sim$ is the cohomologous relation. This is a *pointed set* with distinguished element $[1]$ (the class of the trivial cocycle $c(\sigma) = 1$ for all σ).

When A is abelian, the relation $c' \sim c$ becomes $c'(\sigma) = c(\sigma) \cdot \sigma(a) \cdot a^{-1}$, and the set $H^1(G, A)$ carries an abelian group structure (the product of cocycles is a cocycle). When A is non-abelian, the product of two cocycles need not satisfy the cocycle condition, so $H^1(G, A)$ has no natural group structure. The correct categorical framework is that of pointed sets and morphisms preserving the basepoint.

Proposition 10.3.2: Functoriality

The assignment $A \mapsto H^1(G, A)$ is a functor from the category of G-groups (with G-equivariant group homomorphisms) to the category of pointed sets (with basepoint-preserving maps). For a continuous homomorphism $\varphi\colon G' \to G$ of profinite groups: the restriction $\mathrm{res}\colon H^1(G, A) \to H^1(G', A)$ sending $c \mapsto c \circ \varphi$ is a morphism of pointed sets.

***Proof*:**
A G-equivariant homomorphism $f\colon A \to B$ sends cocycles to cocycles ($f(c(\sigma\tau)) = f(c(\sigma)) \cdot \sigma(f(c(\tau)))$ using G-equivariance) and preserves the cohomologous relation ($f(a^{-1} \cdot c(\sigma) \cdot \sigma(a)) = f(a)^{-1} \cdot f(c(\sigma)) \cdot \sigma(f(a))$). The trivial cocycle maps to the trivial cocycle. For restriction: the composition $c \circ \varphi$ satisfies the cocycle condition for G' whenever c satisfies it for G.

The central structural result for non-abelian H^1 is the long exact sequence in pointed sets, which generalizes the long exact sequence of abelian cohomology.

Theorem 10.3.3: Exact Sequence of Pointed Sets

Let $1 \to A \xrightarrow{i} B \xrightarrow{\pi} C \to 1$ be an exact sequence of G-groups (with i injective, π surjective, and $\ker \pi = i(A)$). There is an exact sequence of pointed sets:

$$1 \to A^G \xrightarrow{i} B^G \xrightarrow{\pi} C^G \xrightarrow{\delta} H^1(G, A) \xrightarrow{i_*} H^1(G, B) \xrightarrow{\pi_*} H^1(G, C)$$

where "exactness" means: the fiber of each map over the distinguished point equals the image of the previous map.

***Proof*:**

The sequence at A^G, B^G, C^G is the usual exact sequence of fixed points, using the left-exactness of the fixed-point functor.

The connecting map δ. For $\gamma \in C^G$: choose a lift $b \in B$ with $\pi(b) = \gamma$ (possible by surjectivity of π). For each $\sigma \in G$: $\pi(\sigma(b)) = \sigma(\pi(b)) = \sigma(\gamma) = \gamma = \pi(b)$, so $b^{-1} \cdot \sigma(b) \in \ker \pi = i(A)$. Define $a(\sigma) = i^{-1}(b^{-1} \cdot \sigma(b)) \in A$. This is a 1-cocycle: $a(\sigma\tau) = i^{-1}(b^{-1} \cdot \sigma\tau(b)) = i^{-1}(b^{-1} \cdot \sigma(b) \cdot \sigma(b^{-1}) \cdot \sigma\tau(b)) = i^{-1}(b^{-1} \cdot \sigma(b)) \cdot \sigma(i^{-1}(b^{-1} \cdot \tau(b))) = a(\sigma) \cdot \sigma(a(\tau))$ (using that $i(A)$ is normal in B and G-stable). A different lift $b' = b \cdot i(a_0)$ replaces a by the cohomologous cocycle $a'(\sigma) = a_0^{-1} \cdot a(\sigma) \cdot \sigma(a_0)$. So $\delta(\gamma) = [a] \in H^1(G, A)$ is well defined.

Exactness at C^G. The image of $\pi\colon B^G \to C^G$ consists of those $\gamma \in C^G$ that lift to a G-fixed element $b \in B^G$. For such γ: $\sigma(b) = b$ for all σ, giving $a(\sigma) = i^{-1}(b^{-1} \cdot b) = 1$, so $\delta(\gamma) = [1]$. Conversely, if $\delta(\gamma) = [1]$: the cocycle a is a coboundary, $a(\sigma) = a_0^{-1} \cdot \sigma(a_0)$ for some $a_0 \in A$. Then $b' = b \cdot i(a_0)$ satisfies $\sigma(b') = \sigma(b) \cdot \sigma(i(a_0)) = b \cdot i(a(\sigma)) \cdot i(\sigma(a_0)) = b \cdot i(a_0^{-1} \cdot \sigma(a_0)) \cdot i(\sigma(a_0)) = b \cdot i(a_0) = b'$, so $b' \in B^G$ lifts γ.

Exactness at $H^1(G, A)$. The image of δ consists of classes $[a]$ arising from some $\gamma \in C^G$. The map $i_*\colon H^1(G, A) \to H^1(G, B)$ sends $[a]$ to $[i \circ a]$. For $[a] \in \mathrm{im}(\delta)$: $(i \circ a)(\sigma) = b^{-1} \cdot \sigma(b)$ is a coboundary in B, so $i_*([a]) = [1]$. Conversely, if $i_*([a]) = [1]$: the cocycle $i \circ a$ is a coboundary in B, i.e., $(i \circ a)(\sigma) = b^{-1} \cdot \sigma(b)$ for some $b \in B$. Then $\gamma = \pi(b) \in C^G$ (since $\pi(b^{-1} \cdot \sigma(b)) = \pi(i(a(\sigma))) = 1$, giving $\sigma(\pi(b)) = \pi(b)$), and $\delta(\gamma) = [a]$.

Exactness at $H^1(G, B)$. The image of i_* consists of classes $[c] \in H^1(G, B)$ with $c(\sigma) \in i(A)$ (up to B-coboundary). The map π_* sends $[c]$ to $[\pi \circ c]$. For $[c] \in \mathrm{im}(i_*)$: $(\pi \circ c)(\sigma) = \pi(i(a(\sigma))) = 1$, so $\pi_*([c]) = [1]$. Conversely, if $\pi_*([c]) = [1]$: the cocycle $\pi \circ c$ is a coboundary in C, i.e., $\pi(c(\sigma)) = \gamma^{-1} \cdot \sigma(\gamma)$ for some $\gamma \in C$. Lifting γ to $b \in B$: the cocycle $c'(\sigma) = b \cdot c(\sigma) \cdot \sigma(b)^{-1}$ is cohomologous to c and satisfies $\pi(c'(\sigma)) = 1$, so $c'(\sigma) \in i(A)$. Thus $[c] = [c'] \in \mathrm{im}(i_*)$.

The exact sequence terminates at $H^1(G, C)$: there is no natural continuation to H^2 in the non-abelian setting (without additional structure). The obstruction to extending the sequence to H^2 is the subject of section 10.4; when A lies in the center of B, the sequence does extend to the abelian $H^2(G, A)$.

Definition 10.3.4: Twisting

Let $c \in Z^1(G, A)$ be a 1-cocycle. The *twisted G-group* ${}_cA$ is the group A equipped with the twisted G-action:

$$\sigma *_c a = c(\sigma) \cdot \sigma(a) \cdot c(\sigma)^{-1}$$

The cocycle condition ensures this is a group action. The *twisting bijection* is:

$$H^1(G, A) \cong H^1(G, {}_cA)$$

sending a class $[c']$ to $[c^{-1} \cdot c']$ (with the cocycle condition verified using the twisted action). Under this bijection, $[c] \in H^1(G, A)$ maps to $[1] \in H^1(G, {}_cA)$: twisting shifts the basepoint from $[1]$ to $[c]$.

Twisting is the mechanism that controls fibers in the non-abelian exact sequence. For the map $\pi_*: H^1(G, B) \to H^1(G, C)$: the fiber $\pi_*^{-1}([c_C])$ over a class $[c_C] \in H^1(G, C)$ is not a subgroup of $H^1(G, B)$ (since $H^1(G, B)$ is not a group), but a torsor under $H^1(G, {}_{c_B}A)$ for any c_B lifting c_C. The "size" of the fiber depends on the twisted group, which varies with the basepoint.

Example 10.3.5: The Brauer Group via Non-Abelian H^1

The exact sequence of G_K-groups:

$$1 \to (K^{\mathrm{sep}})^\times \xrightarrow{\cdot\,\mathrm{Id}} \mathrm{GL}_n(K^{\mathrm{sep}}) \xrightarrow{\pi} \mathrm{PGL}_n(K^{\mathrm{sep}}) \to 1$$

(where $(K^{\mathrm{sep}})^\times$ embeds as scalar matrices) induces the exact sequence of pointed sets:

$$\begin{aligned} K^\times &\to \mathrm{GL}_n(K) \to \mathrm{PGL}_n(K) \xrightarrow{\delta} H^1(G_K, (K^{\mathrm{sep}})^\times) \\ &\to H^1(G_K, \mathrm{GL}_n(K^{\mathrm{sep}})) \to H^1(G_K, \mathrm{PGL}_n(K^{\mathrm{sep}})) \xrightarrow{\delta'} H^2(G_K, (K^{\mathrm{sep}})^\times) \end{aligned}$$

Hilbert 90 (theorem 8.1.1) gives $H^1(G_K, (K^{\mathrm{sep}})^\times) = 0$, and generalized Hilbert 90 (exercise 10.2.1 (1)) gives $H^1(G_K, \mathrm{GL}_n(K^{\mathrm{sep}})) = 1$. The sequence collapses to:

$$1 \to H^1(G_K, \mathrm{PGL}_n(K^{\mathrm{sep}})) \xhookrightarrow{\delta'} H^2(G_K, (K^{\mathrm{sep}})^\times) = \mathrm{Br}(K)$$

The connecting map δ' sends a CSA of degree n (viewed as a twisted form of $M_n(K)$ via descent, example 10.2.3) to its Brauer class. The image of δ' is $\mathrm{Br}(K)[n]$ (the n-torsion), and the injectivity of δ' says: a CSA of degree n is determined up to isomorphism by its Brauer class (the Brauer class plus the degree determines the algebra). Passing to the colimit over n: $\mathrm{Br}(K) = \varinjlim_n H^1(G_K, \mathrm{PGL}_n)$, recovering the cohomological description of the Brauer group from theorem 7.2.4 in Chapter **??**.

Exercise 10.3.1

1. Show that $H^1(G_K, \mathrm{GL}_n(K^{\mathrm{sep}})) = \{*\}$ implies, via the exact sequence of pointed sets for $1 \to \mathrm{SL}_n \to \mathrm{GL}_n \xrightarrow{\det} (K^{\mathrm{sep}})^\times \to 1$, that $H^1(G_K, \mathrm{SL}_n(K^{\mathrm{sep}})) \cong K^\times / \det(\mathrm{GL}_n(K)) = K^\times/(K^\times)^n$... verify whether this identification is correct and describe the map explicitly.

2. For $K = \mathbb{R}$: compute $H^1(G_{\mathbb{R}}, O_n(\mathbb{C}))$ by classifying non-degenerate real quadratic forms of rank n up to isometry. (*Hint:* the classification is by signature (p, q) with $p + q = n$.)

3. Let $c \in Z^1(G_K, \mathrm{PGL}_2(K^{\mathrm{sep}}))$ be the cocycle corresponding to a quaternion algebra D over K. Describe the twisted group ${}_c\mathrm{GL}_2(K^{\mathrm{sep}})$ explicitly as the unit group $D^\times$ of D (viewed as a G_K-group), and verify that $H^1(G_K, {}_c\mathrm{GL}_2) = \{*\}$ (Hilbert 90 for $D^\times$).

4. For the connecting map $\delta' \colon H^1(G_K, \mathrm{PGL}_n) \to \mathrm{Br}(K)$: show that the fiber $(\delta')^{-1}([\alpha])$ over a Brauer class $[\alpha]$ is either empty (if $[\alpha]$ has order not dividing n) or a single point (if $[\alpha] \in \mathrm{Br}(K)[n]$).

10.4 Non-Abelian H^2 and Gerbes

The exact sequence of theorem 10.3.3 terminates at $H^1(G, C)$: the non-abelian H^1 has no natural continuation to H^2 for arbitrary coefficient groups. A partial extension is possible when the kernel A is *central* in B (i.e., $A \subseteq Z(B)$), in which case the connecting map extends one step further into the abelian $H^2(G, A)$. This section describes this extension, introduces the non-abelian H^2 in its limited form, and connects the theory to the broader categorical framework of gerbes.

Theorem 10.4.1: Obstruction Exact Sequence

Let $1 \to A \xrightarrow{i} B \xrightarrow{\pi} C \to 1$ be an exact sequence of G-groups with $A \subseteq Z(B)$ (i.e., A is central in B, and in particular A is abelian). The exact sequence of theorem 10.3.3 extends by one term:

$$\cdots \to H^1(G, B) \xrightarrow{\pi_*} H^1(G, C) \xrightarrow{\delta'} H^2(G, A)$$

where $H^2(G, A)$ is the ordinary (abelian) H^2. The map δ' is the *obstruction*: a class $[c] \in H^1(G, C)$ lifts to $H^1(G, B)$ if and only if $\delta'([c]) = 0$ in $H^2(G, A)$.

***Proof*:**
Construction of δ'. Let $c \colon G \to C$ be a 1-cocycle. For each $\sigma \in G$, choose a lift $\tilde{c}(\sigma) \in B$ with $\pi(\tilde{c}(\sigma)) = c(\sigma)$ (possible by surjectivity). The map $\tilde{c}$ need not satisfy the cocycle condition in B; the failure is measured by:

$$\alpha(\sigma, \tau) = \tilde{c}(\sigma) \cdot \sigma(\tilde{c}(\tau)) \cdot \tilde{c}(\sigma\tau)^{-1}$$

Since $\pi(\alpha(\sigma, \tau)) = c(\sigma) \cdot \sigma(c(\tau)) \cdot c(\sigma\tau)^{-1} = 1$ (the cocycle condition for c), the element $\alpha(\sigma, \tau)$ lies in $\ker \pi = i(A)$. The map $\alpha \colon G \times G \to A$ is a 2-cocycle: the associativity of the group operation in B gives the cocycle identity $\sigma(\alpha(\tau, \rho)) \cdot \alpha(\sigma, \tau\rho) = \alpha(\sigma, \tau) \cdot \alpha(\sigma\tau, \rho)$ (using $A \subseteq Z(B)$ to commute α past elements of B). The centrality of A is essential here: without it, α would depend on the conjugation action of $\tilde{c}$ on A, and the cocycle identity would fail.

Well-definedness. A different choice of lifts $\tilde{c}'(\sigma) = \tilde{c}(\sigma) \cdot i(\beta(\sigma))$ (with $\beta \colon G \to A$) modifies α by a coboundary: $\alpha'(\sigma, \tau) = \alpha(\sigma, \tau) \cdot \beta(\sigma) \cdot \sigma(\beta(\tau)) \cdot \beta(\sigma\tau)^{-1} = \alpha(\sigma, \tau) \cdot (d\beta)(\sigma, \tau)$. A cohomologous cocycle $c'(\sigma) = \gamma^{-1} \cdot c(\sigma) \cdot \sigma(\gamma)$ (with $\gamma \in C$) also modifies α by a coboundary (lift γ to $\tilde{\gamma} \in B$ and adjust). So $\delta'([c]) = [\alpha] \in H^2(G, A)$ is well defined.

Exactness. The class $[c] \in H^1(G, C)$ lies in $\mathrm{im}(\pi_*)$ if and only if $c = \pi \circ b$ for some 1-cocycle $b\colon G \to B$ (up to coboundary). Taking $\tilde{c} = b$: the obstruction $\alpha(\sigma,\tau) = b(\sigma)\cdot\sigma(b(\tau))\cdot b(\sigma\tau)^{-1} = 1$ (the cocycle condition for b). So $\delta'([c]) = 0$. Conversely, if $\delta'([c]) = 0$: the 2-cocycle α is a coboundary, $\alpha(\sigma,\tau) = \beta(\sigma)\cdot\sigma(\beta(\tau))\cdot\beta(\sigma\tau)^{-1}$ for some $\beta\colon G \to A$. Replacing $\tilde{c}(\sigma)$ by $\tilde{c}(\sigma)\cdot i(\beta(\sigma))^{-1}$: the new lift satisfies the cocycle condition in B (the obstruction vanishes), providing a lift $b \in Z^1(G, B)$ with $\pi \circ b = c$.

Example 10.4.2: The Brauer Group as an Obstruction

The central extension $1 \to (K^{\mathrm{sep}})^\times \to \mathrm{GL}_n(K^{\mathrm{sep}}) \to \mathrm{PGL}_n(K^{\mathrm{sep}}) \to 1$ has $(K^{\mathrm{sep}})^\times$ central in $\mathrm{GL}_n(K^{\mathrm{sep}})$, so the obstruction exact sequence applies:

$$H^1(G_K, \mathrm{GL}_n) \to H^1(G_K, \mathrm{PGL}_n) \xrightarrow{\delta'} H^2(G_K, (K^{\mathrm{sep}})^\times) = \mathrm{Br}(K)$$

Since $H^1(G_K, \mathrm{GL}_n) = 1$ (exercise 10.2.1 (1)), the connecting map δ' is injective. The obstruction class $\delta'([A]) \in \mathrm{Br}(K)$ of a CSA A (viewed as a class in $H^1(G_K, \mathrm{PGL}_n)$) is its Brauer class. The content of the obstruction: A splits (i.e., $A \cong M_n(K)$) if and only if its Brauer class is trivial.

The 2-cocycle $\alpha(\sigma,\tau)$ from the proof has a concrete description: choose a K^{sep}-basis of $A \otimes_K K^{\mathrm{sep}} \cong M_n(K^{\mathrm{sep}})$ and express the G_K-action on A as $\sigma \mapsto \tilde{c}(\sigma) \in \mathrm{GL}_n(K^{\mathrm{sep}})$ (the semilinear action). The failure of $\tilde{c}$ to be a homomorphism is measured by $\alpha(\sigma,\tau) = \tilde{c}(\sigma)\cdot\sigma(\tilde{c}(\tau))\cdot\tilde{c}(\sigma\tau)^{-1} \in (K^{\mathrm{sep}})^\times$, which is exactly the 2-cocycle representing the Brauer class of A in $H^2(G_K, (K^{\mathrm{sep}})^\times)$. This recovers the crossed product construction of **??** from Chapter **??**.

Example 10.4.3: Lifting Projective Representations

A *projective representation* of G_K of dimension n is a continuous homomorphism $\bar{\rho}\colon G_K \to \mathrm{PGL}_n(K^{\mathrm{sep}})$ (equivalently, a 1-cocycle with values in PGL_n). The obstruction to lifting $\bar{\rho}$ to a linear representation $\rho\colon G_K \to \mathrm{GL}_n(K^{\mathrm{sep}})$ is the class $\delta'([\bar{\rho}]) \in H^2(G_K, (K^{\mathrm{sep}})^\times) = \mathrm{Br}(K)$. This class is the *Schur obstruction*: it measures whether the projective representation arises from a central simple algebra that splits. For K a local field: $\mathrm{Br}(K) \cong \mathbb{Q}/\mathbb{Z}$ (corollary 8.1.4 from Chapter 8), so the obstruction is a rational number modulo 1. For $K = \mathbb{Q}$: the Brauer group exact sequence (corollary 9.1.3 from Chapter 9) constrains which projective representations can be lifted, and the obstruction vanishes if and only if the associated CSA splits at every place.

The obstruction exact sequence handles the case where A is central in B. The centrality condition is essential: without it, the "2-cocycle" $\alpha(\sigma,\tau) = \tilde{c}(\sigma)\cdot\sigma(\tilde{c}(\tau))\cdot\tilde{c}(\sigma\tau)^{-1}$ does not commute with elements of B, and the cocycle identity fails. For a non-central kernel, the obstruction takes values not in an abelian H^2 but in a more general object: the non-abelian H^2 of Giraud.

Definition 10.4.4: Non-Abelian H^2 (Giraud)

For a G-group A (possibly non-abelian), the *non-abelian* $H^2(G, A)$ is the set of equivalence classes of *G-gerbes bound by A*: categorical objects that generalize torsors from groups to groupoids. A G-gerbe bound by A is a category $\mathcal{G}$ equipped with a G-action such that: (i) $\mathcal{G}$ is locally non-empty (it has objects over some open cover in the G-topology), and (ii) the automorphism group of any object is isomorphic to A (with the isomorphism compatible with the G-action up to inner automorphisms). Two gerbes are equivalent if they are connected by a G-equivariant equivalence of categories. The set $H^2(G, A)$ is pointed (the trivial gerbe is the category of A-torsors with the standard G-action).

When A is abelian, this construction recovers the ordinary $H^2(G, A)$ (an abelian gerbe bound by A is equivalent to a 2-cocycle $\alpha\colon G \times G \to A$). When A is non-abelian, $H^2(G, A)$ depends on the "band" (or "lien") of A—the outer automorphism class of the G-action on A—and does not admit a group structure. The dependence on the band is a genuinely non-abelian phenomenon: two G-groups with the same underlying group but different G-actions can have different H^2, even though their underlying abstract H^2 (ignoring the G-action) would be the same. The full theory is developed in Giraud [13] using the language of stacks and fibered categories; the construction above gives the essential idea without the full categorical apparatus.

The relationship between the obstruction exact sequence and the gerbe framework is as follows. For a central extension $1 \to A \to B \to C \to 1$: the obstruction $\delta'([c]) \in H^2(G, A)$ of a class $[c] \in H^1(G, C)$ corresponds, under the Giraud construction, to the gerbe of "liftings of c to B." This gerbe has objects (over K^{sep}) that are 1-cocycles $b\colon G \to B$ with $\pi \circ b = c$, and the automorphism group of any such object is A (acting by the coboundary relation). The gerbe is trivial (has a global object) if and only if the lifting exists, i.e., $\delta'([c]) = 0$.

The non-abelian H^2 has no further extension: there is no general "non-abelian H^3." The obstruction theory stops at H^2, reflecting the structural fact that torsors (classified by H^1) and gerbes (classified by H^2) exhaust the naturally arising "non-abelian" cohomological objects in classical algebraic geometry. Higher cohomology with non-abelian coefficients requires the theory of higher stacks and ∞-categories, which lies beyond the scope of this book.

10.4.5 Connections to the Langlands Program The non-abelian $H^1(G_K, \hat{G})$ for the Langlands dual group $\hat{G}$ of a reductive group G parametrizes *L-parameters*: continuous homomorphisms from the Weil group W_K (or the conjectural Langlands group L_K) to $\hat{G}(\mathbb{C})$, up to conjugacy. The local Langlands correspondence predicts a bijection between L-parameters and "L-packets" of irreducible representations of $G(K_v)$ for local fields K_v. The global Langlands correspondence extends this to automorphic representations. The cohomological framework of this chapter—non-abelian H^1, twisting, descent, and obstruction classes in H^2—provides the language for formulating and studying these correspondences. The Langlands program remains one of the central research programs in modern number theory; see [18] for the connection between Galois representations and automorphic forms.

The non-abelian H^1 and H^2 developed in this section and section 10.3 provide a classification framework that extends far beyond abelian Galois cohomology: H^1 classifies twisted forms and torsors (theorem 10.2.2), the obstruction map in H^2 controls lifting problems (theorem 10.4.1), and the gerbe framework gives a categorical interpretation of the obstructions. The final section of this chapter places these constructions within the broader context of étale cohomology, where the passage from fields to schemes unifies Galois cohomology with the cohomology of algebraic varieties.

Exercise 10.4.1

1. Verify the obstruction cocycle identity: for the 2-cocycle $\alpha(\sigma, \tau) = \tilde{c}(\sigma) \cdot \sigma(\tilde{c}(\tau)) \cdot \tilde{c}(\sigma\tau)^{-1}$ (with $\tilde{c}$ a set-theoretic lift of a 1-cocycle $c\colon G \to C$), show that $\sigma(\alpha(\tau, \rho)) \cdot \alpha(\sigma, \tau\rho) = \alpha(\sigma, \tau) \cdot \alpha(\sigma\tau, \rho)$ using the centrality of A in B.
2. Show that when A is abelian, the non-abelian $H^2(G, A)$ (in the sense of definition 10.4.4) reduces to the ordinary $H^2(G, A)$: a gerbe bound by an abelian group A is determined up to equivalence by a 2-cocycle $\alpha\colon G \times G \to A$, and the equivalence relation on gerbes corresponds to the coboundary relation on 2-cocycles.
3. For $K = \mathbb{Q}$ and the exact sequence $1 \to \mu_2 \to \mathrm{SL}_2 \to \mathrm{PGL}_2 \to 1$: the obstruction map $\delta'\colon H^1(G_{\mathbb{Q}}, \mathrm{PGL}_2) \to H^2(G_{\mathbb{Q}}, \mu_2) = \mathrm{Br}(\mathbb{Q})[2]$ sends a quaternion algebra D to its Brauer class. Show that D lifts to $H^1(G_{\mathbb{Q}}, \mathrm{SL}_2)$ (i.e., D has a "reduced norm 1 form") if and only if $[D] = 0 \in \mathrm{Br}(\mathbb{Q})$ (i.e., $D \cong M_2(\mathbb{Q})$).
4. Describe the gerbe associated to a Brauer class $[\alpha] \in \mathrm{Br}(K)$: it is the category whose objects over K^{sep} are K^{sep}-vector spaces V of dimension n with an isomorphism $\mathrm{End}(V) \cong A \otimes_K K^{\mathrm{sep}}$ (where A is the CSA representing $[\alpha]$), and whose morphisms are isomorphisms of vector spaces compatible with the endomorphism structure. Show that this category is non-empty over K^{sep} but has no global object over K (unless $[\alpha] = 0$).

10.5 Connections to Étale Cohomology

The Galois cohomology of a field K is the cohomology of the profinite group G_K acting on discrete modules. A natural question: can this framework be extended from fields ($\operatorname{Spec} K$) to more general geometric objects (schemes, varieties)? The answer is étale cohomology, developed by Grothendieck and his collaborators in the 1960s. The étale cohomology of a scheme X recovers Galois cohomology when $X = \operatorname{Spec} K$ is a field, but extends to arbitrary schemes, providing the cohomological tools needed to prove the Weil conjectures and to formulate arithmetic duality over rings of integers and curves. This section gives an overview of the étale theory and its relationship to the Galois cohomology developed in this book, without developing the full machinery of sites and toposes.

Definition 10.5.1: Étale Site (Overview)

For a scheme X: the *étale site* $X_{\text{ét}}$ is the category whose objects are étale morphisms $U \to X$ (flat, unramified morphisms of finite presentation), equipped with the étale topology: a family $\{U_i \to U\}$ is a *covering* if $\bigsqcup U_i \to U$ is surjective. A *sheaf* $\mathcal{F}$ on $X_{\text{ét}}$ assigns an abelian group $\mathcal{F}(U)$ to each étale morphism $U \to X$, satisfying the sheaf condition with respect to étale coverings. The *étale cohomology groups* are:

$$H^r_{\text{ét}}(X, \mathcal{F}) = R^r\Gamma(X_{\text{ét}}, \mathcal{F})$$

the right derived functors of the global sections functor $\Gamma(X_{\text{ét}}, \mathcal{F}) = \mathcal{F}(X)$.

The definition parallels the construction of sheaf cohomology in algebraic topology (replacing the classical topology with the étale topology), and the formal properties are analogous: long exact sequences from short exact sequences of sheaves, Mayer–Vietoris sequences from coverings, Leray spectral sequences from morphisms of sites. The key advantage of the étale topology over the Zariski topology is that it is fine enough to detect arithmetic information (the Zariski topology on $\operatorname{Spec} K$ is trivial, while the étale topology recovers the full profinite structure of G_K).

The fundamental comparison theorem identifies étale cohomology of a field with Galois cohomology.

Theorem 10.5.2: Comparison Theorem

For $X = \operatorname{Spec} K$ (a field): the category of étale sheaves on $X_{\text{ét}}$ is equivalent to the category of discrete G_K-modules, via the functor sending a sheaf $\mathcal{F}$ to its stalk $\mathcal{F}_{\overline{K}} = \varinjlim_{L/K \text{ finite sep.}} \mathcal{F}(\operatorname{Spec} L)$ with the natural G_K-action. Under this equivalence:

$$H^r_{\text{ét}}(\operatorname{Spec} K, \mathcal{F}) \cong H^r(G_K, \mathcal{F}_{\overline{K}})$$

for all $r \geq 0$.

***Proof*:**
An étale morphism $U \to \operatorname{Spec} K$ is a disjoint union of spectra of finite separable extensions $\operatorname{Spec} L_i \to \operatorname{Spec} K$ (since étale over a field means separable). The étale coverings of $\operatorname{Spec} K$ are therefore the families $\{\operatorname{Spec} L_i\}$ with $\bigsqcup \operatorname{Spec} L_i \to \operatorname{Spec} K$ surjective, i.e., at least one $L_i = K$. A sheaf $\mathcal{F}$ on $(\operatorname{Spec} K)_{\text{ét}}$ assigns to each finite separable L/K a group $\mathcal{F}(\operatorname{Spec} L)$, functorially in L, satisfying the sheaf condition for étale covers. The stalk $\mathcal{F}_{\overline{K}} = \varinjlim_L \mathcal{F}(\operatorname{Spec} L)$ is a discrete G_K-module (the action of $\sigma \in G_K$ comes from the functoriality: $\sigma^* \colon \mathcal{F}(\operatorname{Spec} L) \to \mathcal{F}(\operatorname{Spec} \sigma(L))$). The functor $\mathcal{F} \mapsto \mathcal{F}_{\overline{K}}$ is an equivalence of categories (with inverse sending a discrete G_K-module M to the sheaf $\mathcal{F}_M(\operatorname{Spec} L) = M^{\operatorname{Gal}(K^{\text{sep}}/L)}$). Under this equivalence: the global sections functor $\Gamma(\operatorname{Spec} K, \mathcal{F}) = \mathcal{F}(\operatorname{Spec} K) = M^{G_K}$ coincides with the G_K-fixed points functor, and the derived functors of both are the same: $H^r_{\text{ét}}(\operatorname{Spec} K, \mathcal{F}) = R^r\Gamma = H^r(G_K, M)$.

The comparison theorem says that Galois cohomology *is* étale cohomology for the scheme $\operatorname{Spec} K$. The entire content of this book—profinite group cohomology, Tate duality, the Brauer group, Poitou–Tate, Selmer groups—is the étale cohomology of the one-point scheme $\operatorname{Spec} K$. The power of the étale framework emerges when X is replaced by a more complex scheme.

For a number field K with ring of integers $\mathcal{O}_K$: the scheme $X = \operatorname{Spec} \mathcal{O}_K$ is a one-dimensional arithmetic scheme, and its étale cohomology interpolates between the Galois cohomology of K (at the generic point) and the Galois cohomology of the residue fields $\mathcal{O}_K/\mathfrak{p}$ (at the closed points). The Leray spectral sequence for the inclusion $j \colon \operatorname{Spec} K \hookrightarrow \operatorname{Spec} \mathcal{O}_K$ relates:

$$H^p_{\text{ét}}(\operatorname{Spec} \mathcal{O}_K, R^q j_* \mathcal{F}) \Longrightarrow H^{p+q}_{\text{ét}}(\operatorname{Spec} K, \mathcal{F}) = H^{p+q}(G_K, M)$$

The higher direct images $R^q j_* \mathcal{F}$ are supported on the closed points (the primes of $\mathcal{O}_K$), and their stalks are the local cohomology groups $H^q(G_v, M)$ at each prime v. The spectral sequence assembles these local contributions into the global cohomology, providing a geometric reinterpretation of the localization maps and the Poitou–Tate sequence from Chapter 9.

Example 10.5.3: Étale Cohomology of $\operatorname{Spec}\mathbb{Z}[1/n]$

For $X = \operatorname{Spec}\mathbb{Z}[1/n]$ and the constant sheaf $\mathcal{F} = \mathbb{Z}/n\mathbb{Z}$: the étale fundamental group is $\pi_1^{\text{ét}}(X) = G_{\mathbb{Q},S}$ where $S = \{\infty\} \cup \{p : p \big| n\}$ (the primes dividing n and the archimedean place). The comparison gives $H^r_{\text{ét}}(\operatorname{Spec}\mathbb{Z}[1/n], \mathbb{Z}/n\mathbb{Z}) = H^r(G_{\mathbb{Q},S}, \mathbb{Z}/n\mathbb{Z})$, recovering the S-ramified Galois cohomology of definition 9.2.2. For $n = p$ prime and $r = 1$: this counts cyclic p-extensions of $\mathbb{Q}$ unramified outside $\{p, \infty\}$ (subfields of $\mathbb{Q}(\zeta_{p^\infty})$). For $r = 2$: the group $H^2_{\text{ét}}(\operatorname{Spec}\mathbb{Z}[1/p], \mathbb{Z}/p\mathbb{Z})$ is controlled by the global EP formula (theorem 9.3.2).

The duality theorems of Chapters 8 and 9 have natural étale generalizations. Local Tate duality (theorem 8.3.2) becomes Poincaré duality for the étale cohomology of local fields (or, more precisely, for the henselization of $\operatorname{Spec}\mathcal{O}_K$ at a closed point). The Poitou–Tate sequence (theorem 9.2.5) becomes the Artin–Verdier duality theorem: a nine-term exact sequence for the étale cohomology of $\operatorname{Spec}\mathcal{O}_{K,S}$ (the S-integer ring), with the local terms contributed by the closed points and archimedean places.

For smooth projective varieties over finite fields, étale cohomology provides the cohomological framework for the Weil conjectures—the deepest application of the theory. The key idea: the number of $\mathbb{F}_{q^m}$-points of a variety X over $\mathbb{F}_q$ is expressed as an alternating sum of traces of the Frobenius endomorphism on étale cohomology groups (the Lefschetz trace formula), and the properties of these cohomology groups (finite dimensionality, Poincaré duality, purity of Frobenius eigenvalues) translate into properties of the zeta function.

Theorem 10.5.4: Weil Conjectures (Grothendieck–Deligne)

Let X be a smooth projective variety of dimension d over $\mathbb{F}_q$. The zeta function:

$$Z(X,t) = \exp\Big(\sum_{m=1}^{\infty} |X(\mathbb{F}_{q^m})| \cdot \frac{t^m}{m}\Big)$$

satisfies:

1. *(Rationality)* $Z(X,t) = \prod_{r=0}^{2d} \det(1 - t \cdot \mathrm{Frob}_q \big| H^r_{\text{ét}}(X_{\overline{\mathbb{F}}_q}, \mathbb{Q}_\ell))^{(-1)^{r+1}}$ for any prime $\ell \neq \operatorname{char}(\mathbb{F}_q)$.
2. *(Functional equation)* $Z(X, q^{-d}t^{-1}) = \pm q^{d\chi/2} \cdot t^{\chi} \cdot Z(X,t)$, where $\chi = \sum(-1)^r \dim H^r_{\text{ét}}$ is the Euler characteristic.
3. *(Riemann hypothesis)* The eigenvalues of Frob_q on $H^r_{\text{ét}}(X_{\overline{\mathbb{F}}_q}, \mathbb{Q}_\ell)$ have absolute value $q^{r/2}$.

Rationality and the functional equation were proved by Grothendieck using the Lefschetz trace formula in étale cohomology. The Riemann hypothesis was proved by Deligne [12]. The Frobenius Frob_q acts on $H^r_{\text{ét}}(X_{\overline{\mathbb{F}}_q}, \mathbb{Q}_\ell)$ by functoriality (the q-power Frobenius is an endomorphism of $X_{\overline{\mathbb{F}}_q}$), and the rationality formula expresses the point-counting zeta function in terms of this action—a far-reaching generalization of the relationship between the Frobenius and the Artin map in class field theory (see theorem 9.4.2).

The étale cohomology groups $H^r_{\text{ét}}(X_{\overline{\mathbb{F}}_q}, \mathbb{Q}_\ell)$ are the arithmetic analogues of the singular cohomology groups $H^r(X(\mathbb{C}), \mathbb{Q})$ for a complex variety. The comparison theorem of Artin gives $H^r_{\text{ét}}(X_{\overline{\mathbb{F}}_q}, \mathbb{Q}_\ell) \cong$

$H^r(X(\mathbb{C}), \mathbb{Q}) \otimes \mathbb{Q}_\ell$ when X lifts to characteristic zero, connecting the arithmetic (Frobenius eigenvalues, point counts) to the topology (Betti numbers, Hodge structure). This comparison is the foundation of the "philosophy of motives": the cohomology groups H^r should be independent of the particular cohomology theory (étale, de Rham, crystalline), and the relationships between these realizations encode arithmetic information.

The passage from Galois cohomology (this book) to étale cohomology (algebraic geometry) is the passage from $\operatorname{Spec} K$ to general schemes. The tools developed here—profinite group cohomology, Tate duality, the Brauer group, descent, non-abelian H^1—all have étale analogues that form the foundation of modern arithmetic geometry. The Brauer group of a scheme X is $H^2_{\text{ét}}(X, \mathbb{G}_m)$, generalizing $\operatorname{Br}(K) = H^2(G_K, (K^{\text{sep}})^\times)$. The Selmer groups and Sha of section 9.5 are computed via étale cohomology of the Néron model of an abelian variety over $\operatorname{Spec} \mathcal{O}_K$. The reader who has mastered the Galois cohomology of this book is prepared to study the étale theory in its full generality: see Milne [15] for the foundational treatment, and [2] for the scheme-theoretic prerequisites. The cohomological perspective developed here—from profinite groups to global duality to descent—will serve as both foundation and guide.

Exercise 10.5.1

1. Show that the category of étale sheaves of abelian groups on $(\operatorname{Spec} K)_{\text{ét}}$ is equivalent to the category of discrete G_K-modules, by constructing the equivalence explicitly (the functor $\mathcal{F} \mapsto \mathcal{F}_{\overline{K}}$ and its inverse $M \mapsto \mathcal{F}_M$).
2. The étale fundamental group $\pi_1^{\text{ét}}(\operatorname{Spec} \mathbb{Z}[1/n], \overline{\eta})$ (based at a geometric generic point $\overline{\eta}$) is the Galois group $G_{\mathbb{Q},S}$ where $S = \{p : p | n\} \cup \{\infty\}$. Explain why: the finite étale covers of $\operatorname{Spec} \mathbb{Z}[1/n]$ are the spectra of rings of integers $\mathcal{O}_L[1/n]$ for number fields $L/\mathbb{Q}$ unramified outside S, and $\pi_1^{\text{ét}}$ is the profinite group classifying these covers.
3. (If the reader has read [2]) For $X = \mathbb{P}^1_{\mathbb{F}_q}$ (the projective line over $\mathbb{F}_q$): compute $|X(\mathbb{F}_{q^m})| = q^m + 1$ directly, verify that $Z(X, t) = 1/((1-t)(1-qt))$, and identify the étale cohomology groups $H^r_{\text{ét}}(X_{\overline{\mathbb{F}}_q}, \mathbb{Q}_\ell)$ for $r = 0, 1, 2$.
4. Using the comparison theorem and the computation of $H^1(G_{\mathbb{Q},S}, \mathbb{Z}/p\mathbb{Z})$ from example 9.3.3, compute $H^1_{\text{ét}}(\operatorname{Spec} \mathbb{Z}[1/p], \mathbb{Z}/p\mathbb{Z})$ for an odd prime p and describe the corresponding étale covers of $\operatorname{Spec} \mathbb{Z}[1/p]$.

A

Adèles and Idèles

The global duality theorems of Chapter 9 require Galois modules that assemble local information at every place of a number field simultaneously. The adèle ring and idèle group provide the correct framework: they are restricted products that combine the completions K_v across all places v of a number field K, with a topology that remembers the integral structure at almost all places. This appendix develops the definitions and essential properties, assuming the theory of places and completions from [10] and the classical finiteness theorems (finite class number, Dirichlet unit theorem) from [4]. The treatment is independent of [17]; readers of that book will find a parallel development there with different emphasis.

A.1 Restricted Products and the Adèle Ring

The completions K_v of a number field K are locally compact groups, and the ring of integers $\mathcal{O}_v \subseteq K_v$ at each finite place v is a compact open subgroup. The full product $\prod_v K_v$ carries the product topology, but this is too coarse for number-theoretic purposes: it contains elements with unbounded denominators at infinitely many places, and its unit group has the wrong topology. The restricted product resolves both problems by imposing the constraint that almost all components lie in the designated compact open subgroup.

Definition A.1.1: Restricted Product

Let $\{G_v\}_{v\in V}$ be a family of locally compact topological groups, and for all but finitely many v let $K_v \le G_v$ be a compact open subgroup. The *restricted product* of (G_v, K_v) is the subgroup:

$$\prod_{v\in V}{}'(G_v, K_v) = \left\{ (g_v)_{v\in V} \in \prod_v G_v \;\middle|\; g_v \in K_v \text{ for all but finitely many } v \right\}$$

equipped with the topology generated by sets of the form $\prod_{v\in S} U_v \times \prod_{v\notin S} K_v$, where $S \subseteq V$ is a finite set containing all v for which K_v is undefined, and each $U_v \subseteq G_v$ is open.

An element of the restricted product is a tuple (g_v) that is "integral at almost all places": only finitely many components are allowed to escape the compact open subgroup K_v. The topology is finer than the subspace topology from $\prod_v G_v$—the basic open sets require the components outside S to lie *exactly* in K_v, not just in any open subset of G_v.

Proposition A.1.2: Local Compactness of Restricted Products

The restricted product $\prod'_v(G_v, K_v)$ is a locally compact Hausdorff topological group.

***Proof*:**
For any finite set S of places (containing all places where K_v is undefined), the subgroup $G_S = \prod_{v\in S} G_v \times \prod_{v\notin S} K_v$ carries the product topology, which is locally compact (a finite product of locally compact groups times a product of compact groups, the latter compact by Tychonoff's theorem). The restricted product is the union $\bigcup_S G_S$ over all finite S, and the topology on the restricted product restricts to the product topology on each G_S. Since G_S is open in $G_{S'}$ for $S \subseteq S'$ (the factor $\prod_{v\in S'\setminus S} K_v$ is open in $\prod_{v\in S'\setminus S} G_v$), the restricted product is locally compact: every point has a compact neighborhood (contained in some G_S). The Hausdorff property follows from the Hausdorff property of each G_v.

The notion of restricted product applies equally to rings: if each G_v is a locally compact ring and each K_v is a compact open subring, the restricted product inherits a ring structure with componentwise operations.

Throughout this appendix, K denotes a number field with ring of integers $\mathcal{O}_K$. The set of places of K is denoted V_K, decomposed as $V_K = V_K^{\text{fin}} \cup V_K^{\infty}$ into finite and archimedean places. For each finite place v, the completion K_v is a non-archimedean local field with ring of integers $\mathcal{O}_v$ and maximal ideal $\mathfrak{m}_v$; for each archimedean place v, the completion K_v is $\mathbb{R}$ or $\mathbb{C}$. The reader is referred to [10] for the construction of completions and their structural properties, and to [4] for the arithmetic of $\mathcal{O}_K$ (prime splitting, class group, unit group).

Definition A.1.3: Adèle Ring

The *adèle ring* of K is the restricted product:

$$\mathbb{A}_K = \prod_{v \in V_K}{}' (K_v, \mathcal{O}_v)$$

where the restricted product is taken with respect to the compact open subrings $\mathcal{O}_v \subseteq K_v$ at each finite place v, and the archimedean completions K_v ($v \in V_K^\infty$) appear without restriction (i.e., K_v serves as its own "compact open subgroup" for the purpose of the restricted product construction). The ring operations are componentwise addition and multiplication.

An adèle $a = (a_v)_v \in \mathbb{A}_K$ is a tuple giving an element $a_v \in K_v$ at each place v, subject to the constraint $a_v \in \mathcal{O}_v$ for all but finitely many finite places v. The *ring of integral adèles* is the compact open subring $\mathbb{A}_K^\circ = \prod_{v|\infty} K_v \times \prod_{v\nmid\infty} \mathcal{O}_v$. The finite adèles are the restricted product over finite places only: $\mathbb{A}_K^{\mathrm{fin}} = \prod'_{v \in V_K^{\mathrm{fin}}} (K_v, \mathcal{O}_v)$.

Proposition A.1.4: The Adèle Ring Is Locally Compact

$\mathbb{A}_K$ is a locally compact Hausdorff topological ring.

***Proof*:**
Immediate from proposition A.1.2: $\mathbb{A}_K$ is a restricted product of the locally compact fields K_v with respect to the compact open subrings $\mathcal{O}_v$.

The number field K embeds into $\mathbb{A}_K$ diagonally.

Proposition A.1.5: Diagonal Embedding

The diagonal map $\iota\colon K \hookrightarrow \mathbb{A}_K$, defined by $\iota(a) = (a, a, a, \ldots)$ (viewing $a \in K_v$ via the canonical embedding $K \hookrightarrow K_v$ at each place), is a ring homomorphism embedding K as a discrete subring of $\mathbb{A}_K$.

***Proof*:**
The map ι is a ring homomorphism (componentwise operations). The image is well defined as an adèle: for any $a \in K^\times$, the absolute value $|a|_v = 1$ for all but finitely many v (since $a \in \mathcal{O}_v^\times$ for all v not dividing the numerator or denominator ideal of a), so $\iota(a) \in \mathbb{A}_K$. Injectivity is immediate ($a = 0$ if and only if $a_v = 0$ at any place).

For discreteness: the open neighborhood $U = \left\{(a_v) \in \mathbb{A}_K \;\middle|\; |a_v|_v < 1 \text{ for all } v \in V_K^\infty,\ a_v \in \mathcal{O}_v \text{ for all } v \in V_K^{\mathrm{fin}}\right\}$ of $0 \in \mathbb{A}_K$ satisfies $\iota(K) \cap U = \{0\}$: an element $a \in K$ with $|a|_v < 1$ at all archimedean places and $|a|_v \leq 1$ at all finite places satisfies $\prod_v |a|_v < 1$ if $a \neq 0$, contradicting the product formula $\prod_v |a|_v = 1$ (in logarithmic form, [10, Arakelov Degree of Principal Divisors]). So $a = 0$.

From this point forward, K is identified with its image $\iota(K) \subseteq \mathbb{A}_K$ and treated as a discrete subring of $\mathbb{A}_K$.

Theorem A.1.6: Compactness of $\mathbb{A}_K/K$

The quotient $\mathbb{A}_K/K$ is compact.

***Proof*:**
Step 1: Reduction to a fundamental domain. Let $n = [K : \mathbb{Q}]$. A $\mathbb{Z}$-basis $\omega_1, \ldots, \omega_n$ of $\mathcal{O}_K$ determines an isomorphism of topological groups $\mathbb{A}_K \cong \mathbb{A}_{\mathbb{Q}}^n$ (via the basis). Under this isomorphism, K embeds as a discrete subgroup of $\mathbb{A}_{\mathbb{Q}}^n$, and it suffices to show $\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}$ is compact (the general case follows by taking the n-fold product).

Step 2: The case $K = \mathbb{Q}$. Here $\mathbb{A}_{\mathbb{Q}} = \mathbb{R} \times \mathbb{A}_{\mathbb{Q}}^{\text{fin}}$ and $\mathbb{A}_{\mathbb{Q}}^{\text{fin}} = \prod'_p(\mathbb{Q}_p, \mathbb{Z}_p)$. The subring of finite integral adèles is $\widehat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$, which is compact. The claim is that the compact set $[0,1) \times \widehat{\mathbb{Z}}$ maps surjectively onto $\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}$. Given an adèle $(a_\infty, (a_p)_p)$: first, since the a_p are integral at almost all p, there exists $N \in \mathbb{Z}$ with $Na_p \in \mathbb{Z}_p$ for all p (take N divisible by all primes where $a_p \notin \mathbb{Z}_p$). The Chinese Remainder Theorem and strong approximation for $\mathbb{Q}$ produce $q \in \mathbb{Q}$ with $a_p - q \in \mathbb{Z}_p$ for all p and $a_\infty - q \in [0, 1)$ (first approximate the finite components, then adjust the real component by an integer). Then $(a_v)_v - \iota(q) \in [0,1) \times \widehat{\mathbb{Z}}$.

Step 3: Compactness. The image of the compact set $[0,1] \times \widehat{\mathbb{Z}}$ under the continuous projection $\mathbb{A}_{\mathbb{Q}} \to \mathbb{A}_{\mathbb{Q}}/\mathbb{Q}$ is compact. Since this image is all of $\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}$ (by Step 2), the quotient is compact.

The compactness of $\mathbb{A}_K/K$ is the adèlic reformulation of two classical finiteness theorems: the finiteness of the class number and the finiteness of the unit group modulo roots of unity. The adèlic language packages these into a single topological statement.

Example A.1.7: Adèles of $\mathbb{Q}$

1. The integral adèles of $\mathbb{Q}$ are $\mathbb{A}_{\mathbb{Q}}^\circ = \mathbb{R} \times \widehat{\mathbb{Z}}$. The diagonal image of $\mathbb{Z}$ in $\mathbb{A}_{\mathbb{Q}}^\circ$ is the set $\{(n, (n, n, \ldots)) : n \in \mathbb{Z}\}$, which is dense in $\{0\} \times \widehat{\mathbb{Z}}$ (by the density of $\mathbb{Z}$ in $\widehat{\mathbb{Z}}$ from exercise 1.1.1 (1)) but discrete in the full adèle topology.
2. Strong approximation for $\mathbb{Q}$ takes the following form: the image of $\mathbb{Q}$ in $\mathbb{A}_{\mathbb{Q}}^{\text{fin}} = \prod'_p \mathbb{Q}_p$ is dense. Stated differently: given elements $a_{p_1} \in \mathbb{Q}_{p_1}, \ldots, a_{p_r} \in \mathbb{Q}_{p_r}$ and an error bound $\epsilon > 0$, there exists $q \in \mathbb{Q}$ with $|q - a_{p_i}|_{p_i} < \epsilon$ for all i and $|q|_p \leq 1$ for all other primes p. The proof uses the Chinese Remainder Theorem ([6, Chinese Remainder Theorem For Integers]) and is standard.
3. (If the reader has read [2]) The adèle ring is the ring of global sections of a certain sheaf on $\operatorname{Spec} \mathcal{O}_K$ with generic fiber K. More precisely, for a smooth group scheme G over $\mathcal{O}_K$, the restricted product $\prod'_v G(K_v)$ (restricted with respect to $G(\mathcal{O}_v)$) is the group of adèlic points $G(\mathbb{A}_K)$. The case $G = \mathbb{G}_a$ recovers $\mathbb{A}_K$, and $G = \mathbb{G}_m$ will produce the idèle group in appendix A.2.

Exercise A.1.1

1. Show that the restricted product topology on $\prod'_v (G_v, K_v)$ makes it a topological group (i.e., the group operations are continuous).
2. Verify the fundamental domain claim for $\mathbb{A}_\mathbb{Q}/\mathbb{Q}$ directly: show that every element of $\mathbb{A}_\mathbb{Q}$ is equivalent modulo $\mathbb{Q}$ to an element of $[0,1) \times \widehat{\mathbb{Z}}$.
3. Let $S \subseteq V_K$ be a finite set of places containing all archimedean places. Show that the S-adèles $\mathbb{A}_{K,S} = \prod_{v\in S} K_v \times \prod_{v\notin S} \mathcal{O}_v$ form an open subring of $\mathbb{A}_K$, and that $\mathbb{A}_K = \bigcup_S \mathbb{A}_{K,S}$ as S ranges over all such finite sets.
4. Let L/K be a finite extension of number fields. Show that $\mathbb{A}_L \cong \mathbb{A}_K \otimes_K L$ as topological rings, where the right side carries the topology making $\mathbb{A}_K \otimes_K L \cong \mathbb{A}_K^{[L:K]}$ (via a K-basis of L) a product of copies of $\mathbb{A}_K$.
5. Use the product formula $\prod_v |a|_v = 1$ for $a \in K^\times$ to show that the diagonal embedding $K \hookrightarrow \mathbb{A}_K$ has discrete image, without appealing to the explicit fundamental domain argument.

A.2 The Idèle Group and Idèle Class Group

The multiplicative group $\mathbb{A}_K^\times$ of the adèle ring, equipped with the subspace topology from $\mathbb{A}_K$, fails to be a topological group: inversion is not continuous (an adèle can have components approaching zero at infinitely many places, making the inverse escape every compact set). The idèle group corrects this by using the restricted product topology on the units, treating the local unit groups $\mathcal{O}_v^\times$ as the compact open subgroups.

Definition A.2.1: Idèle Group

The *idèle group* of K is the restricted product:

$$\mathbb{I}_K = \prod_{v\in V_K}{}' (K_v^\times, \mathcal{O}_v^\times)$$

where $\mathcal{O}_v^\times$ is the group of local units at each finite place v, and the archimedean factors $K_v^\times$ ($v \in V_K^\infty$, with $K_v^\times = \mathbb{R}^\times$ or $\mathbb{C}^\times$) appear without restriction. The group operation is componentwise multiplication.

An idèle $\alpha = (\alpha_v)_v$ is a tuple of local elements $\alpha_v \in K_v^\times$ with $\alpha_v \in \mathcal{O}_v^\times$ (i.e., $|\alpha_v|_v = 1$) for all but finitely many finite places. The set of places where $\alpha_v \notin \mathcal{O}_v^\times$ is the *support* of the idèle. The *finite idèles* are $\mathbb{I}_K^{\mathrm{fin}} = \prod'_{v\in V_K^{\mathrm{fin}}} (K_v^\times, \mathcal{O}_v^\times)$.

Proposition A.2.2: Topology of the Idèle Group

The idèle group $\mathbb{I}_K$ is a locally compact Hausdorff topological group. The natural inclusion $\mathbb{I}_K \hookrightarrow \mathbb{A}_K$ (viewing $K_v^\times \subseteq K_v$) is continuous, but the subspace topology on $\mathbb{I}_K$ inherited from $\mathbb{A}_K$ is strictly coarser than the idèle topology.

***Proof*:**
The first claim follows from proposition A.1.2: the groups $K_v^\times$ are locally compact and the subgroups $\mathcal{O}_v^\times$ are compact and open in $K_v^\times$ (at a finite place, $\mathcal{O}_v^\times = \{x \in K_v \mid |x|_v = 1\}$ is the preimage of $\{1\} \subseteq \mathbb{R}_{\geq 0}$ under the continuous open map $|\cdot|_v$, hence compact and open).

For the inclusion: the map $\mathbb{I}_K \hookrightarrow \mathbb{A}_K$ sends $(\alpha_v) \mapsto (\alpha_v)$, and a basic open set in $\mathbb{A}_K$ restricts to an open set in $\mathbb{I}_K$ (since $\mathcal{O}_v^\times \subseteq \mathcal{O}_v$ is an intersection of open-in-$\mathbb{I}_K$ with open-in-$\mathbb{A}_K$).

For the strict coarseness: the map $\alpha \mapsto \alpha^{-1}$ is continuous in the restricted product topology on $\mathbb{I}_K$ (since inversion is continuous on each $K_v^\times$ and preserves $\mathcal{O}_v^\times$), but not continuous in the subspace topology from $\mathbb{A}_K$. To see this, consider a sequence of idèles $\alpha^{(n)}$ in $\mathbb{A}_\mathbb{Q}$ with $\alpha^{(n)}_{p_n} = p_n$ (a uniformizer at the nth prime) and $\alpha^{(n)}_v = 1$ for all other v. In the subspace topology from $\mathbb{A}_\mathbb{Q}$, $\alpha^{(n)} \to 1$ (each component eventually equals 1). But $(\alpha^{(n)})^{-1}_{p_n} = p_n^{-1}$, and the sequence $(\alpha^{(n)})^{-1}$ does not converge to 1 in $\mathbb{A}_\mathbb{Q}$ (the p_n-component $p_n^{-1} \notin \mathbb{Z}_{p_n}$ for each n). So inversion is discontinuous in the subspace topology, confirming it is strictly coarser.

The idèle group carries a natural "size" function that measures the total effect of an idèle across all places.

Definition A.2.3: Idèle Norm

The *idèle norm* (or *content*) is the continuous homomorphism:

$$|\cdot|_K \colon \mathbb{I}_K \to \mathbb{R}_{>0}, \qquad |(\alpha_v)|_K = \prod_{v \in V_K} |\alpha_v|_v$$

The product converges because $|\alpha_v|_v = 1$ for all but finitely many v, reducing it to a finite product.

Continuity of the idèle norm follows from the fact that on each basic open subgroup $\prod_{v\in S} K_v^\times \times \prod_{v\notin S} \mathcal{O}_v^\times$, the norm is the finite product $\prod_{v\in S} |\alpha_v|_v$, a continuous function of finitely many variables.

The number field K embeds into $\mathbb{I}_K$ diagonally, and the product formula constrains this embedding.

Theorem A.2.4: Product Formula and Discreteness

The diagonal embedding $K^\times \hookrightarrow \mathbb{I}_K$, sending $a \mapsto (a, a, a, \ldots)$, has discrete image. For every $a \in K^\times$:

$$|a|_K = \prod_v |a|_v = 1$$

In particular, $K^\times \subseteq \ker |\cdot|_K$.

***Proof*:**
The product formula is [10, Arakelov Degree of Principal Divisors] in logarithmic form. For discreteness: the open neighborhood $U = \{(\alpha_v) \in \mathbb{I}_K \mid |\alpha_v - 1|_v < 1 \text{ for all } v \in V_K^\infty,\ \alpha_v \in \mathcal{O}_v^\times \text{ for all } v \in V_K^{\text{fin}}\}$ of the identity in $\mathbb{I}_K$ satisfies $K^\times \cap U = \{1\}$. If $a \in K^\times \cap U$, then $|a|_v \leq 1$ at all finite places (so $a \in \mathcal{O}_K$) and $|a - 1|_v < 1$ at all archimedean places. The condition $|a-1|_v < 1$ at archimedean

places and $a \in \mathcal{O}_K$ leaves only finitely many possibilities for a (the ring $\mathcal{O}_K$ is discrete in the archimedean completions). Combined with $|a|_v = 1$ at all finite v (from $a \in \mathcal{O}_v^\times$), the element a is a unit of $\mathcal{O}_K$ satisfying $|a-1|_v < 1$ at all archimedean places. By Dirichlet's unit theorem (see [4, Dirichlet Unit Theorem]), $\mathcal{O}_K^\times$ is finitely generated, and the archimedean constraint forces $a = 1$ (the only unit in $\mathcal{O}_K$ within distance 1 of 1 at every archimedean place, for the neighborhood U chosen sufficiently small).

From this point forward, $K^\times$ is identified with its image in $\mathbb{I}_K$.

Definition A.2.5: Idèle Class Group

The *idèle class group* of K is the quotient:

$$C_K = \mathbb{I}_K / K^\times$$

equipped with the quotient topology. The idèle norm descends to a continuous homomorphism $|\cdot|_K \colon C_K \to \mathbb{R}_{>0}$ (since $|a|_K = 1$ for $a \in K^\times$). The *norm-one idèle class group* is:

$$C_K^0 = \ker\big(|\cdot|_K \colon C_K \to \mathbb{R}_{>0}\big)$$

The idèle class group is the global analogue of $K_v^\times$ for a local field. The local reciprocity map of theorem 8.4.4 sends $K_v^\times \to G_{K_v}^{\mathrm{ab}}$; global reciprocity (Chapter 9) will send $C_K \to G_K^{\mathrm{ab}}$.

The following theorem is the idèlic repackaging of two classical finiteness results: the finiteness of the class group $\mathrm{Cl}(K)$ and the Dirichlet unit theorem.

Theorem A.2.6: Compactness of C_K^0

The norm-one idèle class group C_K^0 is compact.

***Proof*:**
Step 1: Connection to the class group and units. Define the subgroup of "S-unit idèles" $\mathbb{I}_K^S = \prod_{v\in S} K_v^\times \times \prod_{v\notin S} \mathcal{O}_v^\times$ for a finite set of places S containing V_K^∞. The composition $\mathbb{I}_K \to C_K \to \mathrm{Cl}(K)$ (sending an idèle to the class of its associated fractional ideal $\prod_v \mathfrak{p}_v^{v(\alpha_v)}$) is surjective, and its kernel is $K^\times \cdot \mathbb{I}_K^{V_K^\infty}$ (an idèle whose associated ideal is principal differs from a diagonal element by a unit idèle). The class group exact sequence is:

$$1 \to K^\times \cdot \mathbb{I}_K^{V_K^\infty} / K^\times \to C_K \to \mathrm{Cl}(K) \to 0$$

Since $\mathrm{Cl}(K)$ is finite (see [4, Class Number]), compactness of C_K^0 reduces to showing that $K^\times \cdot \mathbb{I}_K^{V_K^\infty} / K^\times \cap C_K^0$ is compact.

Step 2: The unit idèles. The subgroup $\mathbb{I}_K^{V_K^\infty}$ consists of idèles that are units at all finite places. An element of $\mathbb{I}_K^{V_K^\infty} \cap K^\times \cdot \ker|\cdot|_K$ is determined by its archimedean components (the finite components are all 1). The image of $\mathcal{O}_K^\times$ in the archimedean component $\prod_{v|\infty} K_v^\times$ is a discrete cocompact subgroup of the norm-one surface $\left\{ (x_v)_{v|\infty} \in \prod_{v|\infty} K_v^\times \;\middle|\; \prod_v |x_v|_v = 1 \right\}$ by Dirichlet's unit theorem (see [4, Dirichlet Unit Theorem]): the unit group $\mathcal{O}_K^\times$ has rank $r_1 + r_2 - 1$

(where r_1 and r_2 are the numbers of real and complex places), which equals the dimension of the norm-one surface. The quotient of this surface by the image of $\mathcal{O}_K^\times$ is therefore compact.

Step 3: Assembly. The group C_K^0 fits into an exact sequence:

$$1 \to (K^\times \cdot \mathbb{I}_K^{V_K^\infty} \cap C_K^0) \to C_K^0 \to \mathrm{Cl}\,(K) \to 0$$

The term on the left is compact by Step 2 (a compact quotient of a locally compact group, pulled back along a continuous map). The term on the right is finite. An extension of a compact group by a finite group is compact (the preimage of each element of $\mathrm{Cl}\,(K)$ is a coset of the compact subgroup, and a finite union of compact sets is compact). So C_K^0 is compact.

The compactness of C_K^0 is the single most important structural fact about the idèle class group. It enters the global theory through the following consequence: the Herbrand quotient of C_L^0 for a cyclic extension L/K is well defined and computable, which is the starting point for the cohomological approach to global class field theory.

Proposition A.2.7: Structure of C_K

The idèle norm gives a split short exact sequence:

$$1 \to C_K^0 \to C_K \xrightarrow{|\cdot|_K} \mathbb{R}_{>0} \to 1$$

The splitting is given by $t \mapsto (\alpha_v)$ where $\alpha_{v_0} = t$ at a fixed archimedean place v_0 and $\alpha_v = 1$ for $v \neq v_0$. In particular, $C_K \cong C_K^0 \times \mathbb{R}_{>0}$.

***Proof*:**
Surjectivity of $|\cdot|_K$: for any $t \in \mathbb{R}_{>0}$, the idèle with $\alpha_{v_0} = t$ at a real place v_0 and $\alpha_v = 1$ elsewhere has $|(\alpha_v)|_K = t$. Its image in C_K maps to t. The section $s\colon \mathbb{R}_{>0} \to C_K$ is a continuous homomorphism (it is the composition of the inclusion $\mathbb{R}_{>0} \hookrightarrow K_{v_0}^\times$ with the projection $\mathbb{I}_K \to C_K$). The identity $|\cdot|_K \circ s = \mathrm{id}$ gives the splitting.

Example A.2.8: Idèles of $\mathbb{Q}$

1. The idèle group of $\mathbb{Q}$ is $\mathbb{I}_\mathbb{Q} = \mathbb{R}^\times \times \prod'_p(\mathbb{Q}_p^\times, \mathbb{Z}_p^\times)$. Since $\mathbb{Q}$ is a PID, the class group $\mathrm{Cl}\,(\mathbb{Q})$ is trivial, so $C_\mathbb{Q} = \mathbb{I}_\mathbb{Q}/\mathbb{Q}^\times$ has no "class group part." The structure is: $C_\mathbb{Q} \cong \mathbb{R}_{>0} \times \widehat{\mathbb{Z}}^\times$. To see this: every idèle $(\alpha_\infty, (\alpha_p)_p)$ is equivalent modulo $\mathbb{Q}^\times$ to one with $\alpha_\infty > 0$ (multiply by -1 if needed) and $\alpha_p \in \mathbb{Z}_p^\times$ for all p (multiply by a suitable power of each bad prime). The residual freedom is multiplication by positive rationals that are p-adic units for all p, i.e., by $\{1\}$. So $C_\mathbb{Q} \cong \mathbb{R}_{>0} \times \widehat{\mathbb{Z}}^\times$. The norm-one subgroup is $C_\mathbb{Q}^0 \cong \widehat{\mathbb{Z}}^\times$ (compact, as required by theorem A.2.6).

2. The local-to-global comparison: for each prime p, the inclusion $\mathbb{Q}_p^\times \hookrightarrow \mathbb{I}_\mathbb{Q}$ (at the p-component, with 1 elsewhere) followed by the projection $\mathbb{I}_\mathbb{Q} \to C_\mathbb{Q}$ gives a continuous homomorphism $\mathbb{Q}_p^\times \to C_\mathbb{Q}$. These local maps assemble the local arithmetic of each completion into the global idèle class group. The local reciprocity maps rec_{K_v} of Chapter 8 will similarly assemble into

the global reciprocity map via these inclusions.

3. For comparison with the classical ideal class group: the map $\mathbb{I}_{\mathbb{Q}}^{\mathrm{fin}} \to \mathrm{Div}(\mathbb{Z}) = \bigoplus_p \mathbb{Z}$ sending $(\alpha_p)_p \mapsto \sum_p v_p(\alpha_p) \cdot (p)$ is surjective with kernel $\prod_p \mathbb{Z}_p^\times$. Passing to the quotient by $\mathbb{Q}^\times$ recovers $\mathrm{Cl}(\mathbb{Z}) = 0$ (every fractional ideal of $\mathbb{Z}$ is principal). For a general number field, the same construction gives the class group as a quotient of C_K.

Exercise A.2.1

1. Show that the subspace topology on $\mathbb{I}_K$ from $\mathbb{A}_K$ is strictly coarser than the restricted product topology by verifying that the set $W = \left\{(\alpha_v) \in \mathbb{I}_K \;\middle|\; |\alpha_v|_v = 1 \text{ for all } v \in V_K^{\mathrm{fin}}\right\}$ is open in the restricted product topology but not in the subspace topology.
2. Verify the claim $C_{\mathbb{Q}} \cong \mathbb{R}_{>0} \times \widehat{\mathbb{Z}}^\times$ by constructing an explicit isomorphism and checking it is a homeomorphism.
3. Show that the idèle norm $|\cdot|_K \colon \mathbb{I}_K \to \mathbb{R}_{>0}$ is surjective, and that the composition $K^\times \hookrightarrow \mathbb{I}_K \xrightarrow{|\cdot|_K} \mathbb{R}_{>0}$ has image $\{1\}$.
4. Let S be a finite set of places containing V_K^∞. Define the S-idèle class group $C_{K,S} = \mathbb{I}_K/(K^\times \cdot \prod_{v \notin S} \mathcal{O}_v^\times)$. Show that there is a natural surjection $C_{K,S} \twoheadrightarrow \mathrm{Cl}_S(K)$ to the S-class group, and identify its kernel in terms of the S-unit group $\mathcal{O}_{K,S}^\times$.
5. Let L/K be a finite extension. Show that the norm map $N_{L/K} \colon \mathbb{I}_L \to \mathbb{I}_K$, defined by $N_{L/K}(\alpha)_v = \prod_{w|v} N_{L_w/K_v}(\alpha_w)$, is a continuous homomorphism satisfying $|N_{L/K}(\alpha)|_K = |\alpha|_L$ for all $\alpha \in \mathbb{I}_L$.

A.3 Galois Action on Idèles

The adèle ring and idèle group become Galois modules when a Galois extension L/K is fixed. The action of $G = \mathrm{Gal}(L/K)$ on $\mathbb{I}_L$ permutes the places of L above each place of K and acts on the corresponding completions. The central result of this section is that the G-module $\mathbb{I}_L$ decomposes as a restricted product of induced modules (proposition A.3.4), reducing the Galois cohomology of idèles to a direct sum of local cohomology groups. This decomposition is the bridge between the local theory of Chapter 8 and the global theory of Chapter 9.

Throughout this section, L/K denotes a finite Galois extension of number fields with $G = \mathrm{Gal}(L/K)$.

Proposition A.3.1: Galois Action on Adèles and Idèles

The group G acts continuously on $\mathbb{A}_L$ and $\mathbb{I}_L$ by:

$$(\sigma \cdot \alpha)_w = \sigma(\alpha_{\sigma^{-1}w})$$

for $\sigma \in G$, $\alpha = (\alpha_w)_w \in \mathbb{A}_L$ (or $\mathbb{I}_L$), and w a place of L. Here $\sigma^{-1}w$ is the place $x \mapsto |\sigma^{-1}(x)|_w$, and σ acts on $L_{\sigma^{-1}w}$ via the canonical isomorphism $\sigma \colon L_{\sigma^{-1}w} \xrightarrow{\sim} L_w$ induced by $\sigma \colon L \xrightarrow{\sim} L$.

***Proof*:**
The map σ permutes the places of L: if w is a place extending v of K, then σw is another place extending v (since σ fixes K). The induced map $\sigma\colon L_{\sigma^{-1}w} \to L_w$ is a continuous isomorphism of locally compact fields, sending $\mathcal{O}_{\sigma^{-1}w}$ to $\mathcal{O}_w$ (the valuation is preserved). The formula $(\sigma \cdot \alpha)_w = \sigma(\alpha_{\sigma^{-1}w})$ defines a ring automorphism of $\mathbb{A}_L$ (componentwise): it permutes the factors and applies a continuous field isomorphism on each. The image of an adèle is an adèle: if $\alpha_w \in \mathcal{O}_w$ for all but finitely many w, then $(\sigma \cdot \alpha)_w = \sigma(\alpha_{\sigma^{-1}w}) \in \mathcal{O}_w$ for all but finitely many w (the set of exceptional places is permuted by σ). Continuity follows from the continuity of each local map $\sigma\colon L_{\sigma^{-1}w} \to L_w$. The same argument applies to $\mathbb{I}_L$ (units are permuted to units, and $\mathcal{O}_w^\times$ maps to $\mathcal{O}_w^\times$).

The Galois action on adèles and idèles recovers the global elements as fixed points.

Theorem A.3.2: Galois Fixed Points

$\mathbb{A}_L^G = \mathbb{A}_K$ and $\mathbb{I}_L^G = \mathbb{I}_K$.

***Proof*:**
An adèle $\alpha \in \mathbb{A}_L$ is G-fixed if and only if $\sigma(\alpha_{\sigma^{-1}w}) = \alpha_w$ for all $\sigma \in G$ and all places w of L. Fixing a place v of K and choosing a place w_0 of L above v: the places above v are $\{\sigma w_0 : \sigma \in G/G_{w_0}\}$ where $G_{w_0} = \{\sigma \in G \mid \sigma w_0 = w_0\}$ is the decomposition group. The G-fixed condition requires $\alpha_{\sigma w_0} = \sigma(\alpha_{w_0})$ for all σ, and the G_{w_0}-fixed condition on the w_0-component requires $\alpha_{w_0} \in L_{w_0}^{G_{w_0}} = K_v$. So a G-fixed adèle has $\alpha_{w_0} \in K_v$ and all components above v determined by this single value via the Galois action. The resulting tuple is exactly the diagonal embedding of an element of K_v into $\prod_{w|v} L_w$. Assembling over all v: $\mathbb{A}_L^G = \mathbb{A}_K$.

The same argument applies to $\mathbb{I}_L$, replacing K_v by $K_v^\times$ and L_w by $L_w^\times$ throughout.

The norm map on idèles assembles the local norms at each place.

Definition A.3.3: Idèle Norm Map

The *idèle norm map* $N_{L/K}\colon \mathbb{I}_L \to \mathbb{I}_K$ is defined by:

$$N_{L/K}(\alpha)_v = \prod_{w|v} N_{L_w/K_v}(\alpha_w)$$

for each place v of K, where w ranges over the places of L above v and N_{L_w/K_v} is the local norm.

The norm map is compatible with the abstract norm $N_G(\alpha) = \prod_{\sigma\in G} \sigma \cdot \alpha$ on the G-module $\mathbb{I}_L$: the product $\prod_\sigma \sigma(\alpha_{\sigma^{-1}w_0})$ at a fixed place $w_0|v$ computes $\prod_{w|v} N_{L_w/K_v}(\alpha_w)$ after accounting for the decomposition group orbits (see exercise A.3.1 (1)).

The key structural result decomposes $\mathbb{I}_L$ as a G-module into induced modules from the local fields.

Proposition A.3.4: Decomposition of Idèles as G-Module

For each place v of K, fix a place w_0 of L above v, and let $G_{w_0} = \mathrm{Gal}(L_{w_0}/K_v) \leq G$ be the decomposition group. Then:

$$\mathbb{I}_L \cong \prod_{v \in V_K}' \mathrm{Ind}_{G_{w_0}}^{G}(L_{w_0}^\times)$$

as G-modules, where $\mathrm{Ind}_{G_{w_0}}^{G}(L_{w_0}^\times) = \mathbb{Z}[G] \otimes_{\mathbb{Z}[G_{w_0}]} L_{w_0}^\times$ is the induced module, and the restricted product is taken with respect to $\mathrm{Ind}_{G_{w_0}}^{G}(\mathcal{O}_{w_0}^\times)$ at finite places.

***Proof*:**
The places w of L above v are in bijection with G/G_{w_0} via $\sigma \mapsto \sigma w_0$. The v-component of $\mathbb{I}_L$ is $\prod_{w|v} L_w^\times = \prod_{\sigma \in G/G_{w_0}} L_{\sigma w_0}^\times$. The G-action permutes these factors: τ sends the σw_0-component to the $\tau\sigma w_0$-component via $\tau\colon L_{\sigma w_0} \xrightarrow{\sim} L_{\tau\sigma w_0}$. This is the defining property of the induced module $\mathrm{Ind}_{G_{w_0}}^{G}(L_{w_0}^\times) = \bigoplus_{\sigma \in G/G_{w_0}} L_{\sigma w_0}^\times$: the G-module that is freely generated from the G_{w_0}-module $L_{w_0}^\times$ by letting G permute the coset summands.

Taking the restricted product over all v: the integrality condition $\alpha_w \in \mathcal{O}_w^\times$ for all but finitely many w translates to the restricted product being taken with respect to $\mathrm{Ind}_{G_{w_0}}^{G}(\mathcal{O}_{w_0}^\times) = \bigoplus_{\sigma \in G/G_{w_0}} \mathcal{O}_{\sigma w_0}^\times$ at each finite v. The isomorphism is G-equivariant by construction.

Applying Shapiro's lemma (**??** from Chapter 3) to this decomposition gives the fundamental reduction from global to local cohomology.

Theorem A.3.5: Cohomology of Idèles

For all $n \geq 0$:

$$H^n(G, \mathbb{I}_L) \cong \bigoplus_{v \in V_K} H^n(G_{w_0}, L_{w_0}^\times)$$

where w_0 is any place of L above v and $G_{w_0} = \mathrm{Gal}(L_{w_0}/K_v)$ is the decomposition group. The direct sum is finite: $H^n(G_{w_0}, L_{w_0}^\times) = 0$ for all unramified v with $n \geq 1$.

***Proof*:**
Step 1: Shapiro's lemma. By proposition A.3.4, $\mathbb{I}_L$ is a restricted product of induced modules. Shapiro's lemma (**??**) gives $H^n(G, \mathrm{Ind}_{G_{w_0}}^{G}(L_{w_0}^\times)) \cong H^n(G_{w_0}, L_{w_0}^\times)$ for each v. Since cohomology commutes with direct sums (and the restricted product is a filtered colimit of finite direct products, each handled by the additivity of cohomology):

$$H^n(G, \mathbb{I}_L) \cong \prod_{v}' H^n(G_{w_0}, L_{w_0}^\times)$$

where the restricted product is taken with respect to the subgroups $H^n(G_{w_0}, \mathcal{O}_{w_0}^\times)$.

Step 2: Vanishing at unramified places. For v unramified in L/K: the decomposition group $G_{w_0} = \mathrm{Gal}(L_{w_0}/K_v)$ is cyclic (generated by the local Frobenius), and L_{w_0}/K_v is an unramified extension of local fields. The cohomology of the unit group $\mathcal{O}_{w_0}^\times$ in an unramified extension vanishes in all positive degrees: $H^n(G_{w_0}, \mathcal{O}_{w_0}^\times) = 0$ for $n \geq 1$ (theorem 8.1.3(1) from Chapter 8). By the long exact sequence associated to $1 \to \mathcal{O}_{w_0}^\times \to L_{w_0}^\times \xrightarrow{v_{w_0}} \mathbb{Z} \to 0$: the cohomology $H^n(G_{w_0}, L_{w_0}^\times)$ for $n \geq 2$ is isomorphic to $H^n(G_{w_0}, \mathbb{Z})$. For $n \geq 2$ and G_{w_0} cyclic: $H^n(G_{w_0}, \mathbb{Z}) \cong H^{n-2}(G_{w_0}, \mathbb{Z})$ by the periodicity of cyclic group cohomology (theorem 4.3.4 from Chapter 4), and $H^0(G_{w_0}, \mathbb{Z}) = \mathbb{Z}$ with $\hat{H}^0(G_{w_0}, \mathbb{Z}) = \mathbb{Z}/[L_{w_0} : K_v]\mathbb{Z}$. In the Tate cohomology, $\hat{H}^n(G_{w_0}, L_{w_0}^\times) = 0$ for all n at unramified places (this is the content of the local computation: Hilbert 90 gives $H^1 = 0$ (theorem 8.1.1), the invariant map gives $|H^2| = [L_{w_0} : K_v]$ (theorem 8.1.5), and the valuation exact sequence accounts for all of it). In particular, $H^n(G_{w_0}, L_{w_0}^\times) = 0$ for $n \geq 1$ at unramified places.

Step 3: Finiteness of the sum. Since only finitely many places of K ramify in L/K (see [4, Counting Ramified Primes]), the vanishing of Step 2 reduces the restricted product to a finite direct sum over the ramified places and the archimedean places. So:

$$H^n(G, \mathbb{I}_L) \cong \bigoplus_{v \in V_K} H^n(G_{w_0}, L_{w_0}^\times)$$

as a genuine (finite) direct sum.

The decomposition $H^n(G, \mathbb{I}_L) \cong \bigoplus_v H^n(G_{w_0}, L_{w_0}^\times)$ is the mechanism by which the local duality theorems of Chapter 8 enter the global picture. Each summand on the right is a local cohomology group computed in **??** and section 8.2, and the global theory of Chapter 9 will analyze the long exact sequence arising from the fundamental short exact sequence of G-modules.

Proposition A.3.6: Fundamental Exact Sequence

The short exact sequence of G-modules:

$$1 \to L^\times \to \mathbb{I}_L \to C_L \to 1$$

induces a long exact sequence in cohomology:

$$\cdots \to H^n(G, L^\times) \to H^n(G, \mathbb{I}_L) \to H^n(G, C_L) \xrightarrow{\delta^n} H^{n+1}(G, L^\times) \to \cdots$$

The terms $H^n(G, L^\times)$ are controlled by Hilbert 90 ($H^1 = 0$, theorem 8.1.1) and the Brauer group ($H^2 \cong \mathrm{Br}(L/K)$, theorem 7.2.4). The terms $H^n(G, \mathbb{I}_L)$ decompose into local cohomology by theorem A.3.5. The terms $H^n(G, C_L)$ are the object of study in Chapter 9.

***Proof*:**
The short exact sequence $1 \to L^\times \to \mathbb{I}_L \to C_L \to 1$ is exact: the diagonal embedding $L^\times \hookrightarrow \mathbb{I}_L$ has discrete image (theorem A.2.4), and the idèle class group is the quotient by definition (definition A.2.5). All three terms are discrete G-modules (the action factors through the finite group $G = \mathrm{Gal}(L/K)$), so the long exact sequence in group cohomology applies (see [9, Long Exact Sequence From a Right-Derived Functor]).

The beginning of this long exact sequence gives the information that Chapter 9 will exploit. At $n = 0$: the sequence $1 \to K^\times \to \mathbb{I}_K \to C_K \to H^1(G, L^\times)$ and Hilbert 90 ($H^1(G, L^\times) = 0$) give the exactness of $1 \to K^\times \to \mathbb{I}_K \to C_K \to 0$, recovering the definition of C_K. At $n = 1$: the connecting map $\delta^0 : C_K/N_{L/K}(C_L) \to H^1(G, L^\times) = 0$ is trivial, and the sequence continues:

$$0 \to H^1(G, C_L) \to H^2(G, L^\times) \to H^2(G, \mathbb{I}_L)$$

The map $H^2(G, L^\times) \to H^2(G, \mathbb{I}_L) \cong \bigoplus_v H^2(G_{w_0}, L^\times_{w_0})$ sends the global Brauer group $\mathrm{Br}(L/K)$ to the direct sum of local Brauer groups $\bigoplus_v \mathrm{Br}(L_{w_0}/K_v)$, via the sum of local invariant maps. The kernel of this map is exactly $H^1(G, C_L)$, which measures the failure of the local-to-global principle for the Brauer group. This is the starting point of the Poitou–Tate sequence in Chapter 9.

Example A.3.7: The Fundamental Sequence for Cyclic Extensions

Let L/K be a cyclic extension of number fields with $G = \langle\sigma\rangle \cong C_n$. The Tate cohomology of $\mathbb{I}_L$ decomposes as:

$$\hat{H}^0(G, \mathbb{I}_L) \cong \bigoplus_v \hat{H}^0(G_{w_0}, L^\times_{w_0}) = \bigoplus_v K_v^\times / N_{L_{w_0}/K_v}(L^\times_{w_0})$$

At unramified finite places, the local norm is surjective on units and $N(\pi_{w_0}) = \pi_v^{f_v}$ (where f_v is the residue degree), so the local norm index $[K_v^\times : N(L^\times_{w_0})] = [L_{w_0} : K_v]$ is accounted for by the valuation. At split places ($f_v = 1$, $G_{w_0} = 1$): the local norm is an isomorphism, contributing a trivial factor. The nontrivial contributions come from the ramified and archimedean places.

The Herbrand quotient of C_L can be computed from this decomposition. Since C_L^0 is compact (theorem A.2.6) and G is finite, the Herbrand quotient $h(G, C_L)$ is well defined. The multiplicativity of the Herbrand quotient (**??** from Chapter 4) applied to the fundamental exact sequence gives:

$$h(G, C_L) = \frac{h(G, \mathbb{I}_L)}{h(G, L^\times)}$$

Each factor is computable: $h(G, \mathbb{I}_L) = \prod_v h(G_{w_0}, L^\times_{w_0})$ (the product over all places, with the unramified places contributing 1), and $h(G, L^\times)$ is determined by the Herbrand quotient of the unit group and the class number (see **??**).

Exercise A.3.1

1. Let $\alpha \in \mathbb{I}_L$ and $\sigma \in G$. Show that the abstract norm $N_G(\alpha) = \prod_{\sigma \in G} \sigma \cdot \alpha$ (computed in the G-module $\mathbb{I}_L$) agrees with the idèle norm map $N_{L/K}(\alpha) \in \mathbb{I}_K$ defined in definition A.3.3. (*Hint:* compare the v-components for a fixed place v of K, using the orbit-stabilizer decomposition $G = \bigsqcup_{\sigma \in G/G_{w_0}} \sigma G_{w_0}$.)
2. Verify $\mathbb{A}_L^G = \mathbb{A}_K$ directly for $L = \mathbb{Q}(i)$, $K = \mathbb{Q}$, with $G = \{1, \sigma\}$ where σ is complex conjugation. Identify the places of L above each place of K and describe the Galois action explicitly.
3. Show that $H^1(G, \mathbb{I}_L) = 0$ by combining the Shapiro decomposition (theorem A.3.5) with Hilbert 90 applied locally (theorem 8.1.1).
4. Let L/K be a cyclic extension of degree n with $G = \langle\sigma\rangle$. Using the surjection $\hat{H}^0(G, \mathbb{I}_L) \twoheadrightarrow \hat{H}^0(G, C_L)$ (from the Tate cohomology long exact sequence and Hilbert 90), show that $[C_K : N_{L/K}(C_L)] \le [L : K]$. Verify this bound explicitly for $L = \mathbb{Q}(i)$, $K = \mathbb{Q}$ by computing

the local norm indices $[K_v^\times : N(L_w^\times)]$ at each place v and identifying the image of $K^\times$ in $\bigoplus_v K_v^\times/N(L_w^\times)$.

5. Show that the norm map on idèle class groups $N_{L/K}\colon C_L \to C_K$ is well defined (i.e., $N_{L/K}(L^\times) \subseteq K^\times$, so the map descends to the quotient), and that $|N_{L/K}(\bar{\alpha})|_K = |\bar{\alpha}|_L$ for the induced norm on C_L.

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Bibliography

[1] Nathanael Chwojko-Srawley. *Everything You Need To Know About Abelian Varieties*. 1st ed. 2026. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_abelian_varieties.pdf.

[2] Nathanael Chwojko-Srawley. *Everything You Need to Know about Algebraic Geometry*. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_algGeo.pdf.

[3] Nathanael Chwojko-Srawley. *Everything You Need To Know About Algebraic K-Theory*. 1st ed.

[4] Nathanael Chwojko-Srawley. *Everything You Need To Know About Algebraic Number THeory*. Vol. 1. URL: https://nathanaelsrawley.com/pdf-viewer?src=.%2F.%2Fassets%2Fpdfs%2Fbooks%2FEYNTKA-algebraic-number-theory.pdf&name=Algebraic%20Number%20Theory&source=books.

[5] Nathanael Chwojko-Srawley. *Everything You Need To Know About Commutative Algebra*.

[6] Nathanael Chwojko-Srawley. *Everything You Need To Know About Core Algebra*. 1st ed. 2025. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_algebra.pdf.

[7] Nathanael Chwojko-Srawley. *Everything You Need To Know About Core Representation Theory*. 1st ed. 2025. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_core_representation_theory.pdf.

[8] Nathanael Chwojko-Srawley. *Everything You Need To Know About Elliptic Curves*. 1st ed. 2025. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_elliptic_curves.pdf.

[9] Nathanael Chwojko-Srawley. *Everything You Need To Know About Homological Algebra*. 1st ed. 2025. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_homological_algebra.pdf.

[10] Nathanael Chwojko-Srawley. *Everything You Need To Know About Local Field Theory*. URL: https://nathanaelsrawley.com/pdf-viewer?src=.%2F.%2Fassets%2Fpdfs%2Fbooks%2FEYNTKA-local-field-theory.pdf&name=Local%20Field%20Theory&source=books.

[11] Nathanael Chwojko-Srawley. *Everything You Need To Know About Topology*.

[12] Pierre Deligne. “La conjecture de Weil. I”. In: *Publications Mathématiques de l’IHÉS* 43 (1974), pp. 273–307.

[13] Jean Giraud. *Cohomologie non abélienne*. Vol. 179. Grundlehren der mathematischen Wissenschaften. Berlin–Heidelberg–New York: Springer-Verlag, 1971.

[14] John P. Labute. "Classification of Demushkin groups". In: *Canadian Journal of Mathematics* 19 (1967), pp. 106–132.

[15] James S. Milne. *Étale Cohomology*. Vol. 33. Princeton Mathematical Series. Princeton University Press, 1980.

[16] Nathanael Chwojko-Srawley. *Everything You Need To Know About Algebraic Topology*. 1st ed. 126 pp. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_algebraic_topology.pdf.

[17] Nathanael Chwojko-Srawley. *Everything You Need To Know About Class Field Theory*. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_class_field_theory.pdf.

[18] Jürgen Neukirch, Alexander Schmidt, and Kay Wingberg. *Cohomology of Number Fields*. 2nd ed. Vol. 323. Grundlehren der mathematischen Wissenschaften. Springer, 2008.

[19] Nikolay Nikolov and Dan Segal. "On finitely generated profinite groups, I: strong completeness and uniform bounds". In: *Annals of Mathematics* 165.1 (2007), pp. 171–238.

[20] Luis Ribes and Pavel Zalesskii. *Profinite Groups*. 2nd ed. Ergebnisse der Mathematik und ihrer Grenzgebiete. Springer, 2010.

[21] Jean-Pierre Serre. *Galois Cohomology*. Springer Monographs in Mathematics. Translated from the French by Patrick Ion. Springer, 1997.